Workload Mobility

Use **VCF Operations HCX** as a VI administrator to migrate workloads efficiently and securely within and across data centers.

VCF Operations HCX provides a comprehensive solution for migrating and managing workloads within and across VCF private clouds. With VCF Operations HCX, organizations can extend, migrate, and rebalance workloads within and across VCF instances without re-architecting applications or modifying network configurations. VCF Operations HCX simplifies and automates these operations at scale, ensuring secure, high-performance connectivity that supports hybrid cloud strategies and data center modernization initiatives.

VCF Operations HCX has the following capabilities:

- Seamless workload migration. Migrate VMs and applications in waves live or in bulk with minimal or zero downtime.
- Network extension. Extend networks while preserving IP addresses and topologies to maintain application connectivity.
- Inter- and intra-VCF mobility. Enable workload mobility within and across multiple VCF instances, including site-to-site and region-to-region migrations.
- Simplified integration. Automate the connection between VCF environments to reduce complexity and manual effort.
- Application placement optimization. Enables high-performance application mobility across different environments, making it easier to optimize workload placement based on business requirements.

In This Chapter

- Getting Started with VCF Operations HCX
- Administering VCF Operations HCX

Getting to Know VCF

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes.

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

Getting Started with VCF Operations HCX

About Getting Started

This information describes how to plan for the installation and operation of VMware Cloud Foundation Operations HCX services in a vSphere data center.

Information includes key concepts necessary for planning a deployment in your environment. The information provides a series of checklists which help you gather the required deployment information by identifying the compute and network details used during the VCF Operations HCX installation and configuration. For actual installation and operational details, see About Administering VMware Cloud Foundation Operations HCX.

Intended Audience

This information is for anyone who wants to deploy VCF Operations HCX. The information is for virtual infrastructure administrators and architects and who are familiar with virtual machine technology and data center operations.

Introduction

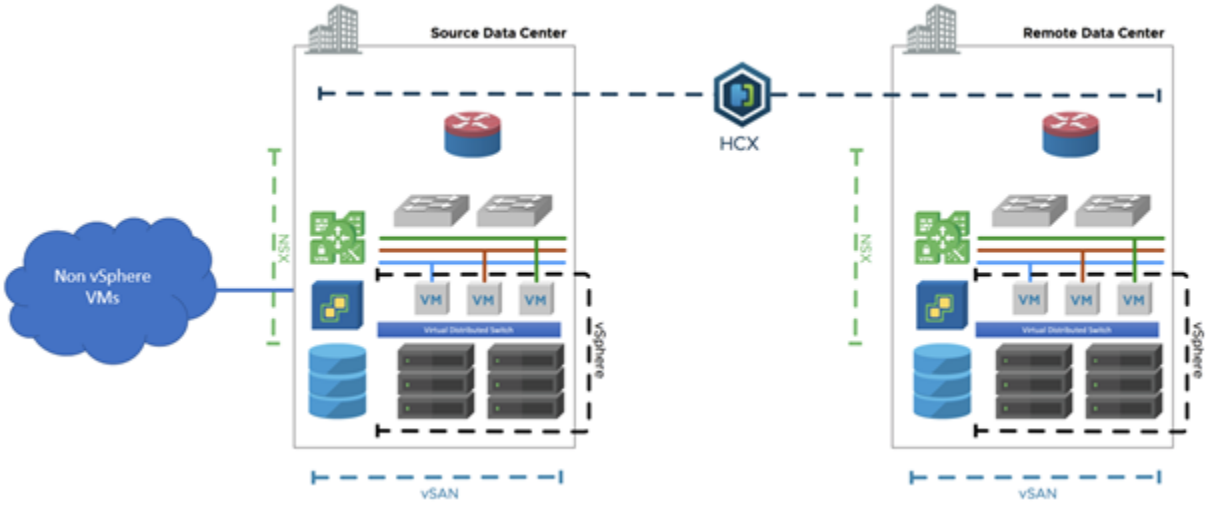
Deploying VCF Operations HCX requires information about your vSphere sites, networks, and configurations. Collecting the required configuration details and making some design choices in advance can help speed deployment time and

manage resources. This document explores deployment concepts, considerations, and practices, and it provides installation checklists to help with configuration planning.

Note:

This document supplements the information found in the About Administering VMware Cloud Foundation Operations HCX. Operational procedures for VCF Operations HCX are not included in this document.

Overview



VCF Operations HCX provides services between two or more distinct environments. The environments include all supported vSphere environments. With the OS Assisted Migration (OSAM) feature, VCF Operations HCX provides services to migrate non-vSphere workloads.

VCF Operations HCX Installations

In VCF Operations HCX, there is a notion of a source and a destination environment. You use the same installation files to install both environments.

The table highlights the differences between the two environments:

	Source site	Destination site
When to use	This is the vCenter Server containing the virtual machines that will be migrated. This site is always the source of site pairing requests, network extensions, and virtual machine migrations.	This is the vCenter Server that is the target of site pairing requests, network extensions, and virtual machine migrations. The destination site can also serve as the source of a site pair in bi-directional installations. **Note:** For single-site OS Assisted Migration, the OSAM service must be deployed at the destination site.
Installer	You download the OVA from the Broadcom support site. From the support site, you select the latest version of the VCF Operations HCX OVA. For hosted VCF deployments, contact your Service Provider for assistance. See About OVAs. **Note:** If you have installed VCF 9.1 and later, see Import Existing VCF Operations HCX to a VCF Domain.	You download the OVA from the Broadcom support site. From the support site, you select the latest version of the VCF Operations HCX OVA. For hosted VCF deployments, contact your Service Provider for assistance. See About OVAs. **Note:** If you have installed VCF 9.1 and later, see Import Existing VCF Operations HCX to a VCF Domain.
Supports vSphere versions within Technical Guidance	Yes	Yes
Requires Generally Available version of vSphere	No	Yes
Requires NSX	No	Yes
Endpoint for Site Pairing	Yes	Yes
Can initiate Site Pairing	Yes	Yes
Licensing/Activation	For detailed information, see VCF Operations HCX Licensing.	For detailed information, see VCF Operations HCX Licensing.

More About Destination Site Environments

The destination environment is the target for HCX appliance site pairing, network extensions, and workload migration. The destination site is always an environment running vSphere and NSX. The HCX appliance destination environment has these characteristics:

- The destination environment is the target for site pairing, network extension, and virtual machine migrations.
- The destination is always deployed using the VCF Operations HCX OVA.
- The destination environment requires the latest vSphere version. See Software Version Requirements.
- The destination environment requires the latest version of NSX that meets at minimum all the requirements in Requirements for VCF Operations HCX Deployments. Additional requirements in Requirements for Network Extension might apply.
- When the destination is a hosted VCF cloud, the provider may install and configure the OVA on behalf of the tenant.
- When the destination is on-premises or private cloud:
  - The destination meets all the environment and licensing requirements for HCX.
    For detailed information, see VCF Operations HCX Licensing.
  - The user installs and configures VCF Operations HCX.

More About Source Site Environments

Source sites have the following considerations:

- HCX Interconnect (HCX-IX) and HCX Network Extension (HCX-NET-EXT) appliances are always the tunnel initiators when a service mesh is created.
- Supports evacuations from older environments within VMware Technical Guidance. See KB 82702.
- Installation does not require NSX to be present, except when extending NSX networks.
- Supports interoperability with all supported environments for migration or evacuation.

Use Cases and POC Success Criteria

Understanding a few key concepts can help you to get started and be successful deploying HCX.

Key Concepts	Examples
Plan the success criteria for the VCF Operations HCX proof of concept in your environment.	Define the success criteria. For example: • Extend 2 test networks. • Live migrate virtual machines. • Test L2 connectivity using over the extended network. • Reverse migrate a virtual machine. • Bulk migrate a virtual machine.
Understand technology-specific restrictions.	For any VCF Operations HCX technologies that are used, have awareness of possible restrictions and requirements. For example, if a zero downtime application must be migrated, use HCX vMotion or HCX Replication Assisted vMotion (RAV). This example has the following restriction: vMotion based migrations require Virtual Machine Hardware Version 9 or later. For a list of restrictions specific to each migration type, see About Administering VMware Cloud Foundation Operations HCX.

Preparing for General Installation

This information is for both the source and destination environments.

This section has the following assumptions:

- The source vSphere contains the existing workloads and networks for migration. This environment can be any currently supported vSphere environment.
- The destination is a currently supported vSphere environment.
- Deployment variations like multiple vCenter Servers, multi-cloud, OS-assisted, or performance-centric implementations are outside the scope of this checklist.
- For checklist items specific to using OS Assisted Migration (OSAM), see the installation checklist for Preparing for Installation with OS Assisted Migration.

Collect vSphere Environment Details

Use the following table as a checklist for capturing the relevant vSphere environment details at both the source and the destination sites.

Environment Detail	Source Environment	Destination Environment
vSphere Version	• vSphere version must be within Technical Guidance.	• vSphere version is Generally Available.
Distributed Switches and Connected Clusters	• Understand the relationships between clusters and the Distributed Switches. • Imported Distributed Switches are not supported for the HCX-NET-EXT service.	• Understand the relationships between clusters and the NSX Transport Zone. VCF Operations HCX only deploys and extends networks to clusters included in the Transport Zone.

Environment Detail	Source Environment	Destination Environment
ESX Cluster Networks	• Identify the ESX Management, vMotion, and Replication (if it exists). VSphere Standard Switch (VSS), Port Group, or Distributed Virtual Port Group (DVPG) names, VLANs, and subnets. • If these networks vary from cluster to cluster, additional configuration is needed. • Identify available IPs (VCF Operations HCX participates in these networks)	• Identify the ESX Management, vMotion, and Replication (if it exists). VSphere Standard Switch (VSS), Port Group, or Distributed Virtual Port Group (DVPG) names, VLANs, and subnets. • If these networks vary from cluster to cluster, additional configuration is needed. • Identify available IPs (VCF Operations HCX participates in these networks)
NSX version and configurations	• NSX is not required at the source but is required for NSX Network Extension. See "NSX Requirements for Source Site Installation."	• The environment must be NSX. See Requirements for Deployments. • NSX T1 is required for Network Extension.
Review and ensure all software version requirements are satisfied.	See Software Version Requirements.	See Software Version Requirements.
vCenter URL	• https://<vcenter_fqdn>	• https://<vcenter_fqdn>
Use the default administrator account or an account with equivalent privileges	administrator@vsphere.local or equivalent account	administrator@vsphere.local or equivalent account
NSX Manager URL	• NSX is optional. It is only required when VCF Operations HCX is used to extend NSX networks.	• https://<nsxmgr-ip-or-fqdn>
NSX Enterprise Admin or equivalent account	• If VCF Operations HCX is used to extend NSX networks, know the administrator account for the NSX registration step.	• A full access Enterprise Admin user is required when registering the NSX Manager.
Destination vCenter Server SSO URL	• Use the vCenter SSO FQDN as seen in the vCenter Server Advanced Configurations (config.vpxd.sso.admin.uri)	• Use the vCenter SSO FQDN as seen in the vCenter Server Advanced Configurations (config.vpxd.sso.admin.uri).
DNS Server	• DNS is required.	• DNS is required.
NTP Server	• NTP is required.	• NTP is required.
HTTP Proxy Server	• If there is an HTTPS proxy server in the environment, add it to the configuration.	• If there is an HTTPS proxy server in the environment, it must be added to the configuration.

Planning for VCF Operations HCX Deployment

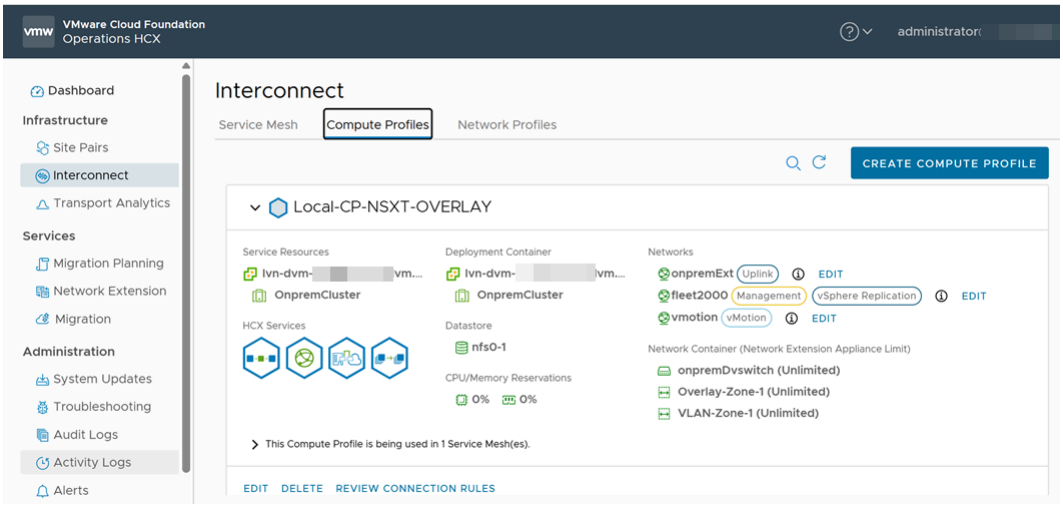
This section identifies information to know before deploying the source and the destination HCX appliances.

	Source Site	Destination Site
HCX appliance placement/zoning	The HCX appliance can be deployed along side other management components (like vCenter or NSX). It does not have to be deployed where the migration workloads reside.	The HCX appliance can be deployed along side other management components (like vCenter or NSX). It does not have to be deployed where the migration workloads reside.
VCF Operation HCX OVA	Download the OVA from `support.broadcom.com`. **Note:** If you have installed VCF 9.1 and later, see Import Existing VCF Operations HCX to a VCF Domain.	Download the OVA from `support.broadcom.com`. **Note:** If you have installed VCF 9.1 and later, see Import Existing VCF Operations HCX to a VCF Domain.

	Source Site	Destination Site
HCX appliance hostname	To create SSL certificates that contain FQDN, use the full FQDN reserved for the HCX Manager.	To create SSL certificates that contain FQDN, use the full FQDN reserved for the HCX Manager.
HCX appliance internal IP Address	The HCX appliance vNIC IP address, which is typically an internal address from the environment's management network.	The HCX appliance vNIC IP address, which is typically an internal address from the environment's management network.
HCX appliance External Name / Public IP Address	• The source HCX appliance initiates the management connection to the destination and does not need a dedicated public IP address. • The source HCX appliance supports outbound connections using Network Address Translation (Source NAT).	• Only required when the paired environments do not have a private connection and connect over the internet. • The external name record resolves to a public IP address. • The destination HCX appliance supports load balanced inbound connections or Network Address Translation (DNAT).
HCX appliance admin / root password		
Verify external access for the HCX appliance	The source HCX appliance makes outbound HTTPS connections to the site paired destination HCX appliance.	The destination HCX appliance receives HTTPS connections from the site paired source HCX appliance.
Proxy requirements	If a proxy server is configured, all HTTPS connections are sent to the proxy. An exclusion configuration is mandatory to allow the HCX appliance to connect to local systems. The exclusions can be entered as supernets and wildcard domain names. The configuration can encompass: • vSphere Management Subnets. • NSX Manager if present. • Internally addressed site pairing targets. • The RFC 1918 IP block, along with the internal domain name, can be used as the exception configuration. For example: `10.0.0.0/8, *.internal_domain.com`	Not applicable.
Configuration and Service limits	Review the VCF Operations HCXconfiguration and operational limits: VMware Configurations Maximum.	Review the VCF Operations HCXconfiguration and operational limits: VMware Configurations Maximum.

Planning the Compute Profile Configurations

A Compute Profile identifies the VCF Operations HCX services and the infrastructure scope when deploying a Service Mesh.



Review the table for a checklist of Compute Profile configuration items. For each row, the table describes how that information is used in the source and the destination Compute Profile.

	Source Compute Profile	Destination Compute Profile
Compute Profile Name	• Using meaningful names simplify operations in multiple Compute Profile deployments.	• Using meaningful names simplify operations in multiple Compute Profile deployments.
Services to activate	• The HCX appliance presents services as a catalog. You can use this catalog to restrict the activation of individual VCF Operations HCX services.	• The HCX appliance presents services as a catalog. You can use this catalog to restrict the activation of individual VCF Operations HCX appliance services.
Service Resources (Data Center or Cluster)	• Every cluster that contains virtual machines is used as a **Service Cluster** in the **Compute Profile**.	• Every cluster that is a valid target should be included as a **Service Cluster** in the **Compute Profile**.
Deployment Resources (Cluster or Resource Pool)	• The **Deployment Cluster** hosts HCX appliances. • It must be connected to VDS for L2 and can reach the service cluster networks for migration.	• The **Deployment Cluster** hosts HCX appliances. • It must be connected to the NSX Transport Zone for L2 and can reach the service cluster networks for migration.
Deployment Resources (Datastore)	• Select the datastore to use with HCX appliance Service Mesh deployments.	• Select the datastore to use with HCX appliance Service Mesh deployments.
Distributed Switches or NSX Transport Zone for Network Extension	• Select the virtual switches, transport zones, or VLANs that contain the virtual machine networks to extend. • The deployment cluster hosts must be connected to the selected switches.	• Select the transport zone or VLAN to use with Network Extension operations.

Planning the Network Profile Configurations

A Network Profile contains information about the underlying networks and allows networks and IP addresses to be pre-allocated before creating a Service Mesh. Review and understand the information in Network Profile Considerations and Concepts before creating Network Profiles for HCX.

Network Profile Type	Source Network Details	Destination Network Details
HCX Uplink	• It is typical for the Management and the Uplink Networks to use the same backing at the source. If a dedicated network is used, collect the following: • VLAN, Port Group • VSS|VDS|NSX Network Name • Gateway IP • Range of available IPs for the HCX appliance to use. For additional network configuration settings, see System Services.	• When connecting environments over Internet, assign the Public IPs networks as the HCX appliance Uplink. • VLAN, Port Group • VSS|VDS|NSX Network Name • Gateway IP • Range of available IPs for the HCX appliance to use.
HCX Management	• Typically, the HCX appliance participates in a designated management network. • VLAN, Port Group • VSS|VDS|NSX Network Name • Gateway IP • Range of available IPs for HCX appliance to use.	• Typically, the HCX appliance participates in a designated management network. • VLAN, Port Group • VSS|VDS|NSX Network Name • Gateway IP • Range of available IPs for HCX appliance to use.
HCX vMotion	• The vMotion network • VLAN, Port Group • VSS|VDS|NSX Network Name • Gateway IP • Range of available IPs for HCX appliance to use.	• The vMotion network • VLAN, Port Group • VSS|VDS|NSX Network Name • Gateway IP • Range of available IPs for HCX appliance to use.
HCX Replication	• The ESX Replication network. This network is the same as the Management network when a dedicated Replication network doesn't exist or when using vSphere Replication Network File Copy (NFC) (required). • VLAN, Port Group • VSS|VDS|NSX Network Name • Gateway IP • Range of available IPs for HCX appliance to use.	• The ESX Replication network. This network is the same as the Management network when a dedicated Replication network doesn't exist or when using vSphere Replication NFC (required). • VLAN, Port Group • VSS|VDS|NSX Network Name • Gateway IP • Range of available IPs for HCX appliance to use.

Isolating the Network Traffic of vSphere Replication in VCF Operations HCX Interconnect Appliance

Inbound Traffic (Source): The HCX Interconnect (IX) appliance supports a dedicated vSphere Replication interface, providing a direct, isolated path for replication data from source hosts.

Outbound Traffic (Destination): Traffic exits the IX appliance via the Management or VR interface to reach the target host's Management (default) or VR NFC VMkernel ports.

When migrating from vSphere 9.1, VCF Operations HCX uses Enhanced vSphere Replication mode. This eliminates the use of VR NFC for outbound traffic. The isolation is enabled using the vSphere Replication tag during Network Profile creation.

Site to Site Connectivity

Underlay requirements for migration operations	• VCF Operations HCX migration services include requirements for bandwidth, latency, and packet loss. See Network Underlay Minimum Requirements.
Public IPs & NAT	• VCF Operations HCX automatically enables strong encryption for site-to-site service mesh communications. It is typical for customers to begin migration projects over the Internet (while private circuits are not available, or will not become available). • VCF Operations HCX supports outbound SNAT at the source. The HCX appliance Uplink can be a private/internal IP address at the source environment. The SNAT of all VCF Operations HCX components to a single Public IP address. • Public IP addresses must be assigned directly in the Uplink Network Profile at the destination VCF Operations HCX configuration. • Inbound DNAT is not supported at the destination.
Source HCX appliance to Destination HCX appliance Network Ports	• The source HCX appliance connects to the destination HCX appliance using port TCP-443. • The source IX connects to the peer IX using port UDP-4500. • The source NE connects to the peer NE using port UDP-4500. • The source HCX appliances initiate the connections.
Other HCX appliance Network Ports	• For a full list of port requirements, VMware Ports and Protocols.

Preparing for Installation with OS Assisted Migration

OS Assisted Migration (OSAM) allows for the migration of guest (non-vSphere) virtual machines from on-premises data centers to private or hosted vSphere destinations.

With OSAM, you can migrate guest workloads that exist within vSphere SDDCs or non-vSphere environments. The OSAM service operates in one of two modes: multi-site and single-site. OSAM multi-site mode applies when migrating non-vSphere workloads from the source to destination sites using the vSphere based source deployment pipeline. Single-site mode applies when migrating workloads from a non-vSphere site to a vSphere site running VCF Operations HCX.

This section supplements the general installation checklist with specific information that is related to OSAM.

Single-Site Mode

To use OSAM in single-site mode, prepare VCF Operations HCX according to the general installation checklist. You can migrate non-vSphere virtual machines from other formats (for example, KVM or Hyper-V) to a vSphere environment with additional preparation.

For single-site mode, OSAM has two components: Sentinel Replication Gateway (SRG) appliance for connecting and receiving the guest workloads from the source environment, and Sentinel software installed on each guest virtual machine.

Multi-site Mode

To use OSAM in multi-site mode, prepare VCF Operations HCX according to the general installation checklist with additional preparation for non-vSphere (KVM or Hyper-V) virtual machine migration.

For multi-site mode, OSAM has three components: Sentinel Gateway (SGW) appliance for connecting and forwarding guest workloads in the source environment, Sentinel Data Receiver (SDR) for receiving guest workloads in the destination environment, and Sentinel software installed on each guest virtual machine.

Collect the non-vSphere Environment Details

This section identifies information about the KVM or Hyper-V environments that is relevant for HCX deployments with OSAM.

Environment Detail	Description
Hypervisor	OS Assisted Migration is available only for KVM or Hyper-V workloads.
Guest Network for OS Assisted Migration	This is the network on which guest virtual machines communicate with the HCX-SGW or HCX-SRG of OS Assisted Migration. This can be the same network used by another HCX appliance function, such as Management.
Guest virtual machines	Identify the set of KVM or Hyper-V virtual machines for migration. You must install Sentinel software on each virtual machine for communication, inventory, and replication processes. For a detailed list of supported guest virtual machine operating systems, see Understanding OS Assisted Migration.
KVM or Hyper-V VLAN for Network Extension (multi-site mode only)	For network extension with OSAM deployments, VLANs in the non-vSphere environment must be made available to the distributed switch used for HCX network extension.

Planning for OS Assisted Migration

This section identifies checklist information that applies to deploying HCX with OS Assisted Migration.

Checklist Item	Source HCX	Destination HCX
Checklist for General Installation	Review the source VCF Operations HCX appliance items provided in General Installation.	Review the VCF Operations HCX appliance items provided in General Installation.
Port requirements	Review the OS Assisted Migration requirements at the source site for firewall access: VMware Ports and Protocol. Filter by Purpose, "OSAM."	Review the OS Assisted Migration port requirements at the destination site for firewall access: VMware Ports and Protocol. Filter by Purpose, "OSAM."
Sentinel software	Install Sentinel software on each KVM or Hyper-V workload requiring migration. Sentinel software is only available for download from the **Sentinel** tab in the VCF Operations HCX appliance service mesh UI.	Sentinel software is not available for download from the HCX appliance.

Planning the Network Profile Configuration

The same items called out in Preparing for General Installation apply to the network profile configuration when using VCF Operations HCX with OS Assisted Migration. For OS Assisted Migration, the following additional items might apply:

Checklist Item	Source Compute Profile	Destination Compute Profile
For multi-site mode, create a Network Profile for Guest Network selection during Compute Profile configuration.	This is the network on which guest virtual machines with Sentinel software communicate with the HCX Sentinel Gateway.	Not applicable in the Destination Compute Profile.
For single-site mode, create a network profile for guest network selection during compute profile configuration	Not applicable. In single-site mode, the source of migrations is a non-vSphere site.	This is the network on which guest virtual machines with sentinel software communicate with the Sentinel Replication Gateway. **Note:** In single-site mode, the destination is a vSphere site running an HCX appliance.

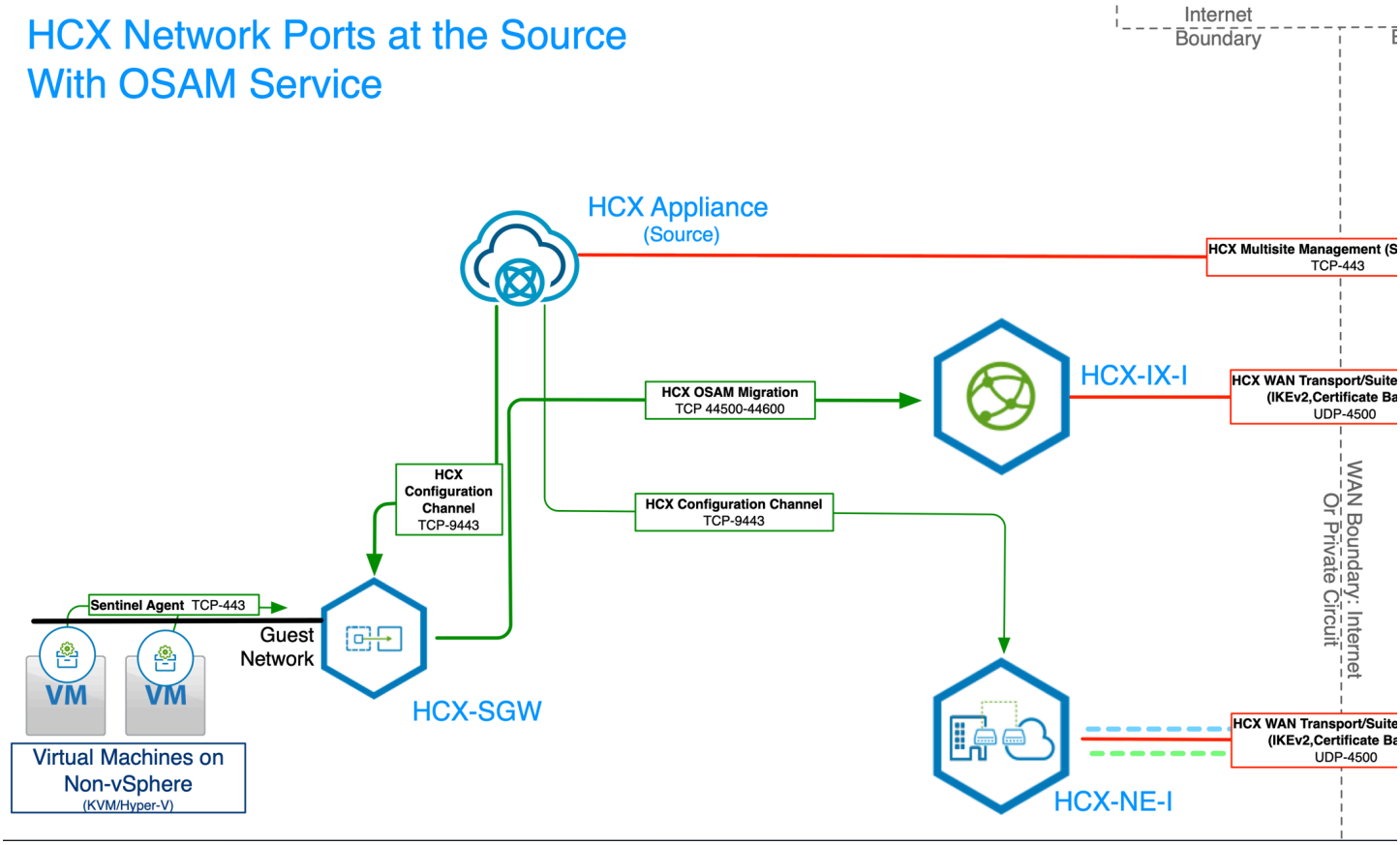
Planning the Compute Profile Configuration

The same items called out in Planning for General Installation apply to the Compute Profile configuration when using VCF Operations HCX with OS Assisted. For OS Assisted Migration, the following additional items might apply:

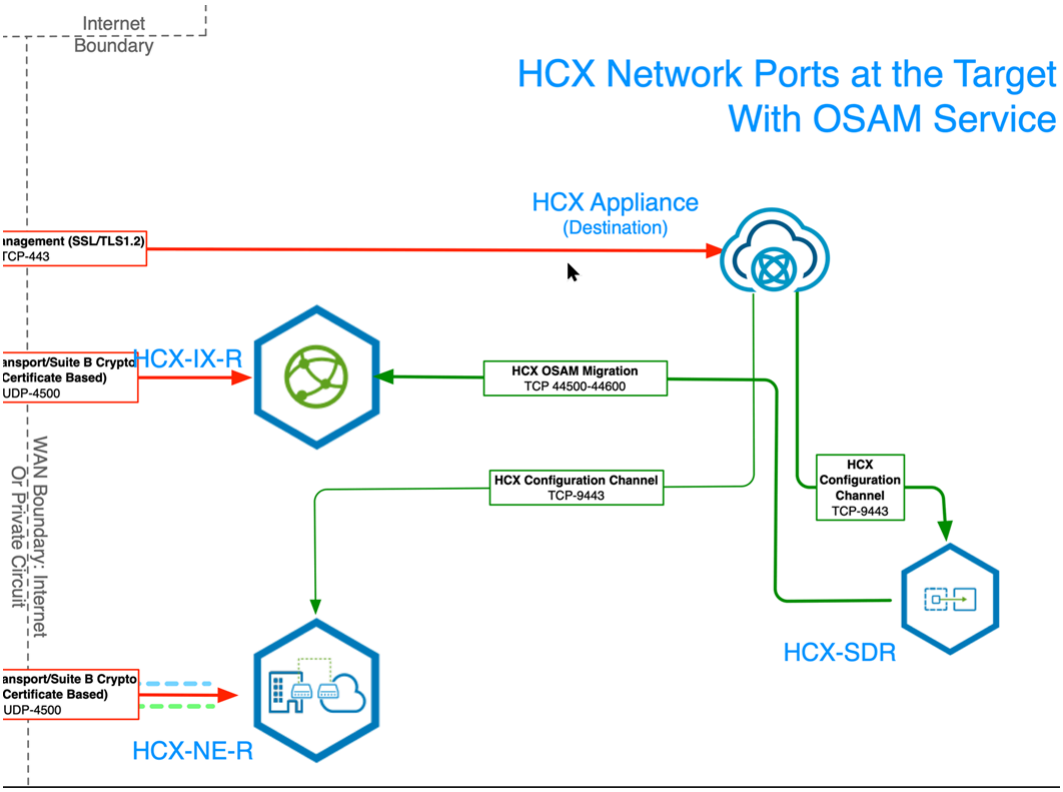
Checklist Item	Source Compute Profile	Destination Compute Profile
Network Extension Services (multi-site mode only)	When migrating KVM or Hyper-V virtual machines to a VMware environment which resides in a different data center geographically, you must activate network extension services in the source compute profile. When migrating KVM or Hyper-V virtual machines to a VMware environment where both the non-vSphere and the vSphere environment reside in the same data center, it might not be necessary to activate network extension.	When migrating KVM or Hyper-V virtual machines to a VMware environment which resides in a different data center geographically, you must activate network extension services in the destination compute profile. When migrating KVM or Hyper-V virtual machines to a VMware environment where both the non-vSphere and the vSphevSpherere environment reside in the same data center, it might not be necessary to activate network extension services.
Guest Network Profile (multi-site mode)	Use the guest network profile when setting up the compute profile. **Note:** If no guest network is set up in the network profile, this can be the same network used by another HCX function, such as Management.	Not applicable.
Guest Network Profile (single-site mode)	Not applicable. In single-site mode, the source of migrations is a non-vSphere sites.	In single-site mode, the destination site for OSAM migrations is a vSphere site running either HCX appliance. For this site, use the guest network profile when setting up the compute profile.

For a list of network ports for OSAM environments at the source and destination, see VMware Ports and Protocols.

Network Ports at the Source with OS Assisted Migration



Network Ports at the Destination with OS Assisted Migration



Example - Completed General Installation

This version of the checklist is prepared using a fictional migration scenario. The entries use the scenario information.

The planning tables in this document are organized assuming there is one source environment and one destination environment:

- It is assumed that the source vSphere contains the existing workloads and networks that are migrated. This environment is any supported version of vSphere. In the case of OSAM, this environment can be a supported non-vSphere format. See Software Version Requirements (Source Requirements).
- It is assumed that the destination is a supported vSphere environment that is the target for network extensions, migrations, and services. See Software Version Requirements (Destination Requirements).

Explanations are included in the regular pre-install checklists. This checklist omits them for brevity.

Scenario - XYZ Migration from a source site to a destination site

The XYZ Widget Company plans to evacuate the XYZ data center into a newly built XYZ SDDC (in a new physical data center). VCF Operations HCX enables the evacuation of all workloads and the decommissioning of EOL hardware and EOS software without upgrades.

The objective for the HCX POC is to test core VCF Operations HCX capabilities that enable the evacuation of the a data center. The proof of concept follows these success criteria:

- Deploy the HCX appliance service mesh.
- Extend the prepared test network (virtual machine VLAN 10 backed DPG).
- Successfully perform HCX vMotion and Bulk migration for a test virtual machine from the existing source data center to the destination SDDC.
  - Understand the time to migrate VM data for each protocol.
  - Understand the ability to use bandwidth for migrations under the POC configuration.
- Test Network Extension:
  - Verify source and destination VM connectivity over the HCX appliance L2 path.

  - Understand source and destination latency.
- Successfully perform reverse migrations from destination to original source.
- Successfully complete the Bulk migration of 3 to 5 VMs in parallel from source to destination data centers.
  - Test the Bulk migration failover scheduler.
  - Upgrade VM Hardware and VM Tools.

Scenario Environment Details

Fictional environment details for the XYZ source site and the XYZ destination SDDC.

Environment Facts	Source	Destination
vSphere	• vSphere 7.0 U1 • Mgmt Cluster • Dev Cluster • Prod Cluster • Existing VDS	• vSphere 9.0 • Mgmt Cluster • Compute-1 Cluster • Compute-2 Cluster • Mgmt VDS • Compute VDS
Cluster Networks	• ESX Management 192.168.100.0/24 VSS-VLAN-100 • ESX vMotion 192.168.101.0/24 VSS-VLAN-101	• ESX Management 10.0.100.0/22 • ESX vMotion 10.0.104.0/22 • ESX Replication 10.0.108.0/22
VM Networking	• Single existing DC VDS. • DVPG Test-VM-NET-10 192.168.10.0/24 VLAN 10. • 5 Test-VMs deployed for the POC.	• NSX Overlay TZ Configured • NSX 1 Router Created.
Storage	• Block Storage Central Array	• vSAN Storage
Site to Site Connectivity	• 1 Gbps Internet / WAN. • No dedicated Public IPs required for VCF Operations HCX (the HCX appliance uses NAT outbound).	• 10 Gbps Internet / WAN. • 3 Public IPs reserved for HCX.

Collect vSphere Environment Details

Collect the relevant environment details in preparation for the installation. The entries can provide context or requirements related to the Environment Detail entry.

XYZ Widget Company scenario information is in brackets.

Environment Detail	Source Environment	Destination Environment
vSphere Version	XYZ existing site is 7.0 U1	XYZ desination site is 9.0
Distributed Switches and Connected Clusters	[Shared VDS: Mgmt, Dev, Prod]	[Mgmt VDS: Mgmt Cluster Compute VDS: Compute-1, Compute-2]
ESX Cluster Networks	[ESX Management 192.168.100.0/24 VSS-VLAN-100 ESX vMotion 192.168.101.0/24 VSS-VLAN-101]	[ESX Management 10.0.100.0/22 ESX vMotion 10.0.104.0/22 ESX Replication 10.0.108.0/22]
NSX version and configurations	[No NSX in existing data center]	[XYZ SDDC is running VMware NSX, with an overlay Transport Zone that includes Compute-1 and Compute-2 clusters]
Verify all software version requirements are satisfied. See Software Version Requirements.	[Verified XYZ existing data center meets all documented version requirements]	[Verified XYZ SDDC meets all documented version requirements]
vCenter Server URL	[https://existing-vcenter]	[https://sddc-1-vcenter.xyz.com]

Environment Detail	Source Environment	Destination Environment
vCenter administrator@vsphere.local or equivalent account	[Verified administrator access to the vCenter Server]	[Verified administrator access to the vCenter Server]
Destination NSX Manager URL	[N/A]	[https://sddc-1-nsxm.xyz.com]
NSX admin or equivalent account	[N/A]	[Verified the NSX admin account]
Destination vCenter Server SSO URL	[embedded]	[https://sddc-1-vcenter.xyz.com]
(Optional) VCF SSO	• Identity Broker Issuer (example): https://<fqdn of VIDB>/acs/t/<tenant name> • Client ID and Client Secret (fetched from VCF Operations Console)	• VCF SSO (example): https://<fqdn of VIDB>/acs/t/<tenant name> • Client ID and Client Secret (fetched from VCF Operations Console)
DNS Server	[existing-dns.xyz.com]	[dns.xyz.com]
NTP Server	[existing-ntp.xyz.com]	[ntp.xyz.com]
HTTP Proxy Server	[proxy.xyz.com]	[Verified xyz does not use HTTP proxy servers]

Planning for the VCF Operations HCX Deployments

XYZ Widget Company Scenario information is in brackets.

	HCX Appliance Deployment at Source	HCX Appliance Deployment at Destination
HCX appliance Placement	[HCX appliance is deployed in the xyz-sddc1]	[HCX appliance is deployed in the XYZ-SDDC-1 Mgmt cluster]
HCX appliance OVA	[The OVA has been downloaded from `support.broadcom.com`]	[The OVA has been downloaded from `support.broadcom.com`.]
HCX appliance Hostname	[source-hcxm.xyz.com]	[sddc-1-hcxm.xyz.com]
HCX appliance Internal IP Address	[192.168.100.50]	[10.0.100.50]
HCX appliance External Name / Public IP Address	[External Name/Pub IP assignment is not applicable]	[sddc1-hcx.xyz.com, Pub IP assignment 192.0.2.50]
HCX appliance admin/root password		
VCF Operations HCX Licensing	[Fetched from paired destination HCX]	[Fetched VCF license from vCenter]

Planning the Compute Profile Configurations

XYZ Widget Company Scenario information is in brackets.

Note:

In the XYZ Widget Company POC scenario, a single Compute Profile is used.

In production deployments, one can create additional Compute Profiles to scale out the VCF Operations HCX services or to achieve connectivity when there are things like per-cluster vMotion or VDS isolation in the environment.

	Source Compute Profile	Destination Compute Profile
Compute Profile Name	[existing-DC-CP]	[sddc-1-CP]
Services to activate	[All services activated]	[All services activated]
Service Resources (Data Center or Cluster)	[existing-dev cluster]	[Compute-1, Compute-2]
Deployment Resources (Cluster or Resource Pool)	[existing-dev cluster]	[sddc-1-compute-1]
Deployment Resources (Datastore)	[existing-block-array]	[sddc-1-vsan-datastore]

	Source Compute Profile	Destination Compute Profile
Distributed Switches or NSX Transport Zone for Network Extension	[existing-shared-dvs]	[sddc-1-nsx-overlay-tz, includes compute clusters]

Planning the Network Profile Configurations

The Network Profiles abstract Network consumption during HCX service deployments. See Network Profile Considerations and Concepts.

XYZ Widget Company Scenario information is in brackets.

Network Profile Type	Source Network Details	Destination Network Details
Uplink	[Using Mgmt]	[xyz-sddc-ext-net 192.0.2.11 - 192.9.2.15]
Management	[existing-mgmt, 192.168.100.0/24, gw: 192.168.100.1 HCX appliance range: 192.168.100.201 - 192.168.100.205]	[xyz-sddc-mgmt, 10.0.100.0/22, gw: 10.0.100.1 10.0.100.201 - 10.0.100.205]
HCX vMotion	[existing-vmotion, 192.168.101.0/24, gw: 192.168.101.1 HCX appliance range: 192.168.101.201 - 192.168.101.205]	[xyz-sddc-vmo, 10.0.104.0/22, gw: 10.0.104.1 10.0.104.201 - 10.0.104.205]
Replication	[Using Mgmt]	[xyz-sddc-repl, 10.0.108.0/22, gw: 10.0.108.1 10.0.108.201 - 10.0.108.205]

Source HCX Appliance to Destination HCX Appliance IP Connectivity

XYZ Widget Company Scenario information is in brackets.

Bandwidth for Migrations	[XYZ existing DC has 1 Gbps Internet uplinks, 500 Mbps can be used for migrations. XYZ-SDDC has 10 Gbps available.]
Public IPs and NAT	[XYZ existing DC VCF Operations HCX components will SNAT. XYZ existing DC Public IP addresses are allocated as follows: One for the HCX appliance (it is configured as an inbound DNAT rule). Two for Uplink Network Profile (one for the IX appliance and one for the Network Extension appliance)]
Source HCX to Destination HCX Network Ports	[XYZ existing DC perimeter firewall has been configured to allow UDP-4500 outbound. XYZ SDDC perimeter firewall has been configured to allow UDP-4500 inbound.]
HCX Network Ports	A full list of port requirements for the HCX appliance is in VMware Ports and Protocols.

HCX Appliance Deployment Considerations

Several aspects of a successful HCX appliance deployment revolve around understanding the importance of the network profile and the compute profile.

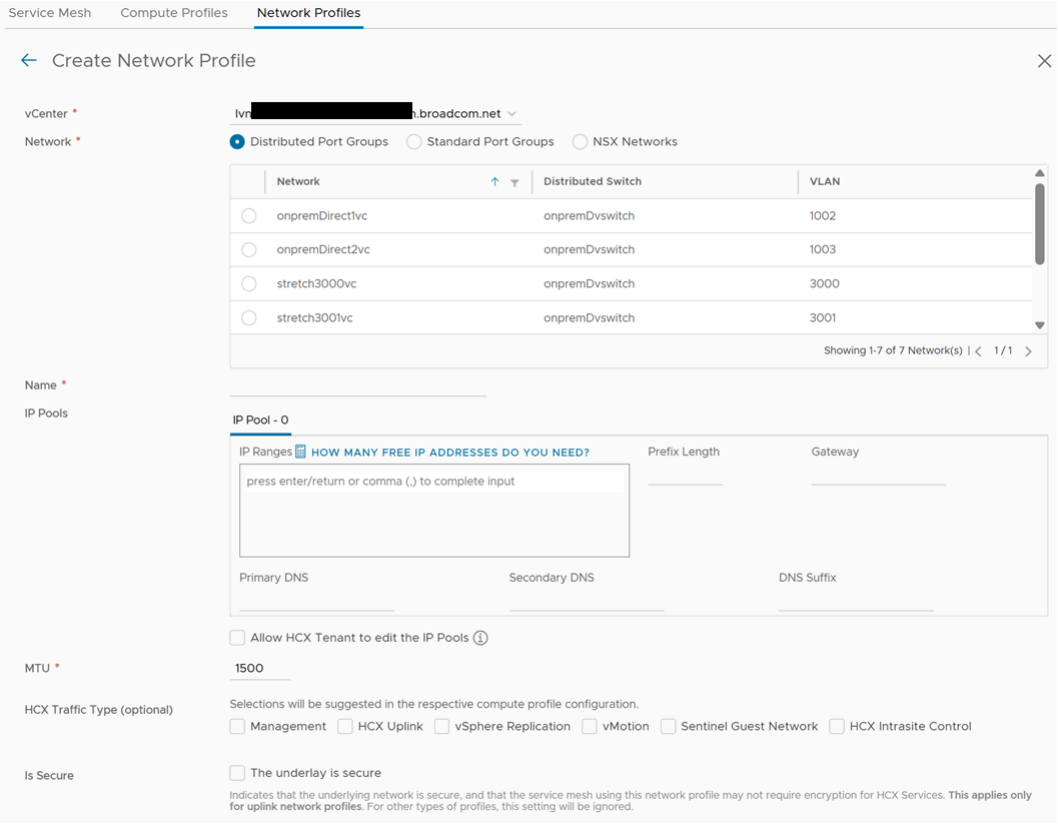
Network Profile Considerations and Concepts

Network profiles are a requirement for completing the compute profile. When a service mesh is created, the network profile configurations are used to connect the deployed HCX appliances.

Introduction to Network Profiles

Network profiles can be pre-created in the **Network Profile** tab or they can be created during the compute profile configuration. A network profile contains:

- One underlying vSphere Port Group (VSS or VDS) or NSX-based network.
- IP address information: gateway IP, the network prefix and MTU, and DNS.
- A pool of IP addresses reserved for the HCX appliance to use during service mesh deployments.



HCX Appliance Traffic Types

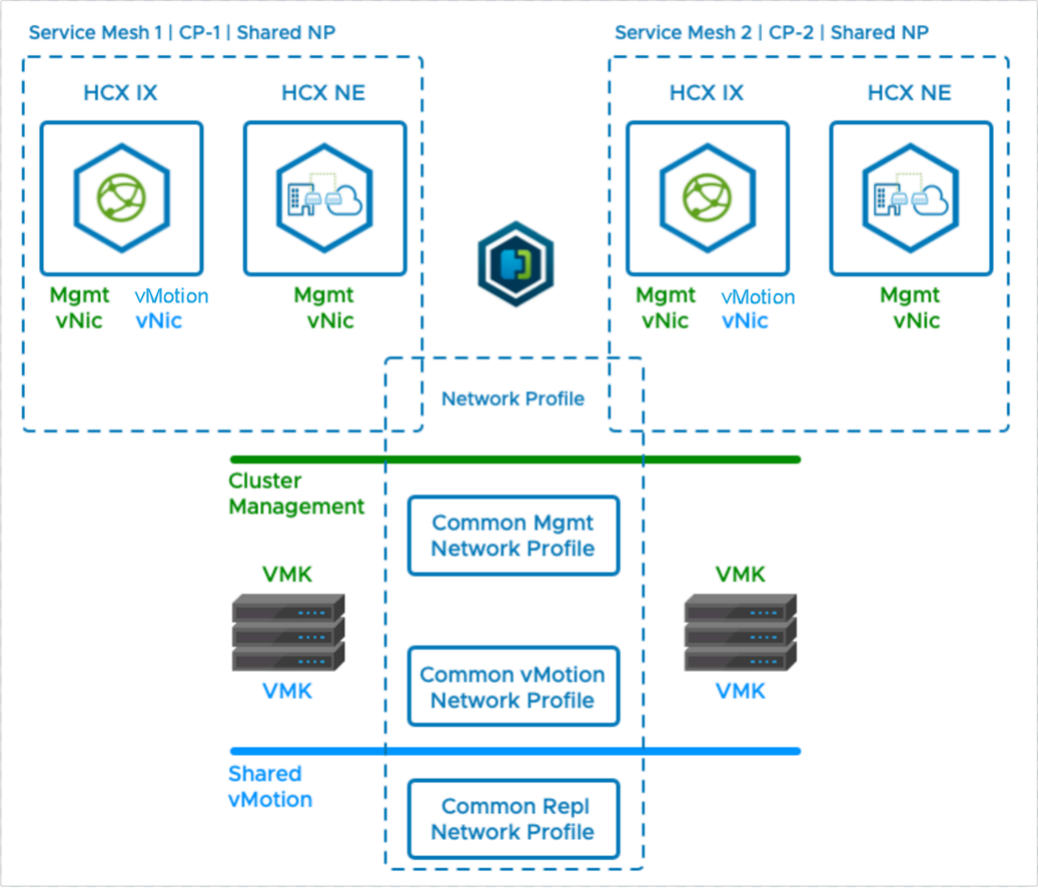
Consider the network profile traffic types are like a router's uplinks and downlinks. The HCX-IX (mobility) and the HCX-NET-EXT (extension) appliances have "uplinks" and "downlinks". The uplink is used to connect the IX or NE appliance to its remote peer. The downlink traffic types (Management, vMotion, Replication) connect the IX or NE appliance to the local environment.

Network Profile Types	Description
Uplink	Used by service mesh components to reach their peer appliances. **Important:** When destination HCX appliances need to be reachable over the Internet, use the Uplink network profile to assign the Public IP addresses. Destination NAT configurations are not supported. The source HCX appliances do not need Public IP addresses and can be configured using traditional SNAT.
Management	Used by service mesh components to connect to HCX appliance, vCenter, NTP, DNS.
HCX vMotion	Used by service mesh components to connect to the ESX cluster for vMotion-based services.

Network Profile Types	Description
Replication	Used by service mesh components to connect to the ESX cluster for Replication-based services. **Note:** This network port type is compatible with ESX vSphere Replication VMkernel traffic but cannot be used for vSphere Replication NFC VMkernel traffic.
Guest Network	In version 9.1.1, in OSAM deployments used by the service mesh, the sentinel agent opens a connection to the SGW appliance over the guest network.

Characteristics of Network Profiles

- Network profile configurations are only used during Service Mesh deployments (IP addresses assigned to the HCX-IX and HCX-NET-EXT , and sentinel appliances).
- The HCX appliance only uses a management interface, it does not use other network profile networks.
- A compute profile always includes one or more network profiles.
- When service mesh is deployed, every network profile that is included in the compute profile configuration is used.
- When a network profile network is used in a service mesh, each HCX appliance consumes a single IP address out of the configured IP pool.
- When a network profile is assigned to a specific HCX appliance traffic type (the traffic types are explained in the next section), a single IP address is used. For example, if the same network profile is assigned for management and Uplink, one IP address is used, not two.
- A network profile can be used with multiple compute profiles.



HCX Traffic Types and HCX Appliances

The table describes which network traffic types are used by the different HCX appliances.

Important:

One IP address is assigned per traffic type.

(For example, if all HCX-IX traffic types are configured to use a single network, a single vNIC with a single IP address is assigned. If a dedicated network is configured for each possible IX traffic type, then the HCX-IX uses four vNICs with an IP in each network. These wiring variations are described in the examples section, after the table.

HCX Appliance	Traffic Types Used
HCX-IX (Migrations)	• Uplink • Management • HCX vMotion • Replication
HCX-NET-EXT (Network Extension)	• Uplink • Management
HCX-SGW (Sentinel Gateway)	• Management • Guest Network
HCX-SDR (Sentinel Data Receiver)	• Management
HCX-SRG (Sentinel Replication Gateway)	• Management • Guest Network

Network Profile Configuration Examples

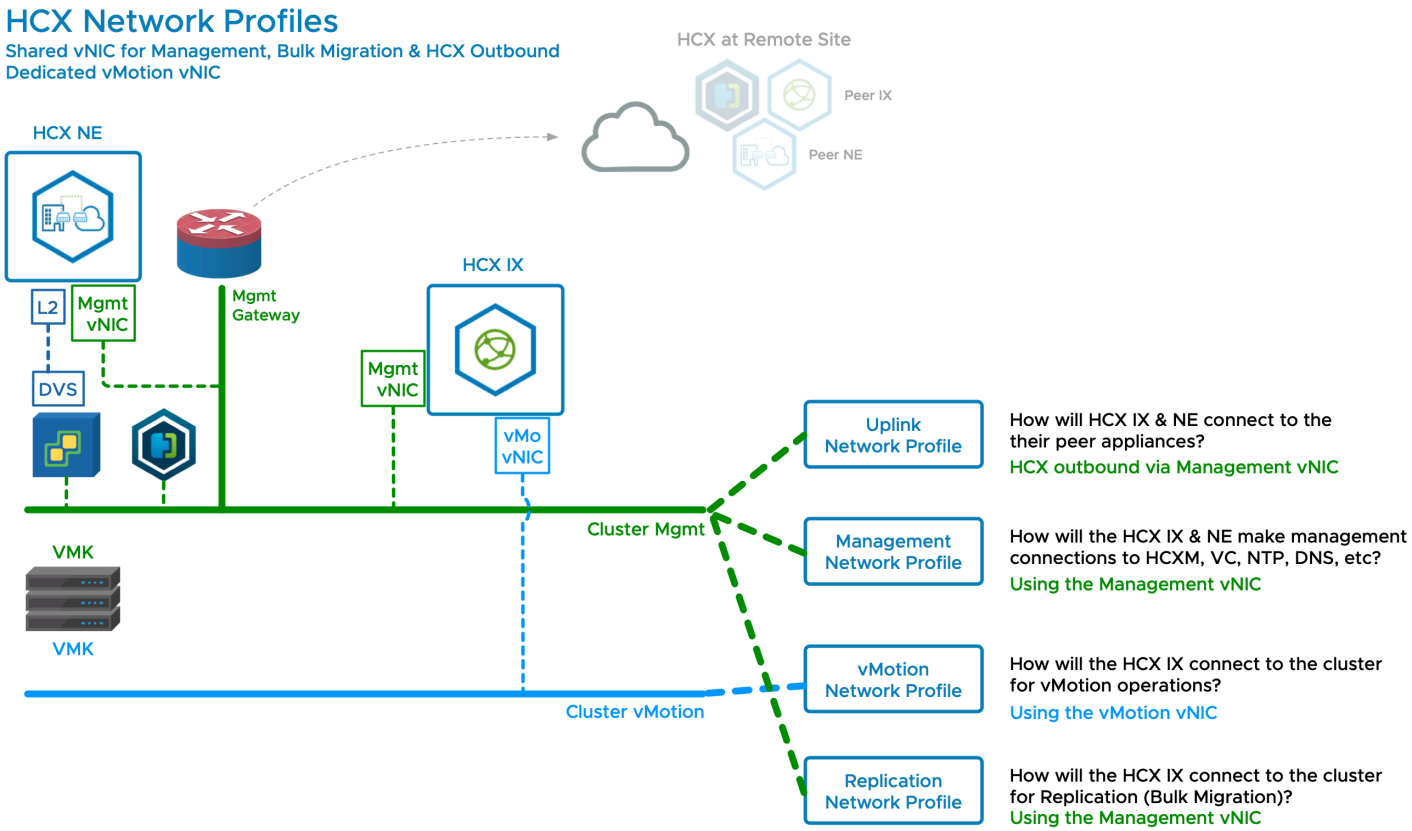
The following examples depict how the service mesh appliances might be wired up.

In fully private VCF Operations HCX deployments where the environments are inside of the same private network, it is typical for the source HCX appliance and the destination HCX appliance network profiles to be structured identically.

It is possible (and common) for the network profile configurations to differ at the source and destination when they are separated by the Internet. The reason for this is that the destination service mesh appliances must have an Uplink network profile with Public IP assignments (this requirement is not true at the source, where internal addresses can use Source NAT for Internet access).

HCX Network Configuration 1 - Shared Management, Replication and Uplink with Dedicated vMotion

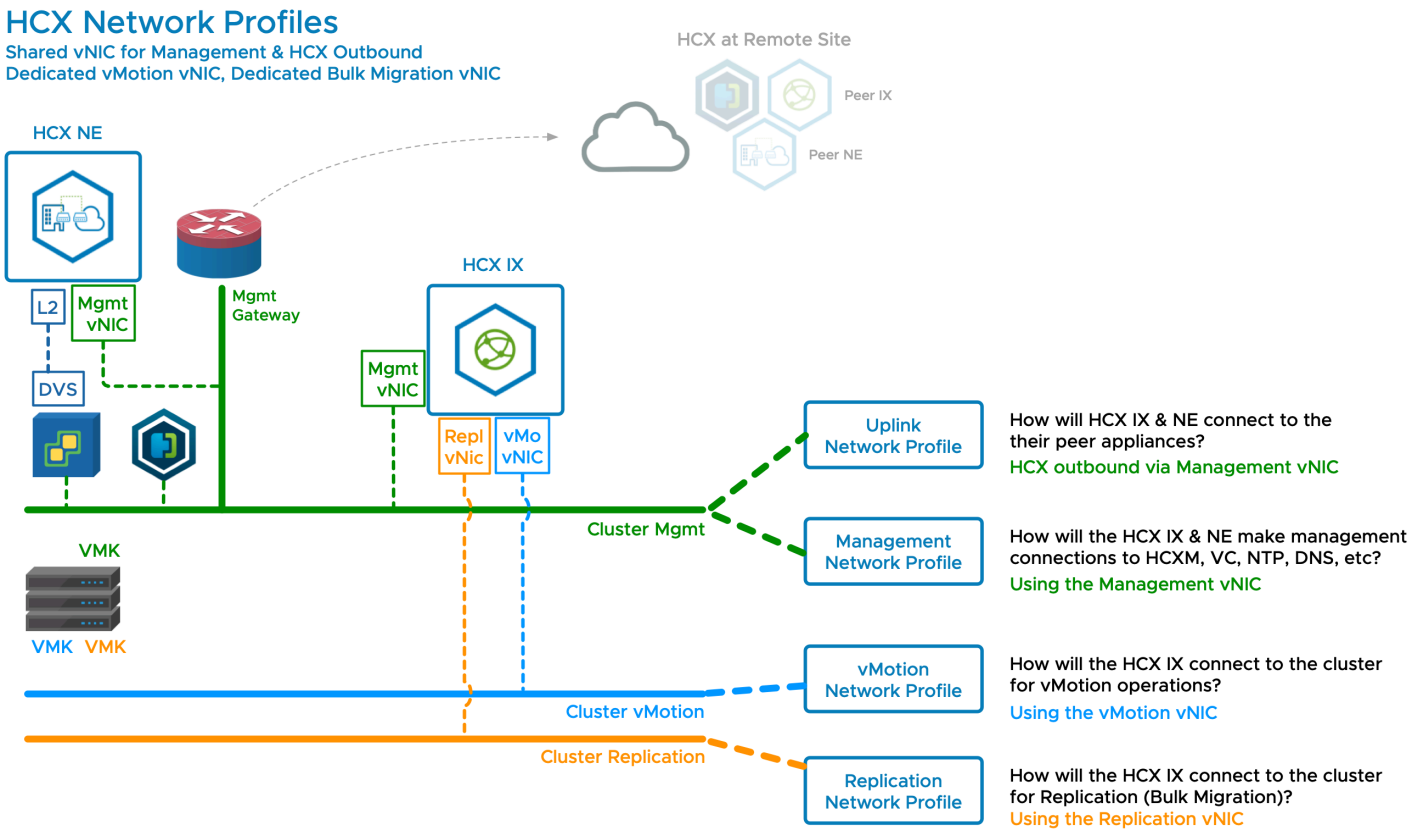
- This configuration trades the benefits gained from separation of traffic to simplify deployments. The same network is selected for Management, Uplink and Replication traffic.
- This configuration requires the management IP addresses assigned to HCX appliances at the destination to be fully reachable from the source HCX appliances without NAT translation. Because of this requirement, this configuration is most typical in HCX appliance deployments fully within a private network.



Network Configuration 2 - Dedicated Replication Network

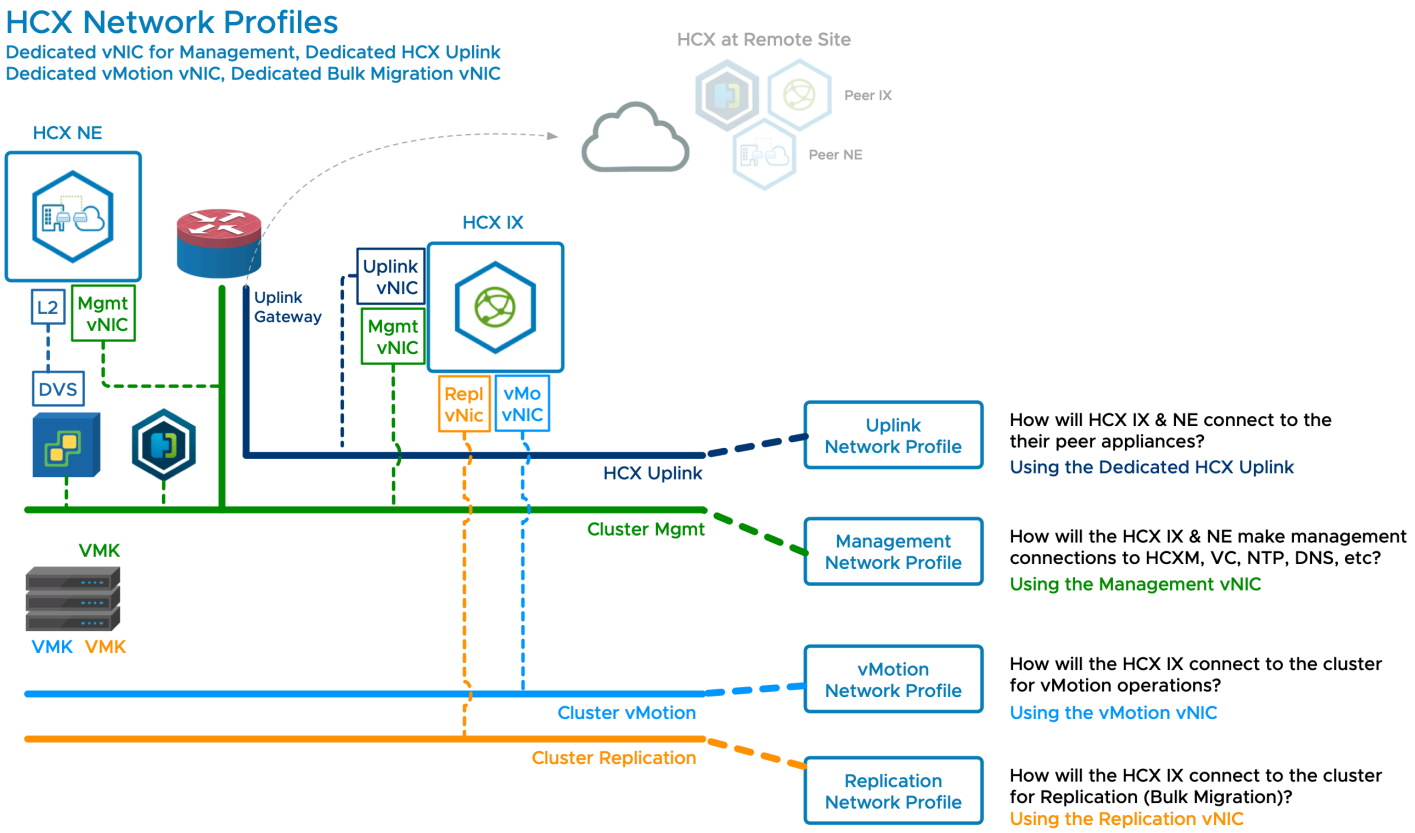
- Configuration 2 adds a dedicated network for the Replication traffic (Bulk Migration).
- This configuration variation is only possible when the cluster hosts use a dedicated Replication VMkernel network.
- Separating the replication traffic is a recommended practice. This configuration is used when a dedicated replication VMkernel interface is available.
- **Important:**

  In deployments where ESX hosts use a dedicated VMkernel configuration for vSphere Replication services, the HCX Interconnect uses a Network Profile configuration dedicated to the vSphere Replication traffic. This configuration does not include the vSphere Replication NFC traffic. The HCX appliance always uses its Management interface for vSphere Replication NFC traffic.



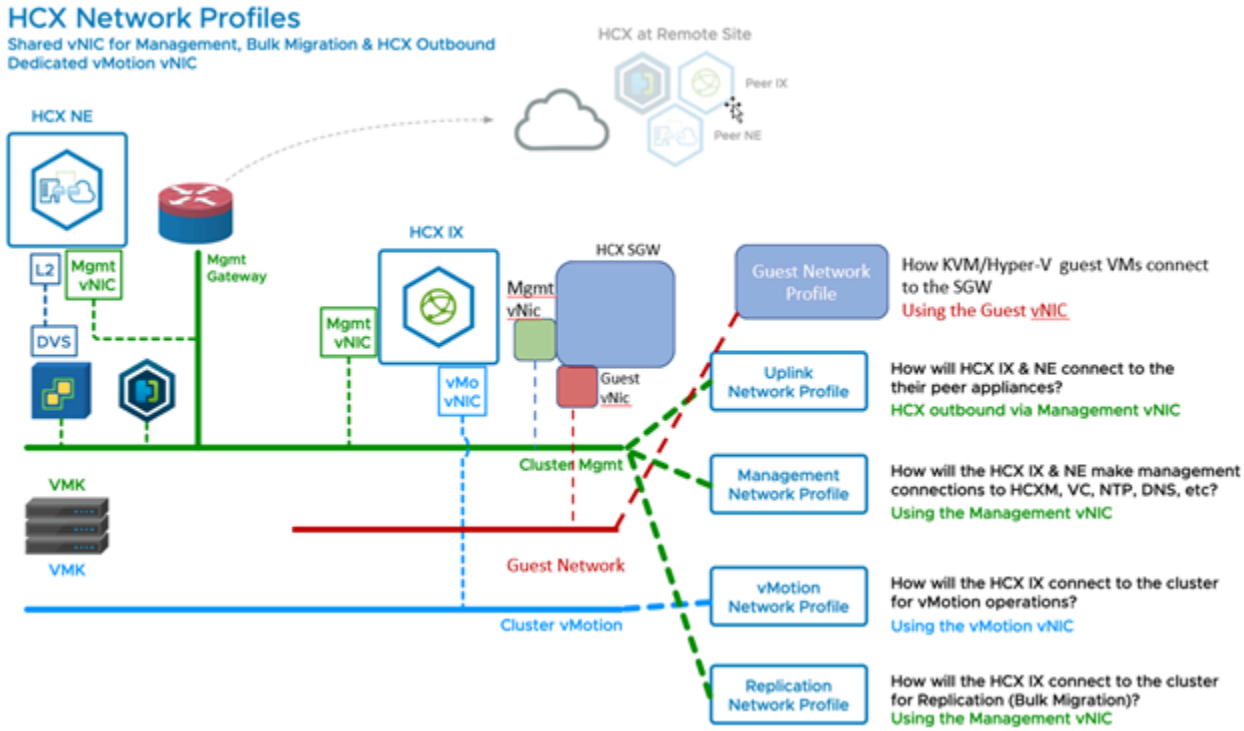
Network Configuration 3 - Dedicated Uplink Network

- Configuration 3 adds a dedicated network for Uplink traffic (Service Mesh Transport traffic).
- This configuration trades simplicity of deployment (see configuration 1) for the benefits of separating the uplink and the management traffic.
- A dedicated Uplink network is a good way to isolate the migration traffic for applying Quality of Service (QoS).
- A dedicated Uplink can be used to consume bandwidth/networks dedicated to the migration project.
- For deployments over the Internet:
  - Public IP addresses are assigned at the destination using the Uplink network profile.
  - The source HCX appliances can use traditional Internet SNAT to securely connect to the destination public IP addresses using strong encryption.
  - Providers leverage this configuration to make VCF Operations HCX services easy to deploy before dedicated private circuits become available.



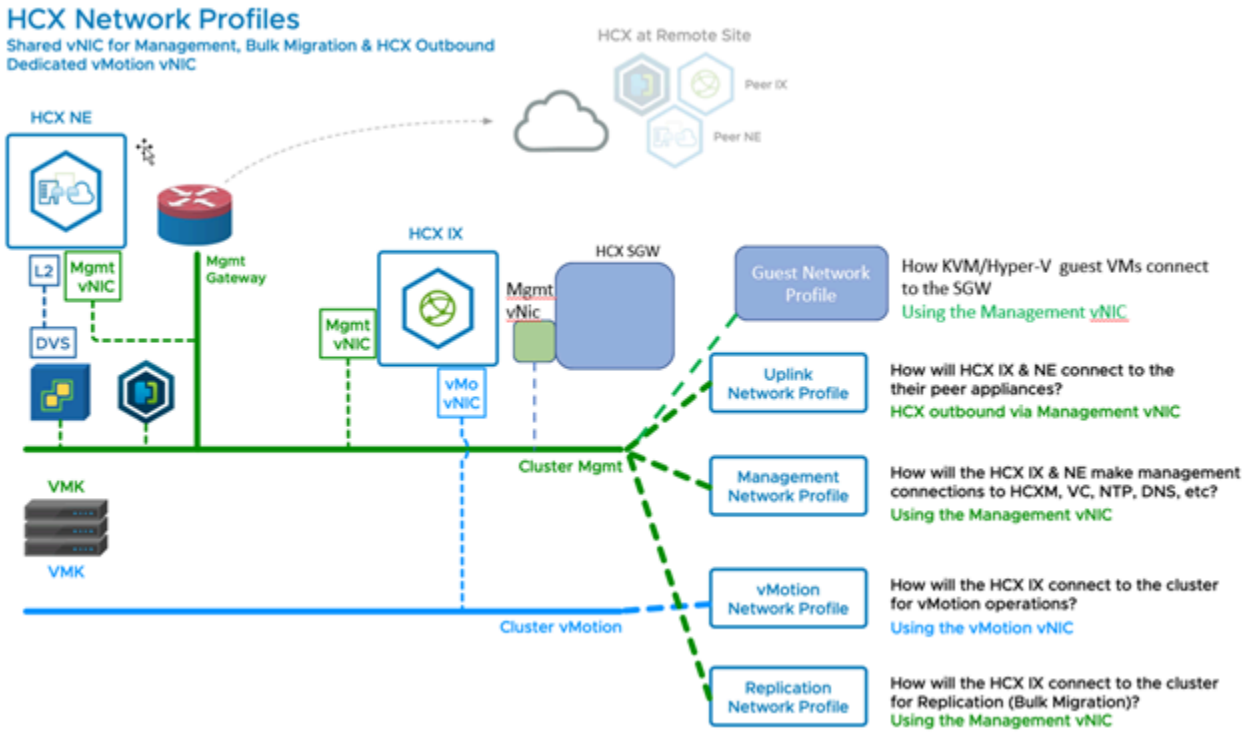
Network Configuration 4 - OS Assisted Migration Using Guest Network

This configuration shows OS Assisted Migration networking when using a dedicated Guest Network for migrations.



Network Configuration 5- OS Assisted Migration Using Management Network

Configuration shows OS Assisted Migration networking when using the Management Network for the Guest Network function.



Compute Profile Considerations and Concepts

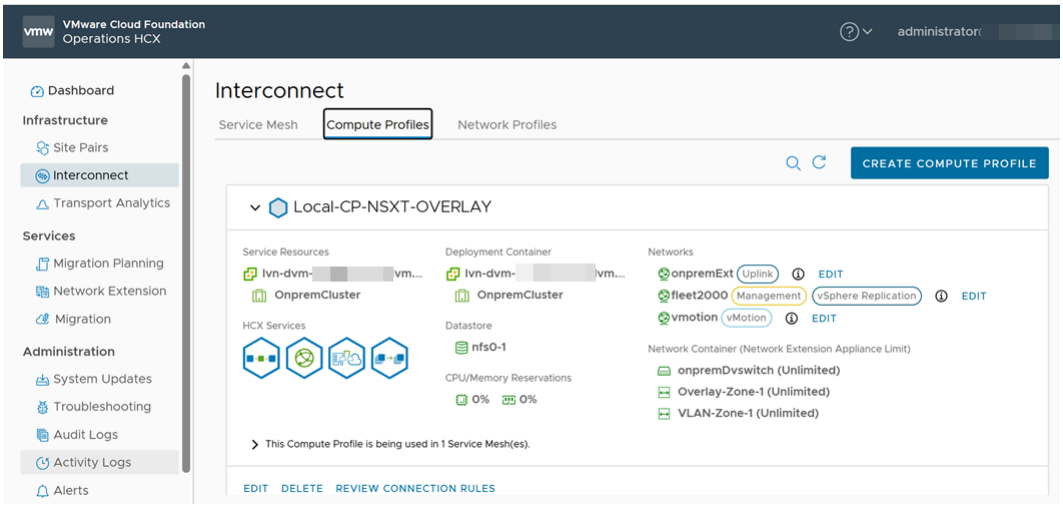
The compute profile is a sub-component of the service mesh. The compute profile describes which HCX services run, and how they are deployed when the service meshh is created.

Introduction to Compute Profiles

A compute profile configuration is required for service mesh deployments. It defines deployment parameters and enables services. For configuration procedures, see Create a Compute Profile. A compute profile is constructed of the following elements:

Services	The VCF Operations HCX services that are activated when a Service Mesh is created.
Service Clusters	At the HCX appliance source, the service cluster hosts contains the virtual machines for migration. For network extension, only distributed switches connected to selected service clusters are displayed. A datacenter container can be used to automatically include clusters within the datacenter container. Clusters are automatically adjusted in the compute profile when clusters are removed or added to the datacenter container. At the HCX appliance destination, the service clusters can be used as the target for migrations.
Deployment Clusters	The cluster (or resource pool) and datastore that host the service mesh appliances. For migrations to be successful, the deployment cluster must be connected such that the service clusters vMotion and Management/Replication VMkernel networks are reachable to the HCX-IX appliance. For network extension to be successful, the deployment cluster must be connected to a distributed switch that has full access to the VM network broadcast domains.
Management Network Profile	The network profile that the HCX appliance uses for management connections.
Uplink Network Profile	The network profile that the HCX appliance uses for HCX appliance to HCX appliance traffic.
vMotion Network Profile	The network profile that the HCX appliance uses for vMotion-based connections with the ESX cluster.

Replication Network Profile	The network profile that the HCX appliance uses for Replication-based connections with the ESX cluster.
Distributed Switches for Network Extension	The distributed switches containing the virtual machine networks for extension.
Guest Network Profile for OSAM	The network profile that the HCX appliance uses to receive connections from the sentinel agents.



Characteristics of Compute Profiles

- An HCX appliance system must have one compute profile.
- The compute profile references clusters and inventory within the vCenter Server that is registered in the HCX appliance.
- Creating a compute profile does not deploy the HCX appliances (compute profiles can be created and not used).
- Creating a service mesh deploys appliances using the settings defined in the source and the destination compute profiles.
- A compute profile is "in use" when it is used in a service mesh configuration.
- Changes to a compute profile profile are not affected in the service mesh until a service mesh Re-Sync action is triggered.

Compute Profiles and Service Appliances

The VCF Operations HCX Services defined in the compute profile identify the appliances that the HCX appliance deploys when creating the service mesh. The compute profile configuration can include the following appliances:

- HCX appliance Interconnect (HCX-IX)
- Network Extension (HCX-NET-EXT)
- Sentinel Gateway (HCX-SGW)
- Sentinel Data Receiver (HCX-SDR)
- Sentinel Replication Gateway (HCX-SRG)

The following compute profile considerations apply to these appliances.

- For the dataplane appliances (HCX-IX, HCX-NET-EXT, HCX-SGW, HCX-SDR, HCX-SRG), it is recommended to change the compute profile if a change in the appliance deployment containers (compute, storage or folder) is required. In such a case, you must resync the service mesh to relocate the appliances.
- If you do not use the compute profile as described above, you can use vSphere vMotion to relocate dataplane appliances to different containers with the following restrictions:
  - The host/cluster must have a port group association with all networks defined under the network profile. For example, this can includes the Uplink, Management, and vMotion networks.

- VCF Operations HCX redeploys appliances into same deployment cluster as configured under the compute profile during Upgrade or Redeploy operations, and removes previously deployed appliances from other clusters.
- The appliances will be relocated/vMotion back to same deployment cluster as configured under compute profile during a service mesh Resync operation. The HCX appliance relocates (vMotion) the appliances back to the same deployment cluster configured in the compute profile.

  Note:
  For Storage vMotion requirements, the target datastore must have an association/reachability with all hosts/clusters to which the appliances relocate.

  When relocating Network Extension appliances, which are carrying data traffic over an extended datapath, the relocation process might experience some packet drops. This applies during relocation (vMotion) to the same cluster or a different cluster depending on the Compute Profile configuration.

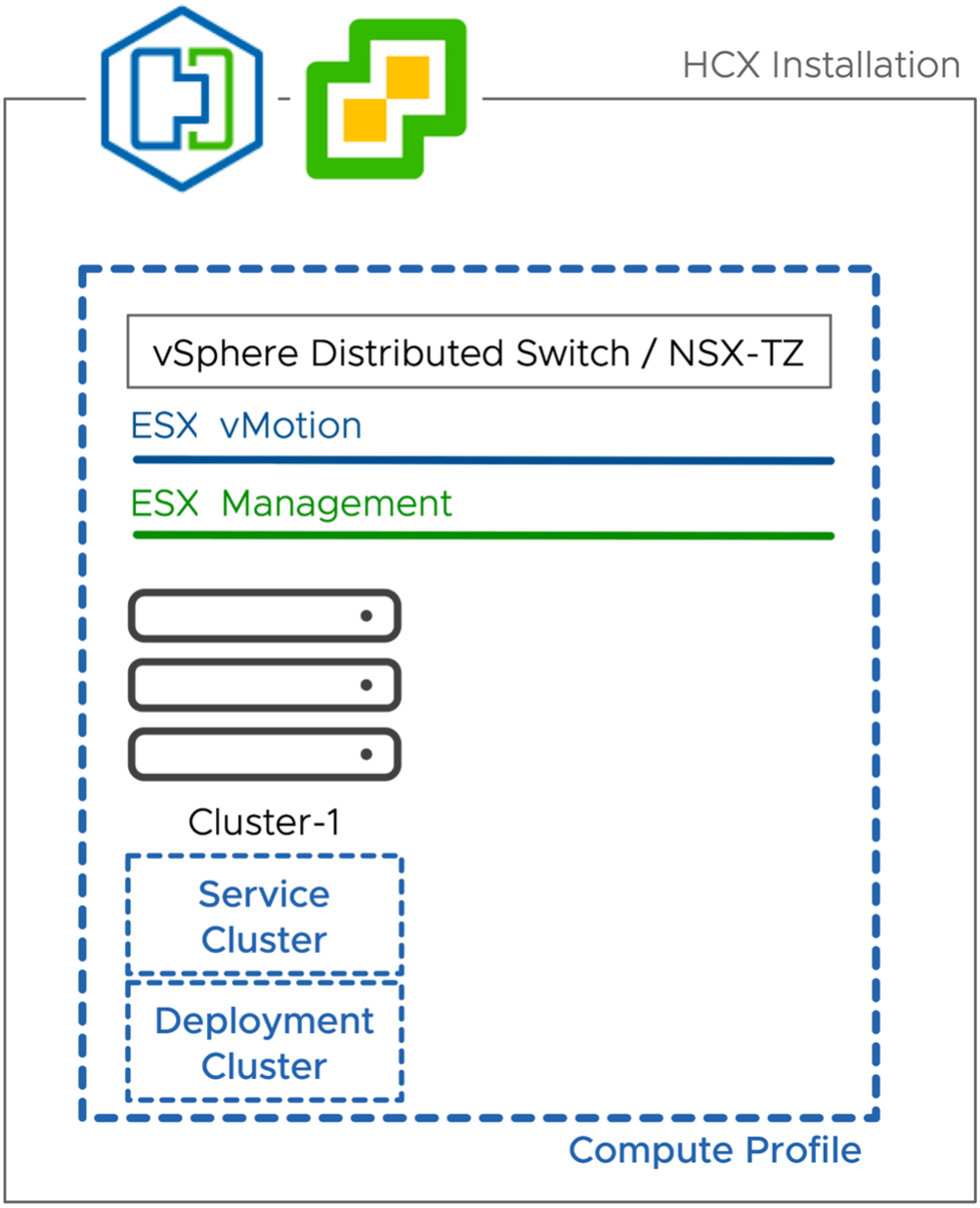
Compute Profiles and Clusters

HCX appliance compute profiles provide flexibility when designing service mesh deployments. The following are examples of different service mesh deployments.

Configuration 1 - Single Cluster Deployments

In the illustrated example, Cluster-1 is both the deployment cluster and service cluster.

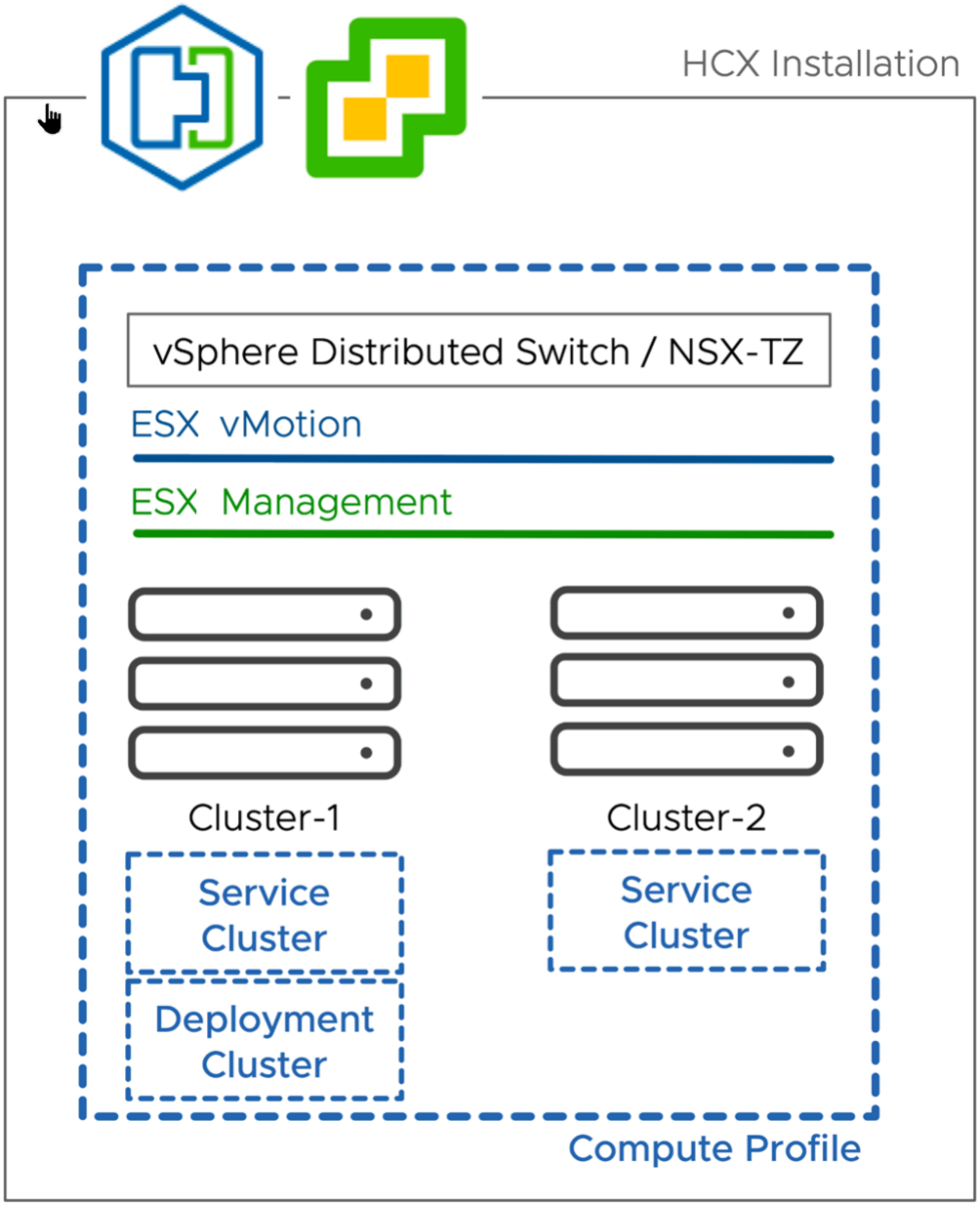
- Single cluster deployments use a single compute profile.
- In the CP, one cluster is designated as a service cluster and as the deployment cluster.



Configuration 2 - Multicluster

In the illustrated example, Cluster-1 is the deployment cluster. Both cluster-1 and cluster-2 are service clusters.

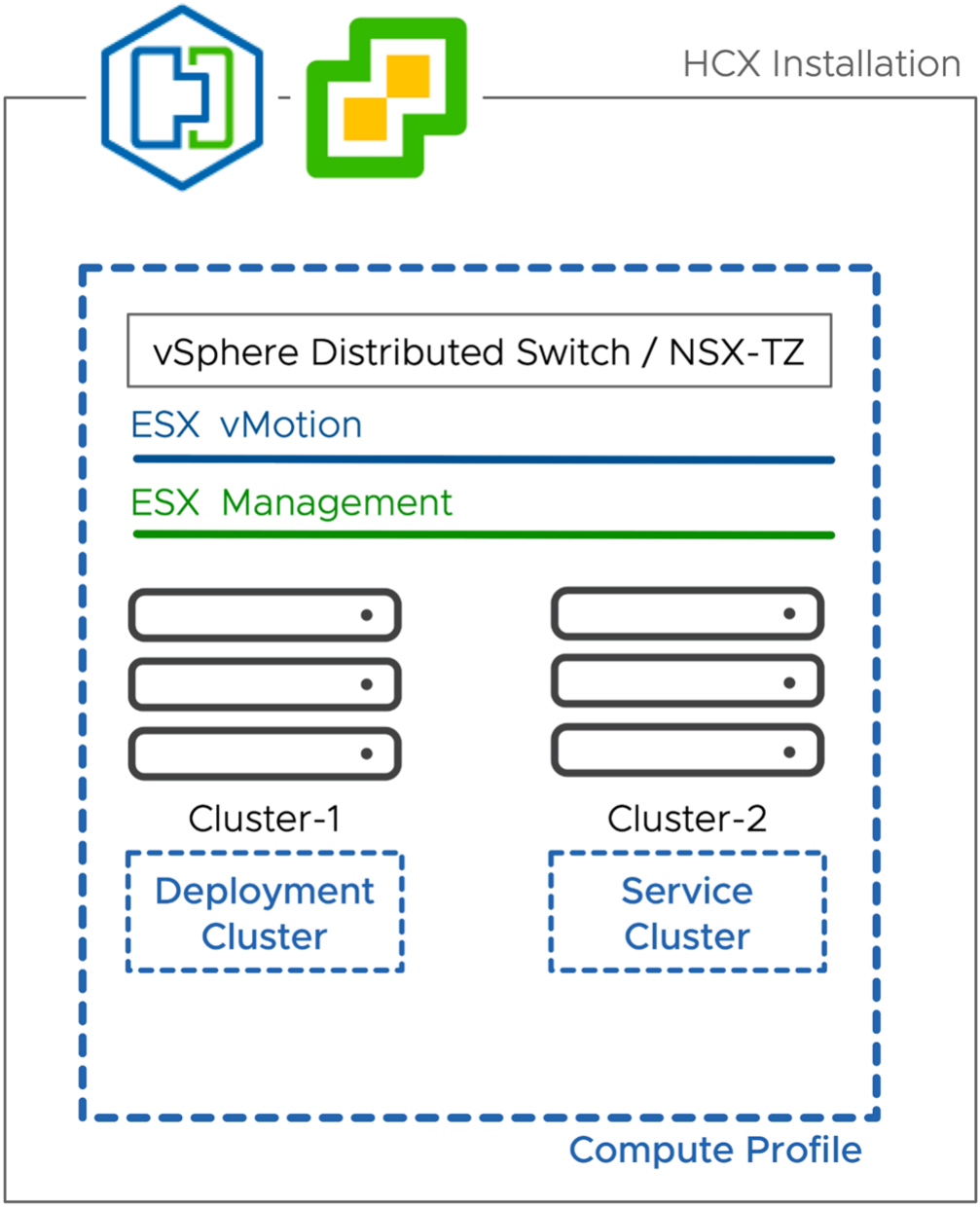
- In this compute profile configuration, one cluster is designated as the deployment cluster, and all clusters (including the deployment cluster) are designated as service clusters.
- All the service clusters must be similarly connected (that is, same vMotion/Replication networks).
- When the service mesh is instantiated, one HCX-IX is deployed for all clusters.
- In larger deployments where clusters might change, a satacenter container can be used (instead of individual clusters) so VCF Operations HCX automatically manages the service clusters.



Configuration 3 - Multicluster (Dedicated Deployment Cluster)

In the illustrated example, cluster-1 is the deployment cluster and cluster-2 is the service cluster.

- In this compute profile configuration, one cluster is designated as the deployment cluster and is not a service cluster. All other clusters are designated as service clusters:
  - This compute profile configuration can be used to dedicate resources to the HCX appliance functions.
  - This compute profile configuration can be used to control site-to-site migration egress traffic.
  - This compute profile configuration can be used to provide a limited scope vSphere distributed switch in environments that heavily leverage the vSphere Standard Switch.
- For VCF Operations HCX migrations, this compute profile configuration requires the service cluster VMkernel networks to be reachable from the deployment cluster, where the HCX-IX is deployed.
- For network extension, this compute profile configuration requires the deployment cluster hosts to be within workload networks' broadcast domain (service cluster workload networks must be available in the deployment cluster distributed switch).
- When the service mesh is instantiated, one HCX-IX appliance is deployed for all clusters.

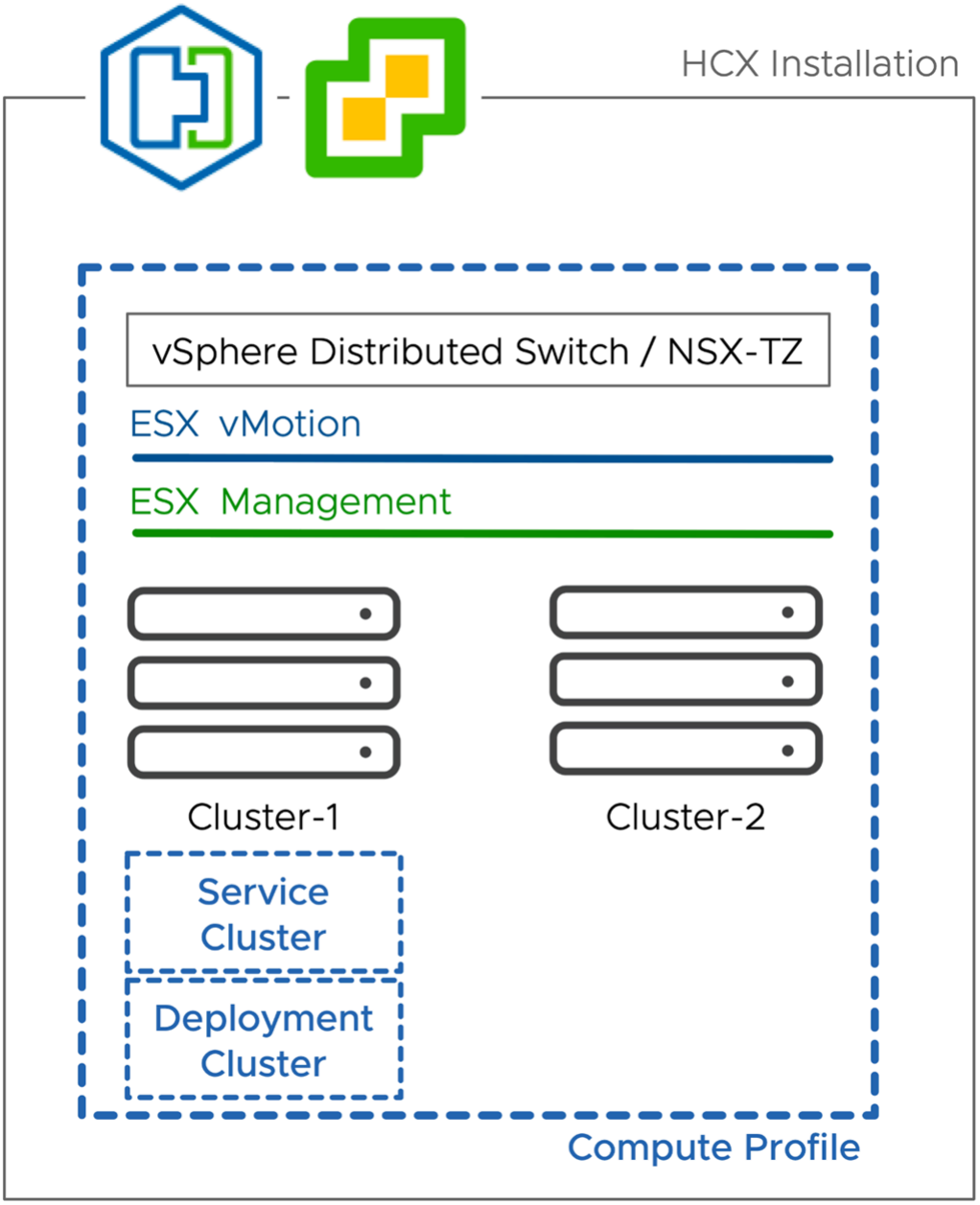


Configuration 4 - Cluster Exclusions

In the illustrated example, cluster-2 is not included as a service cluster.

- In this compute profile configuration, one or more servers have been excluded from the service cluster configuration.
- This configuration can be used to prevent portions of infrastructure from being eligible for VCF Operations HCX services. Virtual machines in clusters that are not designated as a service cluster cannot be migrated using VCF Operations HCX(migrations will fail).

> **Note:**
>
> If vSphere hosts or virtual machines are in a folder outside of all clusters, those will be considered as excluded from the service resources/clusters even if the compute profile allows you to select the entire data center and select folders.



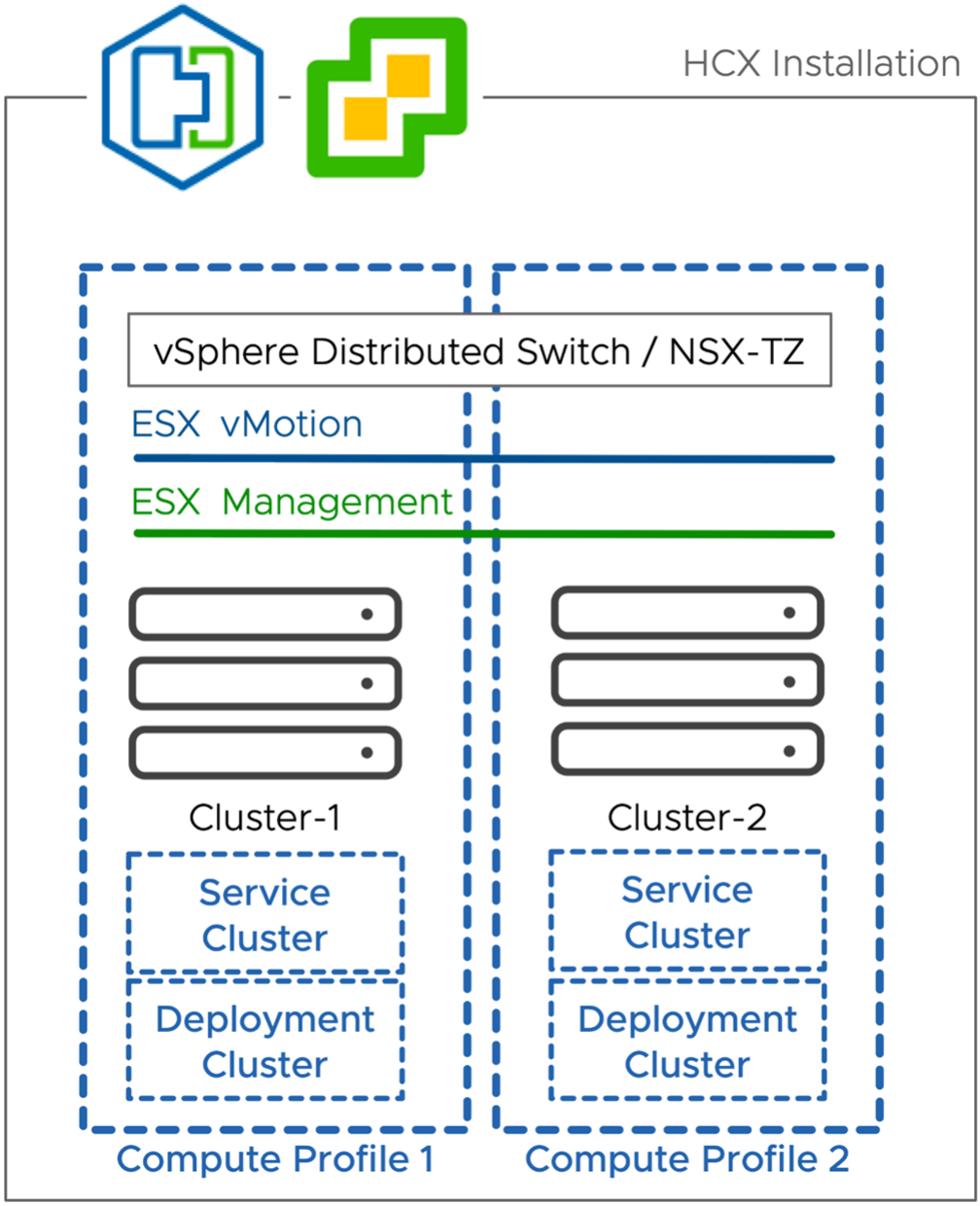
Configuration 5 - Multiple Compute Profiles (Optional, for Scale)

In the illustrated example, compute profile 1 has been created for cluster-1, and compute profile-2 has been created for cluster-2, and the VMkernel networks are the same. Creating additional compute profiles is optional (for scaling purposes).

- In this compute profile configuration, service clusters are separated into compute profiles.
- Every compute profile requires a deployment cluster, resulting in a dedicated service mesh configuration for each compute profile.
- For clarity, if there were 5 clusters in a vCenter Server, you can have dervice clusters separated out in various ways, such as the following:
  - CP-1: 1 Service Cluster, CP-2: 4 Service Clusters
  - CP-1: 2 Service Clusters, CP-2: 3 Service Clusters
  - CP-1: 1 Service Cluster, CP-2: 2 Service Clusters, CP-3: 2 Service Clusters
  - CP-1: 1 Service Cluster, CP-2: 1 Service Cluster, CP-3: 1 Service Cluster, CP-4: 1 Service Cluster, CP-5: 1 Service Cluster

Note:

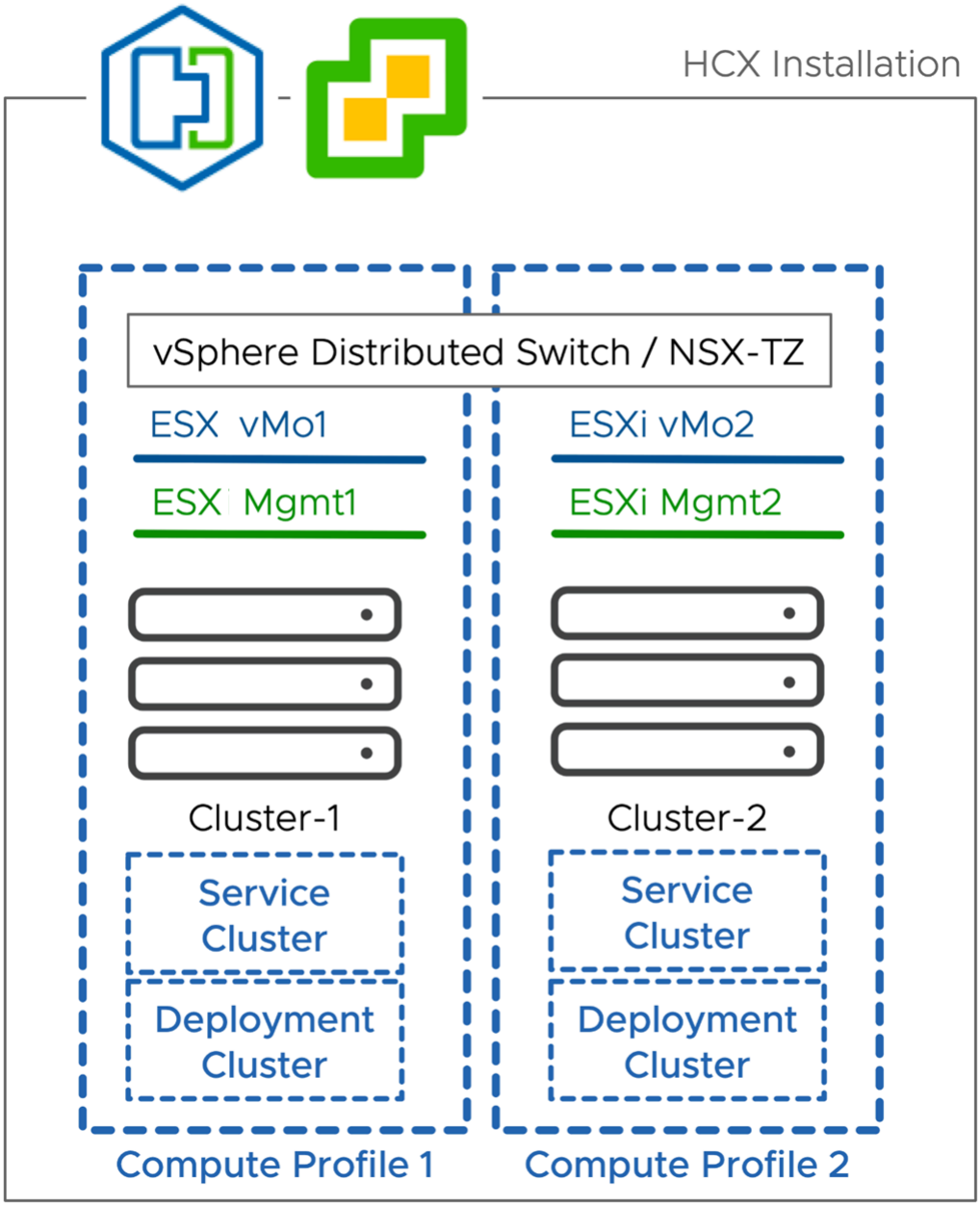
Compute profile configurations can leverage the same network profiles for ease of configuration.



Configuration 6 - Multiple Compute Profiles (with Dedicated Network Profiles)

In the illustrated example, cluster-1 uses vMotion and Mgmt networks 1. cluster-2 uses vMotion and Mgmt networks 2. The VMkernel networks are different, and isolated from each other. Creating dedicated network profiles and dedicated compute profiles is required.

- In this compute profile configuration, the service clusters are separated into distinct compute profiles. The compute profiles reference cluster-specific network profiles.
- When the service mesh HCX-IX appliance connects directly to the cluster's vMotion network, where the Replication and vMotion are different, you must create cluster-specific network profiles and assign them to cluster-specific compute profiles. These profiles will then be used to create a cluster-specific service mesh.



Appendix - Installation Summary Steps

This reference lists all the steps involved when deploying source and destination VCF Operations HCX appliances. The steps are listed here as a quick reference. Requirements are not listed here. Use the checklists in this publication to prepare for the installation.

Install (or Enable) VCF Operations HCX at the Destination

The following steps summarize the initial HCX appliance deployment and configuration in private data center destination environment. For a detailed description, see VCF Operations HCX Licensing.

Note:

If you are using a Provider for the destination site, contact your Provider for enabling HCX.

1. Install the HCX appliance:

a. Use the VCF Operations HCX OVA from `support.broadcom.com` to deploy the HCX appliance in the vSphere Client. See Deploy the OVA in the vSphere Client.

b. Log in to the HCX Manger Appliance Management interface (:9443) .

c. Set the location and system name.

d. Register the vCenter Server and NSX Manager.

e. Restart the VCF Operations HCX services.

f. Define role mapping (this setting defines the groups that can perform mobility operations).

2. Create a network profile:

   a. Browse to the VCF Operations HCX interface UI (:443) to create a network profile. The network profile is a compute profile requirement for service mesh deployment.

3. Create a compute profile:

   a. Browse to the VCF Operations HCX interface UI (:443) to create a compute profile. The compute profile defines how service mesh appliances are deployed in the destination environment.

4. Configure inbound firewall rules for the destination HCX appliance:

   a. Allow TCP-443 inbound from the source HCX appliance to the destination HCX appliance (this might be a NAT Public IP if the environments are separated by Internet).

   b. Allow UDP-4500 inbound from the source HCX-IX and HCX-NET-EXT appliances IP addresses (this might be a NAT IP if the environments are separated by Internet).

5. Configure any site-specific firewalls for traffic between HCX service appliances. Reference ports.broadcom.com for the complete list of HCX appliance network ports.

Next, install the HCX appliance at the source site.

Install VCF Operations HCX at the Source Site

The following steps summarize the initial HCX appliance deployment and configuration in the source environment. For a detailed description, see VCF Operations HCX Licensing.

1. Use the HCX appliance OVA to deploy the HCX appliance system in the vSphere Client. See Deploy the OVA in the vSphere Client.

   After the OVA is deployed and the system is initialized:

   a. Log in to the HCX Appliance Management interface (:9443), and authenticate with the Admin user credentials.

   b. Set the location and system name.

   c. Register the vCenter Server, SSO, and optionally the NSX Manager registered with that vCenter Server.

   d. Restart the VCF Operations HCX services.

   e. Configure Role Mapping (this defines the SSO user groups that can perform mobility operations).

2. Create a site pair:

   a. The VCF Operations HCX interface (:443) registers the destination HCX appliance using the SSO group from the destination environment.

3. Create a compute profile:

   a. Log in to the VCF Operations HCX interface (:443) at the source, and authenticate with a user that is part of the role-mapping group.

4. Create a service mesh:

   a. The VCF Operations HCX interface (:443) uses the service mesh wizard to instantiate services.

      a. In the service mesh interface, you select a compute profile for the HCX appliance source environment, and a compute profile for the destination environments.

      b. Service mesh creation deploys data plane appliances in parallel at the source and the destination environments.

      c. The source HCX appliance service mesh appliances are Initiators, and will automatically attempt to establish tunneling connections to the destination side.

      d. The destination appliance components are Receivers that will only accept tunneling requests from the Initiators.

Administering VCF Operations HCX

About Administering VMware® Cloud Foundation Operations HCX™

This information describes how to plan for, install, and operate VMware Cloud Foundation Operations HCX services in a VMware Cloud Foundation platform. The information includes step-by-step configuration instructions and operational procedures.

Intended Audience

This information is for anyone who wants to install or use VCF Operations HCX. The information is for virtual infrastructure administrators and architects and who are familiar with virtual machine technology and data center operations.

Overview

VCF Operations HCX is a workload mobility platform that is designed for simplifying workload migration, and workload rebalancing across data centers and clouds.

Use cases:

- **Rehosting application workloads**
  You can schedule and migrate vSphere virtual machines within and across data centers without requiring a reboot. Use cases include data center consolidation, moving workloads for efficient host resource utilization, moving workloads from private to public cloud sites or from public to private cloud sites, and flexible migration options for managing large or small migration waves with the least disruption and greatest efficiency.
- **Upgrading and re-platforming applications and infrastructure**
  With VCF Operations HCX, you can migrate workloads from vSphere and non-vSphere (KVM and Hyper-V) environments, within and across private or public clouds, to current vSphere versions without requiring an upgrade. Use cases include moving applications to the latest VMware Cloud Foundation software and hardware platform, and performing platform maintenance without impacting application workloads.
- **Workload rebalancing**
  VCF Operations HCX provides a mobility platform across data centers and providers to allow customers to move applications and workloads at any time to meet scale, cost management, compliance, and vendor neutrality goals. Use cases include balancing application workloads across public and private clouds, moving workloads between providers, and extending the networks serving those workloads between sites.

System Services

To build and maintain a secure optimized transport fabric between paired sites, and to provide feature rich functionality, VCF Operations HCX managed sites run various connectivity and mobility services.

VCF Operations HCX delivers the connectivity and mobility services to enable hybrid interconnect and migration services. These services provide additional functionality for scalability and performance when transforming large data centers or moving large quantities of virtual machines.

The following services are available with VCF Operations HCX.

VCF Operations HCX Services	Description
Interconnect	This service creates and secures connections between VCF Operations HCX installations, supporting management, migration, and replication operations. This service is deployed as a virtual appliance.
WAN Optimization	The WAN Optimization service works with the HCX Interconnect service to improve network performance through a combination of deduplication and compression techniques. This service is optionally deployed as part of the enhanced HCX-IX virtual appliance.
Network Extension	This service extends the virtual machine networks from a VCF Operations HCX source site to an VCF Operations HCX destination site. Virtual machines that are migrated or created on the extended segment at the destination site are Layer 2 adjacent to virtual machines placed on the origin network. This service is deployed as a virtual appliance.

VCF Operations HCX Services	Description
Network Extension High Availability	Network extension High Availability protects extended networks from disruptions that are associated with network extension appliance downtime. Network extension High Availability creates a redundant pair, or group, of connections for network extension appliances. In the event that a network extension appliance fails or is taken offline, the active connection fails over to the standby connection, and the extended network operations continue without interruption.
Bulk Migration	This service uses vSphere Replication protocol to move virtual machines in parallel between sites.
vMotion Migration	This migration method uses the vSphere vMotion protocol to move a single virtual machine between sites with no service interruption.
Mobility Optimized Networking (MON)	MON is a capability of the network extension feature. MON enables optimized virtual machine trafffic for virtual machines that span multiple segmented networks, that have multiple VLAN dependencies, or have hybrid applications during the migration cycle. Migrated virtual machines can be configured to access any destination through an optimized path.
OS Assisted Migration (OSAM)	This migration service moves Linux or Windows-based non-vSphere guest virtual machines from their host environment to a VMware Cloud Foundation platform. The OSAM service operates in one of two modes: multi-site and single-site. • Multi-site mode applies to site pairs where the source site runs both vSphere and non-vSphere environments, and the destination site runs vSphere. In this mode, VCF Operations HCX deploys the HCX-SGW (Sentinel Gateway) appliance at the source site and the SDR (Sentinel Data Receiver) appliance at the destination site. • Single-site mode means vSphere and all VCF Operations HCX services are deployed at the destination site. In this mode, VCF Operations HCX deploys the HCX-SRG (Sentinel Replication Gateway) appliance at the destination site where the non-vSphere workloads will be migrated. Regardless of the mode, every guest workload requires the installation of sentinel software.
Replication Assisted vMotion (RAV)	This service uses both vSphere Replication and vSphere vMotion technologies for large-scale, parallel migrations with no service interruption.
HCX Assisted vMotion (Direct)	The HCX Assisted vMotion (HAV) service works in conjunction with native cross-vCenter vMotion migration to orchestrate migrations between the ESX hosts associated with the source and destination vCenter Servers.
Supervisor Onboarding	VCF Operations HCX supports onboarding traditional virtual machines into specified Supervisor clusters.

VCF Operations HCX Services	Description
Traffic Engineering • Application Path Resiliency • TCP MSS Clamping (TCP Flow Conditioning) • Generic Receive Offload (GRO) • Tunnel Encryption (network extension or migration services)	VCF Operations HCX provides settings for optimizing network traffic for HCX Interconnect and network extension services. • Application Path Resiliency is available for either Interconnect or network extension services. This setting creates multiple tunnel flows for both Interconnect and network extension traffic, which can follow multiple paths across the network infrastructure from the source to the destination data centers. The service then intelligently forwards traffic through the tunnel over the optimal path and dynamically switches between tunnels depending on traffic conditions. Application Path Resiliency forwards traffic over one tunnel at a time and does not load balance across multiple paths. • TCP MSS Clamping is available for network extension service. This setting adjusts the segment size during the TCP connection handshake between end points across the network extension. This optimizes the average packet size to reduce fragmentation and lower the overall packet rate. • Generic Receive Offload is available for the network extension service. VCF Operations HCX reassembles inbound packets into larger ones before delivery to the workload applications, allowing for enhanced application performance. • Tunnel encryption is available for either network extension or migration services. By default, this setting uses transport encryption to protect network extension data and migration traffic between site pairs. Deactivating encryption may increase performance for migration or network extension connections. Deactivating encryption is available only on Uplink networks that have been verified as Secure in the network profile configuration.

System Components

VCF Operations HCX consists of a virtual appliance at the source and destination sites along with associated service virtual appliances for each site.
In the VCF Operations HCX site-to-site architecture, there is a notion of a source and a destination environment with an VCF Operations HCX appliance deployed at each site. In hosted environments, the provider may deploy VCF Operations HCX at the destination. For specific VCF Operations HCX deployment details in a hosted environment, contact your provider.

The source and destination sites are paired together to perform mobility and network extension operations.

In both the source and the destination environments, the VCF Operations HCX virtual appliance is deployed to the same management zone as the site's vCenter Server, enabling deployment of the VCF Operations HCX service virtual appliances across both the source and destination sites.

You initiate migration of virtual machines and establishing service mesh services from the VCF Operations HCX console. During creation of the service mesh, the VCF Operations HCX virtual appliance deploys service virtual appliances supporting mobility and network extension operations.

Interconnect Appliance

The Interconnect service (HCX-IX) appliance provides replication and vMotion-based migration capabilities over the Internet and private lines to the destination site while providing strong encryption, traffic engineering, and virtual machine mobility.

The Interconnect appliance includes deployment of the Mobility Agent service that appears as a host object in the vCenter Server. Mobility Agent is the mechanism that VCF Operations HCX uses to perform vMotion, Cold migration, and Replication Assisted vMotion (RAV) migration to a destination site.

The Interconnect appliance includes the WAN Optimization (WAN-OPT) service. This appliance improves the performance characteristics of private lines or Internet paths by applying WAN optimization techniques like the data de-duplication and compression techniques.

Network Extension Appliance

The network extension (HCX-NET-EXT) appliance provides layer 2 connectivity between sites. A network extension provides the ability to keep the same IP and MAC addresses during virtual machine migrations. When the network extension service is enabled, a pair of virtual appliances is deployed: one in the source and one in the destination site.

Sentinel Gateway Appliance

When using the OS Assisted Migration (OSAM) service in multi-site mode, you migrate guest (non-vSphere) virtual machines between vSphere-based source and destination VCF Operations HCX sites. In multi-site mode, VCF Operations HCX deploys the Sentinel Gateway (SGW) appliance at the source vSphere environment for connecting and forwarding guest workloads.

Sentinel Data Receiver Appliance

When using the OS Assisted Migration (OSAM) service in multi-site mode, VCF Operations HCX deploys the Sentinel Data Receiver (SDR) appliance at the destination vSphere environment for receiving, managing, and monitoring data replication operations.

Sentinel Replication Gateway Appliance

When using the OS Assisted Migration (OSAM) service in single-site mode, you migrate guest (non-vSphere) virtual machines between non-vSphere environments and vSphere-based VCF Operations HCX sites. In single-site mode, VCF Operations HCX deploys the Sentinel Replication Gateway (SRG) appliance at the destination vSphere environment. The SRG manages data replication operations from the non-vSphere environment.

Preparing for Installation

This section describes the system requirements, network ports, and protocols that must be allowed and various other requirements, like software versions and feature interoperability requirements.

System Requirements

Before deploying VCF Operations HCX, consider the required resources for both the source and the destination environment HCX Manager.

Virtual Hardware Requirements for Appliances

Appliance	vCPU	Memory	Disk Space
HCX (appliance) See note for more scaling information.	4	12 GB	60 GB
Interconnect (HCX-IX) See note for required resources when deploying HCX-IX with WAN-OPT.	8	6 GB	6.5 GB
Network Extension (HCX-NET-EXT)	8	3 GB	6.5 GB
HCX WAN Optimization (HCX-WAN-OPT)	16	32 GB	172 GB
Sentinel Gateway (HCX-SGW [source only])	8	8 GB	17.5 GB (disks: 6.5 GB, 8 GB, and 3 GB)
Sentinel Data Receiver (HCX-SDR [destination only])	8	8 GB	17.5 GB (disks: 6.5 GB, 8 GB, and 3 GB)
Sentinel Replication Gateway (SRG)	8	8 GB	17.5 GB (disks: 6.5 GB, 8 GB, and 3 GB)

Note:

- The default VCF Operations HCX virtual appliance sizes are 4 vCPU and 12 GB memory. To support more MON-enabled virtual machines, you can scale up the virtual appliances as described in Knowledge Base article 88401.
- VCF Operations HCX supports scaling up to 1000 concurrent virtual machine migrations. For details, see Knowledge Base article 373010.
- The storage requirement per appliance is doubled during upgrade and redeploy operations, as a second appliance is created for operation.
- For best practice when deploying HCX-IX with WAN-OPT, use 100% CPU and memory reservation.

Migration Host and Datastore Requirements

The requirements in this section apply to vSphere-to-vSphere migrations.

During a virtual machine migration operation, the destination compute and storage resources are selected. These resources must meet the following general requirements:

Destination Host/Datastore Requirements	Description
Host CPU and Memory Capacity	The destination host must support the original vCPU and Memory configurations of the virtual machine.
Host CPU and Memory Usage	The destination host capacity must satisfy admission control requirements.
Host Compatibility with VMware Tools	The destination hosts must support the VMware Tools version of the original virtual machine. For more information, see the VMware Certified Compatibility Guides.
Host Compatibility with VM Hardware	The destination hosts must support the VM Hardware version of the original virtual machine. For more information, see VMware KB 2007240.
Datastore Type	The destination datastore type must be VMFS5, VMFS6, NFS3, NFS4.1, vSAN, vSAN OSA, or vSAN ESA. **Note:** vVOL is not supported for RAV.
Datastore Capacity	The destination datastore capacity must be able to store the following: • Complete original virtual machine data (everything that makes up the virtual machine) • Additional replica instance vmdk file on target datastore (applicable for Bulk or RAV Migration during the delta sync) • Additional data for multiple point in time (MPIT) recovery points • With vSAN, additional data required by the Primary level of failures to tolerate (PFTT) configuration

System Requirements for Scaling Out Deployments

Consider the following requirements when scaling out VCF Operations HCX deployments:

- When scaling VCF Operations HCX, the hardware requirements apply for each additional virtual appliance. For virtual appliance limit information, refer to the VMware Configurations Maximums.
- The migration service, in its default configuration, encompasses all source and destination clusters in the service mesh with a single Interconnect appliance. The migration service can be scaled out by deploying additional service meshes. The additional meshes can be selected during migration operations.
- Multiple network extension appliances can be deployed per vSphere Distributed Virtual Switch (VDS) or NSX Transport Zone.
- When using network extension High Availability (HA), at least two network extension appliances must be deployed per VDS or NSX Transport Zone in order to form an HA group. In the HA group, one appliance is Active and one is Standby.

  Note:

  HA Standby appliances must be considered when calculating the overall hardware resource requirements.

- You must configure the network extension appliance count in the service mesh to include all planned standalone, HA Active, and HA Standby appliances.

  Note:

  For scale-out procedures, use KB article 93605.

Software Version Requirements

VCF Operations HCX appliances must be running supported software versions.

Software Version Requirements for VCF Operations HCX Installations

Note:

Destination environments must be licensed for VMware Cloud Foundation 9.0 and later.

For granular compatibility information, select VCF Operations HCX in the VMware Product Interoperability Matrix.

For information about General Availability and End of Service for VMware software, see VMware by Broadcom Product Lifecycle.

Component Type	Source Environment	Destination environment
VCF Operations HCX	Use the latest HCX Manager version.	Use the latest HCX Manager version.
vSphere version	vSphere versions within Technical Guidance or newer. For details about the evacuation of versions that are no longer generally available, see VMware KB article 82702.	Use the latest HCX Manager version.
NSX	NSX versions within Technical Guidance or newer. See VMware KB article 82702.	Use the latest NSX version supported by the VCF version.

Network Port and Protocol Requirements

VCF Operations HCX deployments require communication over various ports between all HCX appliances at each site and between HCX sites.
You must allow the following connections in VCF Operations HCX deployments:

- The perimeter firewalls must allow VCF Operations HCX connections to vcsa.vmware.com. For systems not participating in CEIP, this is not required.
- The source VCF Operations HCX site firewalls must be configured to allow outbound connections to the destination VCF Operations HCX systems.
- The destination VCF Operations HCX site firewalls must be configured to allow inbound connections from the source VCF Operations HCX system.
- All local connections (within a single VCF Operations HCX site deployment) at the source and destination environment. These connections never traverse from source to destination or from destination to source.
- In environments that rely on a proxy server to centralize outbound HTTPS connnections, the proxy server must be configured on the VCF Operations HCX . Refer to Configure a Proxy Server.

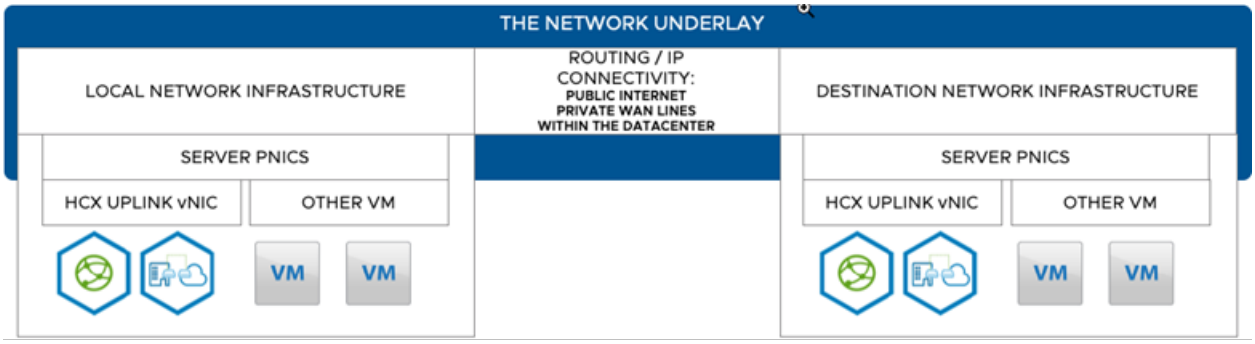
For a complete list of network port and protocol requirements, see VMware Ports and Protocols.

Network Underlay Minimum Requirements

The infrastructure providing network connectivity to a VCF Operations HCX deployment (the underlay) must meet the minimum requirements. The underlay includes any intermediate system that is customer managed, provided by a third party, or part of an Internet service provider network.

What Is the Network Underlay

A network underlay provides the physical or logical connectivity on which the VCF Operations HCX transport packets are tunneled, where a transport packet contains an overlay header. The underlay network does not need to know that it is carrying HCX transport packets. This includes the physical routing infrastructure in the customer data center and (if applicable) any hosted infrastructure, and any physical network services joining the connected locations.

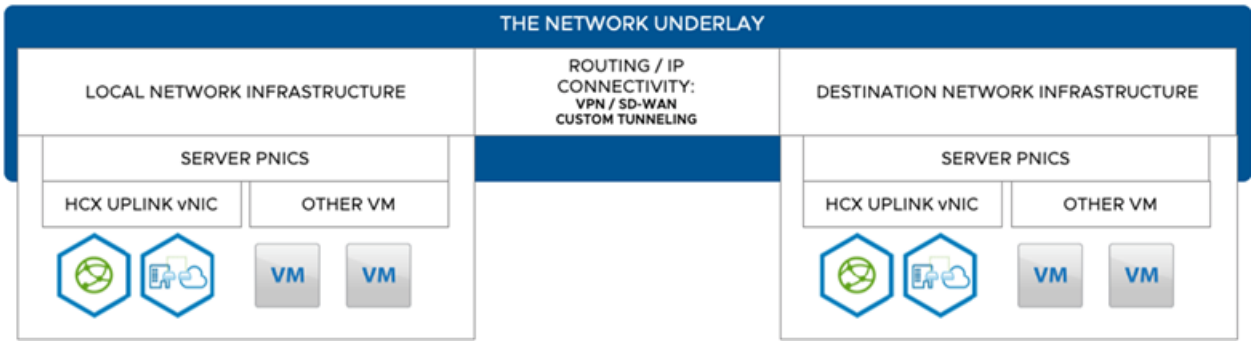


A network underlay can vary from high-bandwidth low-latency private paths between server racks in a data center, to lower-bandwidth higher-latency Internet based connectivity. We use the term "network underlay" to encompass all elements that affect the performance characteristics of an underlay, including the servers and network devices and connections between the vSphere environments. The network underlay requirements must be satisfied when considering all elements of the underlay.

Virtual Private Networks Based Network Underlays

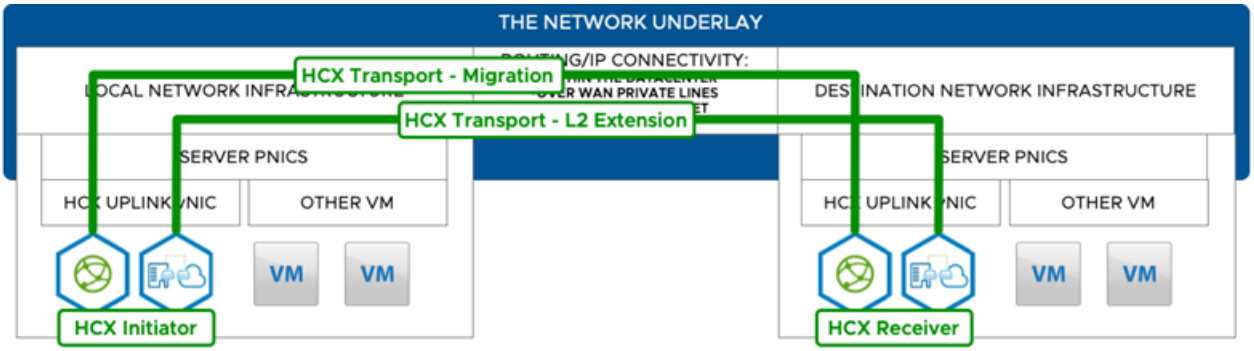
Virtual Private Networks (VPNs) are frequently used for creating secure network connections to private and public vSphere clouds over the Internet. Software-Defined Wide Area Network (SD-WAN) and custom tunneling solutions are used over the internet to improve data traffic transmissions. The SD-WAN, VPN, and other tunneling solutions are collectively referred to as VPN in this document.

The network underlay includes connections with VPN configurations. The network underlay requirements must be satisfied when considering all elements of the underlay.



General Network Underlay Requirements

HCX supports multiple uplinks and each uplink can be connected to a different network underlay. Examples of different network underlays include private line, public Internet, and multi-homed connectivity.



The following table summarizes the requirements from the Network Underlay to use the VCF Operations HCX migration and extension services:

- This table applies to migration and extension overlays (service mesh appliances).
- This table does not apply to HCX Manager or the management connections.

Overlay Requirements	Requirement Details
IP Addressing & IP Reachability	Requires a valid IP address and IP connectivity for end to end communication between the HCX uplink IP at the source site and the destination site.
Bandwidth, Loss and Latency, MTU	All underlays must comply with minimum parameters requirements for services to be supported at the minimum performance level. The minimum requirement applies to all network underlays and is provided in the next table. MTU configuration must be applied to the network profile before deploying the Interconnect appliances. If the MTU is changed on existing appliances, the appliances must be redeployed and the service mesh resynced. MTU 1150 - 9000 is valid for Interconnect.
Source Network Address Translation (SNAT)	SNAT is not required, but can be used to translate HCX Uplink private IP packets to public IP addresses for connections over the Internet. SNAT can only be applied to the HCX Initiator (the HCX source appliances).
Destination Network Address Translation (DNAT) Load Balancing Reverse Proxy	Inbound DNAT, load balancing, or reverse proxy configurations in the underlay are **not** supported for the VCF Operations HCX migration and Extension Transport tunnels.
VPN	Any VPN configuration in the network path is treated agnostically as an underlay, and is supported while the measured underlay parameters meet the documented requirements. Any additional encapsulation and performance degradation, overhead, or cost in addition to the characteristics of the underlay they ride on must be considered when measuring underlay outcomes. VCF Operations HCX does not support VPN configurations where the NSX Tier-0 router provides the VPN termination AND connectivity to HCX Manager uplinks through NSX Service Insertion.

Minimum Network Underlay Requirements for HCX

VCF Operations HCX has network underlay minimums for migration operations. Operations with lesser performance than the minimum values are not supported.

The following table lists the minimum requirements for individual operations at minimum performance.

For vMotion and Replication Assisted vMotion, the bandwidth requirement varies based on whether the WAN-Optimization service is running in the service mesh for the site pair. WAN Optimization can improve the network performance through a combination of deduplication and compression.

In addition to activating the WAN Optimization service in the service mesh, review these additional considerations that are related to network bandwidth performance:

- Minimizing latency, loss, jitter can result in improved migration performance outcomes.

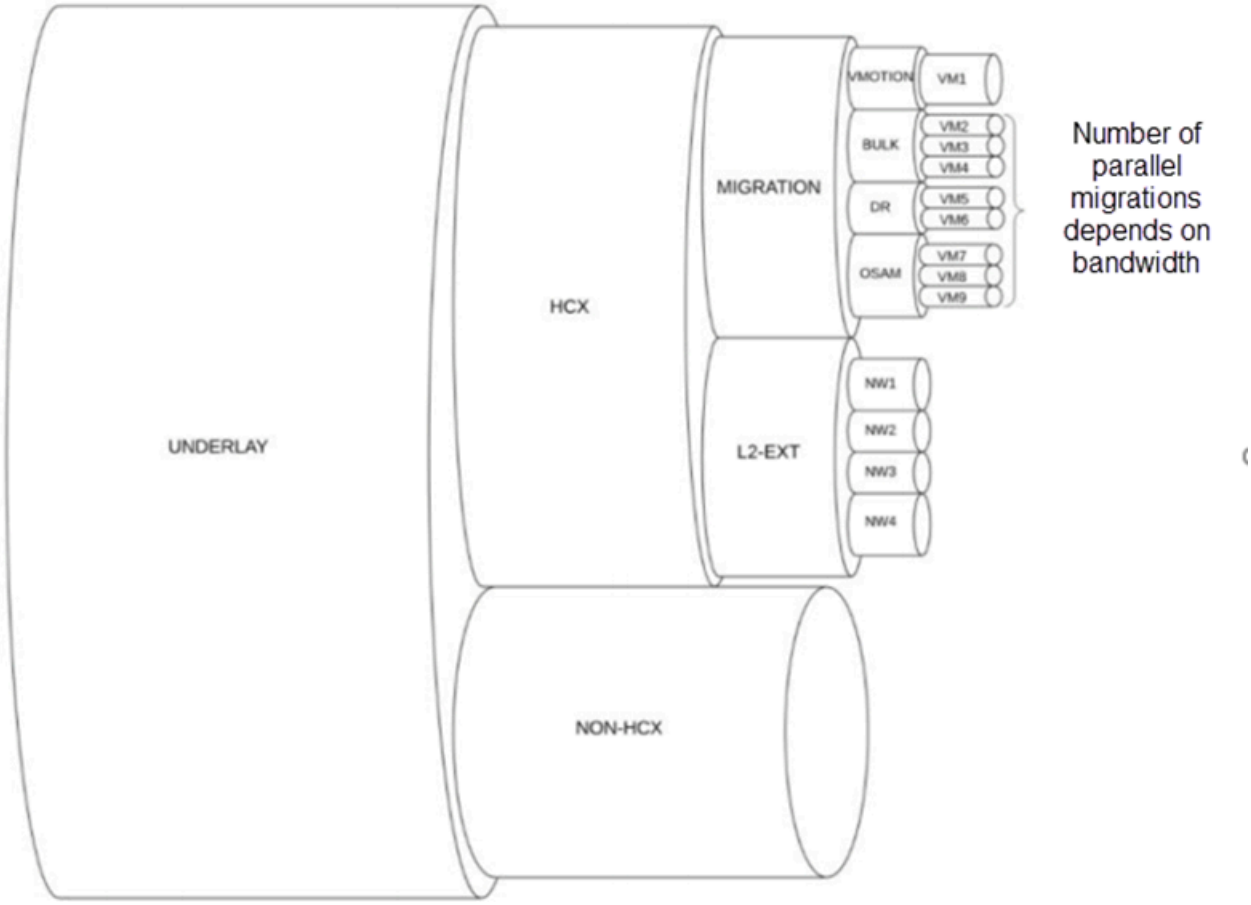
- Parallel operations (migration and extension) can result in increased bandwidth requirements.

Network Parameter	vMotion (IX)	HCX Assisted vMotion (Direct)	Replication Assisted vMotion	Bulk Migration	OS Assisted Migration
Min Bandwidth (Mbps)	150 with WAN optimization 250 without WAN optimization	250	150 with WAN optimization 250 without WAN optimization	50	50
Min MTU (bytes)	1150	1500	1150	1150	1150
Max Packet Loss (%)	0.1	0.1	0.1	1.0	1.0
Max Latency (ms)	150	150	150	150	150

Bandwidth distribution with VCF Operations HCX on a network underlay can be visualized as a set of nested pipes, where the underlay network is the main channel. VCF Operations HCX and non-VCF Operations HCX traffic is carried through main channel. Migration and network extension traffic can be thought of as separate pipelines through the site manager channel. The network extension pipe provides the throughput for all the extended network traffic. The migration pipe handles the vMotion, Bulk, and OS Assisted Migration service traffic.

Note:

The number of parallel migrations that are allowed depends on the bandwidth of the migration pipe.



User Account and Role Requirements

VCF Operations HCX configuration and operation requires an understanding of the various accounts and roles that are involved in deploying, managing, and operating the system.

User Accounts

VCF Operations HCX has the following account requirements for site manager deployments:

Account	Requirements	Additional Information
admin	Both the admin password and the root password must be set.	• Created during the site manager OVA deployment. • Used in HCX Manager Appliance Management (https://*hcx-ip-or-fqdn*:9443) • Used for CLI/terminal shell access.
Account for vCenter Registration	The account must belong to the vSphere administrators group and must have the Migration Admin role assigned.	• The administrator@vsphere.local account is suggested by default, but not required. • vCenter Single Sign-On or VCF Single Sign-on local users that meet the requirements can also be used. • Active Directory service accounts that meet the requirements can be used.
Account for NSX Registration	This account must have the Migration User role assigned.	• The NSX admin account is suggested by default. • Alternate NSX local accounts that meet the requirements can be used. • Active Directory service accounts that meet the requirements can be used. This account is not required for VCF Operations HCX source installations. It is required only when extending NSX Segments, or migrating NSX tags.

Account	Requirements	Additional Information
Accounts for Role Mapping (This account refers to local vCenter and VMware Cloud Foundation SSO user accounts that map to a VCF Operations HCX management role)	A user's group must be included in the VCF Operations HCX Role Mapping configuration before logging into VCF Operations HCX.	VCF Operations HCX supports three user roles: Migration Admin, Migration User, and Migration Appliance Admin. • The Migration Admin role is for users who configure and operate VCF Operations HCX. It allows users to create and modify network profiles, compute profiles, and service meshes. Also allows users to perform migrations and network extensions. • The Migration User role is for Service Provider installations only. It allows users to create and modify compute profiles and service meshes. Also allows users to perform migrations and network extensions. This role does not allow users to add or delete network profiles. • The Migration Appliance Admin role allows users to view and modify VCF Operations HCX Management Appliance configurations. VCF Operations HCX supports groups for local vCenter and for VCF Single Sign-On. • For vCenter users, the vsphere.local \Administrators vSphere SSO Group is added by default to the VCF Operations HCX Migration Admin Role. However, it is not mandatory to use this group. For the Migration User role, no default group is provided. A common practice is to create an hcx-administrators vSphere SSO Group. Local SSO and Active Directory users are populated into this group. The default vsphere.local \Administrators HCX Administrator user group entry in the Role Mapping configuration is replaced with the new hcx-administrators vSphere SSO group • For VCF Single Sign-On users, no default group is provided for either Migration User or Migration Admin roles. Instead, VCF Operations HCX fetches the list of groups from the VCF Single Sign-On identify provider.
VCF Single Sign-On Users (optional)	Single Sign-On across all components of a VMware Cloud Foundation platform.	Need to create separate OAuth Apps (Client ID and Secret): one for the registration of VCF Operations HCX with VCF SSO and another one for site pairing. See Configure VCF SSO.

Account	Requirements	Additional Information
Site Pairing Accounts	The user’s group must be included in the Role Mapping configuration on the remote VCF Operations HCX being paired. The user's group can be in either the Migration User or the Migration Admin group.	The site pairing user is entered along with the destination VCF Operations HCX URL in the site pairing configuration on the source system. The following are typical scenarios: • In a private data center deployment, the site pairing user is traditionally the administrative user for the destination environment. • In a dedicated public cloud deployment, the site pairing user is traditionally the SDDC administrator account provided to the tenant.

Note:

The vCenter and the NSX Manager registration accounts (“service accounts”) must have global object access.

Role Mapping

Access to VCF Operations HCX services and features depends on the assigned user role. User roles are initially assigned during VCF Operations HCX configuration. You can edit these roles later from the Appliance Management.

You can configure VCF Operations HCX for both local vCenter Single Sign-On and VCF Single Sign-On authentication methods. For each method, you can configure Migration Admin and Migration User roles.

Migration Admin	Single Sign-On groups assigned to the Migration Admin role have unrestricted access to perform all VCF Operations HCX configurations and operations.
Migration User	This role is intended for use by Service Providers. Groups assigned to the Migration User role cannot add or delete network profiles.

vSphere Privileges for Migration Operations

User groups that are assigned to the Migration User or the Migration Admin role must have these vCenter privileges to perform migrations.

vCenter Resource Type	User Privilege	Description
ComputeResource	• VirtualMachine.Inventory.Create • VirtualMachine.Interact.PowerOn • VirtualMachine.State.CreateSnapshot • VirtualMachine.State.RemoveSnapshot	Privileges required on the destination compute resource object when performing a migration operation.
HostSystem	• VirtualMachine.Inventory.Create • VirtualMachine.Interact.PowerOn • VirtualMachine.State.CreateSnapshot • VirtualMachine.State.RemoveSnapshot	Privileges required on the destination HostSystem object when performing a migration operation.
ClusterComputeResource	• VirtualMachine.Inventory.Create • VirtualMachine.Interact.PowerOn • VirtualMachine.State.CreateSnapshot • VirtualMachine.State.RemoveSnapshot	Privileges required on the destination ClusterComputeResource object when performing a migration operation.
ResourcePool	• VirtualMachine.Inventory.Create • VirtualMachine.Interact.PowerOn • VirtualMachine.State.CreateSnapshot • VirtualMachine.State.RemoveSnapshot	Privileges required on the destination ResourcePool object when performing a migration operation.

vCenter Resource Type	User Privilege	Description
Folder	• VirtualMachine.Inventory.Create • VirtualMachine.Interact.PowerOn • VirtualMachine.State.CreateSnapshot • VirtualMachine.State.RemoveSnapshot	Privileges required on the destination **Folder** object when performing a migration operation.
Datacenter	• VirtualMachine.Inventory.Create • VirtualMachine.Interact.PowerOn • VirtualMachine.Interact.PowerOff • Resource.HotMigrate • Resource.ColdMigrate • Folder.Create • Folder.Delete	Privileges required on the destination data center objects when performing a migration operation.
Datastore	• Datastore.UpdateVirtualMachineMetadata • Datastore.DeleteFile	Privileges required on the destination datastore objects when performing a migration operation.
DistributedVirtualPortgroup/Network	• Network.Assign	Privileges required on the destination network objects when performing a migration operation.
VirtualMachine	• VirtualMachine.Interact.PowerOn • VirtualMachine.Interact.PowerOff • Resource.HotMigrate • Resource.ColdMigrate • VirtualMachine.State.CreateSnapshot • VirtualMachine.State.RemoveSnapshot • VirtualMachine.Hbr.ConfigureReplication • VirtualMachine.Hbr.MonitorReplication	Privileges required on the source virtual machines when performing a migration operation.
Supervisor	• VirtualMachine.Namespaces.Transfer • Namespaces.Edit	Privileges required on the destination to import a migrated VM into Supervisor Namespace.
Encrypted VM Migration	• Cryptographer.Access • Cryptographer.Migrate • Cryptographer.RegisterVM • VirtualMachine.Supervisor.Transfer	
Supervisor Migration	• Cryptographer.Access • Cryptographer.Migrate • Cryptographer.RegisterVM • VirtualMachine.Supervisor.Transfer	
Supervisor Cluster Namespace for Supervisor Migration	Namespaces.Edit	

Configuration and Service Limits

When you are configuring, deploying, and operating VCF Operations HCX, you must stay within the supported limits.

It is important to be aware of the system and operational limits in the following categories:

- Sites and Service Components
- Migrations
- Network Extension
- Migration-centric Virtual Machine Limits

For a detailed list of the system limits, refer to the VMware Configurations Maximum tool.

Using the Interface

You can access VCF Operation HCX services and system-level operations through one of several interfaces.

Interface	Description
VCF Operations HCX	At both the source or destination site, you perform all operations related to VCF Operations HCX services by logging in to the VCF Operations HCX interface at https://hcx-ip-or-fqdn:443.
HCX Manager Appliance Management	At both the source or destination site, you perform system-level management operations by logging in to the Appliance Management interface at https://*hcx-ip-or-fqdn*:9443.
Central CLI	You access the Central Command Line Interface (CCLI) for debugging or troubleshooting issues with VMware representatives. For access to the CCLI, see Logging in to the Shell.

VCF Operations HCX Licensing

VCF Operations HCX service licensing happens during installation when the system detects a VMware Cloud Foundation (cores) license or the system enters evaluation mode.

About Licensing

VCF Operations HCX has a flexible licensing scheme that allows for multiple use cases and modes of operation. All VCF Operations HCX services are enabled by default for all license types.

License types

VMware Cloud Foundation (cores)

VMware vSphere environments that are licensed for VMware Cloud Foundation are entitled to all VCF Operations HCX services.

VCF Operations HCX systems that register with vCenter version 9.0 and higher automatically discover and inherit the VCF license. VCF Operations HCX systems that have registered with a non-VCF vCenter enter evaluation mode.

VCF Operations HCX polls the vCenter Server periodically for VMware Cloud Foundation license information. If a VMware Cloud Foundation site license is updated, or a vCenter Server becomes licensed for VMware Cloud Foudation, VCF Operations HCX discovers and inherits that license information.

VCenter Entitlement Server

VCF Operations HCX Managers that are registered with vCenter Servers running version 4.x and earlier must retrieve the entitlements from a vCenter Server that runs VCF Operations HCX 9.0 or later.

Evaluation Mode

If no license is discovered during installation, VCF Operations HCX enters Evaluation Mode. Evaluation Mode provides temporary entitlement to all VCF Operations HCX services.

Evaluation Mode is limited to 90 days from the day of installation, with a maximum of 20 virtual machine migrations per migration type. Evaluation mode ends after 90 days or when VCF Operations HCX detects a VMware Cloud Foundation license through monitoring of the vCenter Server.

If VCF Operations HCX is deployed at VMware vSphere Foundation or non-VCF sites without a valid license:

- The system enters into Evaluation Mode by default.
- VCF Operations HCX installations display a banner in the Appliance Management UI (:9443) to indicate the system is running in evaluation mode, which is an expected notification.
- Systems in evaluation mode can get feature inheritance (not the license inheritance) from its paired VCF Operations HCX .
- You can trigger all services from the source site appliance towards the paired destination site appliance which has entitlement to use those services.
- The expiration of a site license generates a warning message letting you know that a site is no longer active. This message can be ignored provided there is a valid license associated with the site pair that continues service inheritance.

	When the license validity expires for an VCF Operations HCX site, or the Evaluation Mode period ends, VCF Operations HCX feature inheritance continues to allow operations as long as one site in the site pair has a valid license.
	When Evaluation Mode ends, VCF Operations HCX enters Restricted Mode.
Grace Period	When a VMware Cloud Foundation license expires, VCF Operations HCX enters a grace period. VCF Operations HCX continues normal operation for the duration of this period. This length of this period varies with each VMware Cloud Foundation license subscription. During this time, users can renew their VMware Cloud Foundation license.
	When the grace period expires, VCF Operations HCX enters Restricted Mode.
Restricted Mode	In this state, the VMware Cloud Foundation license has expired and the grace period has ended. No additional operations are allowed for VCF Operations HCX.

For a detailed description of VCF Operations HCX Services, see System Services.

Service Inheritance

VCF Operations HCX provides a notion of inheritance to provide the most flexibility for sites that do not have VMware Cloud Foundation license entitlement or that are running in evaluation mode. With inheritance, VCF Operations HCX considers the entitlements that are associated with each licensed site during site pairing. Inheritance logic looks at what services have been enabled at each site and compares that with the entitlements associated with the site pair to generate a service mesh supporting those services. See Understanding Service Inheritance.

Installing New Systems

Installation requirements refer to any information required to set up a newly deployed VCF Operations HCX system.

VCF Operations HCX Installation

After you have deployed the VCF Operations HCX OVA file through the vSphere Client, VCF Operations HCX requires you to log in to the Appliance Management interface and perform the initial system configuration. For more information, see Configuring VCF Operations HCX.

> **Note:**
>
> By default, VCF Operations HCX installations participate in the Customer Experience Improvement Program (CEIP), which establishes a connection with Broadcom to collect site telemetry. See About the Customer Experience Improvement Program.

About the Customer Experience Improvement Program

The Customer Experience Improvement Program (CEIP) helps to improve VCF Operations HCX products and services, to identify and fix problems, and to develop best practices and procedures for deploying and using VCF Operations HCX.

Participation in the CEIP is provided by default. With CEIP, VCF Operations HCX periodically collects technical information about your organization's use of VCF Operations HCX and its services and provides that information to Broadcom. This information does not personally identify any individual.

For VCF Operations HCX sites participating in CEIP, perimeter firewalls must allow connections to `vcsa.vmware.com`. See Network Port and Protocol Requirements.

You can choose to leave the program at any time by selecting an option in VCF Operations HCX interface. For more information, see Leave or Join the Customer Experience Improvement Program.

Air-Gap sites do not participate in CEIP. Air-gap sites never transmit information to Broadcom. For more information, see Installing New Systems.

Leave or Join the Customer Experience Improvement Program

An VCF Operations HCX site can leave or join the Customer Experience Improvement Program (CEIP) at any time.

- Obtain the user name and password of the administrator account.
- Sites participating in the CEIP make an Internet connection to vcsa.vmware.com. When leaving or joining CEIP, remove or add firewall rules as necessary.

1. Log in to HCX Manager Appliance Management.
2. Navigate to the **Configuration** tab.
3. Select **CEIP**.
4. Click **Leave CEIP** or **Join CEIP**.

Understanding Service Inheritance

To provide the most flexibility for sites activated with different service entitlements, VCF Operations HCX provides a notion of inheritance.

VCF Operations HCX services operate between paired sites. Before mobility operations can take place, the system creates a service mesh between each site pair. The service mesh deploys the appliances and makes the service connections necessary to allow mobility operations. To complete the service mesh, the required services must be enabled in the compute profile at each site, and each site must have a license entitlement to use those services.

With inheritance, the logic to decide what a site can do is a function of the site pairing along with the entitlements granted to each site during activation. When creating the service mesh, the inheritance logic compares the entitlements applied at each site and then employs the broadest level of entitlement. After establishing the level of service entitlement for the site pair, the inheritance logic looks at each site's compute profile configuration to determine whether the service mesh supports the configured services.

> **Note:**
>
> Regardless of what services are configured in the compute profile, the service mesh for the site pair determines the entitlement to those services. For example, for any VCF Operations HCX site, you can configure the compute profile for all services, but if the service mesh for the site pair provides enablement of only some services, only that subset of services is allowed between sites. For site pairs that have entitlement to all VCF Operations HCX services, the compute profile provides a means to customize services between sites.

For a description of VCF Operations HCX Services, see System Services.

Refer to the following examples.

Example 1:

Site A is paired with Site B and Site C, creating a "V" topology. Site A was activated in Evaluation Mode, providing limited access to all VCF Operations HCX Services. Site B was activated using VCF Solution Licensing, giving entitlement to all VCF Operations HCX services. Site C was also activated in Evaluation Mode. In this example, Site A inherits the entitlements of Site B, but Site A inherits no additional entitlements from Site C. Site pairs A and B will operate in VCF mode. Site pairs A and C will operate in Evaluation Mode.

Example 2:

Sites B and D are paired. Site B was activated using VCF Solution Licensing, which provides entitlement to all VCF Operations HCX services. Site D was activated in Evaluation Mode, which supports all VCF Operations HCX services but with limitations. In this example, even though Site D is activated in Evaluation Mode with capacity limits on migrations, it inherits all the VCF entitlements since it is paired with a site licensed with VCF Solution Licensing and can perform mobility operations without any restrictions.

View Site License Information

You can view site licensing information and entitlement server information in HCX Manager Appliance Management.

1. Log in to the HCX Manager Appliance Management.
2. Select the **Configuration** tab.
3. Select **Licensing**.

   The system displays the VCF Operations HCX license status based on the vCenter license, including the license validity, or expiration date.

   The License status can have the following values:

License Status	Description
Active	Indicates VCF Operations HCX has active license entitlement based on the VMware Cloud Foundation license for the vCenter Server registered with VCF Operations HCX.
Evaluation Period	Indicates VCF Operations HCX is registered with a vCenter Server that does not have a VMware Cloud Foundation license, and VCF Operations HCX is running in Evaluation Mode.
Grace Period	Indicates VCF Operations HCX has reached the validity date for an active license, and is operating in the grace period.
Restricted Mode	Indicates the VCF Operations HCX has reached the end of the grace period for an active license or for the evaluation period, and no further mobility or network extension operations are permitted.

   Active license entitlements from a registered vCenter server configured with HCX are automatically displayed. If the vCenter Server registered with VCF Operations HCX does not have valid VMware Cloud Foundation entitlements, an alternate mechanism is to add a vCenter Entitlement Server that has active VMware Cloud Foundation entitlements.

Add a vCenter Entitlement Server

VCF Operations HCX provides an option to add a VCF 9.0 vCenter Server as a vCenter Entitlement Server in environments where no registered vCenter Server has an active VMware Cloud Foundation 9.0 license.

VCF Operations HCX automatically discovers the active VMware Cloud Foundation license entitlements from a registered vCenter Server that is configured in VCF Operations HCX. If the registered vCenter Server does not have valid VMware Cloud Foundation entitlements, VCF Operations HCX operates in Evaluation Mode with limited migration capability. In some scenarios, such as evacuating a legacy data center, you might not want to update the vCenter license, but need the extra migration capability provided by VCF Operations HCX with active VMware Cloud Foundation license entitlements. In this case, you can associate VCF Operations HCX with a vCenter Server that has an active VMware Cloud Foundation license.

VCF Operations HCX inherits the VMware Cloud Foundation license entitlements from the vCenter Entitlement Server, enabling VCF Operations HCX with the full range of mobility features and services.

> **Note:**
>
> Adding a vCenter Entitlement Server provides license entitlement only. Inventory objects that are associated with the vCenter Entitlement Server are not visible to VCF Operations HCX.

1. Log in to HCX Manager Appliance Management.
2. Navigate to the **Configuration** tab.
3. Select **Licensing**.

   If the VCF Operations HCX license status is Evaluation Mode, the system provides the option to add a vCenter Entitlement Server.
4. Click **Add**, and enter the vCenter Entitlement Server URL and credentials.
5. Click Save.

   The license status is updated to Active, and VCF Operations HCX is updated for full mobility operations.

NSX Requirements for VCF Operations HCX Deployments

In VCF Operations HCX installations connecting private environments, NSX must be installed and configured at the destination before deploying VCF Operations HCX. This section details the requirements.

NSX Requirements for Deployments

For environments requiring NSX virtual networking, you must install and configure NSX, including integration with vCenter, before deploying VCF Operations HCX.

- In the destination environment, the NSX Manager must be installed and integrated with vCenter. For a list of supported NSX versions, see the VMware Product Interoperability Matrix.
- The NSX Manager must be registered during VCF Operations HCX installation with the admin user.
- If the NSX Manager IP or FQDN uses self-signed certificates, it might be necessary to trust the NSX system manually using the Import Cert by URL interface in the HCX Appliance Management .
- VCF Operations HCX requires an NSX Manager registered with the same vCenter Server used to register with HCX Manager.
- When NSX is registered, both Overlay and VLAN segments can be used during the network profile creation.
- Deployment of VCF Operations HCX in environments with multiple data centers that are prepared for NSX and a single vCenter Server (sharing transport zones and datastores) is not supported.

Note:

The network extension service has additional NSX requirements. See Requirements for Network Extension.

NSX Requirements Source Site Installation

NSX is not required for VCF Operations HCX installation at the source site, but requirements apply if you are extending NSX networks. When extending any source network, the destination site must be configured with NSX.

For more information, see NSX Requirements for Deployments.

Installing VCF Operations HCX

This section describes how to install and configure HCX Manager at the source or destination sites.

The source and destination sites work together to provide VCF Operations HCX services. VCF Operations HCX requires installation of HCX Manager at both the source and destination sites.

Installation Workflow

A fully private VCF Operations HCX installation, where both the destination/modernized environment and the source/existing environments must be considered involves the following steps.

1. Prepare for deployment as described in Preparing for General Installation
2. Download the VCF Operations HCX OVA. See Download the OVA.
3. Deploy VCF Operations HCX in both the source and destination environment using the VCF Operations HCX OVA. See Deploy the OVA in the vSphere Client.
4. Configure firewall rules at source and destination sites to allow inbound and outbound connectivity based on the VCF Operations HCX services or features used. See https://ports.broadcom.com/home/VMware-HCX.
5. Configure VCF Operations HCX at both the source and destination sites. See Configuring VCF Operations HCX
6. Pair source and destination sites. See Add a Site Pair.
7. Configure the network profile and compute profile for both the source and destination VCF Operations HCX installations. See Configuring and Managing the Interconnect.
8. Configure VCF Operations HCX. See Configuring VCF Operations HCX.
9. Create the service mesh at the source site for the site pair. See Configuring and Managing the Interconnect.

If you have VCF 9.0 and later installed and are deploying VMware Operations HCX using VCF Operations, see https://techdocs.broadcom.com/us/en/vmware-cis/vcf/vcf-9-0-and-later/9-0/deployment/deploying-a-new-vmware-cloud-foundation-or-vmware-vsphere-foundation-private-cloud-.html.

About OVAs

You use the same OVA file to deploy VCF Operations HCX at the source and destination sites.

For private cloud deployments, you download the OVAs from the Broadcom Support Portal. From the Downloads site, you select the latest version of the VCF Operations HCX OVA.

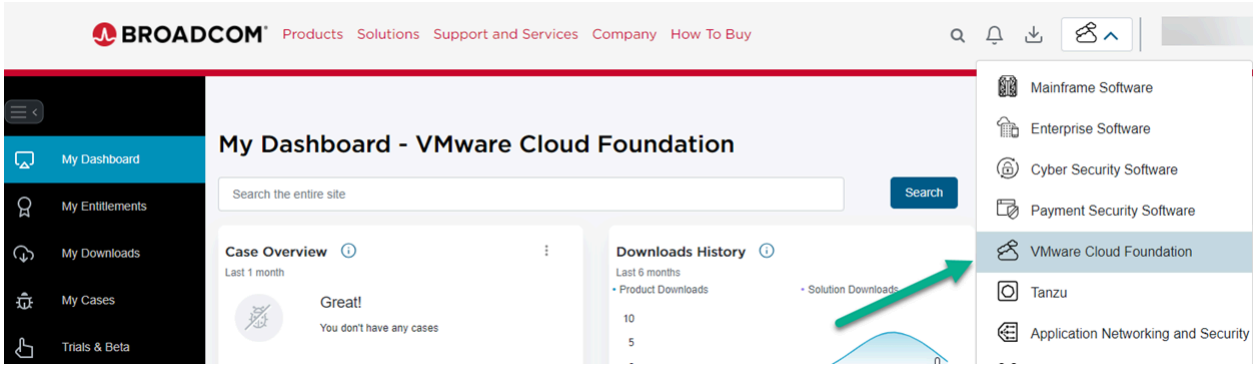
In hosted environments, the provider may deploy and install VCF Operations HCX.

Download the OVA

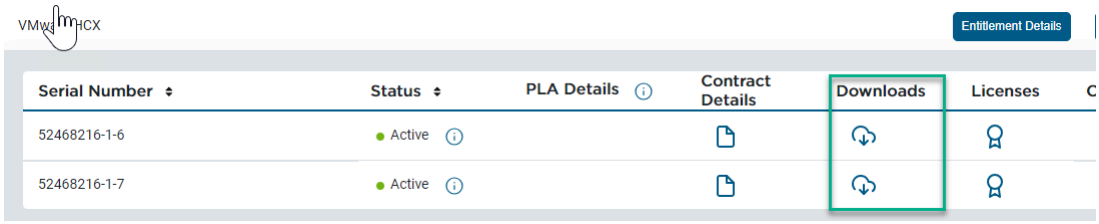
The OVA file is used for deploying VCF Operations HCX.

Use the following procedure to download the VCF Operations HCX OVA.

1. Log in to `https://support.broadcom.com`.
2. From the drop-down menu, select **VMware Cloud Foundation**.



3. In the navigation pane, select **My Entitlements**.
4. Expand your site and navigate to the VCF Operations HCX entitlement.
   Under entitlement, the screen displays an option to download binaries.



5. Select a download option to display the list of bundles available for download.
6. Select a release, and download the OVA file.

Deploy the downloaded OVA in the vCenter instance.

Deploy the OVA in the vSphere Client

Use this procedure to deploy the downloaded VCF Operations HCX OVA using a standard OVA template installation through the vSphere Client.

Use the downloaded OVA to install the latest version of VCF Operations HCX at the sites that will serve as the source and destination of mobility operations.

1. Log in to the vSphere Client.
2. Right-click any inventory object that is a valid parent object of a virtual machine, such as a data center, folder, cluster, resource pool, or host, and select **Deploy OVF Template**.
3. Enter a URL to download and install the OVF package from the Internet. Alternatively, browse to a location accessible from your computer, such as a local hard drive or a network share for a local file, and click **Next**.
4. Enter a unique virtual machine name and the inventory location, and click **Next**.
5. Select a compute resource location, and click **Next**.
6. On the Review details page, verify the OVA template details, and click **Next**.
7. On the License agreements page, read and accept the VMware End User License Agreement, and click **Next**.
8. On the Select storage page, select the virtual disk format, storage policy, storage name, and then click **Next**.
9. On the Select networks page, select the destination network, and click **Next**.
10. On the Customize Template page, set the appropriate deployment properties:
    - **Passwords** - Configure the CLI and Appliance Management admin password and the root user password.
    - **Network Properties** - Enter the network properties such as IP address, netmask, and default gateway.
    - **Static Routes** - Add a static route for a destination subnet or host.
    - **DNS**
      - **DNS Server List** - Enter the list of DNS servers for this virtual machine.
      - **Domain Search List** - Domains that you enter are searched in the order you list them, and the search stops when a valid name is found.

      Note:

      Add the .local subdomain in environments without multicast DNS correctly configured as in RFC-6762.
    - **Services Configuration**
      - **NTP Server List** - Enter the list of NTP servers and ensure that the NTP server can be reached from the virtual machine. If the NTP time is out of sync, services fail to start.

      Note:
      During OVA deployment, use “space” or “comma” separation for multiple NTP/DNS/Domain Name servers.
11. Click **Next**.
12. Review the deployment settings and click **Finish**.

Allow up to 5 minutes for initialization, then log in to the Appliance Management for the initial configuration.

To complete the VCF Operations HCX installation, see Configuring VCF Operations HCX.

Support for NSX Federation

NSX Federation provides a centralized management plane that is known as an NSX Global Manager (GM) which provides consistent networking across physical sites, and an NSX Local Manager (LM). The LM is a local NSX instance in each data center that receives policies from the GM. This set-up is known as a federated NSX environment.

When you configure VCF Operations HCX, you register with an LM. Registration with a GM is not supported. However, the NSX LM that you select during configuration can be registered to an NSX GM.

In federated NSX environments, the NSX GM populates the NSX LM with global network segments. In VCF Operations HCX, these global segments have the following uses and restrictions:

- With network extension, such global networks cannot be extended.
- You can create network profiles using global segments. However, network profiles that are backed by global segments can only be used in the compute profile for assignment as the Guest Network. They cannot be used to configure Management, Uplink, vMotion, Replication Assisted vMotion services.
- You can perform a migration that uses global segments as the destination network.

When the NSX LM is not registered to an NSX GM, Interconnect constructs such as network profiles, compute profiles, and service meshes are created on local segments. If the NSX LM is registered at a later point with an NSX GM, ownership of the segments transitions to the GM, effectively converting local segments to global segments. As a result:

- Backing for the network profile becomes inconsistent, affecting both the network profile and the compute profiles. This inconsistency means that the only service mesh operation that can be performed on existing service meshes is service mesh deletion.
- Established network extensions between sites continue to work, but you cannot extend a new network or cannot enable/disable Mobility Optimized Networking (MON) for an already extended network. Also, the only operation that you can perform on extended networks in a federated environment is Force Unextend.

Configuring VCF Operations HCX

After you have deployed the OVA file in the vSphere Client, perform the initial configuration immediately when you next open HCX Manager Appliance Management.

- Deploy the OVA in the vSphere Client. Allow up to five minutes after the OVA deployment for the services to initialize.
- Configure firewall rules at source and destination sites to allow inbound and outbound connectivity based on the VCF Operations HCX services or features used. See https://ports.broadcom.com/home/VMware-HCX.

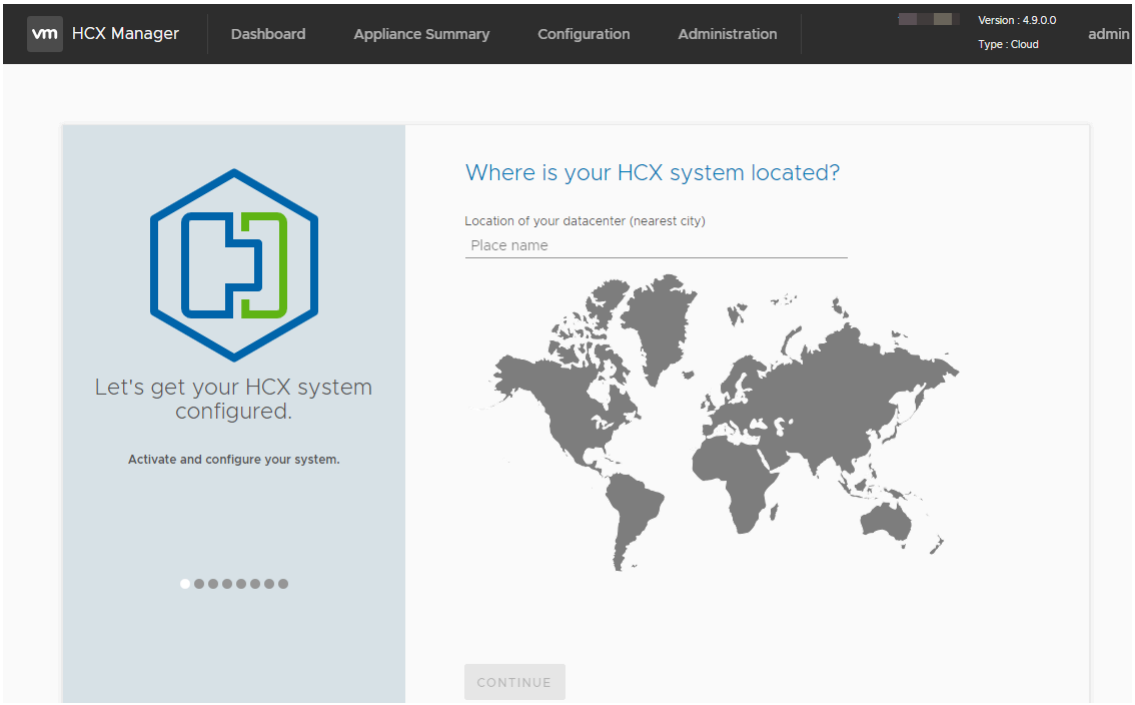
During configuration, you register with NSX. You can only register an NSX Local Manager and not an NSX Global Manager. You might find it useful to review how VCF Operations HCX supports NSX federation. See, Support for NSX Federation

1. Log in to the Appliance Management using admin user credentials.
2. (Optional) If there is a proxy server in the environment in the path for outbound HTTPS connections, configure it now:
   a) Select the **Administration** tab, and navigate to **Network Settings** > **Proxy Configuration**.
   b) Click **Edit**.
   c) Enter the proxy server information, including any exclusions.

      Important:

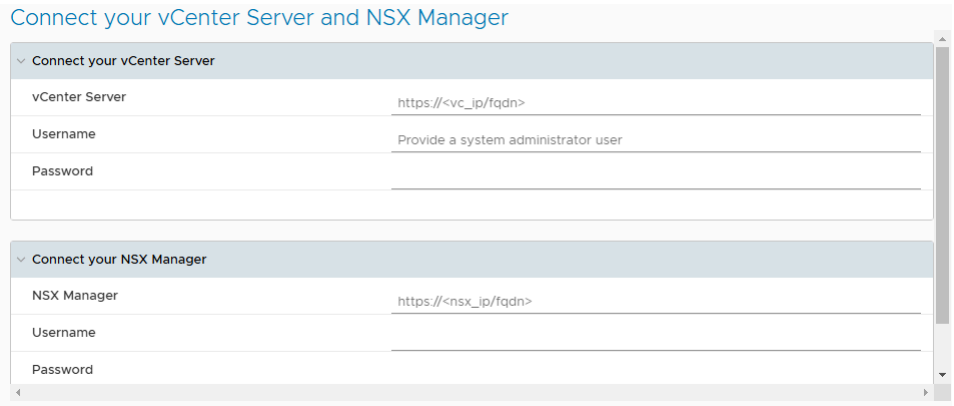
      Add the local vCenter, ESX, NSX, SSO, VCF SSO, and service mesh appliances as exceptions not to be sent to the proxy server.
   d) Make any required changes to the system settings specified when deploying VCF Operations HCX. These changes can include NTP and DNS settings. For more information see, Deploy the OVA in the vSphere Client.
   e) Select the **Configuration** tab to continue the activation procedure.
3. Enter the location where you are deploying the system.

   Select the nearest major city to where the system is geographically located. VCF Operations HCX sites are represented visually in the Dashboard.



4. Click **Continue**.
5. Enter the system name, and click **Continue**.
6. Select the cloud instance to which to connect VCF Operations HCX . This is an extended network on the destination site.

   VCF Operations HCX can connect to only one cloud instance per deployment.
7. Click **Continue**, and enter the required information.
8. If you selected the vSphere instance, enter the configuration details:
   a) Enter the vCenter Server and NSX Manager details and click **Continue**.



   b) When prompted for a certificate, click **Import Certificate**.
   c) Enter the vCenter Single Sign-On Platform Services Controller (PSC) URL to provide local login authentication, and click **Continue**.
   d) Configure VCF SSO by entering identity broker information and client information. This step is optional during the initial VCF Operations HCX installation and can be done at any time after installation. See, Configure VCF SSO.
   e) Click **Submit**.
   f) Enter the Public Access URL.

      This is typically the VCF Operations HCX IP address: https//*hcx-fqdn-or-ip*.

9. Click **Continue**.

   The system verifies the configuration and then displays a configuration summary screen.

10. Click **Restart** to reinitialize the system.

    It can take several minutes to reinitialize the system completely. During this process, the Appliance Management is not available.

11. Continue setup and Configure VCF SSO and VCF Operations HCX roles.

    For either local authentication or VCF SSO authentication, you must assign roles and user groups. Role mapping is mandatory to log into VCF Operations HCX. See Configure HCX Role Mapping

Configure any other VCF Operations HCX sites, and then access Creating and Managing Site Pairs and follow the procedure to add a site pair.

Configure VCF SSO

Configuring VCF Single Sign-on (SSO) provides login access to VCF Operations HCX using the VMware Identity Provider credentials.

Prior to configuring VCF SSO, Admins must create an OAuth application in VCF Operations that specifies the Client ID and Client Secret. To manage the identity source, go to the VMware Cloud Foundation Operations console. See, https://techdocs.broadcom.com/us/en/vmware-cis/vcf/vcf-9-0-and-later/9-1/fleet-management/managing-identity-and-access-using-vcf-single-sign-on/what-is/setting-up-sso/cofigure-vmware-cloud-foundation-identity-provider/configure-identity-provider-for-okta/configure-the-identity-provider.html

VCF Operations HCX allows for two authentication methods: local and VCF SSO. By default, installing VCF Operations HCX requires registration with vCenter. VCF Operations HCX uses vCenter for local authentication.

Optionally, you can configure VCF Operations HCX for login authentication using the VMware Identity Provider, which provides single sign-on access for VCF Operations HCX and for other VCF components.

> **Note:**
>
> You can add or delete VCF SSO authentication at any time without affecting VCF Operations HCX or mobility operations.

1. Log in to the Appliance Management.
2. Select the **Configuration** tab and navigate to **VCF SSO**.
3. Click **Configure** and enter the configuration details:
   - Identity Broker Issuer (example):
     https://<fqdn of VIDB>/acs/t/<tenant name>
   - Client ID (fetched from VCF Operations Console)
   - Client Secret (fetched from VCF Operations Console)
4. Click **Test Connection** to validate the inputs.
5. Click **Save**.

   If prompted for the certificate, click **Import Certificate**.

6. To view the VCF SSO configuration, click the **Dashboard** tab and review the VCF SSO tile.

After configuring VCF SSO, you must assign or map Identity Provider User Groups to HCX Roles. To map these user groups, see Configure HCX Role Mapping.

Configure HCX Role Mapping

Access to VCF Operations HCX requires user authentication at log in, which requires configuration of user roles and groups.

During VCF Operations HCX configuration and activation, you must configure authentication with the local vCenter Server. Optionally, you can configure VCF Operations HCX for VMware Cloud Foundation SSO (VCF SSO) authentication. Regardless of the authentication method, you must assign user groups and roles.

For an overview of VCF Operations HCX user accounts and roles, see User Account and Role Requirements.

1. Log in to the Appliance Management.
2. Select the **Configuration** tab and navigate to **HCX Role Mapping**.
3. Select the authentication type and configure the roles and groups:
   a) For local authentication, select the **vCenter Users** tab and click **Edit.**

      Assign the HCX Roles to the vCenter groups and users that are allowed to perform VCF Operations HCX operations. By default, the Migration Admin role is mapped to the local vSphere administrator group. The Migration User role is intended for use by Service Providers. Groups assigned to this role cannot add or delete network profiles.
   b) For VCF SSO authentication (if configured), select the **VCF SSO Users** tab and click **Edit**.

      Assign roles to the groups that are allowed to perform VCF Operations HCX operations. VCF Operations HCX fetches the list of groups from the configured VCF SSO identify provider. For both the Migration Admin role and the Migration User role, select a list of groups from the dropdown menu.
4. Click **Save**.

The assignment of the roles and user groups is now complete. Log in to HCX Manager at `https://hcx-ip-or-fqdn:443` to verify VCF Operations HCX log-in access. If VCF SSO is configured, follow the prompts to the identity provider's log in page and enter the VCF SSO credentials. At any time, you can use the dropdown menu on the Log in page to select local authentication.

Access Bearer Token Authentication for API Access

As a federated SSO user in VCF Operations HCX, you access APIs by generating an API token and then exchanging the token for an Access Bearer token.

API tokens are generated in VCF Identity Broker, and for ease of access, you can start the process of generating your API token from the VCF Operations HCX UI.

1. In the VCF Operations HCX UI, click the **User Profile** icon and select **Generate API Token** from the dropdown menu.
2. In the VCF Identity Broker page, click **Generate**.
3. In VCF, exchange the token for an OAuth token, see https://techdocs.broadcom.com/us/en/vmware-cis/vcf/vcf-9-0-and-later/9-1/administration-sdks-cli-and-tools/about-vmware-cloud-foundation-development/oauth-token-support-for-api-and-cli-access.html.

For more information about OAuth token support in VCF, see https://techdocs.broadcom.com/us/en/vmware-cis/vcf/vcf-9-0-and-later/9-1/administration-sdks-cli-and-tools/about-vmware-cloud-foundation-development/oauth-token-support-for-api-and-cli-access.html.

Proprietary Header-Based Authentication Access to HCX APIs

You generate an Authorization token to access the HCX APIs by posting a request to the HCX API server.

Verify that you have the log-in credentials for the registered vCenter Server Single-Sign-On (SSO) user.

Verify that the user group to which the SSO user belongs is mapped to an HCX role.

1. POST a request to the authentication URL to acquire a token.

   A successful request returns an "x-hm-authorization" token in the Response Headers that you must include in subsequent API requests.

2. Request header:
   a) `POST https://hcx-1.example.com/hybridity/api/sessions`
   b) `Content-Type: application/json`
   c) `Accept: application/json`
3. Request body in JSON format:
   a) `{ "username": "testuser@vsphere.local", "password": "testpassword"}`

      These credentials should be of the registered vCenter Server SSO user.
4. Response headers:
   a) `x-hm-authorization 13956794:e912:4edd:bbee:fea0fa257691`
5. The Authorization token is valid for 30 minutes after generation. If the token expires, you must regenerate it.

VCF Operations HCX Pairing Interoperability Policy

You must pair VCF Operations HCX on the source site with VCF Operations HCX on the destination site running the same major version. For example, pair 9.0 with 9.1, 9.2 with 9.0, and so on.

In cases where a lower minor version is paired with a higher minor version, the functionality of the service mesh is reduced to the common denominator between versions. For best results and feature set compatibility, run the same minor version on both the source and the destination sites.

Migrating standalone legacy environments to VCF 9

A legacy environment is considered as an environment running standalone legacy versions of vCenter, vSphere, NSX-V, NSX-T, or NSX. VCF Operations HCX 9 is able to migrate workloads from legacy environments to VCF 9.

For the specific versions of these legacy environments, see the Interoperability Matrix.

Support of VCF Operations HCX 9 for legacy environments that are included in the Interoperability Matrix is maintained throughout the lifecycle of VCF 9.

There is a special pairing policy for HCX 9 to assist customers with migration operations from legacy environments to VCF 9.

You deploy and pair VCF Operations HCX 9 on the source (legacy) and destination (VCF 9) sites. After pairing is complete and HCX 9 is licensed with a VCF 9 entitlement, entitlement inheritance enables VCF Operations HCX 9 on the source to inherit the entitlement from the licensed VCF Operations HCX 9 on the destination that is deployed within VCF 9. You don’t need to enter a license on the source VCF Operations HCX.

> **Important:**
>
> VCF Operations HCX 9 is the last release that supports migrations of legacy environments.

Migrating between legacy environments using VCF Operations HCX 9

In cases where your destination site is not VCF 9, you can still use VCF Operations HCX 9 on both sites to migrate workloads.

When you pair VCF Operations HCX 9 across non-VCF 9 sites, you must configure an Entitlement Server on VCF Operations HCX 9 on one of the sites. An Entitlement Server is a vCenter 9 instance that is deployed and licensed within VCF 9. Then, through entitlement inheritance, the other VCF Operations HCX 9 inherits the entitlement from the connected VCF Operations HCX 9.

For the specific versions of environments on the source and destination site, see the Interoperability Matrix.

> **Important:**
>
> VCF Operations HCX 9 is the last release that supports migrations of legacy environments.

Entitlement Inheritance

For sites that have been activated with different entitlements, the VCF 9 license is the active entitlement on both sites. For example, if the source site has an HCX Manager in evaluation mode and the destination site has a VCF 9 entitlement through a VCF 9 depoyment, the VCF license is the active entitlement on both sites.

Entitlement inheritance only works for the duration of the site pairing and is independent for each pairing. Taking from our example, if the HCX Manager in evaluation mode that inherited the VCF 9 entitlement, is also paired with another HCX Manager in evaluation mode, this second pairing will remain in evaluation mode until either HCX Manager reaches 90 days after activation.

Migrating environments post-VCF 9

The support policy for the next major releases after VCF 9 is one of long-term support, meaning support for the current and the LTS release of the previous major version. Under this policy, VCF Operations HCX 9 is the last release that supports legacy environments.

For legacy environment considerations, see the Interoperability Matrix.

Creating and Managing Site Pairs

This section provides the procedures for creating and managing site pairs.

You create, configure, manage, and view site pair information from the site pairs interface in the HCX Manager.

Understanding Sites

VCF Operations HCX installations, or sites, which are used in site pairing can be vSphere-based or non-vSphere based.

Understanding vSphere-Based Paired Sites

A vSphere-based site is a vSphere installation where VCF Operations HCX has been deployed and activated. For these sites, VCF Operations HCX supports a full range of management and mobility services when paired with other vSphere-based sites.

For workload migration within the vCenter instance registered with HCX Manager, VCF Operations HCX provides a local site.

Some VCF Operations HCX sites serve as the source of mobility and network extension operations while other sites serve as the target of VCF Operations HCX operations.

For each vSphere site, you must configure a compute profile (CP) and network profile (NP). The CP and NP define the resources that VCF Operations HCX uses to deploy the appliances for management and data plane connectivity. They also specify the networks to use for the management and the mobility operations. The CP and NP information is used for creating the service mesh, which is required for each site pair.

With VCF Operations HCX deployments, site pairing can be unidirectional or bidirectional:

- In unidirectional site pairing, the HCX Manager containing the virtual machine inventory and networks pairs with the HCX Manager at the destination site. In this type of site pairing, all service mesh operations, Migration, and network extension operations, including reverse migrations, are always initiated from the source system. In this case, an administrator might see the message `URL not available` when viewing site pairing from the destination site. This is expected behavior because the site pairing is unidirectional.
- In bidirectional site pairing, network extension from the destination site can only be initiated if there is a service mesh that is deployed from the destination site. The URLs for the paired sites are visible from either the source or destination.

Understanding vSphere-Based Local Sites

A local site refers to the vCenter instance that is registered with HCX Manager. VCF Operations HCX supports the HCX-Assisted vMotion (HAV) service for local site so you can migrate workloads within a single vCenter Server using the associated HCX Manager. You do not need to configure any other appliances or services to perform a local site migration.

The HCX-Assisted vMotion (HAV) service is the only service that is enabled for the compute profile and service mesh. The local site must be selected as both the source and destination site to run HAV migrations.

VCF Operations HCX supports migrations within local sites when vCenter and NSX are registered with HCX Manager. If the version of the vCenter Server that is registered with HCX Manager is less than 9.0, an entitlement server must be added to HCX to run HAV migrations for the local site. For information about the VCF Operations HCX pairing policy, see VCF Operations HCX Pairing Interoperability Policy.

To view local site pair details, navigate to **Infrastructure** > **Site Pairs**, and select the **Local Site** tab.

A single local site pair, with a compute profile, and service mesh are set up by default.

Understanding non-vSphere based sites

Non-vSphere based sites do not run vSphere and VCF Operations HCX is not installed in those environments. Examples of non-vSphere based sites include sites running on non-vSphere hypervisors like KVM and Hyper-V. A typical scenario for pairing non-vSphere based sites is when you migrate Guest OS workloads to VMware vSphere-based environments using OS Assisted Migration (OSAM).

For non-vSphere based sites, the process of adding a site pair generates a "pseudo site" that appears in the site pairs interface along with any vSphere-based sites. A non-vSphere site has no installed VCF Operations HCX from which to gather management information, so the site pairs interface provides only limited information about the pseudo-site, including the site name and type.

Unlike vSphere-based sites, non-vSphere sites require no compute profile or network profile configuration, and site management operations are not applicable to non-vSphere based sites. Non-vSphere sites are immediately ready for site pairing.

> **Important:**
> The compute profile configuration for any vSphere-based site that is paired with a non-vSphere site must include OSAM. The service mesh configuration selected for pairing a vSphere-based site with a non-vSphere site must include the Guest Network. This network is used for migration of the non-vSphere workloads.

Add a Site Pair

A Site Pair establishes the connection that is needed for management, authentication, and orchestration of VCF Operations HCX services across a source (vSphere or non-vSphere) and destination environment.

- For vSphere sites, VCF Operations HCX is deployed and activated in the source and destination environments.
- Site URL and user roles:
  - When the destination is a vSphere-based private cloud, the Site URL refers to the HCX Manager at the target site:
    `https://hcx-ip-or-fqdn`
    Provide a user from the destination/target site's local SSO or VCF SSO configuration. The user must be included in the VCF Operations HCX Role-Mapping Group configuration.
    The `administrator@vsphere.local` user is included by default.
  - When the destination system is a hosted environment, the Site URL can use a trusted domain name pre-created by the provider or an IP address.
    Use the credentials for a user that holds a cloud admin role in your private cloud.
- For vSphere-based site pairs, the destination site URL must use a CA signed trusted certificate or be manually trusted on the source system. See Managing CA and Self-Signed Certificates.

For vSphere-based sites, site pairing is between a source VCF Operations HCX and destination VCF Operations HCX.

For non-vSphere sites, pairing is between a non-vSphere site and a VCF Operations HCX installation.

1. From the VCF Operations HCX dashboard, navigate to **Infrastructure** > **Sites**.
2. Select the **Paired Sites** tab, and click **New Site Pair**.
3. Select the Site Type for the remote site:
   a) For vSphere-based site pairing, select **VMware**.
   b) For non-vSphere site pairing, select **Other**.
4. Select the authentication type.

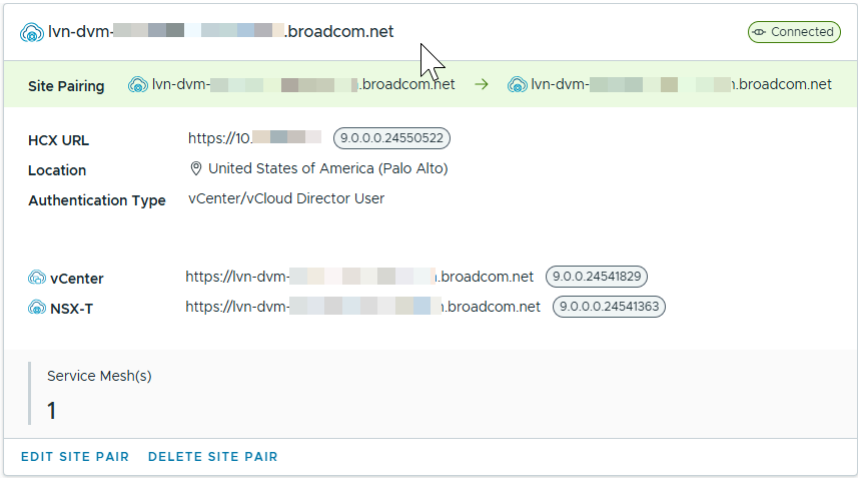
   For local authentication with the vCenter Server, select **vCenter**.

   For authentication with the VMware identity Provider as configured for VMware Cloud Foundation, select **VCF SSO**.
5. Enter the paired site information:
   a) For sites with vCenter authentication, enter the remote VCF Operations HCX site URL, username, and password, and credentials.

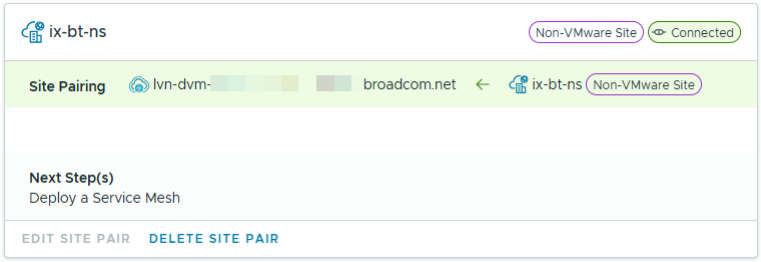
      For sites with VCF SSO authentication, enter the remote VCF Operations HCX URL, client ID, and client secret set up when configuring VCF SSO. This should be a different OAuth application to the one configured on the remote HCX Manager.
   b) For non-vSphere site pairing, enter a pseudo-site name and an optional description.
6. Click **Create**.
7. (Optional) To achieve bidirectional site pairing in vSphere-based VCF Operations HCX installations, repeat this procedure at both sites.

If all validations succeed, the system displays the remote site with the status "Connected". With bidirectional site pairing, both sites show up in the list.

When pairing vSphere-to-vSphere sites, the display indicates that the sites are connected with an arrow indicating the direction of mobility operations is to the destination site. The display includes the destination site URL, location, vCenter, and NSX server information.



When pairing vSphere with non-vSphere sites, the pseudo-site indicates that the sites are connected with an arrow indicating the direction of mobility operations is from the non-vSphere site. Because VCF Operations HCX is not deployed at the non-vSphere site, no other information about the site is available for display.



Create a network profile and compute profile for each vSphere-based site and then apply a service mesh to the site pair. See Configuring and Managing the Interconnect.

Delete a Site Pair

To disconnect paired sites, you must delete the site pair.

- Verify that all mobility operations are complete.
- Unextend networks between sites.
- Delete all service meshes associated with the site pair.

Removing a site pair does not affect individual site configurations. The source and the destination sites comprising the site pair retain compute profile (CP) and network profile (NP) settings.

1. In VCF Operations HCX, click **Site Pairs**.
2. Select a paired site, and click **Delete Site Pair**.
3. Click **Delete**.

Edit a Site Pair

VCF Operations HCX provides an option for editing the credentials of a paired site, including changing the authentication type.

Editing the site pair credentials has no effect on mobility or network extension operations.

> **Note:**
>
> You cannot edit pseudo sites that are associated with non-vSphere site pairs.

1. In the HCX console, click **Sites**.
2. Select a paired site, and click **Edit Site Pair**.
3. (Optional) Change the authentication type: vCenter User or VCF SSO.

   The current authentication type is selected by default.
4. Enter the new credentials for destination VCF Operations HCX .

   For sites with vCenter authentication, enter the username and password.

   For sites with VCF SSO authentication, enter the client ID and client secret setup when configuring VCF SSO. This should be a different OAuth application than is configured on the remote HCX Manager.
5. Click **Edit**.

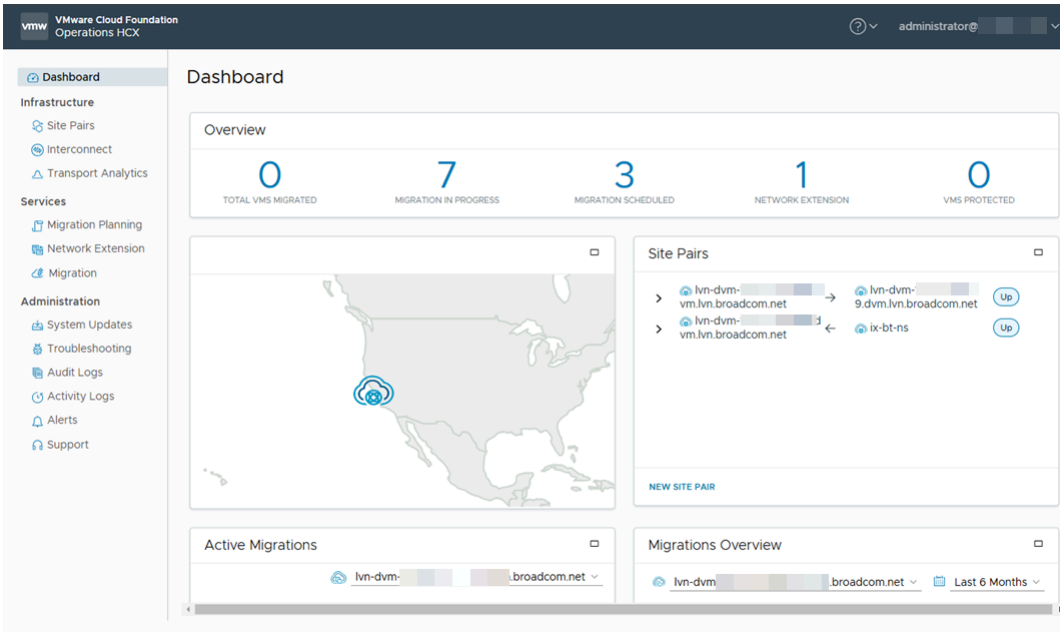
   VCF Operations HCX displays a message indicating the update is in progress.

Viewing Site Pair Information

Information regarding site pairs appears in the VCF Operations HCX Dashboard, site pairs interface, and Interconnect interface.

View Site Pair Information in the Dashboard

When you launch VCF Operations HCX, the Dashboard is the default display that opens in the VCF Operations HCX console. The Dashboard provides at-a-glance visibility into VCF Operations HCX operations, including a summary listing of paired sites. For more information, Understanding the VCF Operations HCX Dashboard.



View Summary Information for a Site Pair

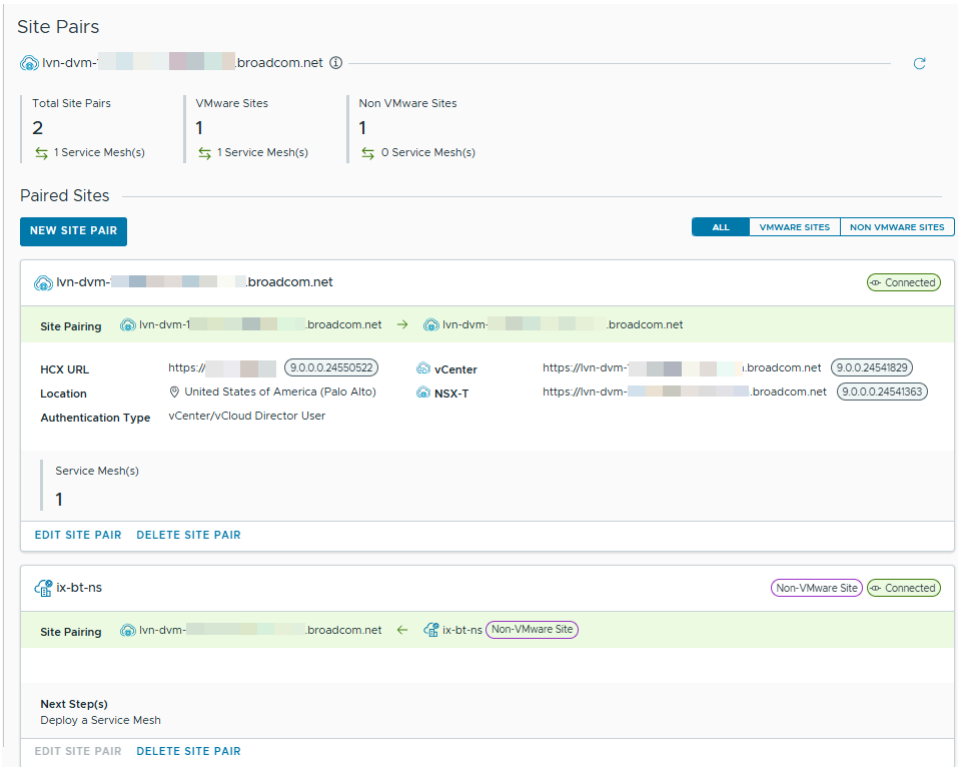
The site pairs interface displays summary information for HCX Manager indicating the total number of paired sites, the number of vSphere-based paired sites, non-vSphere paired sites, and associated service meshes. To access the site pairs interface, navigate to **Infrastructure** > **Site Pairs**.

The site pairs interface provides a separate card for each paired site. Each card lists general information about each site including the Site URL, physical location, registered vCenter version, registered NSX version, Authentication Type, and number of service meshes established for the Site Pair.

> **Note:**
>
> Because a non-vSphere site has no installed HCX manager from which to gather information, the site pairs interface provides only limited information about the pseudo-site, including the site name and type.

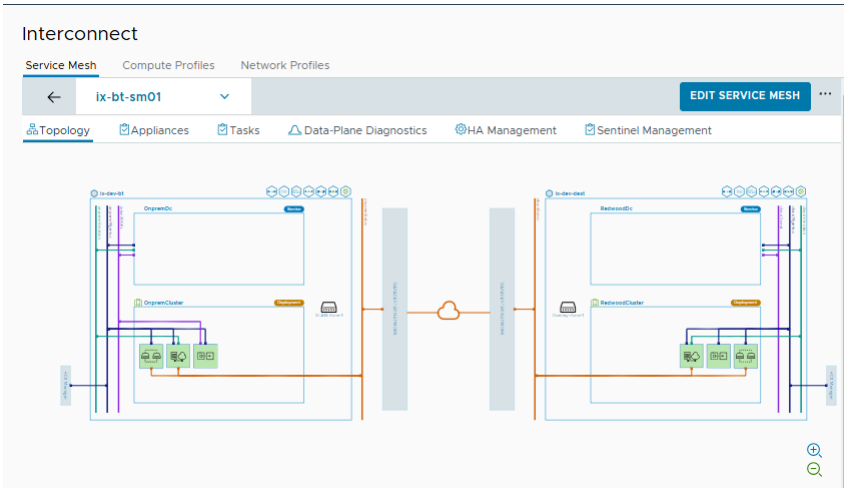
A directional arrow on each site card indicates the direction of mobility operations relative to the VCF Operations HCX .

Each card includes the connection status between sites: Connected or Disconnected.



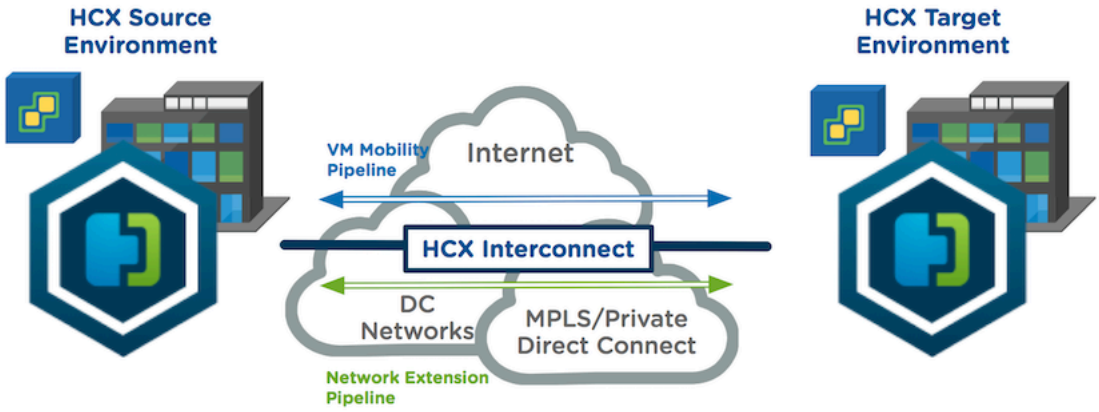
View the Service Mesh Topology for a Site Pair

For a topology view of the service mesh configuration and connections that are associated with a Site Pair, navigate to **Infrastructure** > **Interconnect** and expand the service mesh selection to display the **Topology** tab.



Configuring and Managing the Interconnect

The VCF Operations HCX Interconnect provides a secure pipeline for migration and extension between two connected VCF Operations HCX sites.



Overview of the Interconnect Services Deployment

VCF Operations HCX services are deployed and managed using the service mesh.

The HCX Interconnect appliance creates and secures connections between paired sites, and deploys the appliances supporting management, mobility, and network extension operations. All of these connections and appliances are established and deployed between paired sites using the service mesh.

Configuring the Service Mesh

The service mesh is used to create a secure optimized transport fabric between two paired sites.

The procedures for creating a service mesh vary based on whether the site pairing is between two vSphere-based VCF Operations HCX installations, or between a vSphere-based site and a non-vSphere site. See About the Service Mesh.

About the Service Mesh

When VCF Operations HCX Migration and network extension services are enabled, HCX Manager deploys the associated virtual appliances in the source site and corresponding "peer" virtual appliances on the destination site. The service mesh activates the configuration, deployment, and serviceability of these Interconnect virtual appliance pairs.

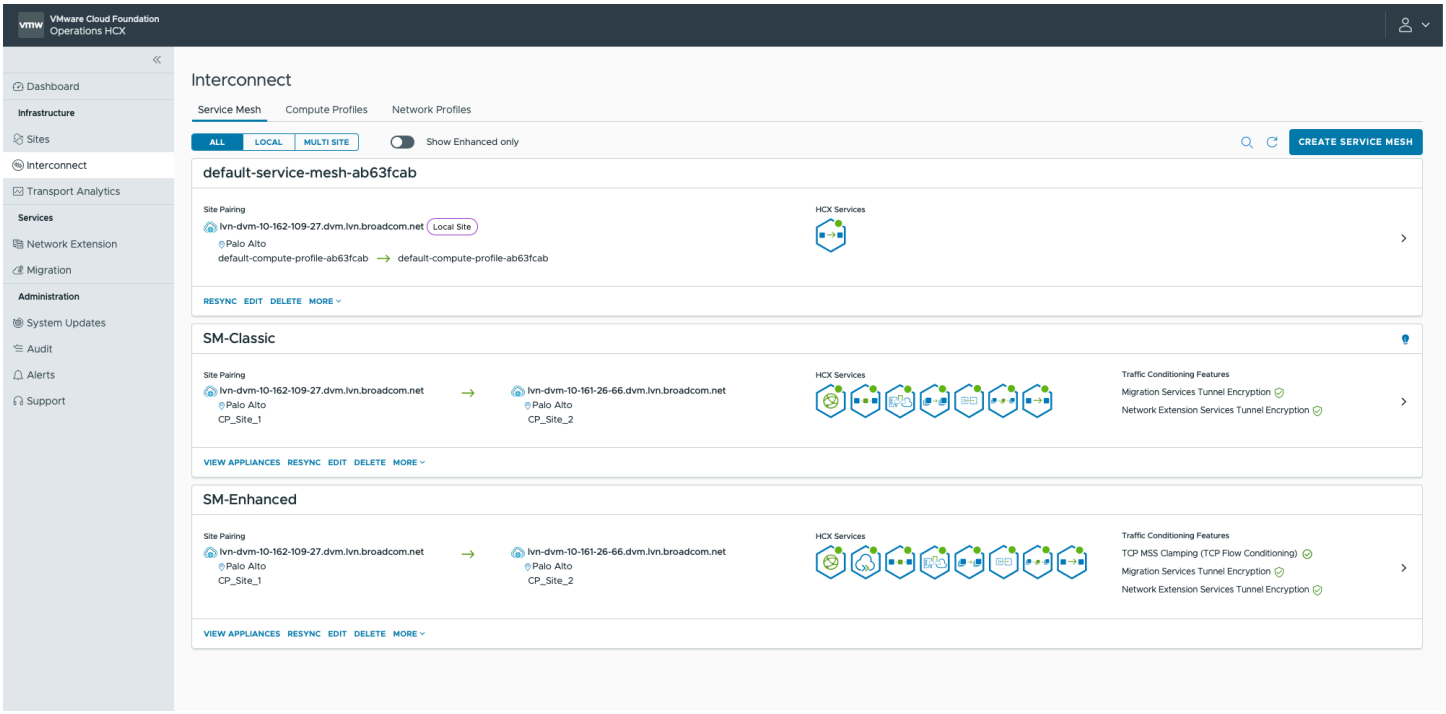
Service Mesh Components

The service mesh logic determines the transport fabric and defines what services are associated with a site-pair based on a number of component factors:

Network Profile	Defines what networks are used for connecting virtual appliances and for VCF Operations HCX management and data plane operations.
Compute Profile	Provisions the infrastructure at the source and the destination site. Defines the placement details (Resource Pool, Datastore) where the system places the virtual appliances.

Note:
The integrated compute profile creation wizard can be used to create the network profiles (or network profiles can be pre-created).

The service mesh, compute profile, and network profile information is accessible by selecting **Interconnect** in the HCX Manager UI.


VMware Cloud Foundation
Operations HCX
Dashboard
Infrastructure
Sites
Interconnect
Transport Analytics
Services
Network Extension
Migration
Administration
System Updates
Audit
Alerts
Support
Interconnect
Service Mesh
Compute Profiles
Network Profiles
ALL
LOCAL
MULTI SITE
Show Enhanced only
CREATE SERVICE MESH
default-service-mesh-ab63fcab
Site Pairing
lvn-dvm-10-162-109-27.dvm.lvn.broadcom.net
Local Site
Palo Alto
default-compute-profile-ab63fcab
default-compute-profile-ab63fcab
HCX Services
RESYNC EDIT DELETE MORE
SM-Classic
Site Pairing
lvn-dvm-10-162-109-27.dvm.lvn.broadcom.net
Palo Alto
CP_Site_1
lvn-dvm-10-161-26-66.dvm.lvn.broadcom.net
Palo Alto
CP_Site_2
HCX Services
Traffic Conditioning Features
Migration Services Tunnel Encryption
Network Extension Services Tunnel Encryption
VIEW APPLIANCES RESYNC EDIT DELETE MORE
SM-Enhanced
Site Pairing
lvn-dvm-10-162-109-27.dvm.lvn.broadcom.net
Palo Alto
CP_Site_1
lvn-dvm-10-161-26-66.dvm.lvn.broadcom.net
Palo Alto
CP_Site_2
HCX Services
Traffic Conditioning Features
TCP MSS Clamping (TCP Flow Conditioning)
Migration Services Tunnel Encryption
Network Extension Services Tunnel Encryption
VIEW APPLIANCES RESYNC EDIT DELETE MORE

To show additional service mesh selections click the right navigation arrow to expand the display.

You can also click **More** and select an option from the dropdown menu. The available options depend on your setup.

View Topology	Displays a graphical representation of the service mesh connections and services. Click an entry in the display for additional details.
Appliances	Lists details about the source and destination appliances that are deployed for each service mesh. For information regarding the management options available after selecting this tab, see Manage Service Mesh Appliances.
View Tasks	Provides a detailed list of tasks associated with service mesh operations.
Run Diagnostics	Run data plane diagnostics to troubleshoot issues in your service mesh. See, Understanding Data Plane Diagnostics.
Migrate Service Mesh	Migrate the service mesh to access enhanced services such as support for WAN Optimization and improved performance for the network extension service. Selecting this option opens the migration wizard which runs a series of checks, creates a new, enhanced service mesh, migrates the existing network extensions, and removes the existing service mesh. Migrating the service mesh is possible when HCX Manager in both the local and remote location is updated to the latest version of VCF Operations HCX. This option is also accessed by clicking the light bulb icon located on the right side of the pane containing service meshes that can be migrated.
HA Management	Displays information and appliances related to network extension High Availability groups. See, Understanding Network Extension High Availability.
Sentinel Management	Provides options for downloading and managing the Sentinel Agent software, which must be installed on guest workloads migrated using OS Assisted Migration. This option appears when the OS Assisted Migration service is selected in the service mesh configuration. See, Sentinel Management.

Creating a Service Mesh

The process for creating a service mesh varies depending on whether the site pairing is with vSphere-based sites or with non-vSphere sites.

For vSphere-based site pairs, the service mesh is always applied at the source site. Follow these general steps for creating a service mesh for paired vSphere-based sites:

1. Pair VCF Operations HCX sites. See Creating and Managing Site Pairs.
2. Create a network profile for the source and destination sites. See Create a Network Profile.
3. Create a compute profile for the source and destination sites. See Create a Compute Profile.
4. Create the service mesh for the Site Pair. See Create a Service Mesh for vSphere-Based Site Pairs.

HCX Interconnect service appliances are only deployed when a service mesh is created.

> **Note:**
>
> Installing VCF Operations HCX requires registration with vCenter (source and destination sites) and VMware NSX Manager (mandatory for the destination site). After VCF Operations HCX installation, configuring the service mesh deploys the HCX data plane appliances necessary to run VCF Operations HCX services: HCX migration (HCX-IX) and HCX network extension (HCX-NET-EXT) services.

For non-vSphere site pairs, apply the service mesh at the vSphere-based VCF Operations HCX site for the site pair. Follow these general steps for creating a service mesh for non-vSphere site pairs:

1. Pair the non-vSphere site. See Creating and Managing Site Pairs.
2. Create a network profile for the vSphere-based site. See Create a Network Profile.
3. Create a compute profile for the vSphere-based site. See Create a Compute Profile.
4. Create the service mesh for the Site Pair. See Create a Service Mesh for Non-vSphere Site Pairs.

Create a Network Profile

The network profile is an abstraction of a distributed port group, standard port group, or NSX Logical Segment, and the Layer 3 properties of that network. A network profile is a sub-component of a complete compute profile.

- VCF Operations HCX is deployed and configured for each site.
- Use the planned network configurations prepared using the checklist that is described in the *Getting Started with VCF Operations HCX* guide.

Create a network profile for each network you intend to use with the VCF Operations HCX services. You select these network profiles when creating a compute profile.

For information about how VCF Operations HCX interaction with NSX federation might affect your network profile, see Support for NSX Federation

Note:

Although a network profile can be assigned any of the functions during the compute profile configuration, consider creating a separate profile for each function as a best practice.

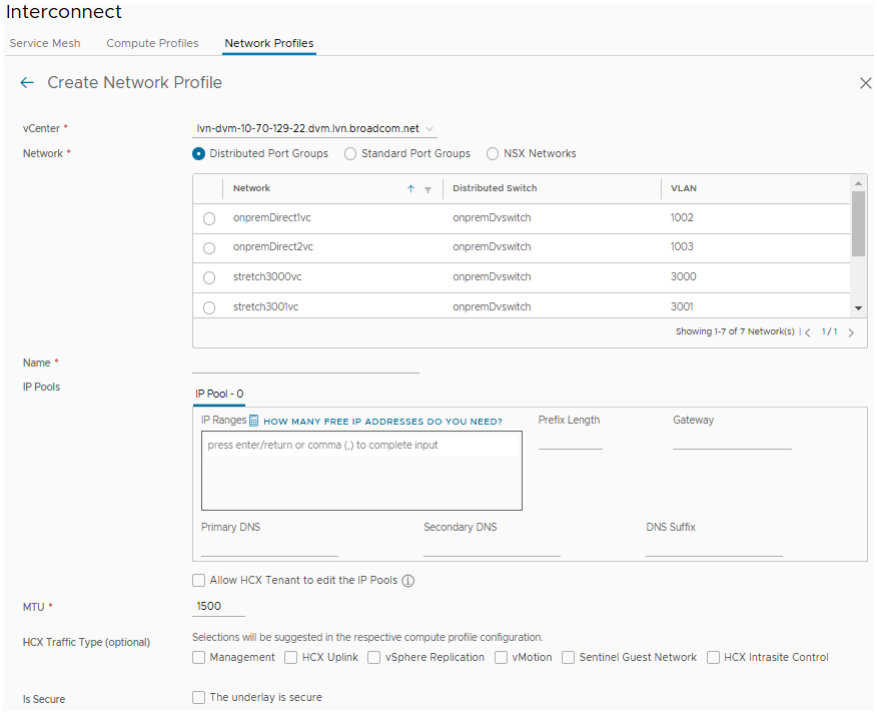
Network	Description
Management	VCF Operations HCX data plane appliances use this network to communicate with management systems like HCX Manager, vCenter, ESX Management, NSX Manager, DNS, and NTP.
Uplink (External)	VCF Operations HCX interconnect (HCX-IX) appliances use this network for WAN communication between source and destination sites.
vMotion	VCF Operations HCX interconnect (HCX-IX) appliances use this network for the traffic exclusive to vMotion protocol operations. **Note:** Configuring this network does not include the vMotion NFC traffic. HCX always uses its Management interface for vMotion NFC traffic.
vSphere Replication (Bulk Migration Service)	VCF Operations HCX interconnect (HCX-IX) appliances use this network for the traffic exclusive to vSphere Replication. **Note:** In deployments where ESX use a dedicated VMkernel configuration for vSphere Replication services, the HCX Interconnect appliance uses a network profile configuration dedicated to the vSphere Replication traffic. This configuration does not include the vSphere Replication NFC traffic. VCF Operations HCX always uses its Management interface for vSphere Replication NFC traffic.
Sentinel Guest Network (for OS Assisted Migration)	The Sentinel Gateway appliances use this vSphere network to connect with non-vSphere virtual machines.

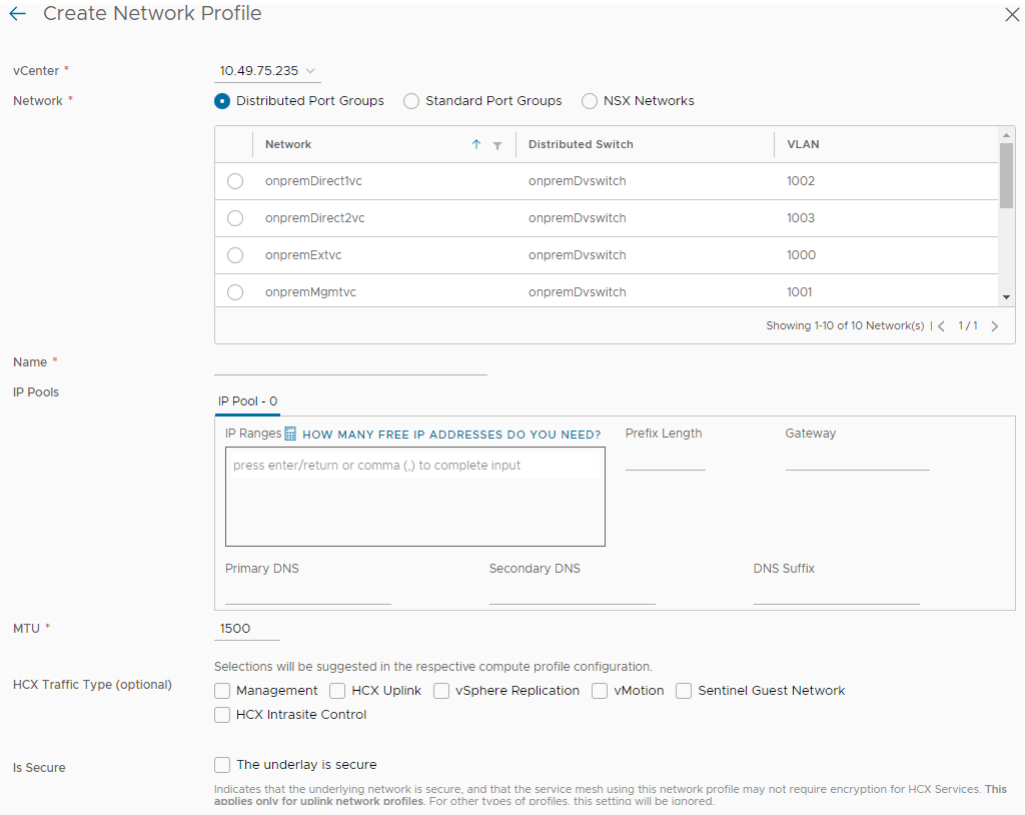
Important:

When creating a separate network profile for vMotion or vSphere Replication services, although the option is available to configure a GW as a standard network profile, traffic for those services use only the default GW in the Management network profile to attempt to access resources in a different subnet. If ESX resources are not L2 adjacent to the IX appliance on those networks, there is a requirement to configure “Static Routes” as part of the Advanced Configurations option in the compute profile to ensure traffic is directed to the default GW on those networks.

1. Log in to VCF Operations HCX using an account with Administrator privileges.
2. Navigate to **Interconnect** > **Network Profiles**.
3. Click **Create Network Profile**.
4. Select a vCenter instance and existing Network.
   a) Select a vCenter instance from the drop-down menu.
   b) Select **Distributed Port Groups**, **Standard Port Groups**, or **NSX Networks**, to filter the available networks by type.
   c) Select one of the available networks.

      To view additional networks, use the pagination at the bottom of the displayed list.





5. Name the network profile.
6. Provide the IP address pool details for the network profile.
   a) Provide an IP address range available for the HCX appliances. Use a comma to separate multiple discontiguous ranges within the same subnet.
   b) Enter the **Prefix Length** for the network containing the IP ranges provided.
   c) Enter the **Default Gateway Address** for the network.
   d) Specify the DNS server information.
7. (Optional) Allow users with VCF Operations HCX Tenant role to edit the IP Pools section of the network profile.
8. Enter the MTU value.
9. (Optional) Associate one or more suggested traffic types with the network selection: Management, HCX Uplink, vSphere Replication, vMotion, Sentinel Guest Network.

   The traffic type selection appears as a suggestion of which networks to use when creating the compute profile. It does not prevent the network from being used for other types of network traffic.
10. (Optional) Verify the underlying network is secure.

    This setting applies to Uplink networks only. For other types, this setting is ignored. It indicates that the underlying network might not require encryption for a given VCF Operations HCX service. You might check this box if the Uplink network is part of a fully private data center or when using direct connect networks.

    This setting works with the service mesh Traffic Engineering selections and is required before deactivating encryption for network extension and migration services. If not selected, network extension and migration data traffic are encrypted by default.
11. To complete the network profile configuration, click **Create**.

Go to Create a Compute Profile.

Note:
To edit an existing network profile, navigate to the specific network profile and click **Edit**.

Create a Compute Profile

A compute profile contains the compute, storage, and network settings that VCF Operations HCX uses on this site to deploy the Interconnect-dedicated virtual appliances when creating a service mesh.

- Install and configure VCF Operations HCX.
- To obtain the optimum system usage, assign resource configurations based on VCF Operations HCX deployment considerations.
- Use the planned configurations collected using Preparing for General Installation.
- To deploy WAN-OPT, make sure that you are using enhanced service meshes. Enhanced service meshes are created when VCF Operations HCX and the HCX Manager on both the local and remote site is at version 9.1.0 and later.
- To deploy the HCX NET-EXT appliance in NSX-enabled environments, the NSX Transport Zone must be associated to all hosts that are part of the deployment cluster. Perform this action in addition to the NSX Overlay or NSX VLAN network extension configuration part of the Network Containers section.

VMware Operations HCX provides a default compute profile which includes only the HAV service and has all data centers grouped into a single service boundary. This compute profile, together with a default service mesh is provided to support local site migration. For more information, see Understanding Sites.

Create a compute profile in both the source and the destination VCF Operations HCX environments using the planned configuration options for each site.

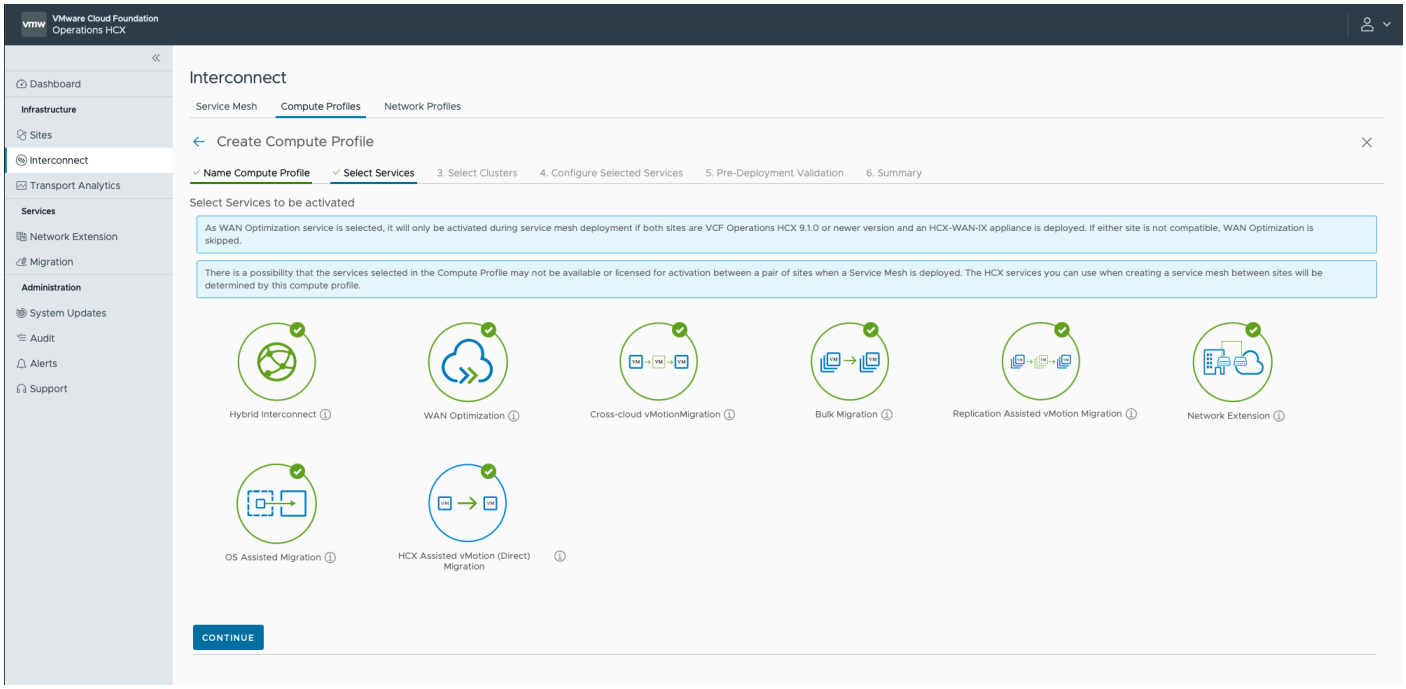
You might find it useful to read the content in the Create a Network Profile topic for more in depth information about some of the steps in this procedure.

> **Note:**
>
> You cannot configure the network extension service in the compute profile if one of the VCF Operations HCX sites is federated. See, Support for NSX Federation

1. Log in to VCF Operations HCX with an account with Administrator privileges.
2. Navigate to **Interconnect** > **Compute Profiles**.

   The system displays all the defined **compute profiles**. If no profiles have been configured, the system highlights the **Create Compute Profile** option.

3. Click **Create Compute Profile**.
4. Enter a name for the compute profile, and click **Continue**.
5. Select the services to be activated, and click **Continue**.

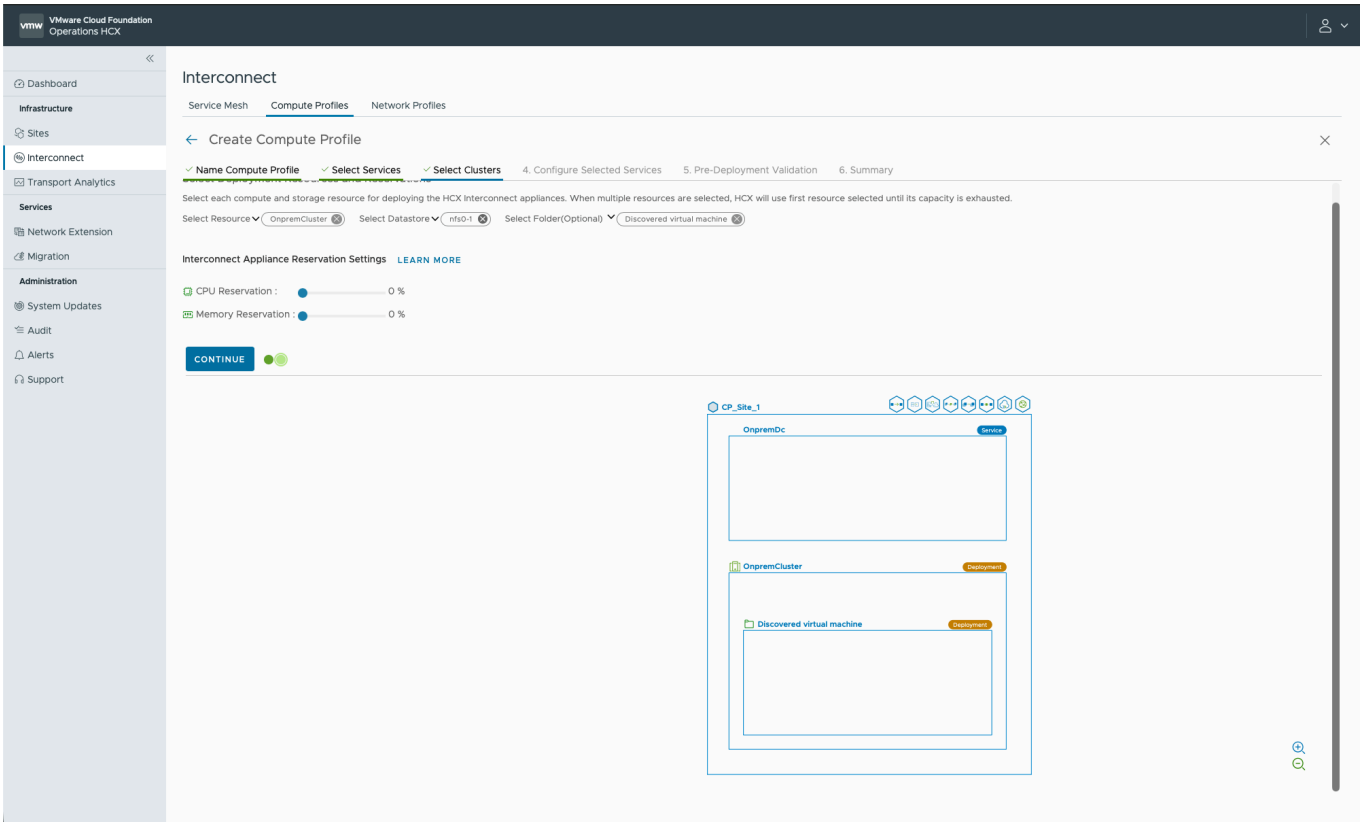


6. Select the Service Resources:
   a) Click the **Select Resources** drop-down menu.
   b) Select each cluster to be activated for VCF Operations HCX services.

      If there is only one cluster, it is selected automatically.
   c) Click **Continue**.
7. Make your resource and resource reservation selections.
   a) From the **Select Resource** drop-down menu, select each cluster or resource pool to be used when deploying HCX Interconnect appliances.
   b) From the **Select Datastore** drop-down menu, select the datastore to be used when deploying HCX Interconnect appliances.

      When multiple compute resources or datastores are selected, VCF Operations HCX uses the first selection until its capacity is exhausted.
   c) (Optional) From the **Select Folder** drop-down menu, specify a folder in which to deploy the HCX appliances.
   d) Use the slider to select the amount of CPU and memory to reserve for VCF Operations HCX operations, and click **Continue**.

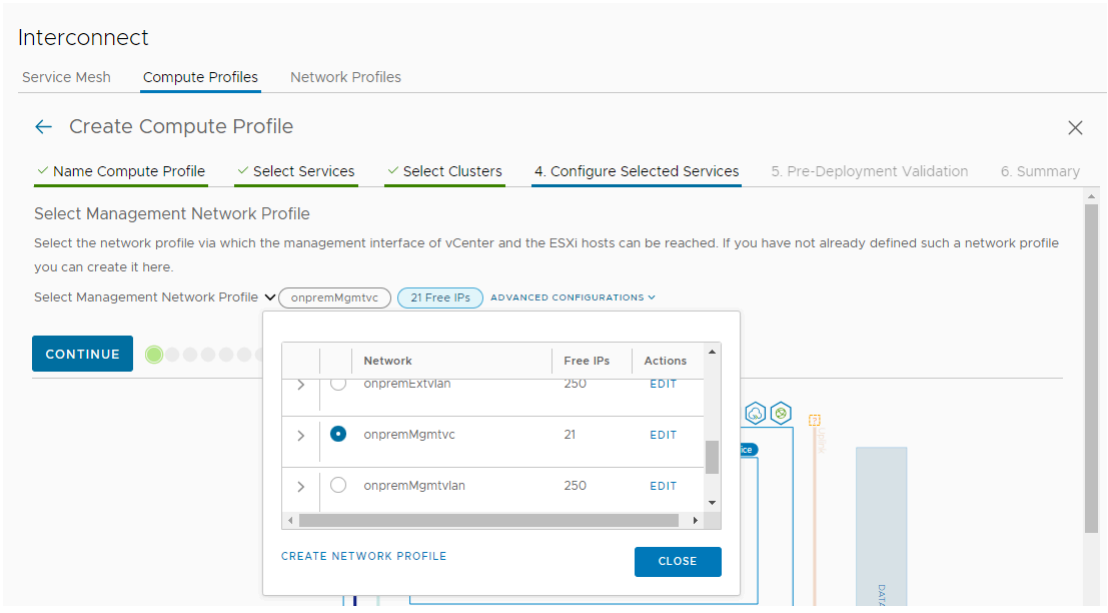
      As a best practice, set the CPU and memory reservation to 100 percent.For example, setting **Memory Reservation** to 100 percent ensures that all of the memory allocated for HCX appliances is always available for VCF Operations HCX operations.



e) Click **Learn More** to display information about the appliance resource requirements.

8. Select the Management network profile:

a) Click the **Select Management Network Profile** drop-down menu.



b) Select an existing network profile or click **Create Network Profile** to create it.

Networks identified with an information icon have been suggested for use with this type of network in the network profile. This is only a suggestion, and you can select other networks for this traffic type.

c) Expand the selected Management network profile to view its details and free IP Addresses. Click **Close** when done reviewing.
d) Click **Continue**.

9. Select the Uplink network profile:

   - The network profile previously selected for another function, like Management can also be assigned as the Uplink network profile.

   a) Click the **Select Uplink Network Profile** drop-down menu.
   b) Select one or more existing network profiles, or click **Create Network Profile** to create to create a new one.
   c) Expand the selected Uplink network profile to view its details and free IP Addresses. Click **Close** when done reviewing.
   d) Click **Continue**.

10. Select the vMotion network profile:
    a) Click the **Select vMotion Network Profile** drop-down menu.
    b) Select an existing network profile or click **Create Network Profile** to create one.
    c) Expand the selected vMotion network profile to view its details and free IP Addresses. Click **Close** when done reviewing.
    d) Click **Continue**.

11. Select the vSphere Replication network profile:

    Assigning a vSphere Replication network profile is useful when there is a VMkernel interface for the network traffic that is exclusive to vSphere Replication operations. If the Management network profile is used for Replication operations, click **Continue** to skip this step.

    a) Click the **Select vSphere Replication Network Profile** drop-down menu.
    b) Select an existing network profile or click **Create Network Profile** to create it.
    c) Expand the selected vSphere Replication network profile to view its details and free IP Addresses. Click **Close** when done reviewing.
    d) Click **Continue**.

12. If HCX OS Assisted Migration is activated, select the Guest network profile for OS Assisted Migration:

    - This is the network on which guest virtual machines communicate with the HCX sentinel appliances OS Assisted Migration.
    - The **network profile** previously selected for another function, like Management can also be assigned as the **Guest Network Profile**.
    - A global segment backed network profile can be used as the Guest Network.

      Note:

      This step appears only in the source compute profile interface, and only if you have selected OS Assisted Migration from the list of available services.

    a) Click the **Select Guest Network Profile** drop-down menu.
    b) Select a network, or click **Create Network Profile** to create it.
    c) Expand the selected **Guest Network Profile** to view its details and free IP Addresses, and click **Close**.
    d) Click **Continue**.

       The Guest Network is depicted in the topology.

13. Select **Network Containers Eligible for Network Extension**:

    This setting defines the networks which are part of selected network container that can be selected for network extension.

Note:

VCF Operations HCX supports configuring these source container types: NSX Overlay Zone, NSX VLAN Zone, and VDS. These source networks can be extended to any of these destination network types: NSX Overlay Segment, NSX VLAN Segment, and VPC Subnet.

VCF Operations HCX supports configuring these destination container types: NSX Overlay Zone and NSX VLAN Zone.

For more information see Requirements for Network Extension.

a) Click the **Select Network Containers** drop-down menu.

The system displays the list of network containers found in the service resources selected in a previous step.

b) Select entries with virtual machine networks. Click **Close**.

You can select VDS, Transport Zones, or some combination for network extension. If only one network container is found in the previously selected service resources, it is pre-selected.

Note:

Only the virtual machine networks on the selected switches or Transport Zones can be used during network extension operations.

c) Click **Advanced Configurations** and set the number of network extension appliances that can be deployed for this container.

If this number is not known, select Unlimited. This number impacts the number of network extension appliances that can be deployed through the service mesh. If the number is too low, it can prevent the service mesh from deploying needed appliances. If the number is too high, it is possible to deploy more network extension appliances than the planned resources can support.

d) Click **Continue**.

14. Pre-Deployment Validation:

VCF Operations HCX checks whether the selected configurations are valid for interconnect deployments.

a) Address any errors reported by the validation.
b) To see which firewall rules might be required if the service mesh uses this service profile, click **Review Connection Rules**.
c) To export the rules, click **COPY ALL** to copy them to the clipboard in the JSON format.
d) Click **Continue**.

15. Ready to Complete:

a) Review the configuration. The topology diagram depicts the selected configurations.
b) To create the compute profile, click **Finish**.

A compute profile is created, and can be used when creating a service mesh.

Once there are valid compute profiles in the source and destination environments, create the service mesh.

Create a Service Mesh for vSphere-Based Site Pairs

A service mesh provides the services configuration for a Site Pair. This section describes the procedure for creating a service mesh for a vSphere-based VCF Operations HCX site pair.

Creating a service mesh requires:

- A connected site pair.
- A valid compute profile at the source and destination site.
- Virtual Switches (for network extension) and network profiles selected in each compute profile must span across all hosts in every selected deployment cluster. If these objects do not span all hosts, it is possible that the appliances will

be deployed in a host missing the needed networks. In this case, the service mesh deployment can fail or services may not function properly.

Note:

To create a service mesh between a vSphere-based VCF Operations HCX site and a non-vSphere site, see Create a Service Mesh for Non-vSphere Site Pairs.

VMware Operations HCX provides a default service mesh which includes only the HAV service and has all data centers grouped into a single service boundary. This service mesh, together with a default compute profile is provided to support local site migration. For more information, see Understanding Sites.

Adding a service mesh initiates the deployment of HCX Interconnect virtual appliances on both of the sites. A service mesh is always created on the source site.

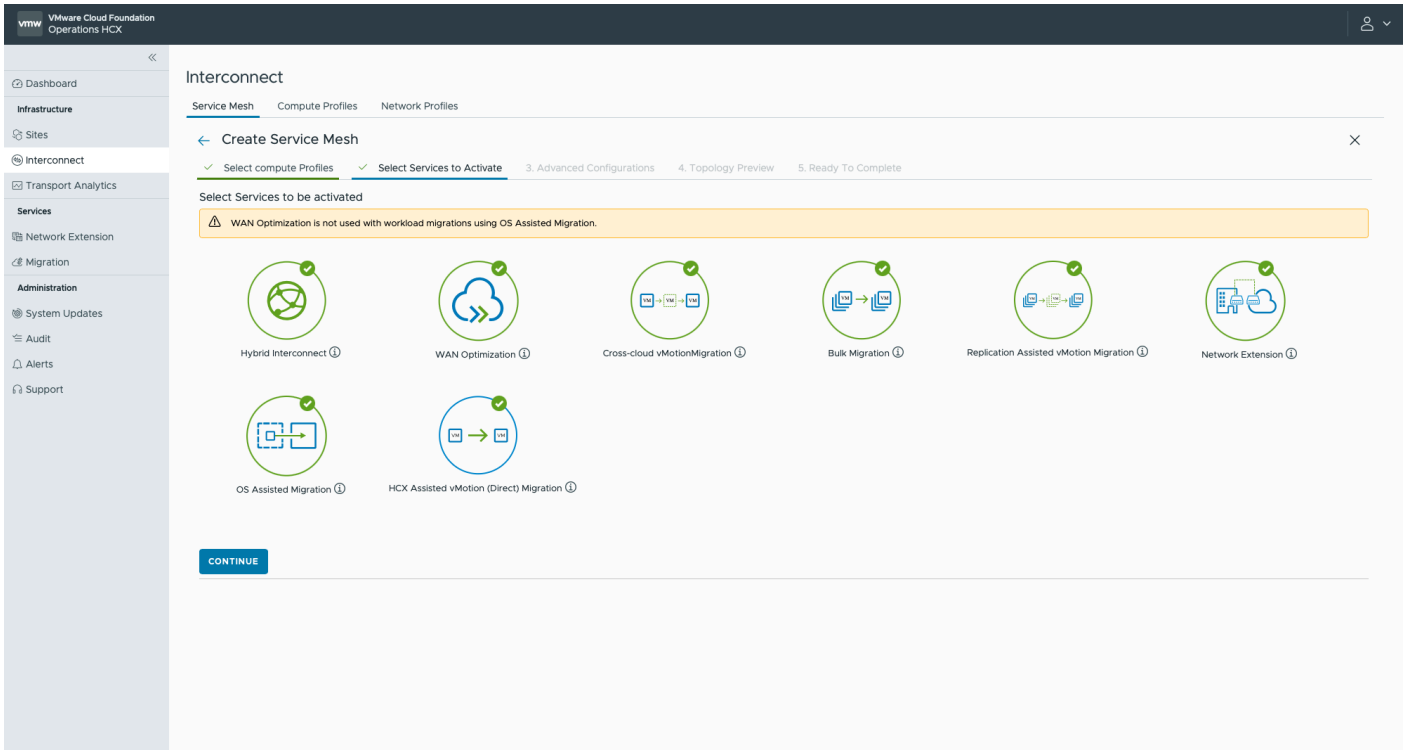
Note:

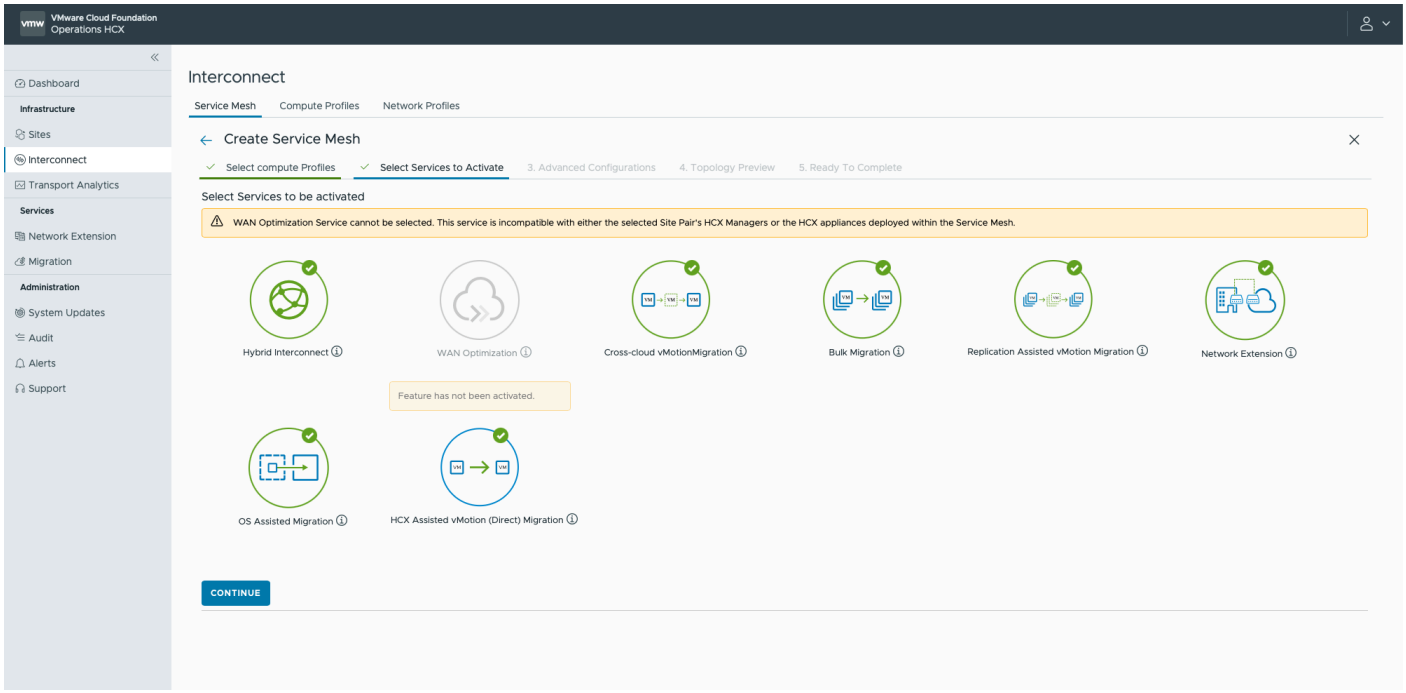
VCF Operations HCX sites require registration with NSX Local Manager at installation time. Selecting a compute profile for the service mesh requires that network profiles used in compute profile configuration use local segments. This affects both the network profile and the compute profile, which in turn disrupts service mesh operations except for service mesh deletion.

You cannot configure the network extension service in the service mesh if one of the VCF Operations HCX sites is federated.

1. Log in to VCF Operations HCX using an account with Administrator privileges.
2. Navigate to **Interconnect** > **Service Mesh.**
3. Click **Create Service Mesh**.
4. Select a site pair:
   a) From the drop-down menu, select a site. The paired site is automatically populated.
   b) Click **Continue**.
5. Select compute profiles:
   a) Select a compute profile from the **Select Source Compute Profile** drop-down menu.
   b) Select a compute profile from the **Select Remote Compute Profile** drop-down menu.
   c) Click **Continue**.
6. Select the services to be activated, and click **Continue**.

   Service meshes created in VCF Operations HCX 9.1 and later display enhanced services.


VMware Cloud Foundation
Operations HCX
Dashboard
Infrastructure
Sites
Interconnect
Transport Analytics
Services
Network Extension
Migration
Administration
System Updates
Audit
Alerts
Support
Interconnect
Service Mesh
Compute Profiles
Network Profiles
Create Service Mesh
Select compute Profiles
Select Services to Activate
3. Advanced Configurations
4. Topology Preview
5. Ready To Complete
Select Services to be activated
WAN Optimization is not used with workload migrations using OS Assisted Migration.
Hybrid Interconnect
WAN Optimization
Cross-cloud vMotionMigration
Bulk Migration
Replication Assisted vMotion Migration
Network Extension
OS Assisted Migration
HCX Assisted vMotion (Direct) Migration
CONTINUE


VMware Cloud Foundation
Operations HCX
Dashboard
Infrastructure
Sites
Interconnect
Transport Analytics
Services
Network Extension
Migration
Administration
System Updates
Audit
Alerts
Support
Interconnect
Service Mesh
Compute Profiles
Network Profiles
Create Service Mesh
Select compute Profiles
Select Services to Activate
3. Advanced Configurations
4. Topology Preview
5. Ready To Complete
Select Services to be activated
WAN Optimization Service cannot be selected. This service is incompatible with either the selected Site Pair's HCX Managers or the HCX appliances deployed within the Service Mesh.
Hybrid Interconnect
WAN Optimization
Cross-cloud vMotionMigration
Bulk Migration
Replication Assisted vMotion Migration
Network Extension
Feature has not been activated.
OS Assisted Migration
HCX Assisted vMotion (Direct) Migration
CONTINUE

The WAN Optimization service is only available for enhanced appliances.

The VCF Operations HCX services available for activation are based on your selections in the source and remote site compute profiles, and based on the service entitlements for each site. In cases where the source and remote sites have been activated with different entitlements, the service mesh can inherit entitlements from either site. For more information, see Understanding Service Inheritance.

7. (Optional) Override the default Uplink network profile:

   By default, the HCX interconnect appliance uses Uplink network profiles defined in the compute profile for the source and the destination sites. You can override the default.

   a) Select one or more networks from the **Select Source Site Uplink Network Profile** drop-down menu, and click **Close**.

      If the service mesh in running in standard mode, it can use up to three Uplinks, adding network path failover and improving overall resiliency for VCF Operations HCX services. Multiple Uplinks are not aggregated for increased throughput capacity. The following specific behaviors apply:

      - VCF Operations HCX attempts to load balance traffic on the network extension (HCX-NET-EXT) appliance based on characteristics of the flow and the performance of the uplinks.
      - VCF Operations HCX does not load balance migration traffic on the Interconnect (HCX-IX) appliance. Additional uplinks might or might not be used.

      Enhanced service meshes are limited to a single uplink network.

   b) Click **Select Destination Site Uplink Network Profile**, and repeat the selections.

   c) Click **Continue**.

8. (Optional) Modify the uplink path MTU value for the source and destination site, and click **Continue**.

   This sets the MTU value for the Uplink networks. After creating the service mesh, you can test the MTU for the Uplink networks using Transport Analytics. See Verifying Underlay Network Performance for service mesh Uplinks.

9. Configure the network extension appliances that are deployed per switch or Transport Zone.

   VCF Operations HCX creates one network extension appliance minimum for each network container that is configured in the source compute profile. For example, if the compute profile specifies an NSX Overlay Transport Zone or VLAN Transport zone as the source containers for network extension, the service mesh will create one network extension appliance for the Overlay Transport Zone and another for the VLAN Transport Zone.

   You can use the advanced configuration settings to deploy additional appliances to distribute or dedicate L2 forwarding load or when configuring network extension High Availability.

   a) In **Advanced Configuration - Network Extension Appliance Scale Out**, review the default Extension appliances per network container.

   b) For each entry, set the number of network extension appliances that VCF Operations HCX deploys when it activates the service mesh configuration.

      Extended network service can be carried by a single (standalone) network extension appliance at each site, or an HA group that consists of two network extension appliances at each site. For example, to create a service mesh that has two non-redundant network extensions and one high-availability network extension, you must set the number of network extension appliances deployed by the service mesh to 2.

      The default setting is 1. This setting restricts the service mesh to deploying one network extension appliance.

      Note:

      You must configure the network extension Appliance limit in the compute profile at both the source and remote sites to equal or exceed the number of scale-out appliances set in the service mesh.

      For the system resource considerations, see System Requirements.

c) In the network extension mode for appliances column, specify whether you want a single extension per vNIC or multiple extensions per vNIC. Single extension uses a separate port for each network extension. This depends on your infrastructure and networks, you might want a multiple mode but your infrastructure might not support this.

   The option to specify multiple extensions per vNIC is only available if your service mesh has been upgraded to enhanced mode.

   The option to specify a single extension per vNIC is not available when both source and destination transport zones are Overlay transport zones.

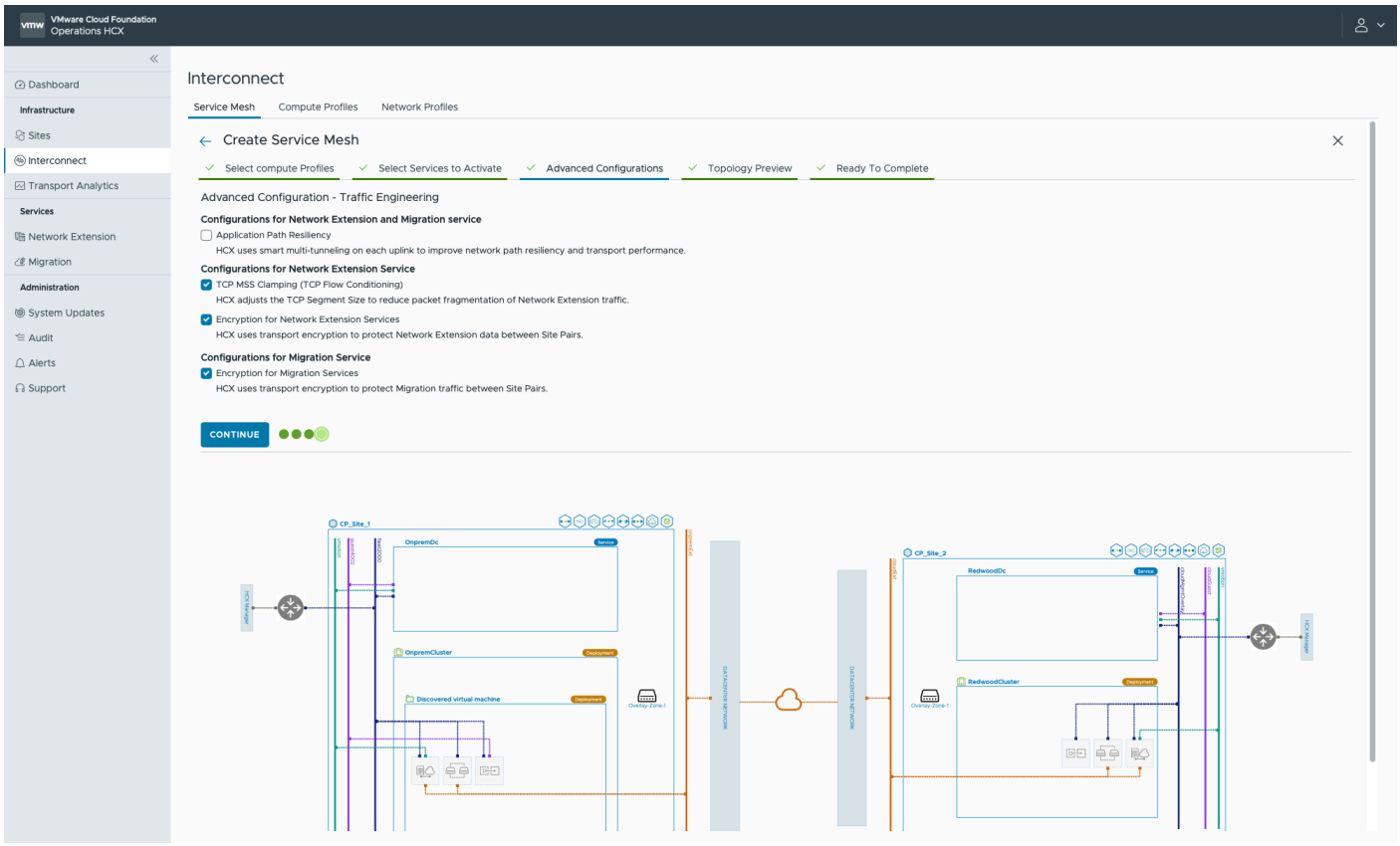
d) In the Configure DHCP profile for network extension column, select **Configure** to automatically configure your environment so that DHCP traffic will flow back and forth when you have completed service mesh setup.

   Configuring DHCP is not applicable when both source and destination switches are DVS.

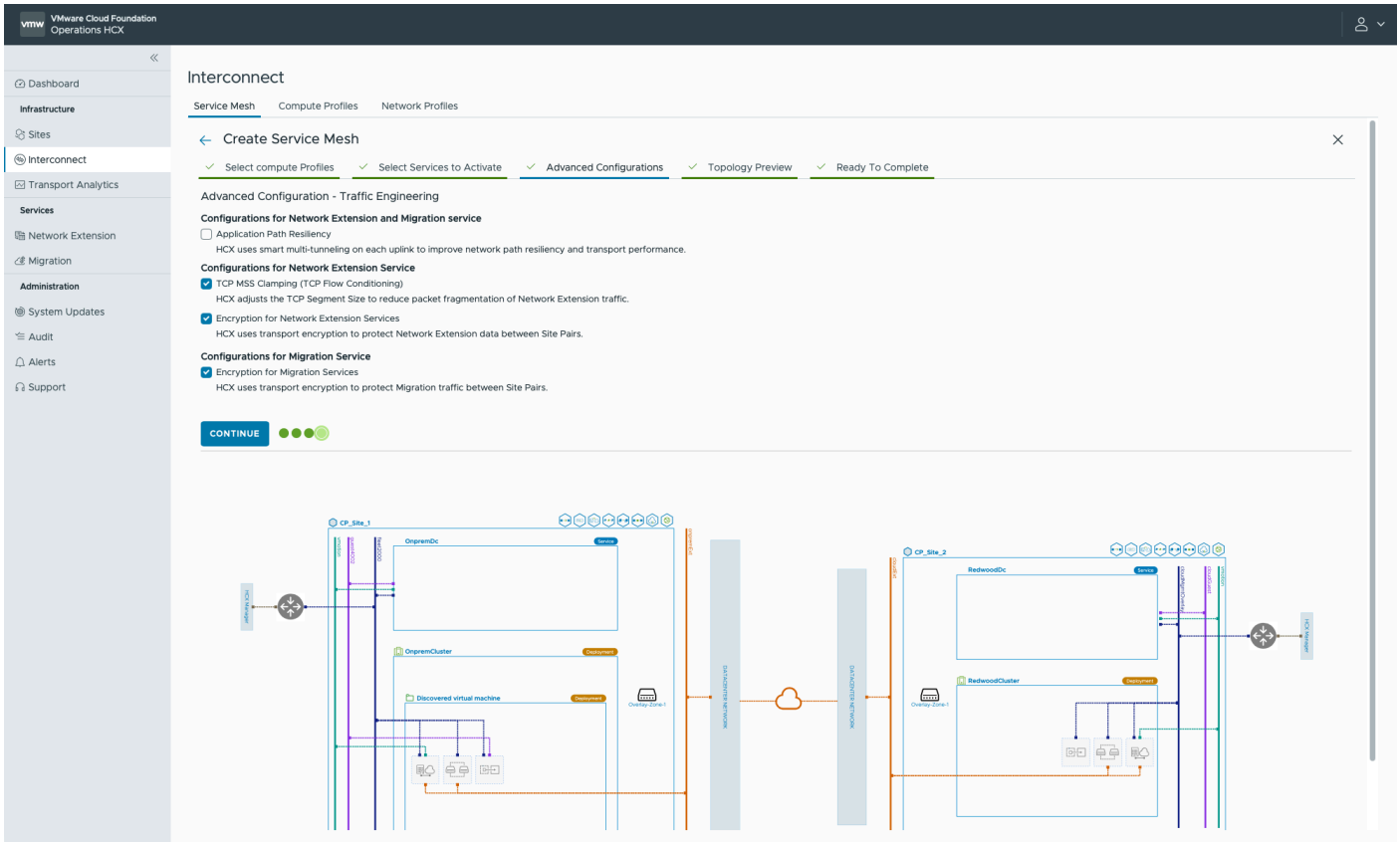
e) Click **Continue**.

10. (Optional) Configure Traffic Engineering features:

    Enhanced:



    Non-enhanced:



a) To create multiple transport tunnels for directing the traffic to a destination site, check **Application Path Resiliency**.

Enabling Application Path Resiliency (APR) creates up to eight transport tunnels between each Interconnect and network extension appliance uplink interface IP address pair between sites. If a few tunnels fail, there is no impact to the data traffic as one transport tunnel out of eight is always used to provide secure data transfer across the Wide Area Network (WAN) or Internet connection.

Application Path Resiliency forwards traffic over one tunnel at a time and does not load balance across multiple paths.

Note:

To view the available tunnels after completing the service mesh configuration, navigate to **Interconnect** > **Service Mesh** > **View Appliances** and expand the HCX-WAN-IX appliance.

Important:
For additional dynamic tunnel requirement, the source Interconnect (IX) and network extension appliances use a random source UDP port in the 4500 – 4628 range and target UDP port as 4500 to create a different flow for each subsequent tunnel. The reverse tunnel originated by target IX/NE appliances have source port as UDP 4500 and destination ports from same random ports used by source appliances for the forward direction in the range 4500 – 4628.

Ensure the firewall settings on either side allow for that connectivity.

b) To dynamically manage the TCP segment size and optimize the transport performance for the HCX network extension service traffic, check the **TCP MSS Clamping** checkbox.

By default, this option is enabled for enhanced service meshes and disabled for standard service meshes, and is available only after activating the HCX network extension service.

c) Select the **Generic Receive Offload** checkbox to reassemble network extension input packets into larger ones before delivery to the workloads.

   This option is not available for service meshes upgraded to enhanced mode.

d) To activate transport encryption for network extension data between site pairs, select the**Encryption for Network Extension Services** checkbox.

   Transport encryption to protect network extension data is activated by default. You can deactivate transport encryption only on Uplink networks that have been verified as secure in the network profile configuration.

e) To activate transport encryption migration traffic between site pairs, select the **Encryption for Migration Services** checkbox.

   Transport encryption to protect migration traffic is activated by default. You can deactivate transport encryption only on Uplink networks that have been verified as secure in the network profile configuration.

11. Review Topology Preview:
    a) Review the selected clusters, resources, and networks.
    b) Click **Continue**.
12. Ready to Complete:
    a) To view a summary of the service mesh selections, click the **here** link.
    b) Name the service mesh.

       The service mesh name has a limit of 50 characters.

    c) To create the service mesh, click **Finish**.

    VCF Operations HCX begins deploying the appropriate appliances and displays a topology view of the service mesh. To monitor the progress of the service mesh deployment, select **Tasks**, and expand the operation.

After the service mesh configuration is complete, verify the underlay network performance for each Uplink Network. The underlay network performance must meet the minimum requirements for VCF Operations HCX services. See Understanding HCX Transport Analytics.

If you are migrating guest workloads using OS Assisted Migration, download and install the sentinel software on each guest workload. See Sentinel Management.

If it is necessary to edit an existing service mesh, such as activating or deactivating services and overriding uplinks, select **Interconnect** > **Service Mesh** > **Edit**. The editing workflow includes a preview screen, listing the changes and describing the impact of those changes on related services prior to finishing the procedure. You can select to complete or cancel the update.

Create a Service Mesh for Non-vSphere Site Pairs

A service mesh provides the services configuration for a Site Pair. This section describes the procedure for creating a service mesh between a vSphere-based VCF Operations HCX site and a non-vSphere site.

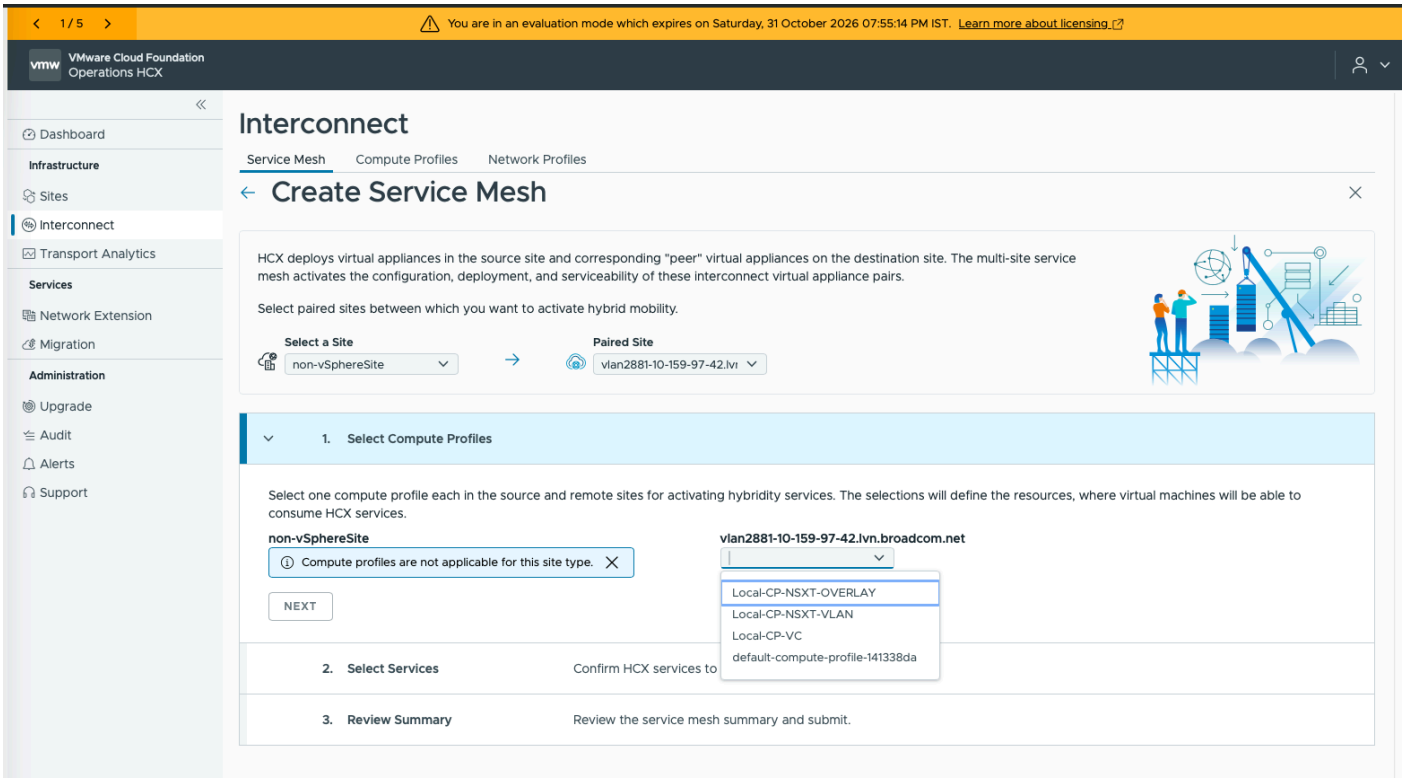
Creating a service mesh requires:

- A connected Site Pair.
- A compute profile for the vSphere-based site. The compute profile service selections must include OS Assisted Migration.

> **Note:**
>
> To create a service mesh between two vSphere-based VCF Operations HCX sites, see Create a Service Mesh for vSphere-Based Site Pairs.

Adding a service mesh initiates the deployment of HCX Sentinel Resource Gateway (SRG) appliance on the vSphere-based site.

1. Log in to VCF Operations HCX with an account with Administrator privileges.
2. Navigate to **Interconnect** > **Service Mesh**.
3. Click **Create Service Mesh**
4. From the drop-down menu, select a non-vSphere site.
5. Click **Continue**.
6. Select a compute profile which has OSAM configured in the services for the vSphere-based HCX site, and click **Next**.



   VCF Operations HCX displays a list of services selected in the compute profile.
7. Select OS Assisted Migration, and click **Next**.
8. Review the configuration summary.
9. Enter a name for the service mesh, and then click **Create Service Mesh**.

   VCF Operations HCX begins deploying the appropriate appliances and displays a topology view of the service mesh. To monitor the progress of the service mesh deployment, select **Tasks**, and expand the operation.

Download and install the sentinel software on each guest workload for migration. See Sentinel Management.

Managing the Service Mesh

After creating a service mesh, you can perform operations on the service mesh and on the HCX appliances deployed by the service mesh.

Update and Synchronize the Service Mesh

The service mesh is the effective service configuration between a source and a destination site. It must be kept updated between the pair following changes in the compute profile, network profile, or service mesh itself.

An update to the compute profile, network profile, or service mesh can have an impact on the system configuration, including possible downtime. The VCF Operations HCX UI provides two operations for addressing service mesh changes: **Edit** and **Resync**. The Edit operation edits the service mesh. The Resync operation takes changes to the compute profile and network profile, and applies them to the service mesh.

If the service mesh configuration includes network extension High Availability (HA) groups, a Resync or Edit operation reconfigures or redeploys all network extension appliances in the HA group.

To change the service mesh setup, navigate to **Interconnect** > **Service Mesh** > **Edit**, and click **Edit** . Click **Update Service Mesh** to save your changes.

To resync, navigate to **Interconnect** > **Service Mesh** > **Resync** and click **Resync**.

When the source is a vSphere environment, editing and synchronizing the service mesh is available from the source site VCF Operations HCX UI where the service mesh was created. These update operations are not available at the destination site.

When the source is a non-vSphere environment, operations must be done from the destination site HCX manager UI.

> **Note:**
>
> For information about upgrading the service mesh appliances, see Upgrade the HCX Service Mesh Appliances.

Renaming Service Mesh Appliances

You can update the service mesh appliance names in the VCF Operations HCX .

The service mesh, which can comprise the HCX-WAN-IX, HCX-NET-EXT, and OSAM appliances, provides the interconnect services between paired source and destination sites. You can change the appliance names in your environment.

> **Note:**
> Do not use the same name for appliances in a site pair.

1. Log in to VCF Operations HCX with an account with Administrator privileges.
2. Navigate to **Interconnnect** > **Service Mesh.**
3. Click **View Appliances**.
4. Select an appliance from the list, and click the **Rename Appliance** tab.
5. Enter the new appliance name, and click **Rename**.

   VCF Operations HCX initiates the name change operation and syncs the changes with the peer site. To monitor the status of the rename operation, click the **Task** tab.

Repeat the procedure for any further name changes.

Manage Service Mesh Appliances

Through the Interconnect interface at the local or remote HCX site, you can perform various operations related to managing the appliances deployed with each service mesh

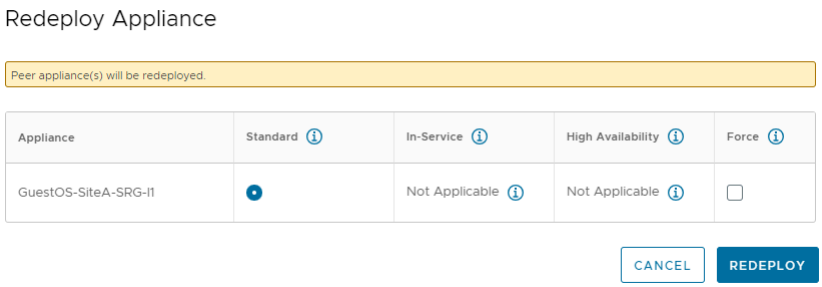
HCX uses a service called the HCX Mobility Agent, which plays an integral role in the Interconnect (IX) appliance functionality. This agent facilitates workload migrations in vCenter. As a result, it cannot be patched, serviced or upgraded using standard ESX tools or images.

The HCX Mobility Agent is displayed in the vCenter inventory as if it were an ESX host, but it is not. The ESX version of the Mobility Agent displayed in the vCenter inventory matches the version of the ESX host in terms of version/family, for example 8.x, but the build number shown does not represent a real ESX build number.

Any CVE/VMSA released by Broadcom for ESX is not applicable to these devices. Remediation must be done by HCX Manager. Any settings, such as vSwitch, Port group or datastore objects related to the ESX host seen in vCenter Inventory are not genuine configuration and have no practical impact on the underlying infrastructure.

VCF Operations HCX provides the following operations for managing service appliances:

Redeploy	Redeploys a specifc appliance at both the source and destination sites. For example, you may need to redeploy as part of a troubleshooting step. **Note:** VCF Operations HCX automatically redeploys the appliances for a service mesh during an upgrade procedure. For more information about upgrading appliances, see About Upgrades. Based on the type of appliance, VCF Operations HCX provides several reployment options.



- **Standard**: Redeploys Interconnect (HCX-WAN-IX) or network extension (HCX-NET-EXT) appliances using the same IP address. With this option, VCF Operations HCX disconnects the current appliances and re-establishes the service tunnels, which can take up to a minute or more.
- **In-Service**: Available only for HCX-NET-EXT appliances. In-Service upgrades allow for minimally disruptive redeployment of the appliances as compared to the Standard option by deploying a new appliance using a separate IP address. For detailed information about this option, see In-Service Upgrade for Network Extension Appliances.
- **High-Availability**: Available only for HCX-NET-EXT appliances configured for High Availability (HA). Redeployment of appliances using the High-Availability option takes advantage of the HA group failover process to complete the redeploy operation. In this case, the Standby pair is upgraded first. After the Standby upgrade finishes, a switchover occurs and the Standby pair takes on the Active role. At that point, the previously Active pair is upgraded and takes on the Standby role. For more information about network extension HA, see Understanding Network Extension High Availability.
- **Force**: Forces the replyment operation to proceed if conflicting operations are encountered. If the HCX-IX appliance is in use - for example, serving migrations - then the appliance redeployment occurs only if the Force option is applied.

Select this option in conjunction with other selections.

Force-Sync	Syncs the appliance configuration in cases where the appliance is not in the expected configured state.
Change Password	Updates the password for the deployed appliance.
Rename Appliance	Renames the deployed appliance.
Activate High Availability	Applies only when selecting network extension appliances.
Migrate Network Extensions	Migrate network extensions. Applies to VCF Operations HCX version 9.1 and later.
Update	Upgrade data plane appliances.

1. Log in to VCF Operations HCX with an account with Administrator privileges.
2. Click **Sites**.
3. Identify a service mesh, and click **View Appliances**.
4. Select the checkbox next to an appliance to view the available management actions.
5. Select an action and respond to any prompts to update the appliance.

Monitor Service Mesh Operations
This section describes how to monitor the status of service mesh-related operations.

1. After initiating a service mesh action, click **More** and select **View Tasks** from the dropdown menu.

   VCF Operations HCX displays a list of operations that are associated with the service mesh. The most current operations are listed first. The display includes the operation name, progress, start and stop time, and task description.
2. Expand each entry to view additional tasks associated with the operation.

Understanding Data Plane Diagnostics

VCF Operations HCX provides diagnostic tools that can help you troubleshoot data plane connectivity issues between site pairs.

During the site pairing process, VCF Operations HCX deploys a set of data plane appliances at both the source and destination sites to support migration and network extension services, including:

- Bulk Migration
- Replication Assisted vMotion Migration
- Cross-cloud vMotion Migration
- OS Assisted Migration
- Network Extension

To perform these services, VCF Operations HCX establishes connections between the appliances using the networks that are defined in the service mesh, including the Management and External networks.

If VCF Operations HCX detects a connection issue with any of the services deployed for the site pair, a message appears indicating that the tunnel or connection status is down or degraded. Run data plane diagnostics to troubleshoot.

> **Note:**
>
> For a service mesh with only HCX Assisted vMotion (Direct), data plane diagnostics is unavailable.

Run Data Plane Diagnostics

You can run data plane diagnostics for a service mesh site pair to help troubleshoot data plane connectivity issues or to better understand connections between services for a site pair. Running diagnostics can take several minutes. While the diagnostics tests are in progress, all service mesh operations are blocked.

> **Note:**
> As a best practice, do not run Transport Analytics while running data plane diagnostics. Running these tests simultaneously can skew test results.

1. In the VCF Operations HCX Console, next to the site pairing for which you want to run data plane diagnostics, select **Run Diagnostics** from the **More** dropdown menu.
2. Review the report results. See, Viewing the Data Plane Diagnostics Report.

Viewing the Data Plane Diagnostics Report

After running the data plane diagnostics, VCF Operations HCX prepares a report covering the connections from each site service mesh.

The Data Plane Diagnostics page displays information about the tested service mesh, and the probes that were run on each site in the service mesh as part of the test.

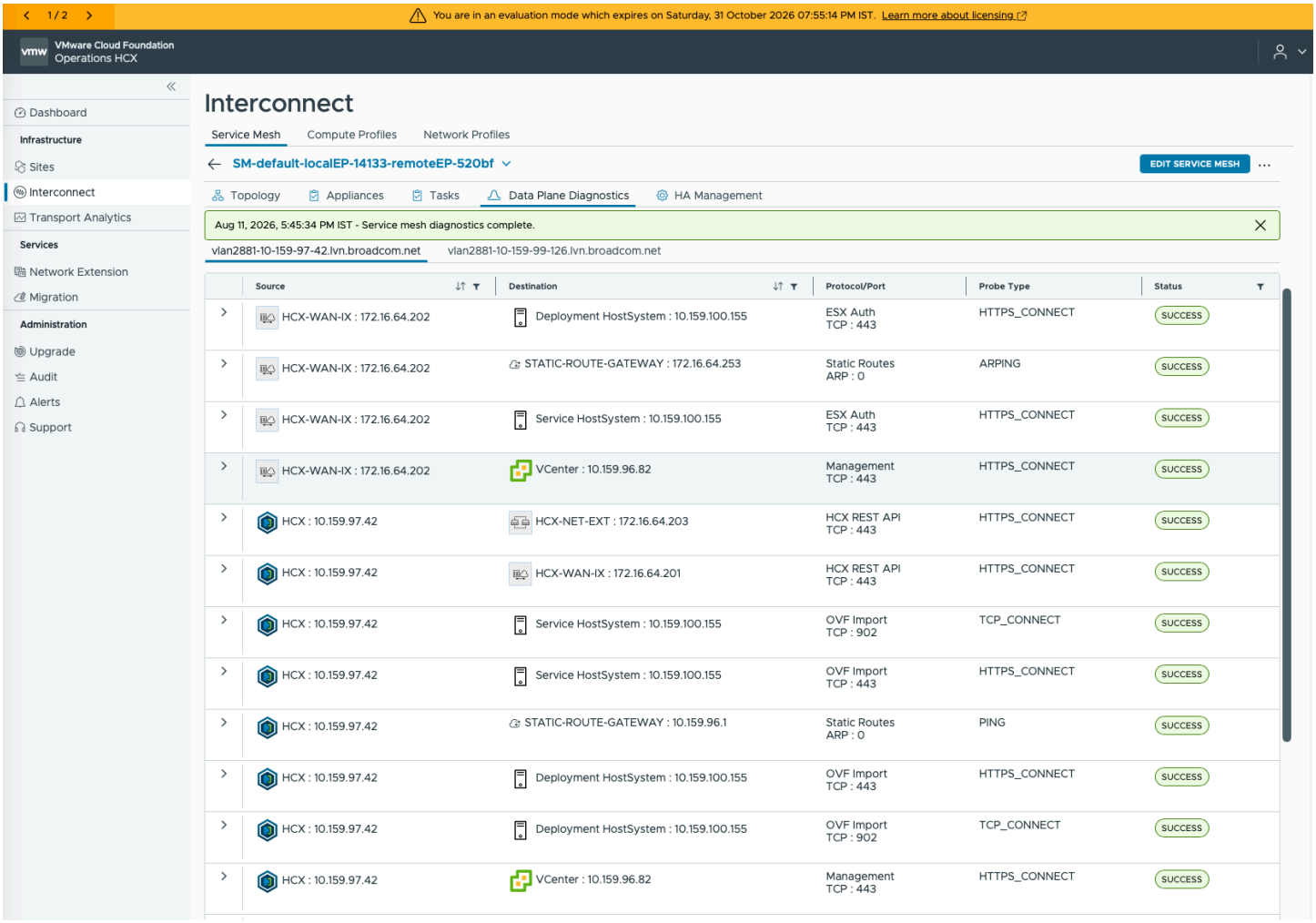
Select a site to see the probe results for each site.

Viewing Test Results

The test results are grouped by the site. For example, if a service mesh has five network extension appliances, the test checks the connectivity for the group of those appliances using the management network and the group of those appliances using the external connectivity (Uplink) network.

You can expand the probe results to see additional details depending on the probe type. For any failed probes, expand the appliance entry for more information.

Figure 488:


1 / 2
You are in an evaluation mode which expires on Saturday, 31 October 2026 07:55:14 PM IST. Learn more about licensing
VMware Cloud Foundation
Operations HCX
Dashboard
Infrastructure
Sites
Interconnect
Transport Analytics
Services
Network Extension
Migration
Administration
Upgrade
Audit
Alerts
Support
Interconnect
Service Mesh
Compute Profiles
Network Profiles
SM-default-localEP-14133-remoteEP-520bf
EDIT SERVICE MESH
Topology
Appliances
Tasks
Data Plane Diagnostics
HA Management
Aug 11, 2026, 5:45:34 PM IST - Service mesh diagnostics complete.
vlan2881-10-159-97-42.lvn.broadcom.net
vlan2881-10-159-99-126.lvn.broadcom.net
Source
Destination
Protocol/Port
Probe Type
Status
HCX-WAN-IX : 172.16.64.202
Deployment HostSystem : 10.159.100.155
ESX Auth TCP : 443
HTTPS_CONNECT
SUCCESS
HCX-WAN-IX : 172.16.64.202
STATIC-ROUTE-GATEWAY : 172.16.64.253
Static Routes ARP : 0
ARPING
SUCCESS
HCX-WAN-IX : 172.16.64.202
Service HostSystem : 10.159.100.155
ESX Auth TCP : 443
HTTPS_CONNECT
SUCCESS
HCX-WAN-IX : 172.16.64.202
VCenter : 10.159.96.82
Management TCP : 443
HTTPS_CONNECT
SUCCESS
HCX : 10.159.97.42
HCX-NET-EXT : 172.16.64.203
HCX REST API TCP : 443
HTTPS_CONNECT
SUCCESS
HCX : 10.159.97.42
HCX-WAN-IX : 172.16.64.201
HCX REST API TCP : 443
HTTPS_CONNECT
SUCCESS
HCX : 10.159.97.42
Service HostSystem : 10.159.100.155
OVF Import TCP : 902
TCP_CONNECT
SUCCESS
HCX : 10.159.97.42
Service HostSystem : 10.159.100.155
OVF Import TCP : 443
HTTPS_CONNECT
SUCCESS
HCX : 10.159.97.42
STATIC-ROUTE-GATEWAY : 10.159.96.1
Static Routes ARP : 0
PING
SUCCESS
HCX : 10.159.97.42
Deployment HostSystem : 10.159.100.155
OVF Import TCP : 443
HTTPS_CONNECT
SUCCESS
HCX : 10.159.97.42
Deployment HostSystem : 10.159.100.155
OVF Import TCP : 902
TCP_CONNECT
SUCCESS
HCX : 10.159.97.42
VCenter : 10.159.96.82
Management TCP : 443
HTTPS_CONNECT
SUCCESS

Understanding HCX Transport Analytics

You can test and review the underlay network performance as seen by VCF Operations HCX appliances for each uplink network using HCX Transport Analytics.

Transport Analytics help to characterize the health and performance of the underlay network. Run bandwidth tests on each uplink network in each service mesh configuration to determine whether the underlay network supporting that uplink meets the minimum requirements for HCX services. Understanding the underlay network performance can help in planning migration wave size and in analyzing data transfer and performance issues.

> **Note:**
>
> HCX Transport Analytics test the end-to-end underlay network supporting each uplink, and not the uplink network itself. The test checks the underlay network between the source and the destination site. Throughout this section, underlay network results are organized and reported against each uplink network in the service mesh.

To support seamless system operation between sites, network extension and migration operations have minimum requirements for underlay network bandwidth, latency, and loss. These services all use underlay networks for transporting data from the source site to the destination site. If the underlay network does not meet the minimum requirements, it can result in degraded service performance or failure. For a complete description of network underlays and the minimum requirements for HCX services, see, Network Underlay Minimum Requirements.

> **Note:**
>
> For service meshes upgraded to enhanced mode, measurements are made with the VCF Operations HCX encapsulations, therefore you might see a difference in the reported numbers. The insights that we provide regarding minimum requirements are adjusted accordingly.

Verifying Underlay Network Performance for service mesh Uplinks

HCX transport analytics help with monitoring the connectivity health between sites for a service mesh and with characterizing the underlay network requirements for HCX services.

- If you have changed the Uplink Networks in the compute profile, navigate to **Interconnect** > **Service Mesh**, and click **Resync** to apply those changes to the Service Mesh.

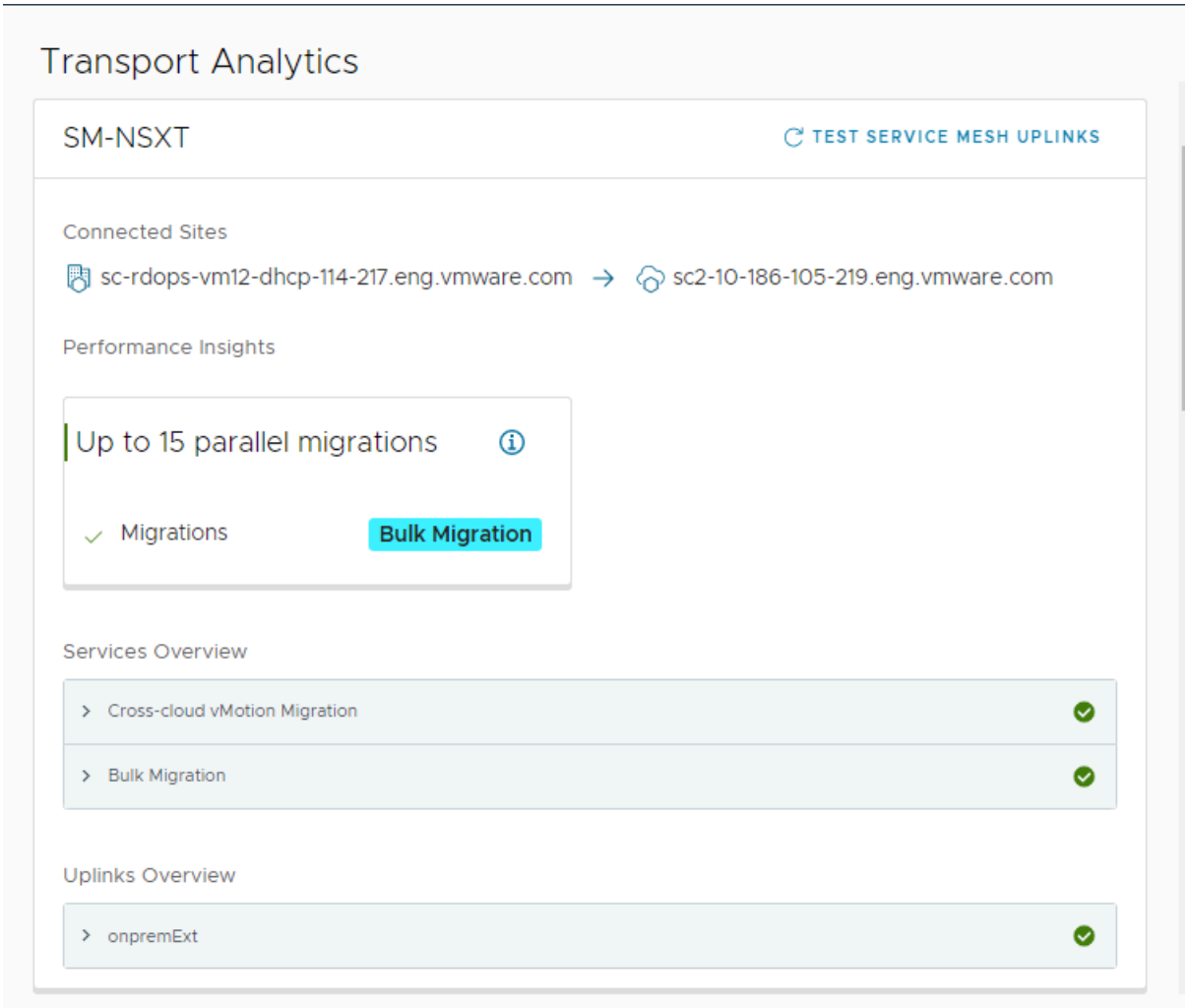
The Transport Analytics page lists all service mesh configurations for a site pair with a separate card for each service mesh. Each service mesh card has the same information:

- Connected Sites
- Performance Insights
- Services Overview
- Uplinks Overview

You can use the search option at the top right of the page to find a specific service mesh.

1. In HCX Manager, navigate to **Transport Analytics**.

   The transport analytics page displays the service mesh inventory.



2. For each service mesh, click **Test Service Mesh Uplinks**.

   Note:
   It is a best practice to test the service mesh uplinks independent of other tests. Running tests like CCLI perftest in parallel can skew test results.

3. Review each section of the service mesh card for health details:

Connected Sites	Identifies the source and the destination sites for the site pair.
Performance Insights (Currently available for Bulk Migration only)	Based on the bandwidth reported by the uplink test and the bandwidth requirement of the HCX service, HCX calculates the underlay network performance associated with the uplink. From that calculation, HCX recommends a number of simultaneous migrations that the underlay network can support at the point when the test was run.
Services Overview	Lists the HCX services enabled in the service mesh. Expand each service entry to view the transport thresholds: Minimum Bandwidth, Maximum Latency, and Maximum Loss. The transport thresholds reflect the underlay network requirements for each service. The health of a service relative to the underlay network is indicated by either a checkmark or an exclamation mark. A checkmark indicates that the underlay network performance meets the minimum requirements for the service, and the Uplink Network is in a healthy state for the specified HCX service. An exclamation mark indicates that the Uplink Network is in a degraded state relative to the underlay network performance, and that Uplink is not recommended to use with the specified HCX service.
Uplinks Overview	Provides the transport analysis and the transport monitoring of the underlay network for each Uplink Network in the service mesh.

- Transport Analysis lists the tested transport thresholds of the underlay network. Expand each Uplink Network entry to view Available Bandwidth, Latency, and Loss metrics. Select **Re-run Test** to start a new test for the selected Uplink Network.

  The health of a service relative to the underlay network is indicated by either a checkmark or an exclamation mark. A checkmark indicates that the service thresholds are met for all the services configured in the service mesh. An exclamation mark is shown if any one of the service thresholds are not met for the services configured in the service mesh.

  If a new network uplink is added to the service mesh, you must run an initial test on that Uplink to generate the transport thresholds. Following the initial test, re-run the test to evaluate the current state of the underlay network.

  Note:
  Available bandwidth is what is currently available at the time of the test. Transport threshold metrics are static and are based on the most recent test. To update the metrics for an Uplink Network, you must re-run the test.

  When an Uplink Network is used in multiple service mesh configurations, each service mesh using that Uplink is updated with new metrics.
- The Transport Monitor provides both path MTU information for each Uplink Network and detailed, time-series bandwidth, latency, and loss information for each Uplink Network. For more information related to minimum underlay requirements, see Network Underlay Minimum Requirements.

  Path MTU has the following values:
- Discovered: This is the value tested at the Uplink interface. If there is a mismatch between the Discovered value and the Current value, it is a best practice to update the Path MTU in the service mesh as a way to improve performance.

  Note:

  VCF Operations HCX automatically checks the Path MTU value every 15 minutes. If there is a mismatch with the Current value, VCF Operations HCX displays a message in **Interconnect** > **Service Mesh** display for the affected service mesh.
- Current: This is the value set in the service mesh for the Uplink network. This value overrides the setting in the network profile. It can be the same as the Configured value.
- Configured: This is the value set in the network profile for the Uplink network.

4. (Optional) To view the detailed metrics, select **Transport Monitor** for each Uplink Network.
5. For each service mesh card listed under Transport Analytics, review the health details to verify that the underlay network performance meets the requirements for the HCX services enabled for that service mesh.

If HCX transport analytics indicate that the Uplink Networks are healthy, proceed to using VCF Operations HCX services. If the Uplink Network status is Degraded, use Transport Monitor to help identify and resolve underlay network performance issues before using VCF Operations HCX services.

Examining Underlay Network Performance Using Transport Monitor
HCX Transport Monitor provides detailed, time-series throughput, latency, and loss information for the uplink networks in a service mesh.

Transport Monitor provides a graphical representation of underlay network performance for each uplink network. Transport Monitor collects the metrics every minute. For each selected uplink, the page provides the same information:

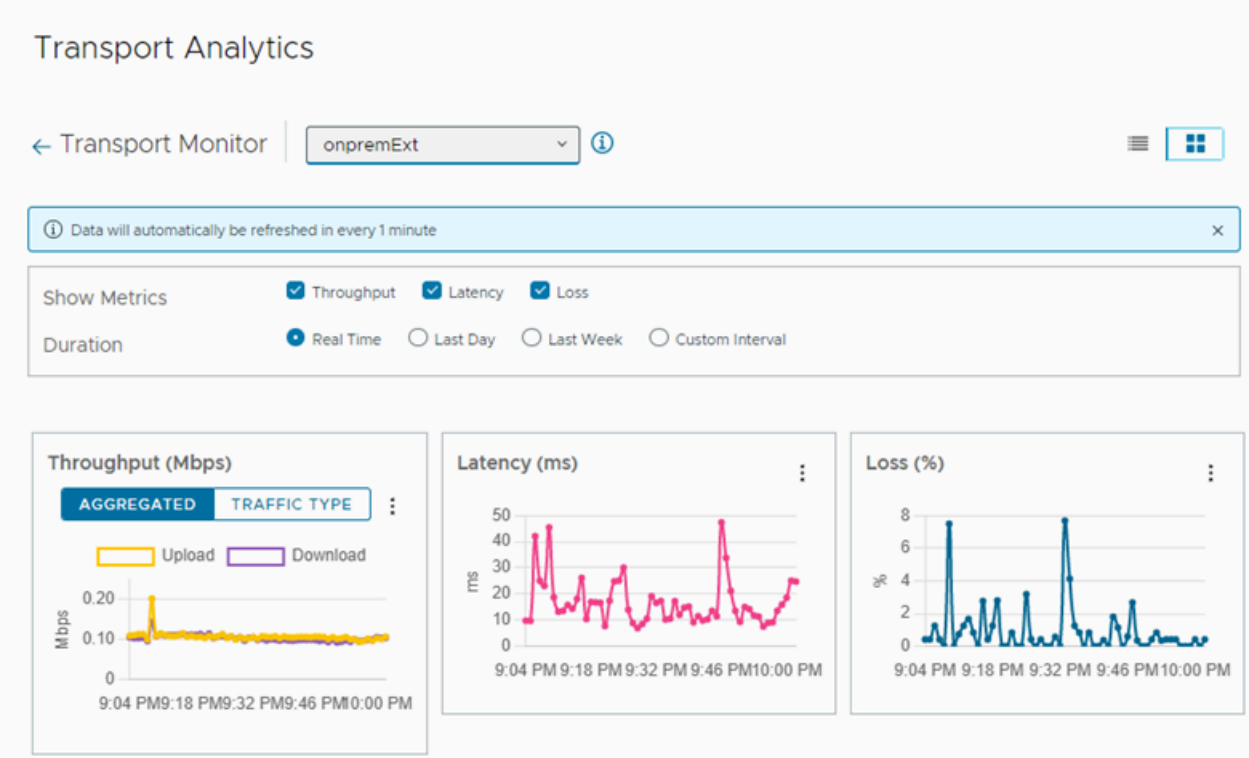
- Metrics selections: Throughput, Latency, Loss
- Duration selections: Real Time, Last Day, Last Week, Custom
- Graphic displays based on selected metrics and duration

In some scenarios, metrics data is not available for an uplink and underlay analytics information is not displayed:

- The service mesh appliance tunnels are down.
- The source environment is using Network Address Translation (NAT).
- The firewall security policies disallow the ICMP traffic.

1. From the Transport Analytics page in the HCX Manager UI, select **Transport Monitor**.

   HCX displays the Transport Monitor page.



2. At the top of the page, use the pull-down menu to select an uplink network.

   The information icon to the right of the pull-down menu specifies which service mesh configurations use the selected uplink network.

3. At the top right of the page, click one of the icons to toggle the orientation for graphs between the inline and the portrait mode.
4. For **Show Metrics**, click the check box in front of each option that you want to display: **Throughput**, **Latency**, **Loss**.
5. For **Duration**, select the time frame to display in the graph:

Duration	Description
Real Time	The system automatically updates the information in the graph every 1 minute.
Last Day	Displays information from the last 24 hours.
Last Week	Displays information from the last 7 days
Custom Interval	Displays information for the selected date range.

6. Review the graphs for each metric type:

Graph	Description
Throughput	Displays the amount data traffic using the uplink network from all HCX services. At the upper-right of the graph, you can select from the Aggregated or the Traffic Type data. Aggregated data combines the traffic throughput for all HCX migration and network extension services using the uplink. Traffic Type displays the data separated by the migration or the network extension traffic. In either case, the graph displays both upload and download throughput. To display the minimum threshold for each migration type, use the **Show Thresholds** pull-down menu in the upper-right corner of the graph.
Latency	Displays the latency of underlay network supporting the uplink. To display the minimum threshold for each migration type, use the Show Thresholds pull-down menu in the upper-right corner of the graph.

Graph	Description
Loss	Displays the percentage of data loss for the underlay network supporting the uplink. To display the minimum threshold for each migration type, use the Show Thresholds pull-down menu in the upper-right corner of the graph.

Sentinel Management

You must download and install HCX sentinel on all guest virtual machines requiring migration using HCX OS Assisted Migration. Sentinel gathers the system configuration from the guest VM and assists with the data replication.

The guest VM information is used by various HCX OS Assisted Migration (OSAM) service processes. In part, the information is used to create an inventory of guest VM systems for migration and to help replication processes prepare the disks on the replica VM for replication and migration.

Sentinel also helps with the data replication by reading data written to the source disks and passing that data to the Sentinel Data Receiver (SDR) appliance at the destination site.

Download the sentinel software in the **Sentinel Management** tab. The tab is available for systems that have the OSAM service selected in the service mesh. For more information about OS Assisted Migration, see Understanding OS Assisted Migration.

Downloading and Installing Sentinel Agent Software

When performing migrations from non-vSphere virtual machines, you must install the HCX sentinel on all guest virtual machines requiring migration using HCX OS Assisted Migration (OSAM). The sentinel agent gathers the system configuration from the guest virtual machine and assists with the data replication.

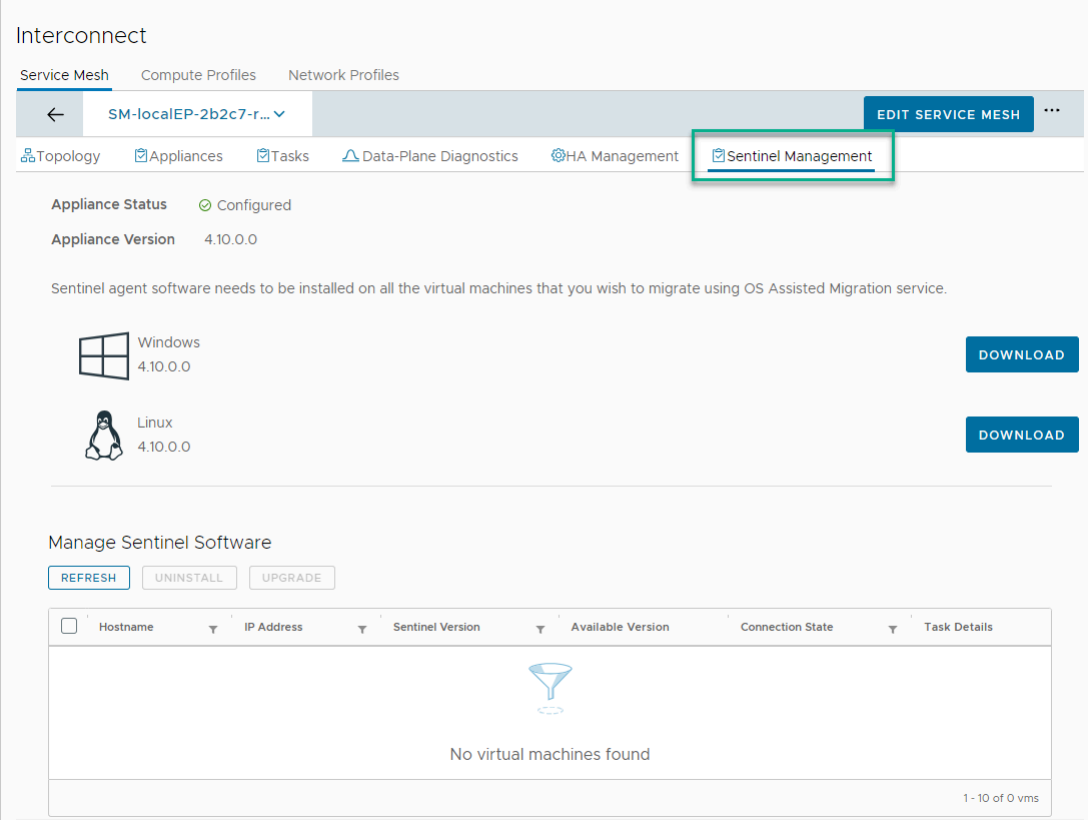
The OSAM service must be enabled in VCF Operations HCX service mesh.

> **Note:**
>
> On Linux systems, if the Polkit (PolicyKit) service is running on the virtual machine where Sentinal software is installed, it can block activation/installation of the sentinel service. In this case, you must disable Polkit, restart the VM, and reinstall the sentinel software.

1. In the source site VCF Operations HCX , navigate to **Interconnect** > **Service Mesh** > **Sentinel Management**.

   > **Note:**
   >
   > In case of a single-site model, this is the destination site VCF Operations HCX.

   VCF Operations HCX displays the sentinel management interface. This interface displays the deployed OSAM appliance version, the sentinel download options, and the inventory of virtual machine workloads where sentinel software has been installed. From the sentinel management interface, one can select the virtual machine workloads and either upgrade or uninstall sentinel. For the list of workloads, you can manage sentinel software upgrades and perform manual uninstall operations.



2. Download the software bundle appropriate for the environment that you are migrating.

   The sentinel software bundle is downloaded to the local machine with the name `<SGW-name>-linux-sentinel-installer.sh` or `<SGW-name>-windows-sentinel-bundle.zip`.

3. For Linux workloads, install the Linux software on all guest VMs that require migration.
   a) Connect to your guest system using SSH.
   b) Copy the linux-sentinel-installer.sh file to the guest system.
   c) At the terminal, enter the command `bash linux-sentinel-installer.sh`.

      The software prompts you for permission to start the installation.
   d) Enter **yes**, and press **Enter**.

4. For Windows workloads, install the Windows sentinel software on all guest VMs that require migration.
   a) Log in to the guest system.
   b) Copy the `windows-sentinel-bundle.zip` file to the guest system.
   c) Unzip the bundle.
   d) To run the installer, double-click **install-sentinel.exe**.
   e) Click **Next** to continue.
   f) Accept the license agreement and click **Next** to continue.
   g) Choose the location where you to install the software and click **Next**.
   h) Click **Finish**.

Review the OSAM migration information. See Understanding OS Assisted Migration.

Uninstalling Sentinel Agent Software

The HCX OS Assisted Migration (OSAM) service automatically uninstalls the sentinel software from the guest system after a successful migration. Alternatively, you can manually remove the software using the sentinel management interface.

Following a successful migration, the OSAM service automatically sends instructions to the guest virtual machine to power off and uninstalling the sentinel agent software upon reboot. The OSAM service then removes the VM from the inventory of non-vSphere virtual machines on VCF Operations HCX .

You can manually uninstall the software from a source VM using the **Uninstall** button in the sentinel management interface. The action taken by the OSAM service to uninstall the sentinel software depends on whether the service has access to the source system:

- OSAM service has a connection to the source VM: OSAM uninstalls the sentinel software from the source VM. Also, OSAM removes the source VM from the inventory of non-vSphere virtual machines on VCF Operations HCX.
- OSAM service has no connection to the source VM: OSAM removes the source VM from the inventory of non-vSphere virtual machines on VCF Operations HCX, but the sentinel software remains installed on the source VM. In this case, if a connection to the source VM is reestablished with the OSAM service, the source VM reappears in the inventory of non-vSphere virtual machines on VCF Operations HCX. To remove the sentinel agent software and delete the source VM from the inventory, repeat the uninstall procedure.

  Note:

  The OSAM service prevents you from uninstalling the HCX sentinel software during the source VM migration.

To uninstall the sentinel agent software manually, use the following procedure.

1. Go to **Interconnect** > **Service Mesh** > **Sentinel Management**.
2. Select the source systems.
3. Click **Uninstall**.
4. Click **Yes**.

   The OSAM service begins the process of uninstalling the software from the source VM.
5. In the sentinel management interface, verify that the entry is removed from the inventory of non-vSphere virtual machines.

Upgrading Sentinel Agent Software

To maintain compatibility with OS Assisted Migration (OSAM) service appliances, update the sentinel agent software on guest virtual machines.

The sentinel appliances are updated to the latest version as described in Upgrade the HCX Service Mesh Appliances.

The VCF Operations HCX upgrade bundle includes the SGW, SDR, and SRG appliance software necessary for upgrade. It includes the sentinel Windows and Linux client software for downloading on each virtual machine. You must upgrade to the latest VCF Operations HCX software to get the latest OSAM updates.

1. Navigate to the VCF Operations HCX console, and select **Interconnect > Service Mesh > Sentinel Management**.

   The system displays the inventory of guest virtual machines. Each entry lists the current sentinel software version and the available software version installed after upgrading the VCF Operations HCX .
2. Select the guest VMs to update:

   Update all guest virtual machines to the same version at the same time as a best practice.

   - To update all guest VMs, check the box at the top of the **Hostname** column.
   - To update individual VMs, check the box next to each VM.
3. Click **Upgrade**.

   The upgrade begins for the selected guest VMs. For the upgrade status, review the Task Details in the Sentinel Management page.

Note:
You can only upgrade the sentinel software when the migration is not in the switchover phase for that guest virtual machine. An attempt to upgrade sentinel on a guest virtual machine that has a switchover in progress results in an upgrade request failure.

Extending Networks

Using the network extension service, you can create Layer 2 networks at the destination site and bridge the remote network to the source network over a multi-gigabit-capable link. The new stretched network is automatically bridged with the network at the source site data center.

About Network Extension

You can bridge networks between VCF Operations HCX activated data centers using network extension.

With network extension, you can extend the virtual machine networks to a VCF Operations HCX remote site. Virtual machines that are migrated or created on the extended network at the remote site behave as if on the same L2 network as virtual machines in the source environment. With network extension, the default gateway for the extended network is only connected at the source site. Traffic from virtual machines in remote sites that must be routed to a different L3 network flow through the source site gateway.

Using network extension, you can:

- Retain the IP and the MAC addresses of the VM, and honor existing network policies.
- Extend VLAN-tagged networks from a VMware vSphere Distributed Switch.
- Extend NSX segments, NSX VPC subnets, and DVPGs.

Requirements for Network Extension

VCF Operations HCX supports extending networks from VMware vSphere Distributed Switches, NSX segments under default project, and NSX Virtual Private Cloud (VPC) subnets. HCX also supports extending networks to vSphere DVPG as the destination.

The following sections list requirements for NSX segments and NSX VPC subnets.

These general requirements apply:

- Never use VCF Operations HCX to extend the vSphere Management network or other VMkernel networks (for example: vMotion, vSAN, replication) to the remote site.
- The upstream switch for an extended network must be configured to accept (not filter out) RARP (reverse ARP) packets.

Networks with VMware NSX

Networks with NSX at the source:

Registering NSX with VCF Operations HCX at the source site is optional, except when extending NSX networks or migrating tags. If NSX is registered, the following requirements apply:

- The NSX Manager must be registered while deploying VCF Operations HCX.
- The NSX Manager must be version 3.2.3 or higher.

The NSX Manager and VCF Operations HCX must be registered with the same vCenter Server.

- The ESX hosts where the NSX segment/VPC subnet originates must be configured as NSX transport nodes.
- NSX Overlay, VPC subnets, and NSX VLAN networks can be extended.

Networks with NSX at the destination:

- Additional network extension appliances are required when extending more than 8 networks. With enhanced service mesh appliances, additional network extension appliances might be required when extending more than 7 networks.

- Enhanced service mesh appliances can be configured with multiple extensions per vNIC wherein a single vNIC of the appliance gets used for as many as 128 extensions.
- NSX Overlay and/or NSX VLAN can be configured in the destination vCenter Server or SDDC.
- For those systems that are updated to the most recent version of VCF Operations HCX, multiple extensions per vNIC mode wherein a single vNIC of an appliance gets used for as many as 128 extensions.

  Note:

  In addition to the preceding requirements, individual requirements for networks with NSX at the source and networks with NSX at the destination also apply.

Networks with NSX VPC subnets

Network extension with VPC subnets at the source:

- Extending a VPC subnet requires both VCF Operations HCX and NSX versions to be at least 9.0.0.0 on source sites.
- You can extend VPC subnets to destination sites having an NSX segment or VPC subnet created for extending the network.

Network extension with VPC subnets at the destination:

- VCF Operations HCX supports Public, Private, and Private Transit Gateway (TGW) access modes when extending to destination VPC subnet.
- Users can opt to connect to gateway (VPC) when removing the extension irrespective of access mode.
- Extending a network to a destination VPC subnet requires NSX 9.0.0.0 and later on the destination site, and VCF Operations HCX 9.0.0.0 and later on both paired sites.
  - Extending a VPC subnet from the source to an Overlay/VLAN segment requires NSX 9.0 and later on the source side.
  - Extending a VPC subnet from the source to a VPC subnet on the destination site requires NSX 9.0 and later on both sides.
  - Extending a DVPG, VLAN segment, or overlay segment to a VPC subnet only requires NSX 9.0 and later on the destination site.

Network extension with VPC subnets at the source and the destination:

Network extension from a source NSX VPC subnet to a destination NSX VPC subnet, requires the following preparation in NSX:

- When creating an NSX Project, make sure that Distributed Firewall Rules do not have DROP or REJECT rules that can affect traffic for the extension networks.
- Provide "Private" VPC IP CIDR when creating "Private" VPC subnets within the CIDR.
- Provide "Public" VPC IP CIDR in case of creating "Public" VPC subnets within the CIDR
- In the source VPC Service Profile, do not configure distributed DHCP.

  Note:

  If the source VPC Service Profile is configured with distributed DHCP, VMs connected to the extended network, either an NSX segment or VPC subnet, do not receive a DHCP IP address from source-side DHCP server.

Restrictions and Limitations for Network Extension

VCF Operations HCX network extension is allowed or prevented under certain conditions.

Detected and Restricted Source Network Types

The VCF Operations HCX network extension service detects and prevents several non-supported network extension scenarios (items are dimmed in the network extension UI):

- vSphere infrastructure networks (ESX VMkernel networks).

- VCF Operations HCX network profile networks - distributed port groups or Segments that are selected in a network profile.
- Untagged distributed port groups - distributed port groups with VLAN type None, ID 0 or NULL.
- Private VLAN (PVLAN) networks.
- vCenter-backed port groups that are configured with ephemeral binding cannot be extended.
- NSX logical switches.

Unsupported Source Configurations

HCX network extension does not support the following source configurations:

- vSphere Standard Switch (VSS) networks.
- Cisco Application Centric Infrastructure (ACI) with VMware Virtual Machine Monitor (VMM).
- Virtual machine networks must only be extended with a single solution. VCF Operations HCX does not support network extension for networks already extended to the same NSX router by an external solution. For example, either network extension or NSX L2 VPN can be used to provide connectivity, but both must not be used simultaneously. Using multiple bridging solutions simultaneously can result in a network outage.
- Virtual machine networks with shared or overlapping VLAN configurations must not be extended to the same destination router. This can result in a network outage.
- Secondary subnets in a single layer-2 network.
- VCF Operations HCX does not integrate with an NSX Global Manager for extending networks (only the non-Federated NSX Local Manager is supported). See Support for NSX Federation.
- Network extensions are not supported between non-vSphere VCF Operations HCX site pairs.

Unsupported Destination Configurations

VCF Operations HCX network extension does not support the following destination configurations:

- VCF Operations HCX does not integrate with an NSX Global Manager for extending networks (only the non-Federated NSX Local Manager is supported). See Support for NSX Federation.
- Mobility Optimized Networking (MON) is not supported when extending networks to a destination NSX VPC subnet, NSX VLAN segment, or vSphere DVPG.
- NSX DVPG is not supported as a destination segment.

Network Extension in NSX Federated Environments

When the local NSX manager registered with VCF Operations HCX at either the source or destination site is later integrated with an NSX Global Manager, established network extensions between sites continue to work. However, you cannot extend a new network or enable/disable Mobility Optimized Networking (MON) for an already extended network. In such cases, existing network extensions can be impacted if the appliance serving the extensions is upgraded or redeployed. See, Support for NSX Federation

Additional Considerations

- HCX supports extending the same network to a maximum of 3 distinct destinations or routers.
- One network extension configuration cannot be extended multiple times to the same destination router.
- A network that has been extended to three environments is only supported to the third location if the sites are physically co-located and traffic does not travel over a Wide Area Network.
  - A three-point extension is not supported with source networks that are based on NSX distributed routing.
  - A three-point extension can lower end-to-end network performance due to the combined latency and additional layer of packet and encryption processing.
- One network extension appliance can only connect to one vSphere Distributed Virtual Switch or NSX Transport Zone.
- Networks can only be extended between one appliance pair (source and destination appliances) per site, and multiple network appliances cannot be used to increase throughput.
- Network extension does not detect or mitigate loops.
- Virtual machine networks that span more than one vCenter Server must not be extended from more than one vCenter Server to the same destination router to avoid a network outage.

- Network extension does not detect or mitigate IP conflicts.
- Network extension does not detect or mitigate MAC conflicts.
- For a cloud/site pair, a given network can be extended through only one appliance and is subject to the resource and the performance limitations of that appliance.
- Network extension connects to an existing segment on the target site if it has the same gateway IP and Prefix configured for the extension, and it disconnects the NSX router interface from the network. If the NSX Tier-1 router interface was previously connected and in service, all communication to the gateway on the extended network on the destination site is disrupted.
  - For destination VPC Subnet, connection is to an existing VPC Subnet on the target site if it has the same gateway IP, prefix length, VPC, and Subnet access mode.
  - For destination NSX VLAN segment, connection is to an existing segment on the target site if it has same Transport Zone and VLAN-ID.
  - For destination vSphere DVPG, connection is to an existing DVPG on the target site if it has the same vSphere DVS and VLAN-ID.
- VMware NSX Traceflow does not work with extended networks.
- Extending DVPG/VLAN segments to NSX DVPG at the destination requires the VCF Operations HCX and NSX versions to be at least 9.1 on both sites.

Extending Networks

Network extension is a layer-two bridging function that is initiated at the source site.

The source and destination site pair are vSphere-based VCF Operations HCX deployments and a service mesh is deployed between them.

You can extend networks by selecting one or more distributed port groups, NSX overlay or VLAN segments, or NSX VPC subnets. When you extend a network, a corresponding network is created at the destination site. If the service mesh does not support extending networks to VLAN-backed segments, see step 9 in this procedure for information about how to modify a service mesh to deploy NE appliances to support VLAN segments.

Network extension is not supported for non-vSphere site pairs.

For a list of supported networks and requirements for network extension, see Requirements for Network Extension.

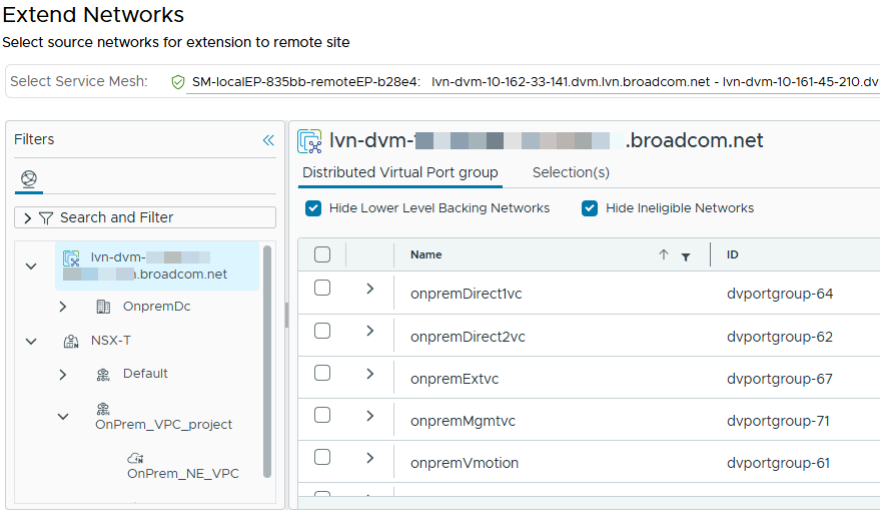
For a list of restrictions regarding network extension, see Restrictions and Limitations for Network Extension.

For the operational limits supported with network extension, see Configuration and Service Limits.

1. In HCX Manager, select **Network Extension**.

   The service meshes associated with the local HCX Manager are displayed. Expand a service mesh to view associated network extensions.

2. At the top of the page, select **Extend Networks**.



3. Select a service mesh.

   If you have only one service mesh, it is selected by default.

4. From the Filters pane, select a network type.

   Use the available filters to hide networks that are ineligible for extension, that do not have virtual machines associated with them, or are networks without extension.

5. In the Networks pane, select one or more source networks to extend.

   The network list changes based on the network type. For vSphere clusters, a list of distributed virtual port groups is displayed. For NSX, NSX segments, VLAN segments, and VPC subnets are displayed.

6. Click **Next**.

7. To extend a source network to an NSX VLAN network:
   a) Select the network extension appliance supporting VLAN.

      The network extension appliance information is based on your service mesh configuration. If your service mesh configuration only supports VLAN, this appliance is selected by default.
   b) Enter the destination VLAN ID.

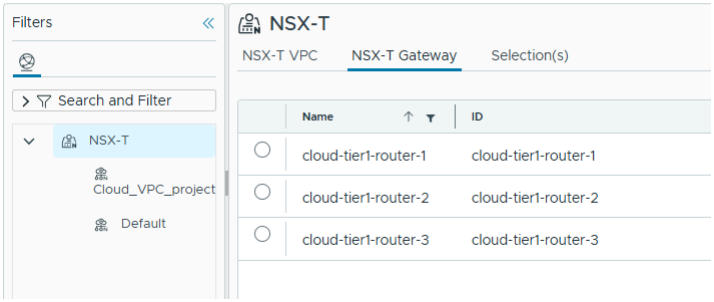
8. To extend a source network to a vSphere DVPG network:
   a) Select the network extension appliance supporting VLAN on DVPG.

      The network extension appliance information is based on your service mesh configuration. If your service mesh configuration only supports VLAN, this appliance is selected by default.
   b) Enter the destination VLAN ID.

9. To extend a source network to an NSX Overlay network:
   a) Select **Destination First Hop Router**.

      A screen appears with the available NSX Gateway selections.

Select Destination First Hop Router



b) Select an NSX gateway.
c) Enter the gateway IP address and Prefix.
d) Select the network extension appliance supporting the NSX Overlay extension.

   The network extension appliance information is based on your service mesh configuration. If your service mesh configuration only supports NSX Overlay networks, this appliance is selected by default.
e) (Optional) Select Mobility Optimized Networking.

   You can select Mobility Optimized Networking (MON) for all workloads that require routing through the local gateway at the target site.

   For more information about MON and other configuration settings, see Understanding Network Extension with Mobility Optimized Networking.
f) (Optional) Expand Settings - Optional for more selections that are related to overlapping networks.

10. To extend a source network to an NSX VPC subnet:
    a) Select **Destination First Hop Router**.
    b) Make an NSX VPC selection and then click **Select**.
    c) Enter the Gateway IP address and Prefix.
    d) Select the network extension appliance supporting the NSX VPC extension.

       The network extension appliance information is based on your service mesh configuration. If your service Mesh configuration only supports NSX VPC, this appliance is selected by default.
    e) Select an access mode from one of the following options:

       - **Public** - to provide public access to and from the VPC subnet.
       - **Private** - to allow network traffic from the VPC subnet but disallows external traffic to the VPC subnet.
       - **Private_TGW** (transit gateway) - to allow access to private subnets when those subnets are all part of the same NSX Project.
11. When extending to an Overlay network, specify the gateway IP address and the Prefix Length CIDR notation <gateway IP/Prefix Length>. For example: 192.168.10.1/24.

    The Gateway IP address and Prefix Length CIDR notation are not mandatory if the destination network is either a VLAN segment and the source is not an Overlay network or a vSphere DVPG. If you are extending to a VLAN network, provide the VLAN-ID.
12. (Optional) For each source network, expand **Settings - optional** and select the appropriate options:

Allow Overlapping Source VLAN	VCF Operations HCX blocks attempts to extend networks that have the same VLAN ID and gateway IP address. Select this option to override the system and allow duplicate VLAN IDs.
Allow Overlapping IP Subnet	By default, VCF Operations HCX blocks attempts to extend the same subnet to the same Tier 1 gateway or the same VPC if extending to a VPC subnet at the destination site. Select this option to override the system and allow overlapping IP subnets.
Allow Overlap with Network Profiles	By default, VCF Operations HCX prevents using a Gateway IP address that conflicts with the IP address range of any of the network profiles. Select this option to override the system and allow the overlap.
Allow Overlapping Destination VLAN	(Available for VLAN only) By default, VCF Operations HCX blocks to use the same destination VLAN ID. Select this option to override the system and allow duplicate VLAN IDs.

13. To finish, click **Submit** to extend the network or **Validate** to perform input validations without starting the network extension.

    To view the task status, navigate to the VCF Operations HCX dashboard and scroll down to the Activity Logs display.

    Note:

    If any conflicting operations on the same appliances are running, then network extension requests are queued. Examples of conflicting operations include service mesh edits or resyncs, appliance operations such as redeploy or force resync, and network extension using the same high availability (HA) group.

Return to the Network Extension page for a summary of information that includes the number of extended networks, extension appliances, mobility optimized networks, and network extension served by an HA group. The HA group name is used in place of an appliance name with an HA tag to indicate the HA activation.

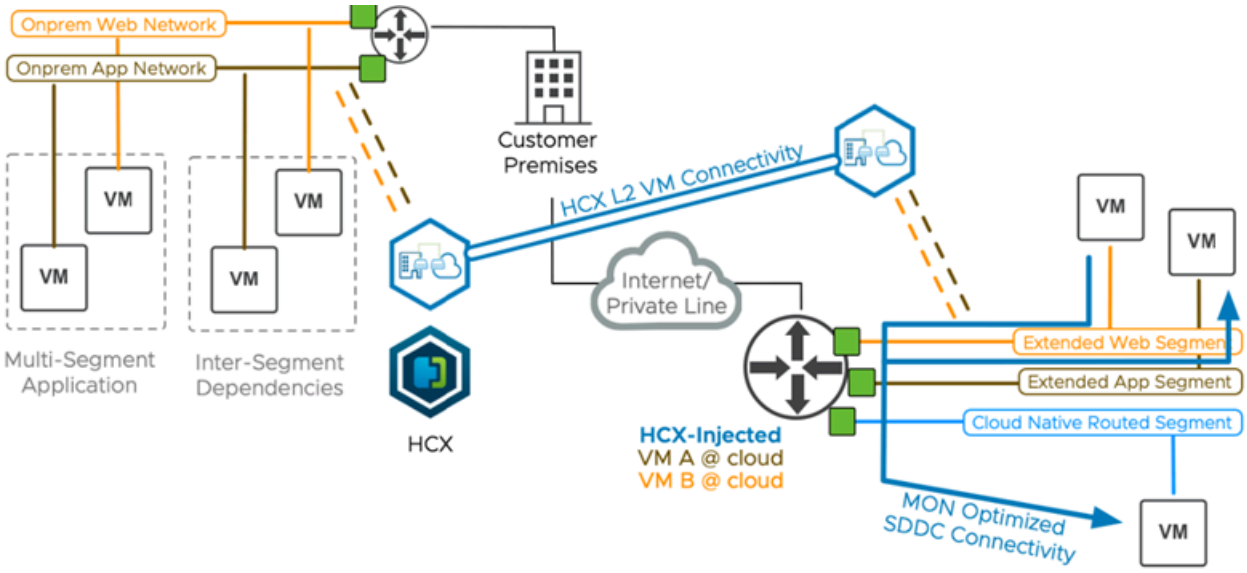
Understanding Network Extension with Mobility Optimized Networking

Mobility Optimized Networking (MON) is feature of the VCF Operations HCX network extension service. MON-enabled network extensions improve traffic flows for migrated virtual machines by enabling selective cloud routing (within the destination environment), avoiding a long round-trip network path through the source gateway.

Note:

MON is not supported for network extension to VLAN networks or VPC subnets.

The behavior of extended networks is such that all routed traffic for migrated workloads is directed back to the source-site gateway. MON allows you to configure the optimal path for migrated workload traffic to other extended network segments, cloud-native network segments, and Internet egress.

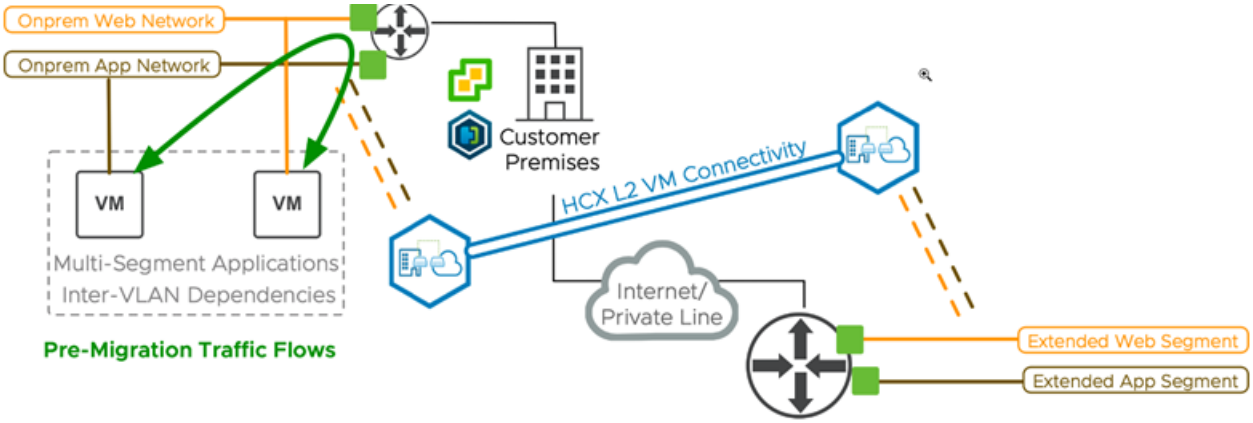


About Mobility Optimized Networking

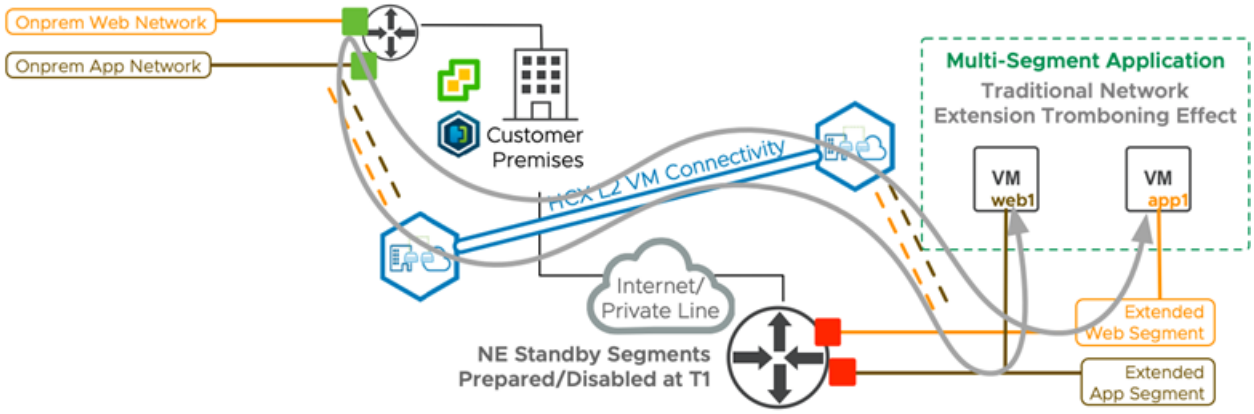
This section provides an overview of workload traffic flows using HCX network extension with and without Mobility Optimized Networking.

Use Cases for Mobility Optimized Networking

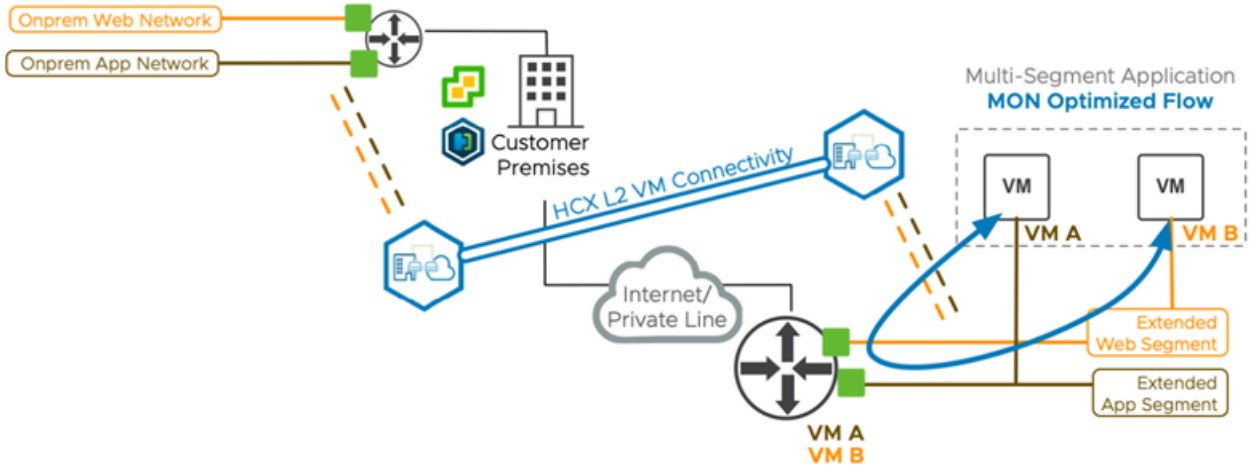
MON improves network performance and reduces latency for virtual machines that have been migrated to the cloud on an extended L2 segment. MON provides these improvements by allowing more granular control of routing to and from those virtual machines in the cloud.



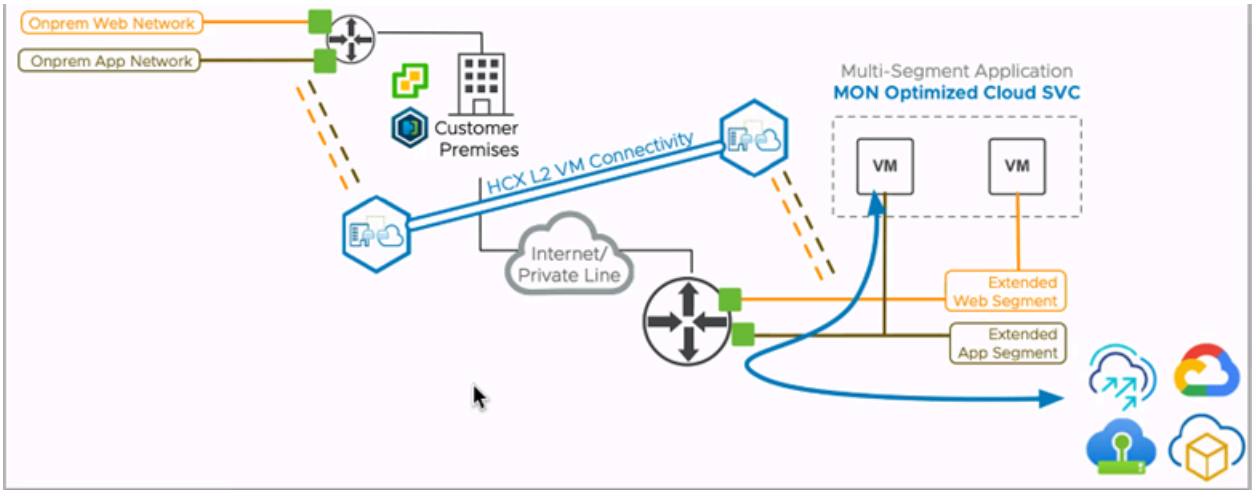
Without MON, HCX network extension expands the on-premises layer-2 network to the cloud SDDC while the default gateway remains at the source. The network tromboning effect is observed when virtual machines in the destination connected to different extended segments communicate.



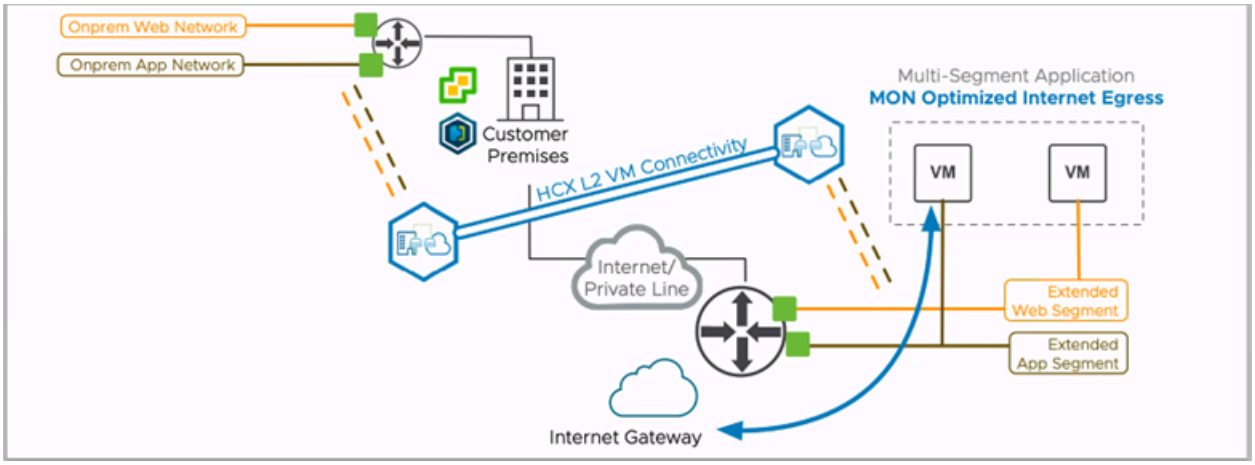
MON enables migrated virtual machines to reach segments within the SDDC without sending packets back to the source environment router.



MON can be configured to allow migrated virtual machines to reach services hosted within a public cloud.



MON enables migrated virtual machines to use the SDDC Internet interface (with SNAT).



Mobility Optimized Networking Outcomes by Migration Type

- HCX Bulk-migrated virtual machines are automatically MON-enabled in the SDDC.
- HCX vMotion and RAV-migrated virtual machines use the on-premises gateway until they are specifically configured to use the cloud gateway in the HCX UI/API.
- Virtual machines attached to the segment prior to enabling MON use the on-premises gateway until they are specifically configured to use the cloud gateway in the MON interface.

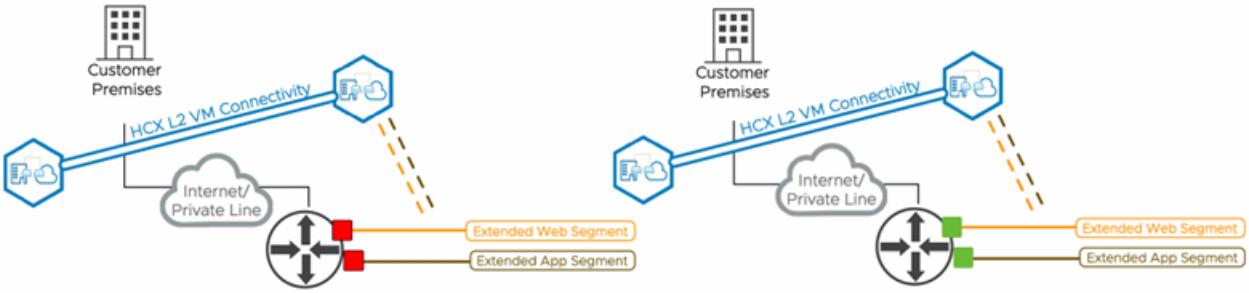
Mobility Optimized Networking Operation

Network extension with Mobility Optimized Networking provides the following functionality:

- Enable or deactivate MON at the time of stretching a network.
- Enable or deactivate MON for already extended networks.
- Enable or deactivate MON on an individual VM basis for VMs residing on extended networks in the SDDC.
- Display which VMs are using MON.
- When using VCF Operations HCX to vMotion a VM, preserve existing network connections while providing the option to activate Mobility Optimized Networking on that VM after migration.
- Configure MON Route Policy to define on-premises (non-SDDC) subnets or exception/deny subnets for local egress.

The following process explains what happens during the various phases of Mobility Optimized Networking.

1. Mobility Optimized Networking is enabled for a VCF Operations HCX extended segment.
   VCF Operations HCX enables the network ID (gateway IP) in the SDDC Compute Gateway. It is enabled with a limited /32 network mask.



2. Static routes are added in the SDDC Compute Gateway for migrated virtual machines on a VCF Operations HCX extended network.
   VCF Operations HCX adds reachability information for the migrated virtual machine (in the form of a virtual machine specific static route) to the SDDC Compute Gateway, allowing reachability within the SDDC. This VM static route is not advertised to the on-premises environment. The HCX L2 path is used to reach subnets not in the SDDC.
3. Using SDDC forwarding technology, the virtual machine uses the SDDC Compute Gateway to reach the SDDC networks.

For reachability outside of the destination Tier-1 gateway, the MON policy configuration is evaluated according to the MON policy configuration. Matching subnets are sent to the original premises router. Nonmatched subnets are sent to the Tier-0 gateway. For more information on MON policy routes, see Mobility Optimized Networking Policy Routes.

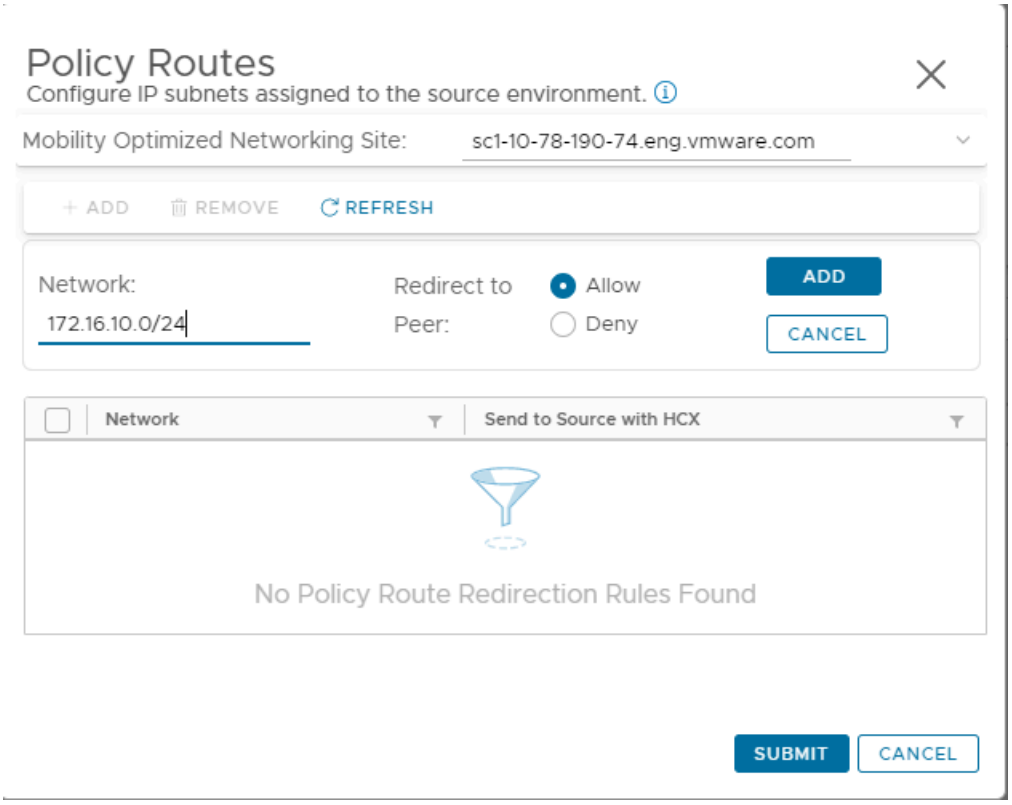
Mobility Optimized Networking Policy Routes

When the destination network for a traffic flow is not within the SDDC, the Mobility Optimized Networking policy is evaluated.

MON policy routes define which traffic is routed through the source gateway versus traffic that is routed through the cloud gateway. The **Advanced** tab in the network extension interface provides an option for configuring policy routes.

When the destination network for a traffic flow is not within the Tier-1 gateway router, the MON policy is evaluated:

- If the destination IP is matched and configured as allow in the MON policy configuration, the packet is forwarded to the premises gateway using the HCX network extension appliance.
- If the destination IP is not matched, or configured to deny in the MON policy, the packet is forwarded to the SDDC Tier-0 to be routed.



Example Policy Route Configurations
Mobility Optimized Networking policy route configuration settings can vary depending on the HCX deployment.

> **Important:**
> The examples in this section are generalized approaches that might not be suitable for all deployments. The policy route configuration defines how routed traffic is forwarded for MON enabled virtual machines. The configurations should be well understood in the context of the site to site routing design. Incorrect configurations can result in disrupted traffic for the MON enabled virtual machines.

Default MON Policy Configuration

The default MON policy includes all RFC-1918 networks. This policy configuration forwards private subnet traffic (not destined to segments within the SDDC) to the on-premises router and sends internet egress traffic to the SDDC tier-0 router.

Policy Configuration for Internet Egress On-premises

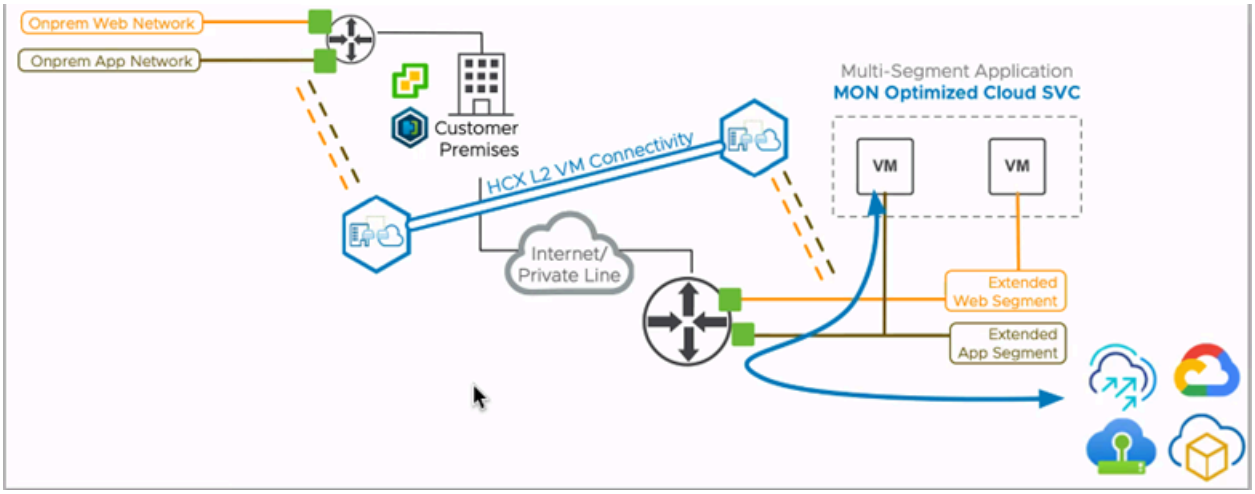
For MON deployments where security policies require internet access on-premises, replace the default MON Policy Configuration:

- Remove the default RFC-1918 entries from the Policy Routes interface.
- Add a single Allow entry for network 0.0.0.0/0.
  This policy configuration forwards private subnet traffic (not destined to segments within the SDDC) to the on-premises router and internet egress traffic, while maintaining routing symmetry.

Policy Configuration for Cloud Services

MON policy routing can be revised to achieve cloud service reachability.

- Configure the IP address ranges for the cloud based service as Deny entries (exclusions) to the MON Policy.
- Deny entries are sent to the Tier-1 router.



Requirements and Limitations for Mobility Optimized Networking Topologies

The HCX Mobility Optimized Networking (MON) feature routes network traffic based on locality of the source and destination virtual machines. MON operation requires the specific configuration of the network extension parameters and the network environment between the source and the destination sites.

Requirements for MON-Enabled Virtual Machines

- VMware Tools must be installed.

Environmental Requirements for MON

- Extended networks meet all NSX requirements. See Requirements for Network Extension.
- An NSX Tier-1 router connected to a Tier-0 router must exist in the destination site prior to enabling this feature.

  Note:

  Layer-2 network extensions can only terminate on a Tier-1 router.
- The default gateway IP and the DHCP Server IP cannot be the same. The gateway for the extended segment can provide DHCP services but must have a different IP address used for the DHCP server.

Limitations of MON

- MON does not provide traffic optimization from MON-enabled virtual machines to virtual machines on other extended networks without MON.
- MON provides optimization in the Cloud. MON does not provide optimization in the source.
- When a virtual machine with multiple NICs is connected to multiple MON enabled extended networks and Target Router Location is changed, that change applies to all the interfaces.

Configuring Mobility Optimized Networking

For data centers using NSX, configure Mobility Optimized Networking for workloads that require routing through the local gateway.

Verify that the destination environment meets the requirements described in Requirements and Limitations for Mobility Optimized Networking Topologies.

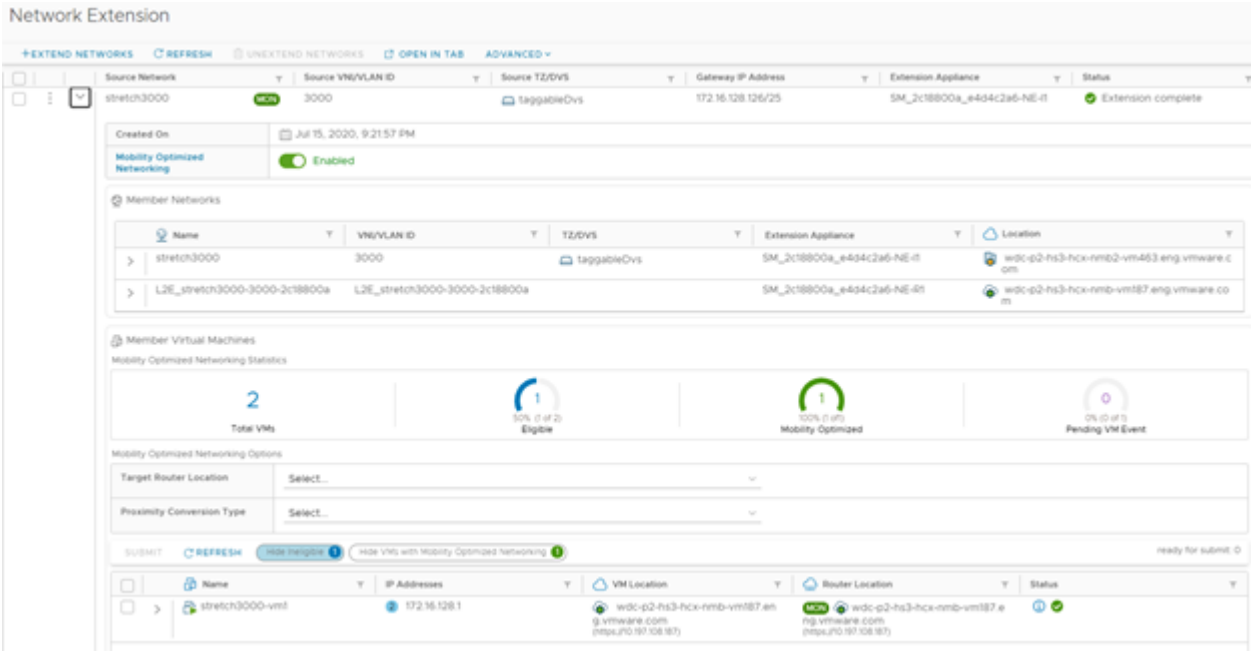
Configuring MON for an extended network is available as one of the Advanced options when creating the extended network.

In some cases, you might not configure MON for an extended network. In other cases, the virtual machines that are members of the extended network are migrated using vMotion or Replication Assisted vMotion, which requires setting MON manually for those VMs.

Use the following procedure to configure MON for an extended network and for virtual machine workloads.

1. In the VCF Operations HCX UI, navigate to **Services** > **Network Extension**.
2. To see the extended networks in the Network Extension screen, expand a site pair.

   The system highlights network extensions activated for MON with an icon.
3. To display network details, expand each extension.



4. If MON is not enabled on the network extension, set the slider for Mobility Optimized Networking, and click **Enable**.

   The Network Extension screen is updated, displaying the member virtual machines and the **Router Location** selection drop-down menu. The **Router Location** identifies the target destination Tier-1 gateway for the member virtual machines.

   Note:
   The Bulk migration operations automatically set the **Target Router Location** as the cloud Tier-1 router. Virtual machines migrated using HCX vMotion and Replication Assisted vMotion (RAV), or VMs attached to the segment prior to enabling MON, continue to use the on-premises router until the **Target Router Location** is set manually in the next step.
5. To enable MON optimization on VMs that have been migrated using vMotion or RAV, or VMs attached to the segment prior to enabling MON, set the Target Router Location for each VM.
   a) Select a VM and expand the row.

      You can select multiple VMs using the check box next to each workload.
   b) Set the Target Router Location for each VM by selecting the destination from the drop-down menu.
   c) Click **Submit**.

   HCX configures all selected VM workloads for Mobility Optimized Networking.

   Note:
   To deactivate MON optimization for a VM, select **Target Router Location** and select the source (**On-premises**) option from the drop-down menu.

If required, configure the policy routes. See Configure Policy Routes for Mobility Optimized Networking.

Change the Virtual Distributed Router MAC Address

You can change the Virtual Distributed Router (VDR) MAC address for extended networks on both the source and destination sites.

If you change the VDR MAC address on the destination site, you must update the changes in HCX Manager. This ensures that communication between workloads on the source and destination sites is not impacted.

When a network is extended, it uses the VDR MAC address which was configured then. Changes to the VDR MAC address anytime later, can impact the traffic on any extended networks, since the VDR MAC address configured is now out-of-sync. To rectify this, VCF Operations HCX provides a way to use the updated VDR MAC address and re-configure the extended networks.

1. Navigate to **Network Extension**.
2. Select the extended network to which you made the changes.

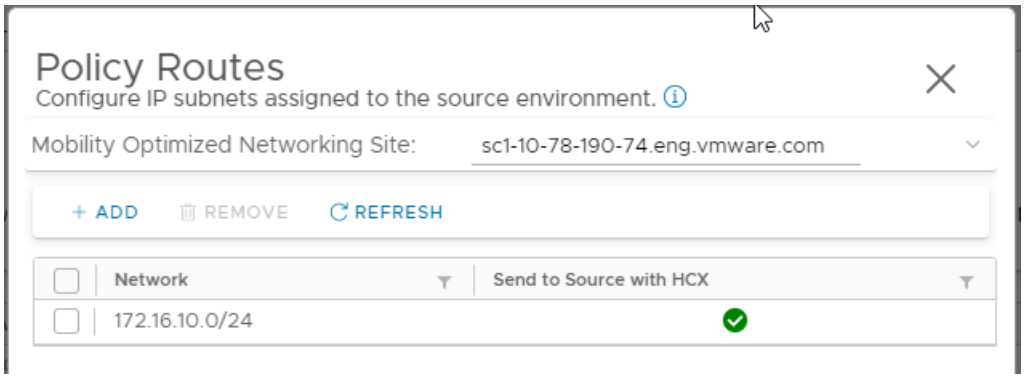
   A message is displayed informing you about a VDR MAC address mismatch.
3. Click the **Actions** button, and select **Update VDR MAC Address** from the dropdown menu.
4. In the Update VDR MAC Address window, click **Update**.

Configure Policy Routes for Mobility Optimized Networking

With Mobility Optimized Networking, you can control which traffic is routed locally using the cloud gateway versus traffic that goes out through the source gateway. Policy routes define which traffic is routed through the source gateway. All other traffic is routed through the cloud gateway.

1. In the VCF Operations HCX UI, navigate to **Network Extension**.
2. Select a Site Pair from the dropdown menu.
3. Click the **Advanced** tab.
4. Click **Policy Routes**.
5. Select a destination site from the dropdown menu.
6. Click **Add**.
7. Complete the entries for **Network IP Address** and **Prefix Length**.
8. Set **Redirect to Peer** to **Allow** or **Deny**.
9. Click **Add**.

   The policy is applied to the MON extended network.



10. To complete the operation, click **Submit**.

Viewing Network Extension Details

VCF Operations HCX provides detailed tunnel state information for the network extension appliance, and connection information for each extended network associated with that appliance.

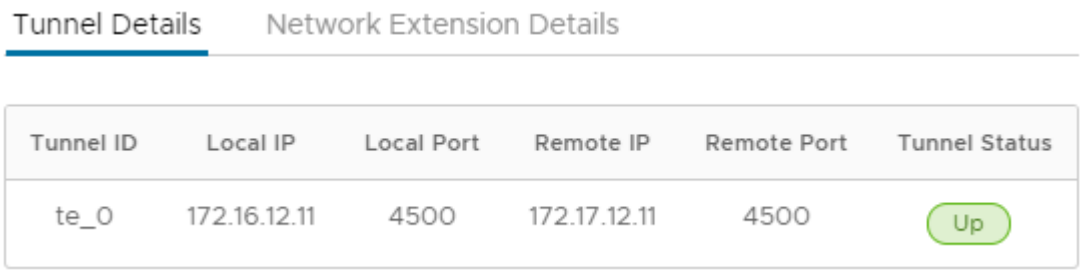
VCF Operations HCX maintains Up or Down state information regarding the tunnel that is used for network extension. The information includes the tunnel ID, local IP address and port number, remote IP address and port number, and tunnel status.

Connection statistics for extended networks includes the bit rate, bytes transferred and received, packet rate, and packets transferred and received. The statistics are updated every 1 minute and stored in the VCF Operations HCX database. The bytes and packets that are transferred and received information reflects the total number since the network extension appliance was powered on.

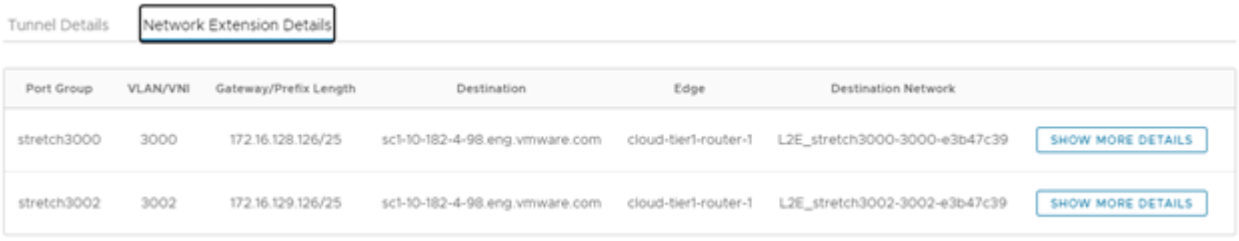
Network extension statistics also detect and display the MAC address of each virtual machine NIC on the extended network. These statistics can be helpful in determining the status of a particular virtual machine on that network. A search option is provided to filter the list of addresses.

1. From the VCF Operations HCX Dashboard, go to **Infrastructure** > **Interconnect** > **Service Mesh**.
2. Click **View Appliances**.
3. From the list of appliances, expand the Network Extension appliance.

   The system displays options for selecting **Tunnel Details** (or in the case of an enhanced service mesh, Network Paths), or **Network Extension Details**. **Tunnel Details** is selected by default.



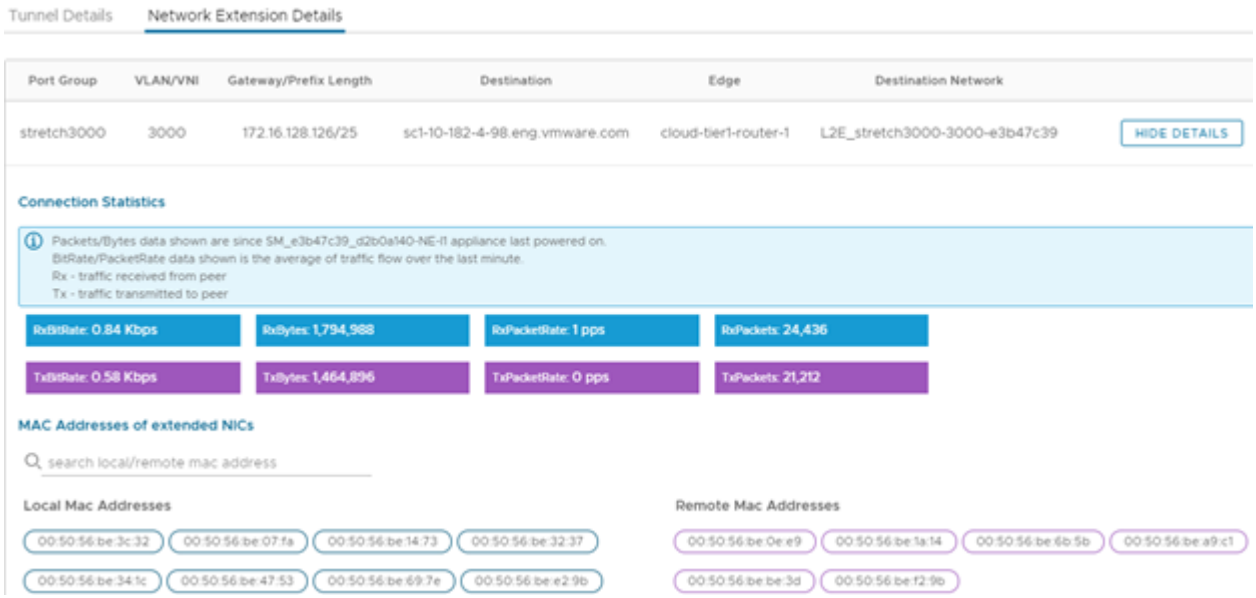
4. To view details regarding extended networks, click **Network Extension Details**.



5. To view the connection statics information for a specific network, click **Show More Details**.

   The following screenshot shows network extension appliance details for service meshes not upgraded to enhanced mode. For the appliances of a upgraded service meshes, the learned MAC addresses are not displayed.

   VCF Operations HCX displays the connection statistics information. To see updated information, click the refresh icon.



6. To close the display, click **Hide Details**.

Log in to the peer VCF Operations HCX site to view connection statistics information from that site.

In-Service Upgrade for Network Extension Appliances

VCF Operations HCX provides options for network extension upgrade or redeployment that help to minimize service downtime and disruptions to on-going L2 traffic.

The following prerequisites apply for In-Service upgrade or deployment.

> **Note:**
>
> In the event that the prerequisites for In-Service mode are not met, use Standard mode to complete Network Appliance upgrade or redeployment operations.

- Appliance tunnels must be in Up state as shown by navigating to **Interconnect** > **Service Mesh** > **Appliances**. Appliances with Down or Degraded tunnels do not support In-Service upgrades.
- HCX Manager must be able to reach network extension appliances via the management network.
- For service mesh appliances not upgraded to enhanced mode:
  - For each Network Appliance, the same number of used IP addresses are available in both the VCF Operations HCX Management and Uplink Network profile address pools during the Upgrade or Redeploy process. For example, upgrading three network extension appliances at the same time requires three available Management IP addresses and three available Uplink IP addresses. Once the process completes, the IP addresses used by the previous appliances are released.
- For service mesh appliances upgraded to enhanced mode:
  - The HCX Intrasite Control network is used between the partner HA network extension appliances. This network is optional in the compute profile only.
    - The Uplink IP addresses are reused, with no additional Uplink IP addresses required.
    - The Intrasite Control IP address is required.
    - To perform an In-Service upgrade: three network extension appliances, three Management IP addresses, and three Intra-site Control IP addresses are required.

The network extension appliances are a critical component of many VCF Operations HCX deployments, during migration and post migration, since extended networks continue to be used after migration in a hybrid environment. VCF Operations HCX operations using extended networks can be impacted during network extension appliance upgrades or when a change to the compute profile or service mesh requires redeploying the network extension appliance.

Network extension appliances are available for **In-Service** or **Standard** (default) upgrade.

Note:

In-Service upgrade is not available for network extension High Availability (HA) groups. HA groups use the failover process to complete the upgrade. In this case, the Standby pair is upgraded first. After the Standby upgrade finishes, a switchover occurs and the Standby pair takes on the Active role. At that point, the previously Active pair is upgraded and takes on the Standby role.

Note:
The network extension In-Service/Standard Upgrade or Redeploy operation will fail if one of the stretched networks has Port Bindings set as “Ephemeral – no binding” for a DVPG.

With an In-Service upgrade or redeployment, the following high-level workflow applies when running in Standard mode.

- New appliances are provisioned at the source and destination site.
- New Uplink and Management IP addresses are reserved for each new network extension appliance.
- NICs on the new appliances are connected, including NICs for extended networks, except the NIC connection state is flagged as Down.
- Secure tunnel connections are established between sites.
- Old appliance Bridge NICs are disconnected. New appliance Bridge NICs are connected.
- The old appliances are deleted. IP addresses used for the old appliances are released and made available.

With an In-Service upgrade or redeployment, the following high-level workflow applies when running in enhanced mode.

- Appliances are provisioned at both the source and destination sites.
- New Intrasite control and management IP addresses are reserved for each new network extension appliance.
- NICs are attached to the newly provisioned appliances.
- High Availability is established between the old and new appliances at each site.
- A High Availability failover is triggered, ensuring the newly deployed appliance becomes active and starts serving traffic.
- High Availability is then broken on both sites, ensuring traffic continues to flow through the new appliances.
- The old appliance is deleted, and its IP addresses are released and returned to the available pool.
- Uplink IP addresses get re-used and are not newly allocated.
- Management IP addresses continue to be additionally allocated.

As a result of this workflow, switchover from the old appliance to the new appliance requires only minimal action and can happen within a few seconds or less. The actual time that it takes to return to forwarding traffic depends on the overall environment.

With a Standard upgrade or redeployment of the network extension appliances, the new appliances use the same Uplink and management IP addresses as the existing appliances. Using the same IP addresses means that VCF Operations HCX must disconnect the existing appliances so that those IP addresses are available for the new appliances. In this case, tunnel connections are established only after switchover happens, requiring 30 seconds or more to re-establish data traffic across extended networks.

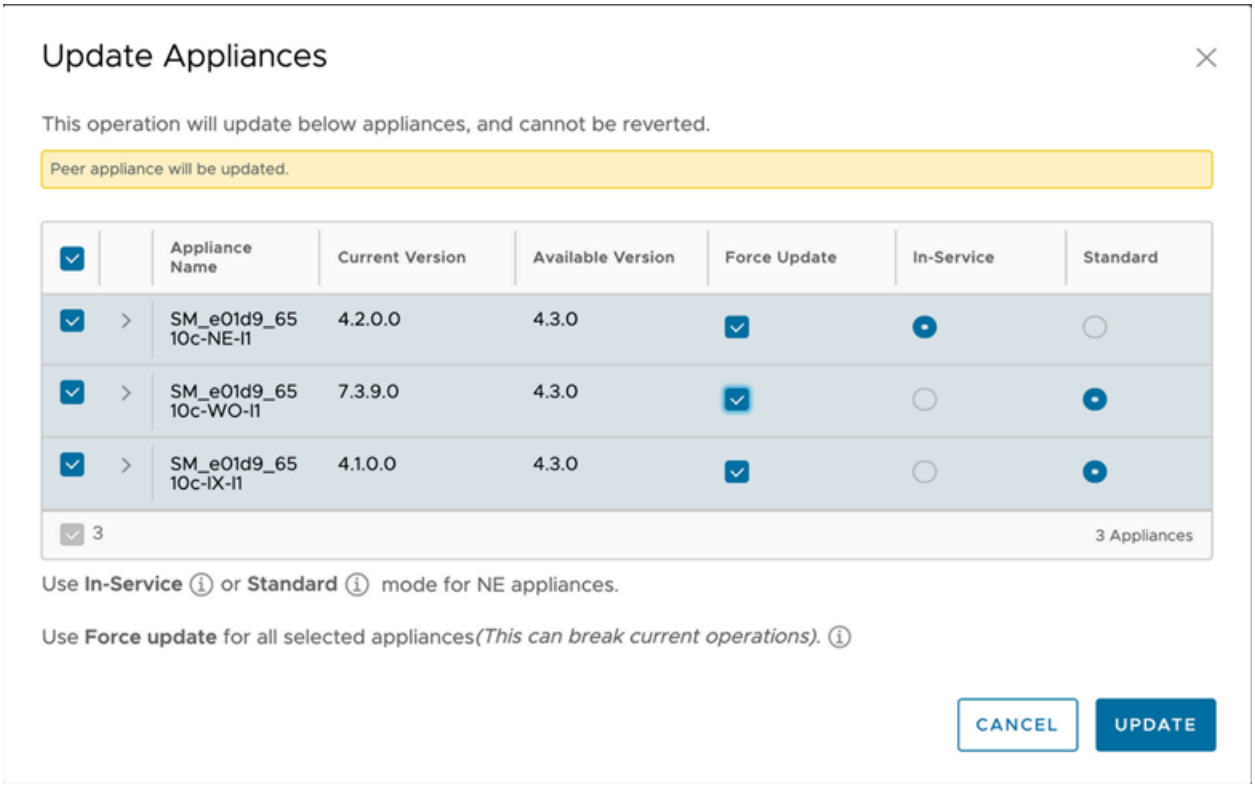
Network extension upgrade or redeployment operations display a pop-up window with the option for In-Service or Standard mode deployment.

For more information about upgrading HCX appliances, see Upgrade VCF Operations HCX.

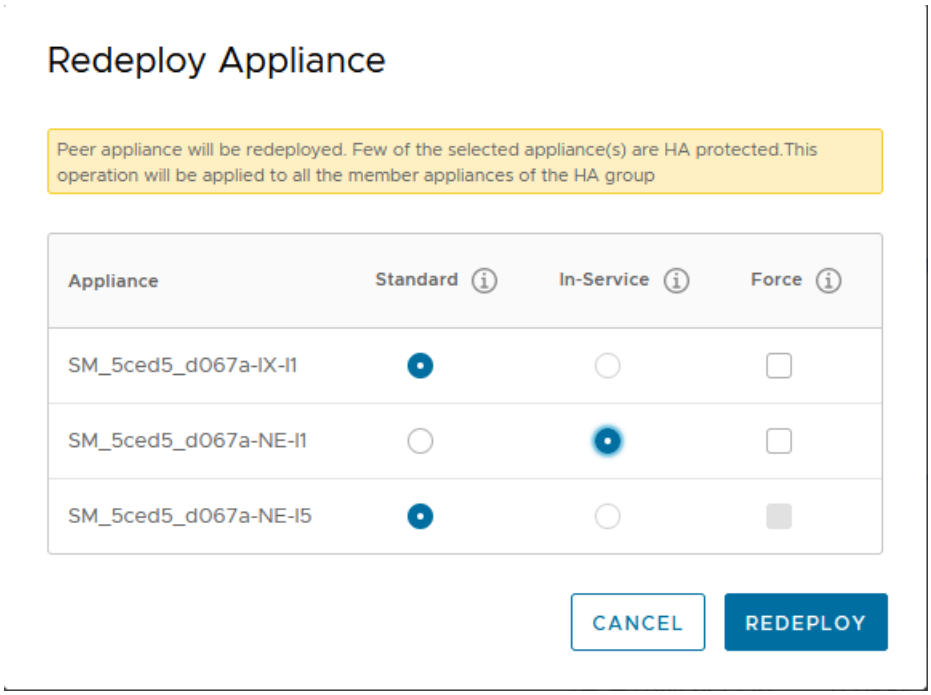
1. Select the network extension appliances for update or deployment.
2. Click **Update** or **Redeploy**.

   The selections that appear depend on the operation being performed. The option for selecting Standard or In-Service mode can appear when updating or redeploying the network extension appliances from either the service mesh View Appliances or View Topology window, or from the service mesh Edit or Resync window.

   In the following Update Appliances screen example, NE-I1 is a standalone appliance that supports In-Service upgrade:



In the following Redeployment Appliance screen example, NE-I1 is a standalone appliance that supports In-Service upgrade, while NE-I5 is part of an HA group with no support for In-Service upgrade.



3. Depending on the operation, click **Update** or **Redeploy**.

Verify the Update or deployment task by navigating to **Interconnect** > **Service Mesh** > **Tasks**. After the task completes, check that the tunnel status is Up.

Removing a Network Extension

Removing a network extension prevents further cross-site communications between virtual machines residing on that network. This operation is typical when the source side network is vacated.

You can remove a network extension at any time, but be aware that removing an extended network can impact the network infrastructure. Migrated virtual machines that use a source environment DHCP server or that use statically assigned network services like DNS or NTP can lose those services after unextending a network.

1. In the VCF Operations HCX UI, select **Network Extension**.

   The system displays a list of site pairs.

2. Select the checkbox next to the network you want to unextend, and from the **Actions** dropdown menu, select **Unextend Networks**.

   If the **Unextend Networks** option fails, select **Force Unextend** in the following cases:

   - Where the **Unextend Networks** option fails due to the state of the network extension appliance. The **Force Unextend** action removes the extension and clears internal operations and processes from the source and the destination sites.
   - Where the source components are no longer available, the **Force Unextend** option removes Extension components from the source and the destination sites.
   - When the on-premises site is not available or powered off, the existing extended networks are required to be unextended from target/Cloud network extension (NE) interface to ensure distributed logical router (DLR) on the destination side is connected for all extended segments where migrated cloud VMs are hosted. See, Restrictions and Limitations for Network Extension.

3. (Optional). To connect the remote side gateway, expand the network entry and select **Connect cloud network to cloud edge gateway after unextending**.

   Dynamic routing can be activated on the Tier 1 gateway as part of the OSPF or the BGP configuration. By default, the cloud segment/VPC subnet is left disconnected from the edge gateway/VPC respectively after removing the network extension. This is done to prevent an edge gateway from advertising a route to the cloud segment and causing a potential routing conflict with the network in the on-premises data center. Selecting this option connects the segment to the Tier 1 gateway after removing the network extension. If dynamic routing is activated, the network is advertised from the Tier 1 gateway.

   Note:

   Unextending a network removes the L2 bridged path without removing the NSX segment or Port group, or NSX interface. If you don't select this option, the NSX router interface remains disconnected.

4. (Optional). To prevent inconsistent network routing from the destination site to the source site when the VPC subnet at the source remains in use, from the expanded network entry, select **Block reuse of subnet CIDR after subnet**.
5. Click **Unextend**.

Migrating Network Extensions

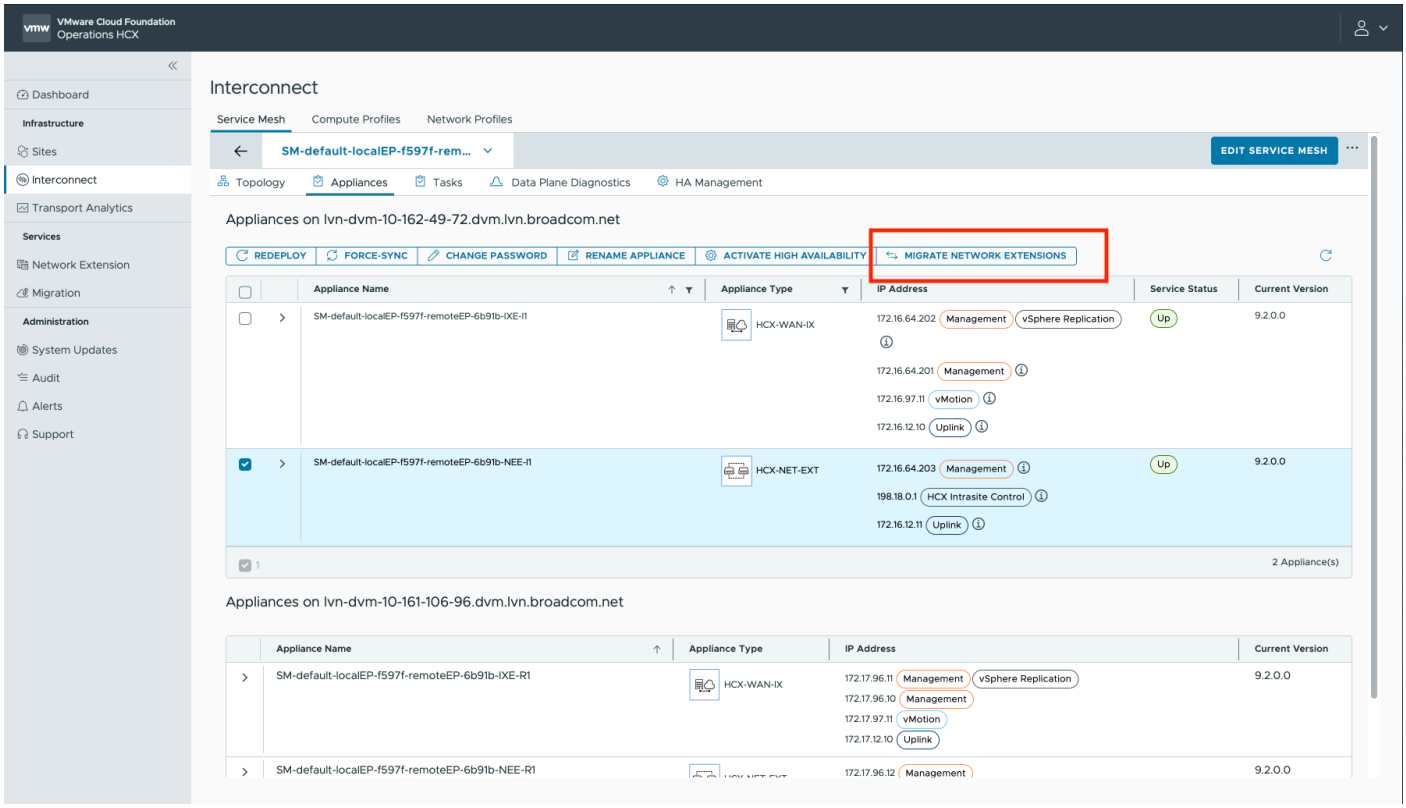
When migrating network extensions, you move existing network extensions from one pair of HCX-NET-EXT appliances to another pair with minimal downtime. This operation is performed to adopt enhanced service meshes from legacy appliances to enhanced appliances and to load balance between enhanced appliances.

Migrating network extensions can be performed on VCF Operations HCX environments that are updated to the latest release. Migration is supported for current network extensions to upgraded systems, and also between upgraded systems.

You can migrate one or more network extensions either at the appliance level or at the network extension level. Meaning, migrate all the network extensions that are part of one appliance, or select specific network extensions that span across multiple appliances.

To migrate a network extension at the appliance level:

1. Navigate to **Interconnect**, and click **View Appliances**.
2. Select an appliance and then select the **Migrate Network Extensions** tab.

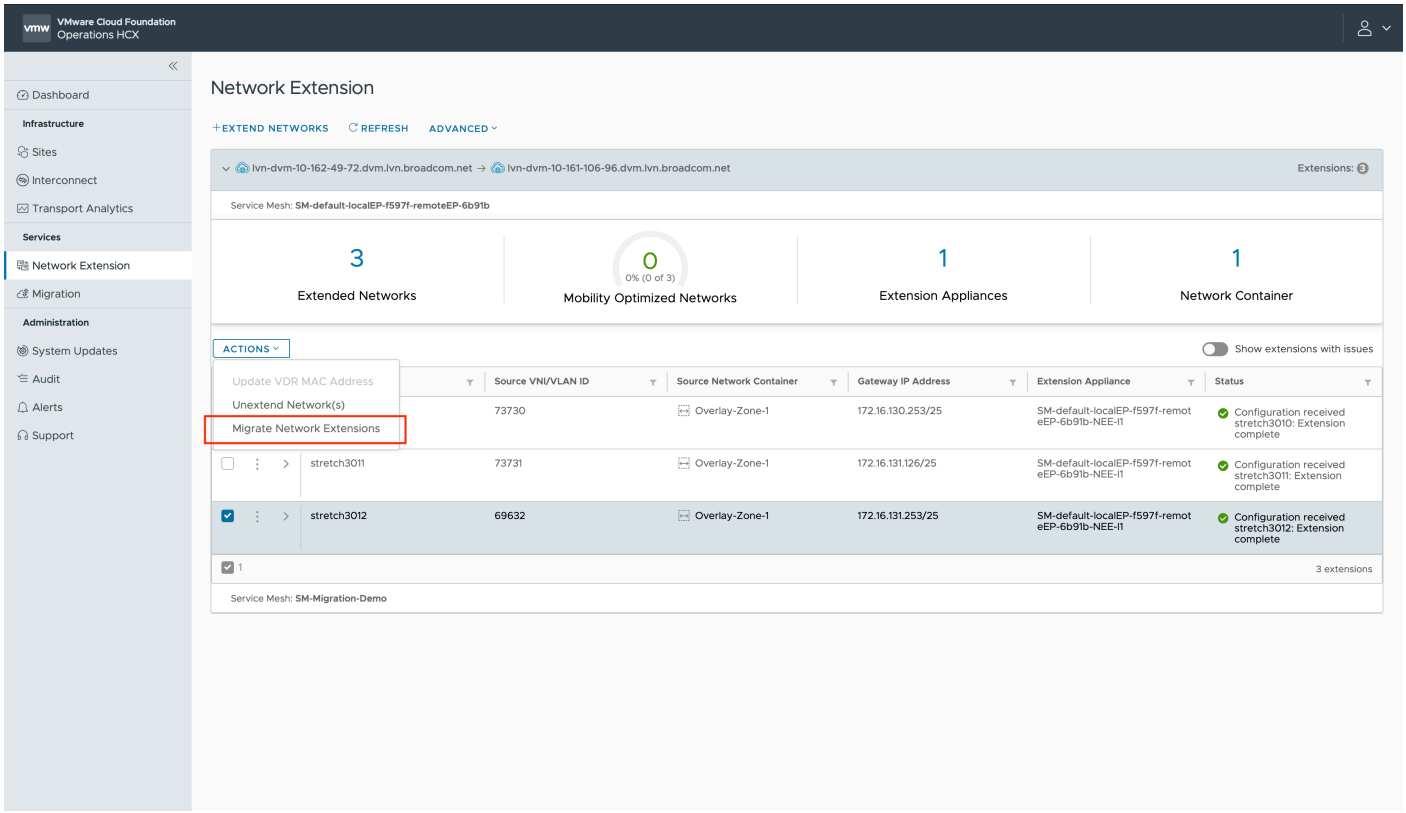


You can migrate one or more network extensions on the appliance.

3. Select the target service mesh and the target extension appliance, and click **Transfer**.
   The extended network is moved from the source appliance to the selected target appliance.

To migrate a network extension at the network level:

1. Navigate to **Network Extension** and select a network.
2. Click the **Actions** button and select **Migrate Network Extensions.**



3. Select the target service mesh and the target extension appliance, and click **Transfer**.
   The extended network is moved from the source appliance to the selected target appliance.

Restrictions to migrating network extensions

The appliance serving the network extension and the selected target appliance must belong to the same network container.

There must be available extension capacity on the appliance you want to migrate. An appliance only appears in the list of appliances if there is available capacity to complete the extension.

Understanding Network Extension High Availability

Network extension High Availability protects extended networks from a network extension failure at either the source or remote site.

Overview

The network extension High Availability (HA) setup requires four network extension appliances, with two appliances at the source site and two at the remote site. Together, these two pairs form the HA Group, which is the mechanism for managing network extension High Availability. Appliances on the same site require the similar configuration and must have access to the same set of resources.

Similar to a standalone network extension deployment, where one network extension appliance at the source site pairs with another standalone network extension appliance at the remote site - each network extension appliance of an HA group pairs with another network extension appliance of the same HA group at remote site. This pairing relationship between the two appliances does not change.

Through a process of role negotiation, appliances of a pair are either Active or both Standby. During a failover event, the Standby pair takes over the network extension service from the Active pair.

The four network extension appliances of an HA group negotiate their roles of Active or Standby automatically after the HA group is formed. Following role negotiation, appliances of a pair are either both Active or both Standby. A heartbeat

signal between the Active pair and the Standby pair at each site synchronizes the two appliances. A loss of heartbeats between the Active pair or the Standby pair at either the source or the remote site triggers a failover.

VCF Operations HCX uses vSphere DRS host anti-affinity to place the Active and the Standby appliances on separate hosts.

When an HA Group is operating normally, the HA state is Healthy. If a problem is detected with the Active appliance, the peer Standby appliance pair starts failover actions and sets its role to Active. At this point, the HA state for the group switches to Degraded. Following failover, if the failed network extension appliance has recovered from its failure, this appliance and its peer appliance in the pair renegotiate their roles to be Standby, and the HA group returns to Healthy state. An HA-enabled network extension Appliance can enter a Failed state when it encounters unexpected conditions during HA related operations. The HA group also enters the Failed state as one of the appliances is in a Failed state. The **Recover** selection in the HA Management tab can recover an HA group from Failed state by redeploying the appliance that is in Failed state.

For a summary of network extension High Availability operational states and roles, see Monitoring Network Extension High Availability.

Considerations for Network Extension High Availability

- Network extension HA provides only appliance level resilience. Appliance Uplink resiliency is achieved using the Application Path Resiliency feature in the service mesh or multiple HCX uplinks.
- Each Active and Standby pair is managed as an HA group, which includes upgrading and redeploying appliances. The process for redeploying and upgrading HA groups is the same as with standalone appliances, except that the operation is applied to both Active and Standby appliances at both the source or remote site. For a list of HA group management operations, see Managing Network Extension High Availability.
- Network extension High Availability protects against one network extension appliance failure in an HA group. More than one appliance failure in the same HA setup at the same time disrupts the network extension service.
- Network extension HA operates in Active/Standby mode.
- Network extension HA operates without pre-emption, with no automatic failback of an appliance pair to the Active role.
- Network extension HA Standby appliances are assigned IP addresses from the network profile IP pool.

Limitations for Network Extension High Availability

- Following a failover event, policy-routed Mobility Optimized Networking (MON) traffic takes longer to recover than non-policy routed traffic. This is due to the time needed for the MON service to re-discover the next hop for the Policy Route traffic after the failover.
- Network extension HA does not support Storage DRS anti-affinity.

Activating Network Extension High Availability

Activate network extension High Availability to provide failover protection for extended networks.

- In the compute profile at all source and destination sites, the network extension appliance limit is configured for at least two times the number of network extension appliances that use networkextension HA, plus the number of appliances required to support existing non-HA extensions. See Create a Compute Profile.

  For example, you plan to create two new HA extensions in addition to having two existing non-HA extensions from a single source to a single destination site. In this case, the compute profile configuration at all source and destination sites must support at least six appliances each, which is two for each HA extension, and two for non-HA extensions.
- At all source and destination sites, all network extension appliances required for the HA pairs are deployed prior to creating network extension HA groups. These appliances must not have any existing network extensions assigned to them.

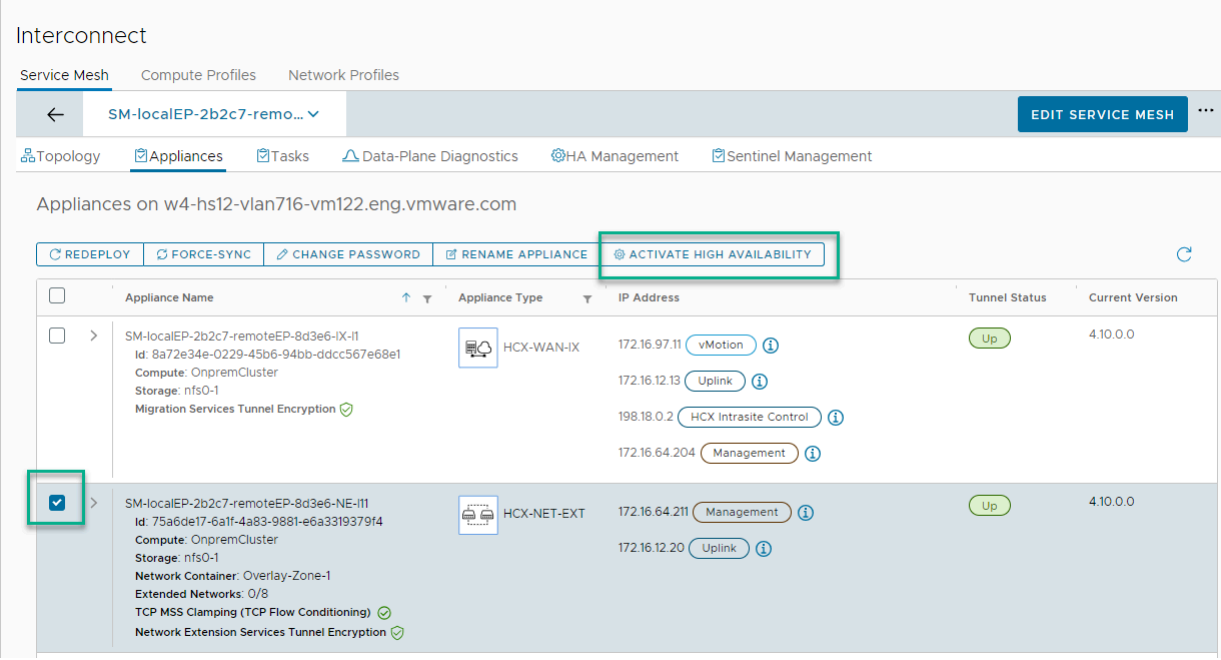
  For example, you plan to create two new HA extensions to take over for two existing non-HA network extensions from a single source to a single destination site. This example requires that you create four new network extension

appliances at the source site and four new appliances at the destination site before creating the two network extension HA groups.

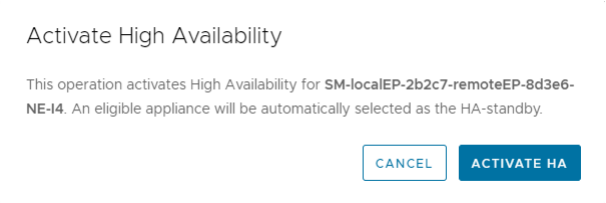
- In the service mesh, the network extension Appliance Scale Out appliance count is set to provide enough appliances to support network extension objectives, including any network extension HA groups. See Create a Service Mesh for vSphere-Based Site Pairs.

Network extension High Availability activation is initiated only from the source site.

1. At the source site, navigate to **Interconnect** > **Service Mesh** > **Appliances**.
2. Select an available network extension appliance and click **Activate High Availability**.



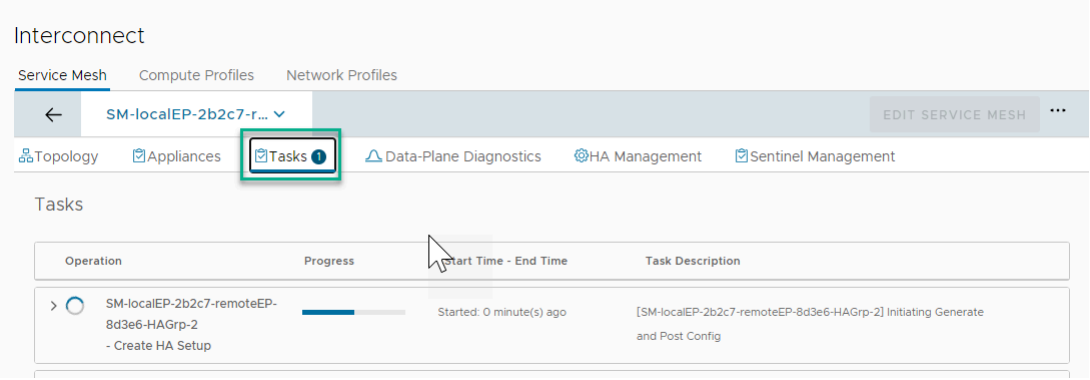
A pop-up window prompts you to confirm activation.



If the network extension appliance is already extending a network, or not enough appliances are available to create an HA group, the pop-up window provides information about the issue.

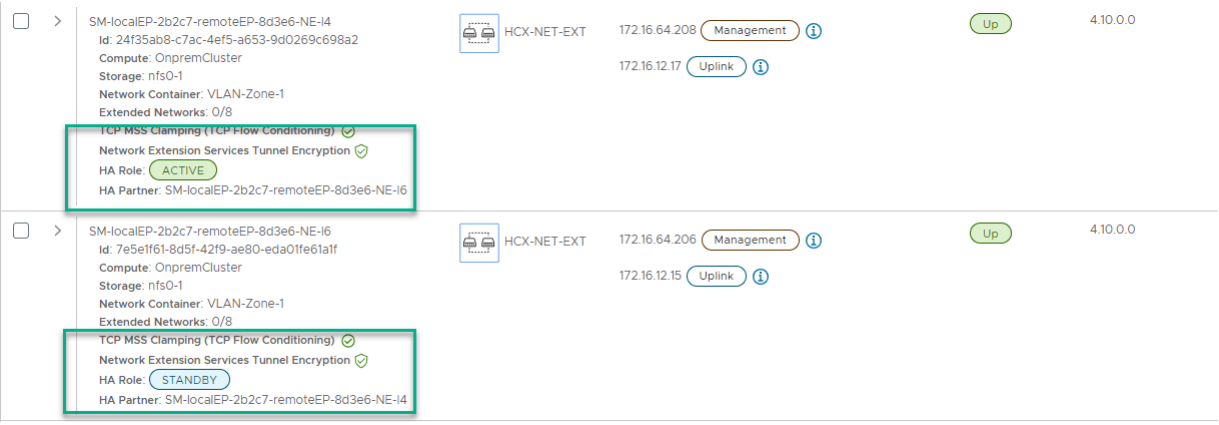
3. Click **Activate HA**.

   The system starts the process of creating the HA group. HCX Manager automatically selects the second pair of network extension appliances to form the HA group. To view the activation progress, select the **Tasks** tab.

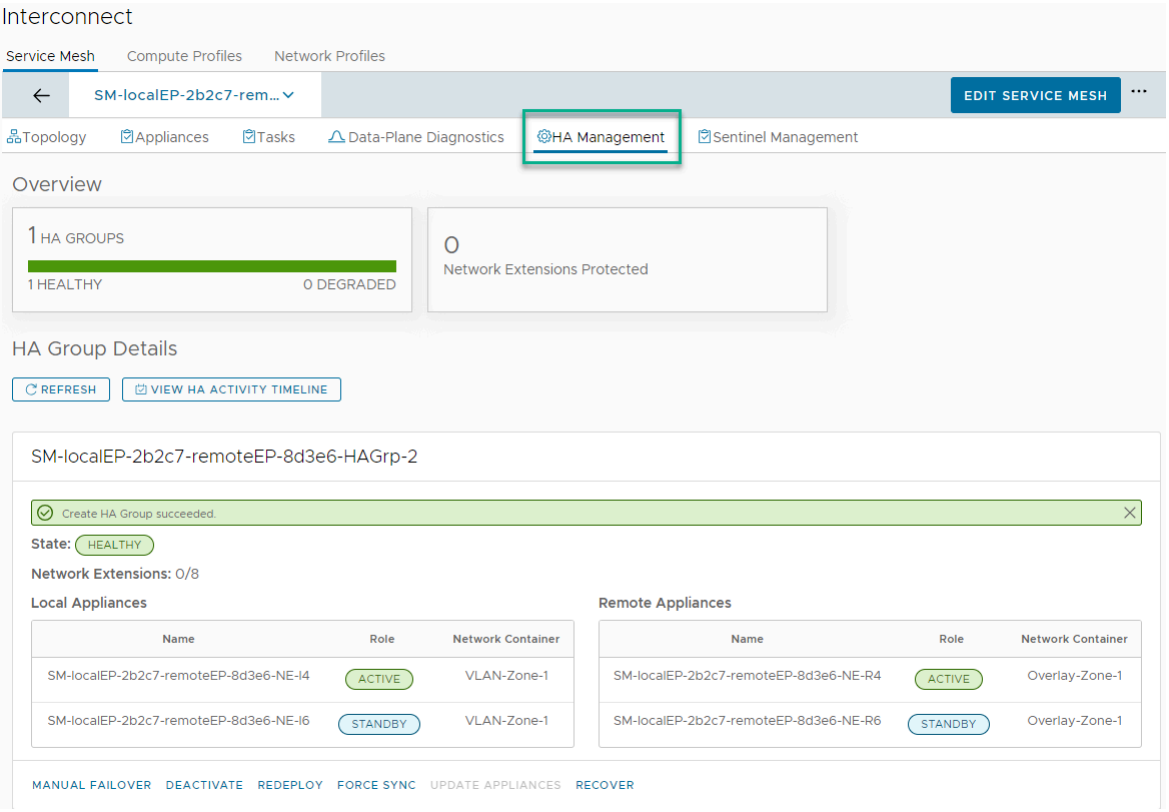


4. Review the list of appliances to verify the Active and Standby assignments.

   The appliance information shows both the HA group role and the partner appliance in the group.



5. To view detailed state and role information about HA groups, click the **HA Management** tab.



After the group is created, the Active and Standby roles for the local and remote appliances display on the HA Management page.

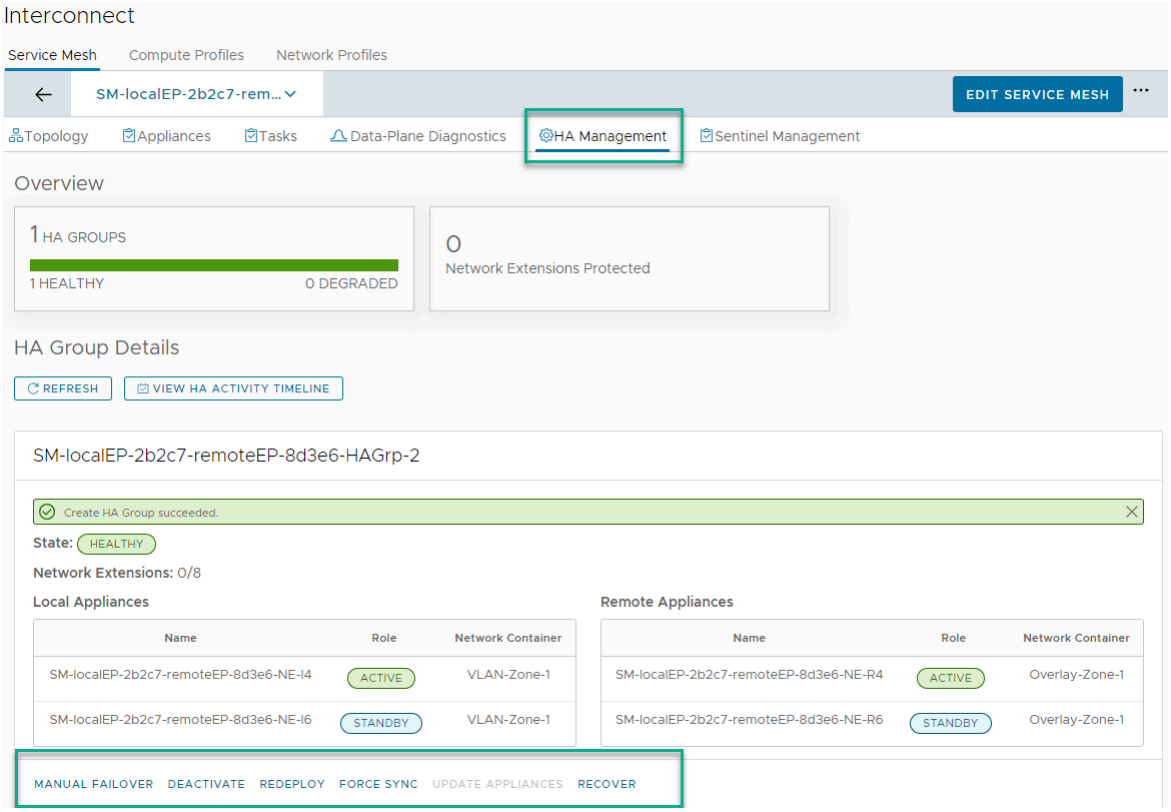
Select a network extension HA group when extending networks. See Extending Networks.

Managing Network Extension High Availability

Each Active and Standby pair, at either the source or remote site, is managed as an HA group.

The process for managing network extension HA groups is the same as with standalone network extension appliances, except that the operation is applied to both Active and Standby appliances at the respective source or remote site.

Perform network extension High Availability management operations in the HA Management interface.



Operation	Description
Manual Failover	The HA Active role transitions to the Standby network extension appliance in the HA Group. Network extensions are unprotected during the operation.
Deactivate	This operation deactivates the HA group. Any association between all the participant appliances is dissolved and the high availability capability for all the associated network extensions is lost. All HCX-NET-EXT appliances return to be standalone HCX-NET-EXT appliances for use in extending networks. When an HA Group is deactivated, any configured network extensions continue to be served by the standalone appliance pair that was in the Active role. This operation includes a **Force Deactivate** option that attempts a best effort to deactivate HA Group even if any error occurred during the process. Using **Force Deactivate** can result in stranded network extensions on the Standby NE appliances.
Redeploy	This operation redeploys the HA group appliances. The operation also redeploys peer appliances at the remote site.

Operation	Description
Force Sync	This operation synchronizes the configuration on HA group appliances, including peer appliances, at the remote site. This operation is applied to all the member appliances of the HA group.
Update Appliances	Use this option to update network extension appliances that are part of an HA group.
Recover	This operation attempts to return an HA group to a Healthy state. This can require redeploying some appliances. To track the state of a recover operation, navigate to **Interconnect** > **Service Mesh** > **View Appliances**, and select the **Tasks** tab.

Monitoring Network Extension High Availability

Information for monitoring network extension High Availability activity includes the HA group health, group roles, and appliance status.

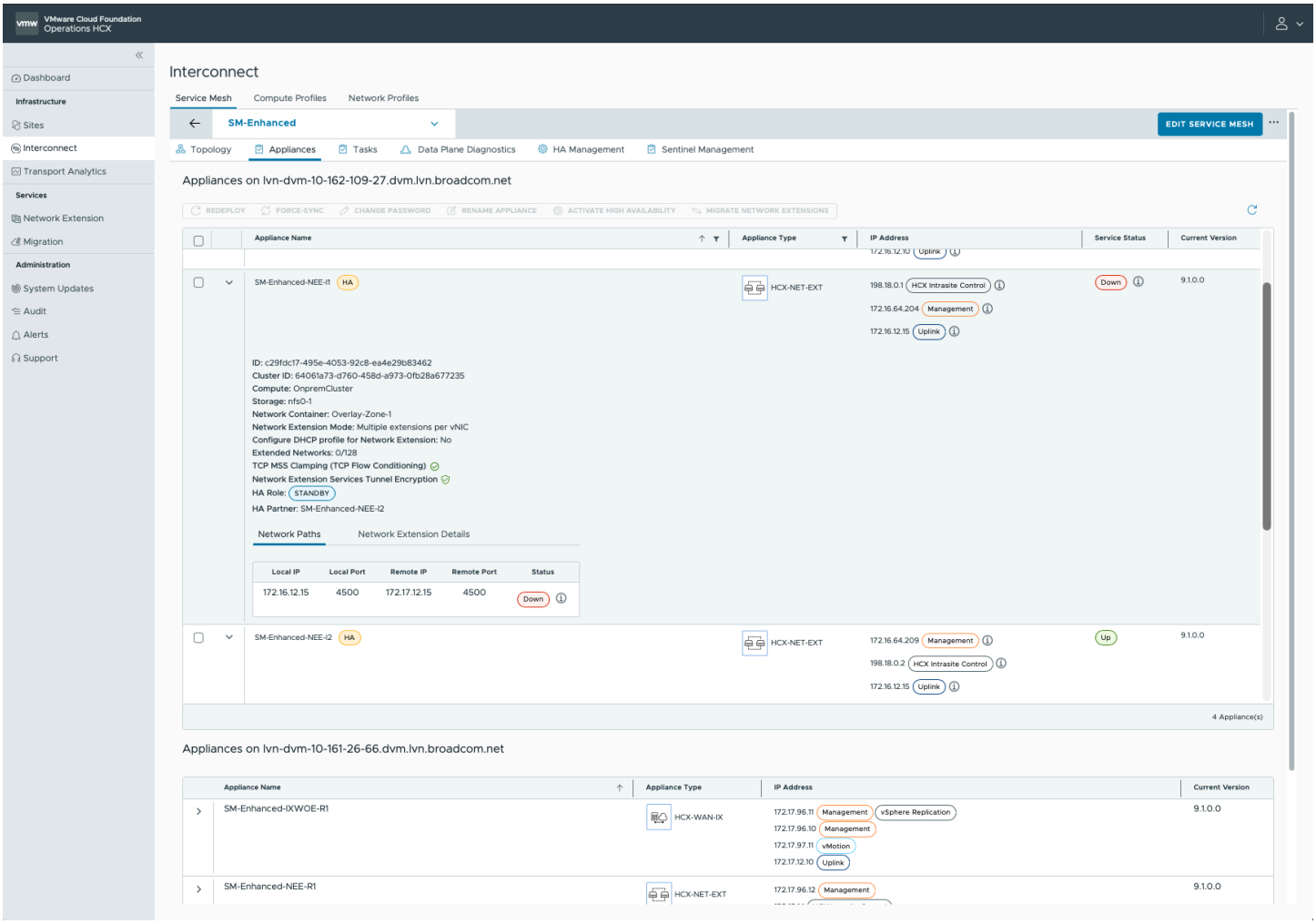
Information that is related to network extension High Availability operation and status appears on multiple HCX interface pages: HA Management, site pairs (service mesh), and network extension.

> **Note:**
>
> Although HA operations take effect within a matter of seconds, it can take a couple of minutes for those changes to propagate to the UI display.

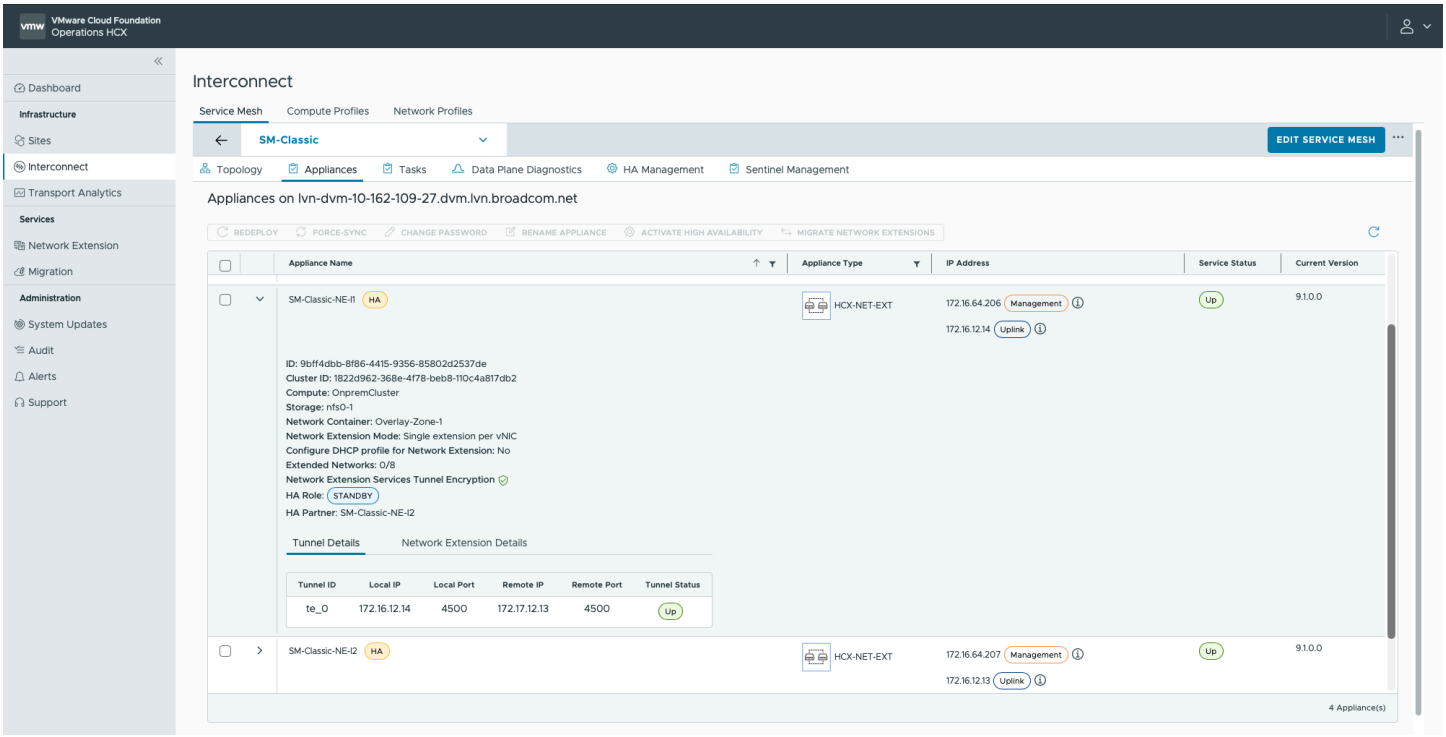
HA Management Information

The HA Management page provides the overall HA group health along with detailed information regarding the condition of individual HA groups at both the source and the destination sites.

With HA in enhanced mode, the appliance pairing can break and result in cross pairing if failover happens at one of the sites.



When HA is in non-enhanced mode, the appliance pair continues to stay the same and can take either an ACTIVE or STANDBY role.



The Overview information indicates the total number of HA groups and the number of those groups that are in a Healthy or a Degraded state. This information also specifies the total number of extended networks that are protected by an HA group.

The HA Group Details information provides the group State and Role information. HA Groups have one of the several states:

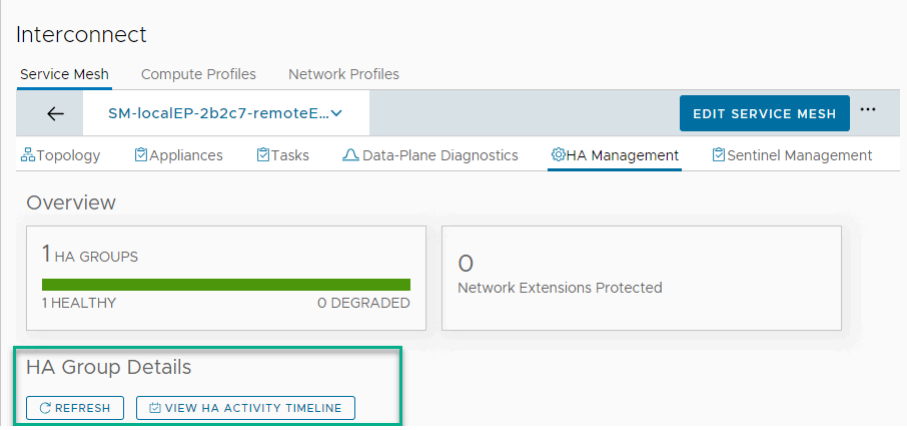
Group State	Description
HEALTHY	The HA group is functioning as expected.
DEGRADED	One or more of the network extension appliances in the HA group are not in a state to perform HA operations. This might not indicate a problem. For example, the HA group state is Degraded when the group is recovering from a failover operation. Administrator intervention might be required to determine the issue. **Note:** The Degraded state only refers to the HA group. The Active appliance continues to serve the network extension service.
FAILINGOVER	A failover operation is in progress for the HA group.
FAILED	The HA group cannot be automatically recovered from a failover. To redeploy the failed appliance pair, click the **Recovery** button.
MAINTENANCE	The HA group is in the process of redeploying or synchronizing as initiated by User.

network extension appliances in an HA group can have one of the several roles:

Group Role	Description
UNINIT	The Network Appliance is pending completion of the HA-specific configuration.
UNDECIDED	The network extension appliance is pending role negotiation with other appliances in the HA group, and its role is not set.

Group Role	Description
ACTIVE	Identifies the network extension appliance that is carrying out the network extension service by handling the network traffic.
STANDBY	Identifies the network extension appliance that is monitoring the ACTIVE appliance but does not provide network extension services and handles no network traffic.

The HA Group Details information includes selections to refresh the page data and to view a timeline of HA activity.

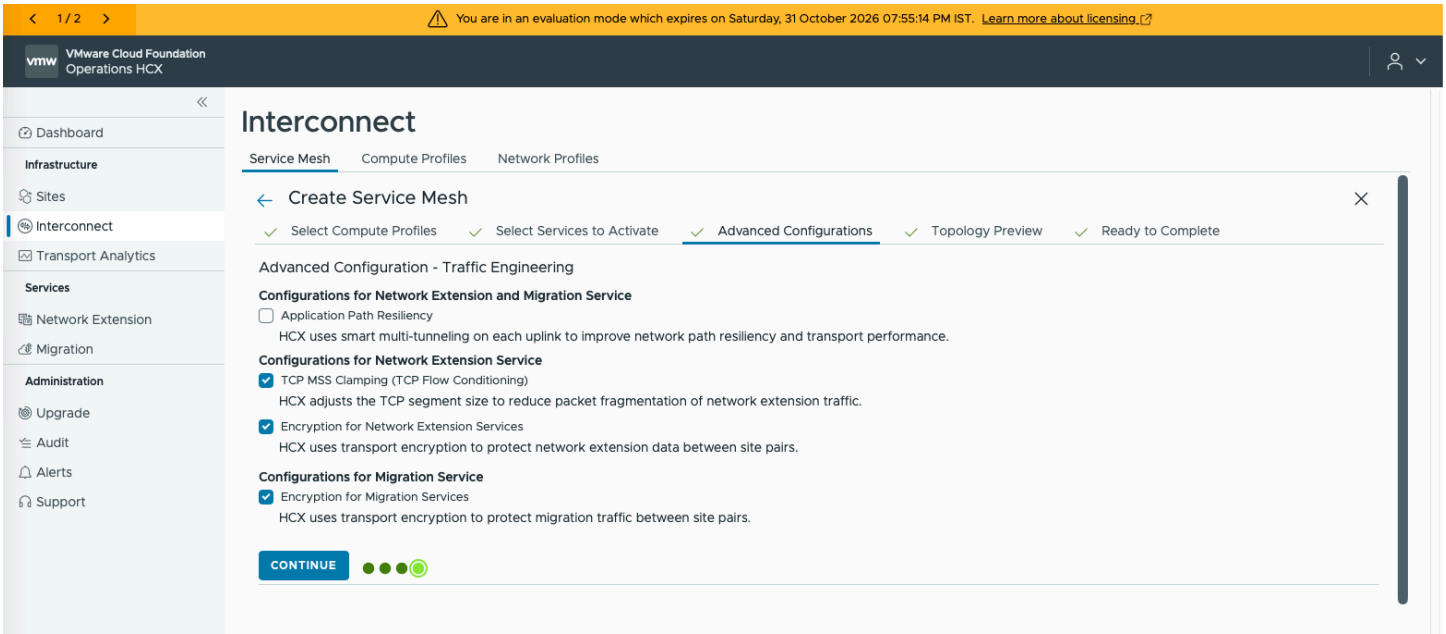


The HA activity timeline provides options selecting the date range and display view, along with an option to filter activity by HA group.

Service Mesh Appliances Information

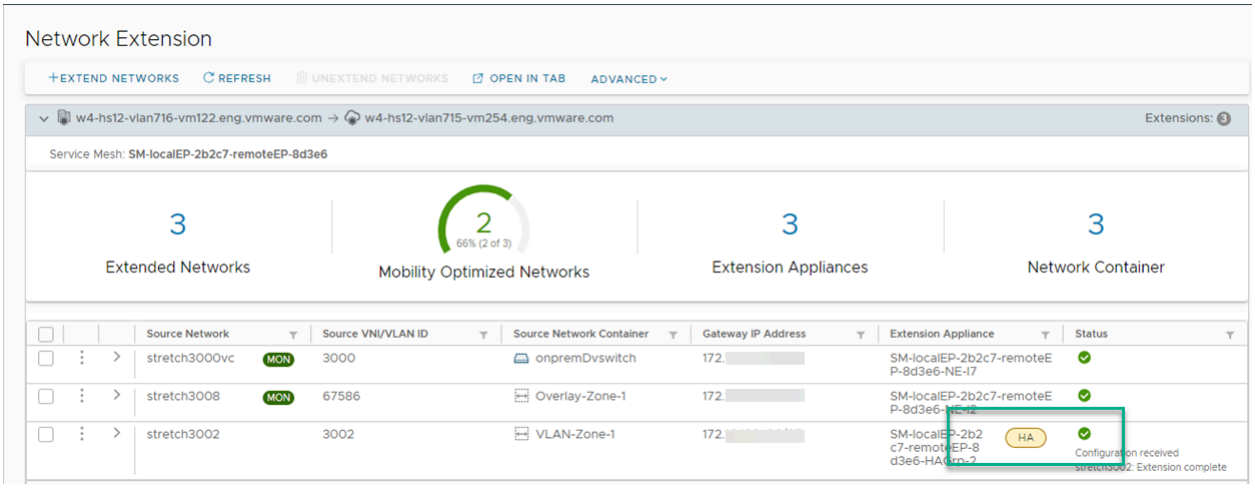
The service mesh Appliances page provides two entries for monitoring network extension HA status: HA Role and HA Partner. The HA Partner entry identifies the other appliance on the same site in the HA group, that they exchange heartbeat signals. The HA Role entry identifies which appliance is operating as the Active or Standby partner in the HA group at the moment.

Figure 489:



Network Extension Monitoring Information

The Network Extension page provides an "HA" tag in the Extension Appliance column to identify networks that are protected using network extension High Availability.



Migrating Virtual Machines

Workloads can be migrated bi-directionally between data centers using various VCF Operations HCX migration technologies.

Organizations migrate application workloads for many reasons. From data center consolidation and evacuation to modernization and maintenance, migrating workloads requires analysis and planning. Administrators identify individual workloads for migration, or waves of workloads based on, for example, cluster, network, or application landscape. VCF Operations HCX provides an array of migration types for moving these workloads including cold, warm, and live migration.

VCF Operations HCX also provides procedures for migrating groups, or waves, of virtual machines. And through integration with VCF Operations for networks, Application Group information can be exported to VCF Operations HCX for migration as mobility groups.

Migration Types

Virtual machines can be moved to and from HCX-activated data centers using multiple migration technologies.

> **Note:**
>
> For information regarding the number of concurrent migrations supported for a specific migration type, see VMware Configuration Maximums.

Bulk Migration

This migration method uses the VMware vSphere Replication protocols to move the virtual machines to a destination site.

- The Bulk migration option is designed for moving virtual machines in parallel.
- This migration type can set to complete on a pre-defined schedule.
- The virtual machine runs at the source site until the failover begins. The service interruption with the bulk migration is equivalent to a reboot.

For more information, refer to Understanding Bulk Migration.

vMotion Migration

This migration method uses the VMware vMotion protocol to move a virtual machine to a remote site.

- The vMotion migration option is designed for moving single virtual machine at a time.
- The virtual machine state is migrated. There is no service interruption during an HCX vMotion migration.

For more information, see Understanding vMotion and Cold Migration.

Cold Migration

This migration method uses the VMware NFC protocol. It is automatically selected when the source virtual machine is powered off.

For more information, see Understanding vMotion and Cold Migration.

Replication Assisted vMotion

Replication Assisted vMotion (RAV) combines advantages from Bulk Migration (parallel operations, resiliency, and scheduling) with vMotion Migration (zero downtime virtual machine state migration).

For more information, see Understanding Replication Assisted vMotion.

VCF Operations HCX OS Assisted Migration

This migration method provides for the bulk migration of guest (non-vSphere) virtual machines using OS Assisted Migration to VMware vSphere on-premises or cloud-based data centers.

For more information, see Understanding OS Assisted Migration.

HCX Assisted vMotion (Direct)

With HCX Assisted vMotion (Direct) migration, HCX orchestrates migrations between HCX Manager source and destination sites leveraging native VMware vSphere cross-vCenter vMotion. For more information, see Understanding HCX Assisted vMotion (Direct).

Overall Migration Restrictions

Each migration technology has its own list of requirements and restrictions as provided in the links before this section. This section provides restrictions for VCF Operations HCX migration no matter the migration technology that is used for the migration.

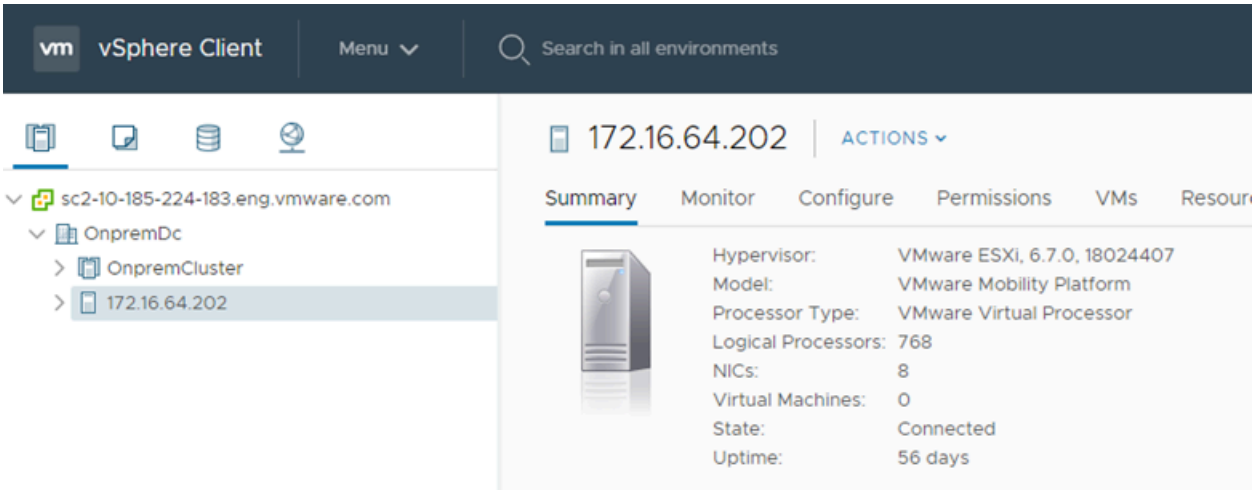
- Virtualization-Based Security (VBS) functionality is not supported with VCF Operations HCX migrations.

Mobility Agent vSphere Host for HCX Migrations

HCX uses the Mobility Agent service to perform vMotion, Cold, and Replication Assisted vMotion (RAV) migrations.

The HCX-IX appliance adds the Mobility Agent service as a host object in the vCenter Server. The processor, memory, storage and networking resources displayed on this object does not represent actual consumption on the physical hypervisor hosting the IX appliance.

> **Important:**
>
> The Mobility Agent host deployment may fail if the vCenter Server is configured with the Certificate Management policy set to custom mode under the advanced parameter `vpxd.certmgmt.mode`. See knowledge article 79003.



CAUTION:

The Mobility Agent host is required. Deleting this host can disrupt HCX Cold, vMotion, and Replication Assisted vMotion (RAV) migrations.

Understanding Bulk Migration

Bulk migration uses host-based replication to relocate a virtual machine between HCX data centers.

To reduce the downtime, the source VM remains online during the replication and is bootstrapped on the destination ESX host after replication completes.

A Bulk Migration request triggers the following actions:

1. Replication begins an initial full synchronization transfer to the remote site. The time it takes to replicate is a function of the size of the VM, the data change rate on the virtual machine disk files (VMDKs), and available bandwidth.
2. Replication bandwidth consumption varies depending on the number of disks for each VM and the number of VMs being migrated concurrently.
3. The switchover can start immediately after the initial synchronization is completed, or it can be delayed until a specific time using the scheduled migration option. By using the scheduled migration option, the switchover can occur during a maintenance window.
4. A delta synchronization with two-hour recovery point objective (RPO) occurs while waiting for the scheduled switchover, after the initial synchronization is completed.
5. Depending upon data churn on source disk, additional snapshots are being created during the RPO cycle. After each RPO cycle, disk consolidation takes place and creates a “hbrdisk.RDID vmdk” called as replica instance vmdk file on target datastore. Refer VMware KB 87028.
6. During switchover, the source VM is powered off to perform a final off-line synchronization, data consolidation, and VM instantiation at the target data center. To perform a Force Power-Off for a bulk migration in progress, ensure a switchover schedule is set at the VM level.
7. Following the switchover, the migrated VM replica is powered on and HCX Manager renames the original VM using a POSIX timestamp suffix to avoid a naming conflict with the migrated VM. If the **Retain MAC** option was not selected, the migrated VM obtains a new MAC address.
8. The migration completes and the original VM is copied to the Migrated VMs folder.

Important:
There must be sufficient resources to power on the VM. If for some reason the new VM cannot power on, it remains powered off, and the original is powered on.

VCF Operations HCX copies the original VM to the Migrated VMs folder. This copy can be accessed in the vSphere Virtual Machines and Templates view, and can be used to recover the migrated VM.

Single vCenter Server Operation

You can use Bulk migration within a single vSphere vCenter (VC) Server. In this case, both HCX Connector and HCX Cloud Manager are deployed to different clusters within the same VC. Other than deployment within the same VC, the process for configuring HCX is the same, which means configuring the compute profile and the service mesh to enable Bulk migration.

> **Note:**
>
> Bulk migration from one cluster to another in a single VC environment does not retain the MAC address of the VM even if the option is explicitly selected.

Requirements for Bulk Migration

- The Hybrid Interconnect Service and the Bulk Migration Service must be activated and in a healthy state in the relevant service mesh.
- The resources to create, power on and use the virtual machine must be available in the destination environment.
- Virtual machines must be running Hardware Version 7 or later.
- Virtual machines must have VMware Tools installed.
- Virtual machines must reside in or be migrated to a Service Cluster (defined in the compute profile).
- Network extension is required for low downtime migration operations.
- Bulk Migration potential throughput can vary depending on bandwidth available for migrations, latency, available CPU/MEM/IOPS, and disk read speed. For a successful switchover phase, the bandwidth and the network conditions must be sufficient to satisfy the operation considering the dataset and virtual machine data change rate.
- When both the source vCenter and HCX Manager versions are 9.1.x or higher the HBR agent must be installed on source ESXi hosts that are running a version earlier than 9.1.x.

For information regarding the migration limits, see VMware Configurations Maximums.

Requirements for Bulk Migration of Encrypted Virtual Machines

Virtual machines can include various forms of encryption:

- VMs using a virtual Trusted Platform Module (vTPM) device.
- VMs with disks encrypted.
- VMs with encryption enabled without disks encrypted.

In addition to the general requirements for Bulk migration, Bulk migration of encrypted VMs has the following requirements:

- The Key Management Server (KMS) or vSphere Native Key Provider (NKP) used to encrypt the source VM is present on the target site for encrypted bulk migration to be successful, regardless of the default KMS at either site.
- The name and IP address of the KMS must match between the source and target sites.
- To support migration of encrypted VMs from the NKP backed site, the NKP configuration must be exported from the source vCenter Server and imported into the target vCenter Server out-of-band to make the migration successful.
- For KMS, the source and target sites must be running vSphere 6.7 or later.
- For NKP, the source and target sites must be running vSphere 7.0 Update 2c or later.
- The HBR agent must be installed and running on the source ESX hosts that are running a version earlier than 9.1.
- To enable the replication of encrypted VMs, one secure listener on the 32032 port is added on vSphere Replication network (IX appliance).
- VCF Operations HCX Firewall rules are updated for secure listener on the 32032 port.

> **Note:**
>
> If a VM's configuration is encrypted, and disks are not encrypted or partially encrypted on source, but the VM encryption storage policy is selected on VCF Operations HCX while migrating the VM, then the complete VM (its configuration and all disks) will be encrypted on target site.
>
> For vTPM enabled VMs, the VMs with encrypted configurations or having one or more encrypted disks would cause the complete VM (its configuration file and all the disks) to get encrypted when migrated to the target site.

Restrictions for Bulk Migration

- Virtual machines with Raw Device Mappings (RDM) in Physical Compatibility mode cannot be migrated. Virtual machines with RDM in Virtual Compatibility mode can be migrated. They are converted to VMDKs at the destination.
- Virtual machines with mounted ISO images cannot be migrated. The HCX Bulk migration operation can be used for force unmount ISO images.
- Virtual machine snapshots are not migrated. The HCX Bulk migration operation has an option to remove the snapshots.
- Virtual machines with DirectPath I/O configurations cannot be migrated without first removing the DirectPath device.
- Virtual machines cannot be migrated while they are using a shared SCSI bus, flagged for multi-writer, enabled for fault tolerance, or configured for shared VMDK disk sharing.
- Virtual machines must be gracefully powered off before being migrated. However, if necessary, HCX can override with the Force Power-off VM option.
- Virtual machines using virtual NVMe (vNVME) Controllers cannot be migrated.
- The Migration interface does not display port groups that are VLAN trunks.
- VMware software appliances (including, but not limited to vCenter and NSX Manager) cannot be migrated with HCX Bulk migration.

Guest OS Customization with Bulk Migration

One characteristic of Bulk Migration is the ability to customize several aspects of the Guest OS.

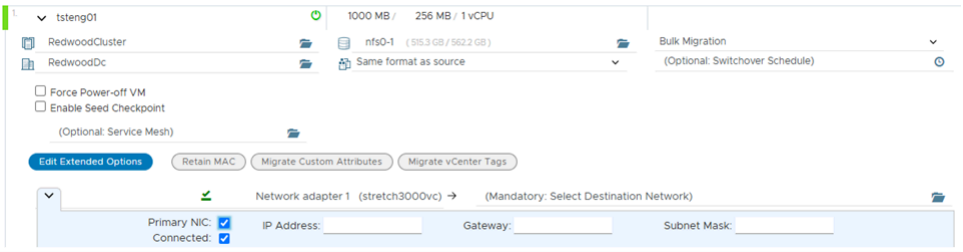
When migrating virtual machines on an extended network, it is best practice to keep the IP address of a virtual machine, MAC address, and overall identity. In some scenarios, it can be beneficial to modify a VM's characteristics. For example, it might be necessary to migrate non-production workloads to free up private network prefixes. Making these changes during the migration can save the effort of manually updating the VM settings after the migration. The following guest customizations are available:

- Guest OS Hostname
- IP Address
- Gateway
- Netmask (Subnet Mask)
- Primary DNS
- Secondary DNS
- Security Identifier (Windows SID)
- Run Pre- or Post-Guest Customization scripts

Guest customization is available only for specific guest OS types. See Guest OS Types for Guest Customization.

Virtual machines with the locale and the UI language other than English US are not supported for guest OS customization.

You select guest customizations using the **Edit Extended Options** in the Migration interface. Certain customization options only appear when you select the **Edit Extended Options** on a virtual machine level.



HCX applies the guest customization options during the switchover before powering on the VM.

This functionality is also supported for reverse migrations.

> **CAUTION:**
>
> When changing Guest OS information, HCX does not store the original settings.

All values must be specified in the wizard, even the values that must remain unchanged.

Values are cleared on the migrated virtual machine for fields left empty if IP Customization is configured.

The Primary NIC option must be selected or checked.

For details about the Extended Options available for Bulk migration, see "Additional Migration Settings."

Note:

Changing the Security Identifier of a Windows machine that is already a member of a Windows domain breaks the domain relationship and requires the machine to be re-joined. On a domain controller, this operation can impact the domain. By default, the **Generate New Security Identifier (SID)** option is not selected.

Guest OS Types for Guest Customization

VCF Operations HCX supports Guest OS customization for specific Windows and Linux operating systems.

For Bulk migrations, VCF Operations HCX supports customizing various aspects of the guest OS on the destination virtual machine. See Guest OS Customization with Bulk Migration.

The virtual machine Guest OS type and guestID are reflected as virtual machine **config** parameters in the vCenter Managed Object Browser (`https://vcenterfqdn/mob`). For IP Customization to work, the virtual machine **guestId** entry must match the supported **Guest OS Types**.

Operating System Types Supported for Guest Customization when the Target vCenter Instance is 9.1 and Earlier

Linux Operating System Types Supported for Guest Customization (64-bit)

Family	Distribution	Operating System	vCenter GuestOS ID (guestID)
Debian	Debian	Debian 10.13	debian10_64Guest
Debian	Ubuntu	Ubuntu 14.04 LTS	ubuntu64Guest
Debian	Ubuntu	Ubuntu 16.04 LTS	ubuntu64Guest
Debian	Ubuntu	Ubuntu 18.04 LTS	ubuntu64Guest
Debian	Ubuntu	Ubuntu 20.04 LTS	ubuntu64Guest
Debian	Ubuntu	Ubuntu 22.04 LTS	ubuntu64Guest
Debian	Ubuntu	Ubuntu 22.10	ubuntu64Guest
Debian	Ubuntu	Ubuntu 23.10	ubuntu64Guest
Debian	Ubuntu	Ubuntu 24.04	ubuntu64Guest
Red Hat	CentOS	CentOS 6.1 - CentOS 6.10	centos6_64Guest
Red Hat	CentOS	CentOS 7.1 - CentOS 7.9	centos7_64Guest
Red Hat	CentOS	CentOS 8.0 - CentOS 8.x	centos8_64Guest
Red Hat	Oracle Linux	Oracle Linux 6.1 - Oracle Linux 6.10	oracleLinux64Guest
Red Hat	Oracle Linux	Oracle Linux 7.1 - Oracle Linux 7.9	oracleLinux64Guest
Red Hat	Oracle Linux	Oracle Linux 8.0 - Oracle Linux 8.9	oracleLinux8_64Guest
Red Hat	Oracle Linux	Oracle Linux 9.1 - Oracle Linux 9.3	oracleLinux9_64Guest
Red Hat	Red Hat Enterprise Linux	RHEL 6.1 - RHEL 6.10	rhel6_64Guest
Red Hat	Red Hat Enterprise Linux	RHEL 7.1 - RHEL 7.9	rhel7_64Guest
Red Hat	Red Hat Enterprise Linux	RHEL 8.0 - RHEL 8.10	rhel8_64Guest
Red Hat	Red Hat Enterprise Linux	RHEL 9.0 - RHEL 9.4	rhel9_64Guest

Family	Distribution	Operating System	vCenter GuestOS ID (guestID)
Red Hat	Rocky Linux	Rocky Linux 8.4 - Rocky Linux 8.9	vCenter 7 and earlier: other4xLinux64Guest vCenter 8 and above: rockylinux_64Guest
Red Hat	Rocky Linux	Rocky Linux 9.0 - Rocky Linux 9.3	vCenter 7 and earlier: other4xLinux64Guest vCenter 8 and above: rockylinux_64Guest
SUSE	SUSE Linux	SUSE Linux Enterprise 11.4	sles11_64Guest
SUSE	SUSE Linux	SUSE Linux Enterprise 12 SP3	sles12_64Guest
SUSE	SUSE Linux	SUSE Linux Enterprise 15 SP6	sles15_64Guest
Photon	Photon OS	Photon 4.0	vmwarePhoton64Guest
Photon	Photon OS	Photon 5.0	vmwarePhoton64Guest

Windows Operating System Types Supported for Guest Customization (64-bit)

Operating System	vCenter GuestOS ID (guestID)
Windows Server 2008 R2	windows7Server64Guest
Windows Server 2012	windows8Server64Guest
Windows Server 2012 R2	windows8Server64Guest
Windows Server 2016	windows9Server64Guest
Windows Server 2019	windows2019srv_64Guest
Windows Server 2022	windows2019srvNext_64Guest
Windows 11	windows11_64Guest

Operating System Types Supported for Guest Customization when the Target vCenter Instance is 9.1.1 and Later

Refer to the VMware Compatibility Guide for supported Guest Operating systems.

Linux Operating System Types Supported for Guest Customization (64-bit)

From the VMware Compatibility Guide select:

- Product Name: **ESXi**
- Product Release Version: **ESXi 9.1**
- OS Family: **Linux**
- OS Arch: **x86(64-bit)**

Windows Operating System Types Supported for Guest Customization (64-bit)

From the VMware Compatibility Guide select:

- Product Name: **ESXi**
- Product Release Version: **ESXi 9.1**
- OS Family: **Windows**
- OS Arch: **x86(64-bit)**

> **Note:**
>
> Operating systems that are **Deprecated** are not supported. Supported operating systems include **Supported**, **Tech Preview**, and **Legacy**.

Understanding vMotion and Cold Migration

VCF Operations HCX integrates with ESX to perform vMotion migrations for powered on virtual machines, and with Cold Migration for powered off virtual machines.

vMotion

VCF Operations HCX vMotion can transfer a live virtual machine from a VCF Operations HCX vCenter Server to a VCF Operations HCX destination site (or from the destination site towards the local site). The vMotion transfer captures the virtual machine active memory, its execution state, its IP address, and its MAC address. The migration duration depends on the connectivity, including both the bandwidth available and the latency between the two sites.

Cold Migration

Cold migration uses the same network path as VCF Operations HCX vMotion to transfer a powered-off virtual machine. During a cold migration, the virtual machine IP address and the MAC address are preserved. Cold migrations must satisfy the vMotion requirements.

Requirements and Limitations for vMotion and Cold Migration

- HCX Interconnect tunnels must be up/active.
- HCX vMotion requires 150 Mbps or higher throughput capability. See Network Underlay Minimum Requirements.
- The virtual machine hardware version must be at least version 9 or higher.
- The underlying architecture, regardless of OS, must be x86.
- VMs with Raw Disk Mapping in compatibility mode (RDM-V) can only be migrated using Bulk Migration.
- VMware NFC is used as the primary protocol during Cold migration and as a secondary protocol during HCX vMotion.

  Note:

  The HCX Interconnect appliance uses a network profile configuration that is dedicated to the vMotion traffic. This configuration does not include the Cold and vMotion NFC traffic. VCF Operations HCX always uses its Management interface for Cold and vMotion NFC traffic. In deployments where ESX uses a dedicated Provisioning VMkernel for NFC traffic, VCF Operations HCX continues to route Cold and vMotion NFC traffic through the Management interface.
- The Migration interface does not display port groups that are VLAN trunks.

For information regarding migration limits, refer to the VMware Configurations Maximums tool.

Virtual Machine Restrictions for vMotion

Virtual machines with the following attributes are not supported for migration.

- Virtual machines cannot be migrated while they are using a shared SCSI bus, flagged for multi-writer, enabled for fault tolerance, or configured for shared VMDK disk sharing.
- Attached virtual media or ISOs.
- Virtual machine hardware version 8 or earlier.
- Although concurrent vMotion migrations can be initiated up to the vSphere limits, VMware HCX vMotion only supports serial vMotion migrations between a source and the destination site per the service mesh. For simultaneous migrations in parallel, select Bulk Migration.
- VCF Operations HCX vMotion defaults to **Opportunistic** mode for per-VM vMotion Encryption if it is set to **Required**. During the migration operation - the mode is changed to Opportunistic during the migration initialization, and then set back to **Required** after the migration is completed.
- Virtual Machines with Change Block Tracking (CBT) can be migrated, but HCX deactivates CBT.
- VMware software appliances (including, but not limited to vCenter and NSX Manager) cannot be migrated with HCX vMotion migration.
- vSphere VM Encryption is not supported.
- Virtual machine workloads with an attached serial port device are not supported. For these workloads, detach such devices prior to starting the migration.

Note:
You can use Bulk Migration to migrate workloads with an attached serial port device.

Understanding Replication Assisted vMotion

HCX Replication Assisted vMotion (RAV) uses the HCX along with replication and vMotion technologies to provide large-scale, parallel migrations with zero downtime.

HCX RAV provides the following benefits:

- Large-scale live mobility: Administrators can submit large sets of VMs for a live migration.
- Switchover window: With RAV, administrators can specify a switchover window.
- Continuous replication: Once a set of VMs is selected for migration, RAV performs an initial sync. After the initial synchronization is complete, a delta synchronization with two-hour recovery point objective (RPO) occurs while waiting for the scheduled switchover.
- Concurrency: With RAV, multiple VMs are replicated simultaneously. When the replication phase reaches the switchover window, a delta vMotion cycle is initiated to do a quick, live switchover. Live switchover happens serially per service mesh.
- Resiliency: RAV migrations are resilient to latency and varied network and service conditions during the initial sync and continuous replication sync.
- Switchover larger sets of VMs with a smaller maintenance window: Large chunks of data synchronization by way of replication allow for smaller delta vMotion cycles.

HCX RAV migration triggers the following events:

1. Replication begins with a full synchronization (replication) of the virtual machine disks to the destination site.
2. Migrated VMs enter a continuous synchronization cycle until a switchover is triggered.
3. Depending upon data churn on the source disk, additional snapshots are created during the RPO cycle. For more information about RPO cycle/Delta Sync refer to VMware KB 87028.
4. You can have the switchover process start immediately following the initial sync or delay the switchover until a specific time using the scheduled migration option. If the switchover is scheduled, the synchronization cycle continues until the switchover begins.
5. The final delta synchronization begins when the switchover phase starts. During this phase, vMotion is engaged for migrating the disk delta data and virtual machine state.
6. As the final step in the switchover, the source VM is removed, and the migrated VM is connected to the network.
   Replication Assisted vMotion creates two folders at the destination site datastore. One folder contains the virtual machine infrastructure definition, and the other contains the virtual machine disk information. This is normal behavior for RAV migrations and has no impact on the functionality of the virtual machine at the destination site.

   Note:

   In some cases, having two folders might impact other applications, such as backup tools, that require access to the virtual machine folders. If necessary, you can consolidate the contents of these two folders datastore vMotion.

Single vCenter Server Operation

You can use Replication Assisted vMotion (RAV) migration within a single vCenter Server. In this case, HCX appliances such as HCX Interconnect, are deployed to cater to different clusters within the same VC. Other than deployment within the same VC, the process for configuring VCF Operations HCX is the same, which means configuring a compute profile and a service mesh to enable RAV.

Requirements for Replication Assisted vMotion

- Interconnect tunnels must be up/active.
- RAV requires 150 Mbps or higher throughput capability. See Network Underlay Minimum Requirements.
- The virtual machine hardware version must be Version 9 or higher.
- The underlying architecture, regardless of OS, must be x86.

- Hybrid Interconnect and Replication Assisted vMotion services must be activated and in a healthy state in the relevant service mesh.
- RAV uses vSphere Replication whose potential throughput can vary depending on bandwidth available for migrations, latency, available CPU/MEM/IOPS, and disk read speed.
- VMware NFC is used as a secondary protocol during Replication Assisted vMotion migration.

  Note:

  VCF Operations HCX always uses its Management interface for NFC traffic. In deployments where ESX hosts use a dedicated Provisioning vmkernel for NFC traffic, VCF Operations HCX continues to route NFC traffic through the Management interface. Provision vmkernels are dedicated network interfaces used for cold migration, cloning, and snapshot transfers, separate from vMotion.
- When both the source vCenter and HCX Manager versions are 9.1.x or higher the HBR agent must be installed on source ESXi hosts that are running a version earlier than 9.1.x.

For information regarding the migration limits, refer to the VMware Configurations Maximums tool.

Restrictions for Replication Assisted vMotion

- Virtual machines with the following attributes are not supported for migration.
  - Virtual machines cannot be migrated while they are using a shared SCSI bus, flagged for multi-writer, or enabled for fault tolerance.
  - Attached virtual media or ISOs.
  - Virtual machine hardware version 8 or earlier.
- Live switchover of concurrent RAV migrations is run serially per service mesh.
- VCF Operations HCX vMotion defaults to **Opportunistic** mode for per-VM vMotion Encryption if it is set to **Required**. During the migration operation, the mode is changed to Opportunistic on the migration initialization, and then set back to **Required** after the migration is completed.
- Replication Assisted vMotion does not support the migration of workloads with Independent persistent and Independent non-persistent disks, as taking a snapshot of such a virtual machine does not produce delta disks, which are required for the underlying RAV migration technology.
- Virtual machines with Raw Device Mappings (RDM) in Physical Compatibility mode cannot be migrated.
- Virtual machines with Raw Device Mappings in compatibility mode (RDM-V) cannot be migrated using RAV and vMotion.
- Virtual machines with DirectPath I/O configurations cannot be migrated without first removing the DirectPath device.
- To migrate Fault Tolerance-enabled VMs, temporarily turn off Fault Tolerance, and perform RAV. When this operation is complete, turn Fault Tolerance back on.
- Migration to or from vVOL datastores is not supported.
- The Migration interface does not display port groups that are VLAN trunks.
- Virtual Machines with Change Block Tracking (CBT) can be migrated, but HCX deactivates CBT.
- RAV migration to VMFS6 target datastores requires the following minimum vSphere version at the target site: vSphere 6.5U3f or vSphere 6.7U3.
- Virtual machines using virtual NVMe (vNVME) controllers cannot be migrated.
- Virtual machine Snapshots cannot be migrated.
- VMware software appliances (including, but not limited to vCenter and NSX Manager) cannot be migrated with HCX RAV migration.
- vSphere VM Encryption is not supported.
- Virtual machine workloads with an attached serial port device are not supported. For these workloads, detach such devices prior to starting the migration.

  Note:
  You can use Bulk Migration to migrate workloads with an attached serial port device.

Understanding HCX Assisted vMotion (Direct)

HCX Assisted vMotion (Direct) provides support for VMware vSphere native cross-vCenter vMotion migration.

With native cross-vCenter vMotion, a direct data path is established using the underlying management network between the VMware ESX hosts associated with the source and destination vCenter Servers. In this case, HCX orchestrates the migration flows between the HCX appliances deployed in each vCenter Server. For more information about native vSphere vMotion, see Migrating Virtual Machines Between vCenter Systems in the VMware vSphere documentation.

Like other HCX migration types, you enable HCX Assisted vMotion (Direct) during the service mesh creation process.

If you configure a site pair with only the HCX Assisted vMotion (Direct) migration type, HCX establishes management plane connectivity between sites, creates a service mesh, and deploys the HCX-IX appliance. In this case, the HCX-IX appliance is only used for VCF Operations HCX control plane communication and not for migration traffic.

How it Works

The workload migration has the following high-level workflow in HCX:

1. During the service mesh configuration, enable HCX Assisted vMotion (Direct).
2. In the Migration interface, create a mobility group for the selected workloads.
3. Select HCX Assisted vMotion (Direct) as the migration type and configure the available transfer, placement, and extended options.
4. HCX validates the configuration.
5. Start the migration.
6. HCX initiates the migration using native cross-vCenter vMotion.
7. HCX displays the status of the migration operation.

Requirements

- A dedicated vSphere vMotion data path must be set up between the source and the target ESX hosts.
- Standard switches at the source and target clusters must have matching security policies and the teaming and failover policies.
- The vSphere Distributed Switch version at the source and the destination sites must be the same. See KB 56991.
- The source and target sites are running vSphere 6.7 or later.
- Migrating encrypted virtual machines:
  - The Key Management Server (KMS) used to encrypt the source VM is present on the target site for encrypted direct vMotion to be successful, regardless of the default KMS at either site.
  - The name and IP address of the KMS must match between the source and target sites.
  - The source and target sites must be running vSphere 7.0 or later.
- Virtual machines must be running Hardware Version 4 or later.
- VMware Tools are installed on the VM.
- Review the port requirements: VMware KB 2106952.
- All the native cross-vCenter vMotion dependencies detailed in KB article 2106952 must be satisfied.

Restrictions

- HCX diagnostics are not available for HCX Assisted vMotion (Direct).
- Virtual machines with mounted ISO images cannot be migrated. The HCX Bulk migration operation can be used for force unmount ISO images, or you can unmount the ISO images manually.
- Virtual machines cannot be migrated while they are using a shared SCSI bus, flagged for multi-writer, enabled for fault tolerance, or configured for shared VMDK disk sharing.
- Migration is not supported when the source VM is on an NSX-backed segment.
- Virtual machines with DvFilter configurations are not supported.
- HCX Assisted vMotion (HAV) supports VMs with I/O filters only if a valid I/O filter policy is active on the source VM. Also the destination environment must have the identical I/O filter provider installed to maintain policy persistence during migration.
- It is a best practice to avoid scheduling VM hardware upgrade on the source vCenter Server while performing migration through VCF Operations HCX.
- VCF Operations HCX applies a maximum of 50 concurrent relocation operations per HCX Manager to avoid overloading the vCenter Server.

- All the native cross-vCenter vMotion limitations and restrictions related to scale and concurrency apply to HCX Assisted vMotion migration.
- HCX Assisted vMotion (Direct) migrations cannot be performed when the source NIC is in a disconnected state, the source VM is attached to NSX segment or VPC subnet, and the target network is mapped to either vCenter Distributed Virtual Port Group (DVPG) or standard VSS network.
- HCX Assisted vMotion (Direct) migration is not supported when there is VDS version mismatch between source and target sites.
- HCX Assisted vMotion (Direct) migration is not supported when the source VM is connected to an NSX segment and the target network is either an NSX Segment or a VPC subnet. Also, migration is not supported when source VM is connected to VPC subnet and target network is either an NSX segment or a VPC subnet and the source NIC is in connected state.
- Adding multiple target datastores on local sites is not supported.

Understanding Supervisor Onboarding for Migrations

When you enable the vSphere Kubernetes Service (VKS) control plane on vSphere clusters and they become Supervisors, the Kubernetes control plane is created inside the hypervisor layer. VCF Operations HCX provides the option to import, or “onboard,” traditional virtual machine workloads into specified Supervisor clusters.

> **Note:**
>
> Supervisor Onboarding supports Bulk migration between vCenter Servers and within a single vCenter Server. For more information, see Understanding Bulk Migration.

Onboarding workloads to the Supervisor is a two-step process. VCF Operations HCX first bulk migrates workloads from the source site to a staging area on the destination site defined by the **Compute Container** setting in the migration configuration wizard. From the staging area, VCF Operations HCX imports the workloads to the Supervisor namespace identified in the migration configuration.

How it Works

1. The user creates the migration groups, selects the Supervisor Onboarding option, and provides the necessary configuration options, including the Supervisor namespace, storage class, and compute container.
2. VCF Operations HCX performs validation checks on the VMs to ensure that they align with the supervisor environment. For example, the Guest OS running on the VM must be supported by the Supervisor.
3. VCF Operations HCX runs additional validation checks on the VMs to verify they meet the Bulk migration parameters, such as the HW version and installation of VM Tools. If there is an issue in the validation process, VCF Operations HCX displays an error. You can address the error and then try the migration again.
4. VCF Operations HCX migrates the VMs to a staging, or layover, area at the destination site.
5. VCF Operations HCX onboards the VMs to the Supervisor and verifies that any applications are running properly.

> **Note:**
>
> If onboarding to the Supervisor is unsuccessful, VCF Operations HCX displays an error and the workload is retained in the layover area. After addressing the issue, you can manually import the workload from the staging area.

Requirements and Limitations

- VCF Operations HCX 9.0 or later must be installed at both the source and target.
- VMs must have VMware Tools installed.
- Support is for Bulk migration only.
- Importing a VM to the Supervisor requires a reboot.

  > **CAUTION:**
  >
  > If an application cannot tolerate downtime, it should not be migrated using the Supervisor onboarding process.
- Onboarding Windows-based VMs has additional migration configuration parameters to be provided by users in the UI: owner name, organization name, computer name, and product key (optional).

Restrictions

Supervisor onboarding does not support the following:

- Workloads that depend on IPv6 cannot be migrated using Supervisor Onboarding.
- VMs with a Guest OS type that does not support Guest OS Customization (GOSC).
- Migration types: OS Assisted Migration (OSAM), HCX vMotion, Replication Assisted vMotion, Cold, HCX Assisted vMotion

For the migration process using Supervisor Onboarding, see Migrating Virtual Machines with Mobility Groups.

Understanding OS Assisted Migration

Together with the sentinel appliances deployed at VCF Operations HCX sites, the HCX OS Assisted Migration (OSAM) service uses the sentinel software that you install on the Linux or Windows-based guest virtual machines to assist with communication and replication from their guest environment to VCF.

When OSAM is activated in your service mesh, VCF Operations HCX deploys the appropriate sentinel appliance depending on the Site Pair. For vSphere-based site pairs (multi-site mode), VCF Operations HCX deploys the Sentinel Gateway (SGW) appliance at the source site and the Sentinel Data Receiver (SDR) appliance at the destination site. These two appliances work together to migrate workloads from a non-vSphere environment to VCF.

For site pairs where the non-vSphere guest environment is paired directly with an VCF Operations HCX site (single-site mode), VCF Operations HCX deploys the Sentinel Resource Gateway (SRG) appliance at the destination vSphere-based site. This appliance alone handles the migration of the guest workloads from the non-vSphere site to the vSphere-based site.

You must install HCX sentinel agent software on all guest virtual machines requiring migration using HCX OS Assisted Migration. The software gathers the system configuration from the guest virtual machine and assists with the data replication. The guest system information is used by various HCX OS Assisted Migration service processes. In part, the information is used to create an inventory of guest virtual machine systems for migration and to help replication processes prepare the disks on the replica virtual machine for replication and migration.

After the sentinel agent is installed, a secure connection is established between the guest virtual machines and the sentinel appliance. You specify the network connections between the guest virtual machines and sentinel appliances by configuring the Guest network profile setting in the compute profile.

VCF Operations HCX builds an inventory of candidates for migration as the sentinel software is installed on the guest virtual machines.

An OS Assisted Migration request triggers the following events:

1. Replication begins a full synchronization transfer to the destination site. The guest virtual machine remains online during replication until the final delta synchronization.
2. Before the final delta synchronization, the OS Assisted Migration service quiesces the guest virtual machine on a best-effort basis.
   For example, it is possible for a Linux service running on the guest virtual machine to start immediately after OS Assisted Migration has quiesced the services and stopped all known processes. If some process starts after quiescing the system, it can potentially lead to final synchronization not completing and appear as though the switchover process is stuck.

   Note:

   As part of both continuous and final synchronization, the system checks for changed blocks or files and replicates them.

   - On a Windows system, the OS Assisted Migration service reads the entire disk to determine the changed blocks to replicate. The process of reading the entire disk can be time consuming.
   - On a Linux system, the OS Assisted Migration service walks the entire file system to determine the changed files to replicate.

1. When the delta synchronization finishes, a switchover is triggered. You can have the switchover process start immediately following the initial sync or delay the switchover until a specific time using the scheduled migration option.

By using the scheduled migration option, the switchover can occur during a maintenance window. The final delta synchronization begins when the switchover phase starts.

During scheduled migrations, HCX sentinel performs continuous synchronization by transferring only the deltas since the previous sync cycle. For Windows HCX sentinel, this synchronization is achieved by identifying the changed file system blocks, whereas for Linux HCX sentinel, this synchronization is achieved by monitoring the changed files. To improve the time it takes to reach that final consistency point for Linux systems, a pre-determined set of files and directories listed in `/opt/vmware/hcx/osam/excluded_paths` is excluded from the continuous synchronization. If you have additional files that do not require monitoring, you can exclude them from continuous synchronization by editing the file. Excluding files requires a restart of the sentinel service named `vmware-hcx-osam-sentinel` using service or systemctl commands.

Note:

Excluded files are always synchronized to the target virtual machine during the initial and final synchronization phases.

2. VCF Operations HCX performs a hardware mapping of the replicated volumes to ensure proper operation, including updates of the software stack on the replica. This fix-up process includes adding drivers and modifying the OS configuration files at the destination. The migrated virtual machine reboots during this process.

   Note:

   When migrating Windows systems, HCX OS Assisted Migration software creates a temporary local user on the migrated Windows system during the switchover phase. This user gets deleted after the fix-up process is completed.

   Note:

   When migrating Linux systems, HCX OS Assisted Migration software uses an independent software stack residing on a separate disk, called the fix-up disk, for the fix-up process. To fix-up 64-bit workload VMs, VCF Operations HCX uses the Photon 5.0 version of the fix-up disk. The fix-up boot disk is detached and deleted at the end of the switchover process.
3. As the final step in the switchover, the source is powered off, the migrated replica is connected to the network, and the switchover completes.

   The vSphere target virtual machine reboots twice during the switchover phase.

   For Linux workloads, if the synchronization process fails for any reason, such as a broken network connection, by default the synchronization is retried for eight hours. To improve the time it takes to reach a final consistency point for switchover to begin, you can shorten the retry period to as little as one hour by editing the file `/opt/vmware/hcx/osam/etc/sync.params` and setting the **max_retry_interval** from one to eight hours. After setting the interval, restart the sentinel service named `vmware-hcx-osam-sentinel` using service or systemctl commands.

   For Windows workloads, the synchronization process retries until the synchronization completes or the operation is canceled.
4. HCX Manager names the replica virtual machine with the host name of the source virtual machine.
5. For Windows workloads, VMware Tools is installed on the migrated virtual machine and migration completes.
6. If the source does not power off, an attempt is made to power off the replica virtual machine.

   If the replica virtual machine successfully powers off, it remains connected to the NICs. In this case, you can manually power off the source and power on the replica. If the replica does not power off, both the guest virtual machine and the replica remain on, but the replica is not connected to the network. In this case, you activate the NICs manually that are attached to the replica virtual machine using vCenter, power-off source virtual machine (if not already), and power on Migrated virtual machine.

Considerations for OSAM Deployment

The OS Assisted Migration (OSAM) service includes several components that work together for connecting and forwarding guest workloads in the source environment.

Refer to the following considerations when deploying and operating OS Assisted Migration in your environment.

- The OSAM service converts non-vSphere guests to vSphere virtual machines. This conversion process involves halting OS services to quiesce the guest virtual machine. The downtime for this conversion process can vary from minutes to hours depending on a virtual machine size and activity.
- OSAM migrations require a guest network configuration in the compute profile.
- The HCX sentinel agent software encrypts all connections to the Sentinel Gateway. The encryption cannot be deactivated.
- The HCX sentinel appliance must be deployed in a vSphere environment, and not within KVM or Hyper-V.
- To use network extension with OSAM deployments, VLANs in the non-vSphere environment must also exist on the source vSphere environment network switches and in vCenter as distributed port groups.
- For vSphere-based site pairs, each service mesh deploys one Sentinel Gateway (SGW) and its peer Sentinel Data Receiver (SDR), and supports up to 50 active replica disks.
- For non-vSphere site pairs, the service mesh deploys one Sentinel Resource Gateway (SRG) that supports up to 50 active replica disks.
- HCX OSAM deployments support 200 concurrent VM disk migrations across a four service mesh scale-out deployment. In this service mesh scale-out model for OSAM, the HCX sentinel download operation is presented per service mesh.
- Guest virtual machines can only be migrated to a datastore that is accessible by the SDR or SRG appliance.
- Redeployment of sentinel appliances is not allowed when any migration is in-progress.
- Only "thin" and "thick" disk provisioning types are supported as the disk provision type for the migrated system. The "Same as Source" option is not supported.
- OSAM Migration using PowerCLI:
  - PowerCLI 11.5 is not supported.
- The OSAM migration service applies the default storage policy to the migrated VMs and their disks. Currently, the OSAM service does not support a user-selected storage policy.
- Changes to source Guest virtual machine configurations while a migration is in progress might not take effect in migrated virtual machines and sometimes might lead to migration failure.
- VMs with Linux GuestOS can have up to 15 hard disks attached.
- VMs with Windows GuestOS can have up to 64 volumes or file-systems.
- Migrations of VMs running on vSphere with OSAM are not supported.

For information regarding migration limits, refer to the VMware Configurations Maximums tool.

Supported Guest Operating Systems

The OS Assisted Migration service supports migration of virtual machines running non-vSphere guest operating systems in Linux or Windows environments.

Linux Environments

VCF Operations HCX supports various Linux-based 64-bit guest operating systems.

Family	Distribution	Operating Sysytem Version	vCenter GuestOS ID (guestID)
Debian	Ubuntu	Ubuntu 14.04 LTS	ubuntu64Guest
Debian	Ubuntu	Ubuntu 16.04 LTS	ubuntu64Guest
Debian	Ubuntu	Ubuntu 18.04 LTS	ubuntu64Guest
Debian	Ubuntu	Ubuntu 20.04 LTS	ubuntu64Guest
Debian	Ubuntu	Ubuntu 22.04 LTS	ubuntu64Guest
Debian	Ubuntu	Ubuntu 22.10	ubuntu64Guest
Debian	Ubuntu	Ubuntu 23.10	ubuntu64Guest
Debian	Ubuntu	Ubuntu 24.04	ubuntu64Guest
Red Hat	CentOS	CentOS 6.1 - CentOS 6.10	centos6_64Guest
Red Hat	CentOS	CentOS 7.1 - CentOS 7.9	centos7_64Guest

Family	Distribution	Operating Sysytem Version	vCenter GuestOS ID (guestID)
Red Hat	CentOS	CentOS 8.0 - CentOS 8.x	centos8_64Guest
Red Hat	Oracle Linux	Oracle Linux 6.1 - Oracle Linux 6.10	oracleLinux64Guest
Red Hat	Oracle Linux	Oracle Linux 7.1 - Oracle Linux 7.9	oracleLinux64Guest
Red Hat	Oracle Linux	Oracle Linux 8.0 - Oracle Linux 8.9	oracleLinux8_64Guest
Red Hat	Oracle Linux	Oracle Linux 9.1 - Oracle Linux 9.3	oracleLinux9_64Guest
Red Hat	Red Hat Enterprise Linux	RHEL 6.1 - RHEL 6.10	rhel6_64Guest
Red Hat	Red Hat Enterprise Linux	RHEL 7.1 - RHEL 7.9	rhel7_64Guest
Red Hat	Red Hat Enterprise Linux	RHEL 8.0 - RHEL 8.10	rhel8_64Guest
Red Hat	Red Hat Enterprise Linux	RHEL 9.0 - RHEL 9.4	rhel9_64Guest
Red Hat	Rocky Linux	Rocky Linux 8.4 - Rocky Linux 8.9	vCenter 7 and earlier: other4xLinux64Guest vCenter 8 and above: rockylinux_64Guest
Red Hat	Rocky Linux	Rocky Linux 9.0 - Rocky Linux 9.3	vCenter 7 and earlier: other4xLinux64Guest vCenter 8 and above: rockylinux_64Guest

Windows Environments

VCF Operations HCX supports various 64-bit Windows guest operating systems on KVM or Hyper-V supervisors.

Operating System	vCenter GuestOS ID (guestID)
Windows Server 2008 R2	windows7Server64Guest
Windows Server 2012	windows8Server64Guest
Windows Server 2012 R2	windows8Server64Guest
Windows Server 2016	windows9Server64Guest
Windows Server 2019	windows2019srv_64Guest
Windows Server 2022	windows2019srvNext_64Guest
Windows Server 2025	windows2022srvNext_64Guest

Guest Operating System Considerations

The OS Assisted Migration service supports a variety of hypervisors and guest operating systems in both Linux and Windows environments.

General Operating System Considerations

Some service limitations are common to Linux and Windows environments where OS Assisted Migration is deployed.

- The HCX-sentinel agents are installed in the guest operating system and automatically make connections to the HCX Sentinel Gateway.
- Guest virtual machines with the locale and the UI language other than English US are not supported.
- UEFI-based source systems using legacy BIOS boot mode are not supported.
- Anti-virus software, or any OS application that is actively accessing file systems, can significantly delay the OSAM switchover phase. Deactivate these types of applications as a best practice before configuring the OSAM migration.

- Network file shares are not supported with OSAM migrations. This includes NFS, SMB, and CIFS, with varying outcomes:
  - Mounted files shares like CIFS may result in migration failures.
  - NFS shares are ignored during the fix-up phase and do not result in migration failure.

Linux Specific Considerations

- Block devices (partitions) with unrecognized content are not migrated to the destination.
  - Unsupported file systems.
  - Unmounted file systems.
  - Unknown content with a partition or a block device.
  - Encrypted file systems.
  - md devices (software RAID).
- Device names for md devices (software RAID) are not retained post migration.
  - It is advised to mount the md devices using UUID or Label in `/etc/fstab`.
  - It is advised to revisit the mdadm.conf and modify the new name and other configuration parameters for the md devices.
- Statics routes are not supported.
- VLAN interfaces are not supported.
- On RHEL/CentOS 7.0, 7.1, and 7.2, the XFS file system UUID is not restored to the original UUID for the file system where /boot resides because mkfs.xfs does not support the functionality (-m option). A new random UUID is generated. Modifying the UUID after the file system is created triggers RHEL bug 1579390.
- /etc/fstab entries for removable media (floppy, CD) are not supported; such entries must be commented out before migration.
- When migrating Linux systems, the OSAM software uses an independent software stack residing on a separate disk for the fix-up process. This fix-up boot disk is detached and deleted at the end of the switchover process.
- The configuration files of Linux system services, such as dhcpd, that reference network interface names are not modified. You must manually modify these files on the migrated system.
- Device names in bonded/teamed interface configuration files are not replaced with the new device names, and this can lead to the bonded interface not showing as UP. You must make sure that the bonded/teamed interface device names match the new name after migration. You might encounter this issue if you use Network Manager tools like nmcli to configure bonded/teamed interfaces.

Windows Specific Considerations

- No support for syncing logical volumes.
- Basic MBR and GPT disk partitions are supported. Dynamic GPT and Dynamic MBR disks partitions are not supported.
  - If the boot disk is dynamic, migration is not supported.
  - If the data disks are dynamic, the data on the disks is not migrated, and the disks appear as raw disks on the migrated system.
- Only NTFS formatted volumes are supported. ESP (EFI system partition) with the FAT32 file system is an exception.
- No support for non-Windows service applications from the system quiescing perspective during the final sync.
- Systems with more than 64 volumes are not supported since vSphere Standard Switch (VSS) allows a maximum of 64 snapshots on a system.
- Any VSS snapshots present on the source Windows system before migration are not usable on the migrated system.
- VLAN interfaces are not supported.
- For Windows systems, in general the pre-requisites for the VMware Tools installation have to be satisfied on the source system. The pre-requisites for the VMware Tools installation can vary based on the target VCenter Server and ESX version, and Windows OS version on the source system. For example, if the target ESX version is 6.5.0 (or higher), VMware Tools version is 10.3.x. To view the list of prerequisites based on different Windows OS versions see VMware KB 55798.
- When migrating Windows systems, the OS Assisted Migration software creates a temporary local user on the migrated Windows system during the switchover phase. This user gets deleted after the fix-up process is completed.

- Do not run Windows updates on the source system during a migration. If Windows updates are in progress, the migration can fail.
- A Windows source system configured as a failover cluster node is not supported.

About Migrating Mobility Groups from Migration Waves

This section describes the process for migrating workloads in migration waves created using VMware Cloud Foundation Operations for Networks.

> **Note:**
>
> For information related to migrating mobility groups that are not created locally in VCF Operations HCX and do not use VCF Operations for networks, see Migrating Virtual Machines with Mobility Groups.

For many use cases, the scope of migration operations can involve hundreds or thousands of virtual machines requiring different migration parameters. Developing a migration plan can help to facilitate the migration of these workloads efficiently. Establishing a migration plan is a service of VCF Operations for networks. VCF Operations HCX works with the migration planning functionality in VCF Operations for networks to configure and run the migration wave.

The high-level process of creating a Migration Plan in VCF Operations for networks has these steps:

1. Add the VM migration attributes (optional).
   Identify and analyze workloads including application flows within your infrastructure.
2. Define the migration scope.
   Create the scope of the migration plan by categorizing workloads based on specific criteria.
3. Create a migration wave.
   Group and organize selected workloads into distinct migration waves according to the migration plan scope.
4. Create mobility groups.
   Use a mobility group to execute migrations concurrently.
5. Sync to VCF Operations HCX or manually export migration waves from VCF Operations for networks.
   Sync migration waves directly to VCF Operations HCX or manually export workload and network CSV files from VCF Operations for networks..

For complete information regarding the development of the migration plan and associated migration waves, see Create a Migration Plan and Export Migration Waves.

The high-level process in VCF Operations HCX for implementing the migration plan and migrating workloads that are defined in the associated migration wave has these steps:

1. Sync from VCF Operations for networks or import a migration wave manually.
   Create a migration wave by syncing directly from VCF Operations for networks or by manually importing the workload and network CSV files.
2. Commit the migration wave (for manual imports).
   Validate the workloads in each mobility group in the migration wave and commit those mobility groups for migration.
3. Configure and schedule the mobility groups.
   Configure the mobility groups for migration, including setting the migration and placement options, and scheduling the transfer and switchover windows.
4. Run pre-migration checks.
   Run the pre-migration checks to validate workload connectivity and service mesh health.
5. Run migrations.
   Run the migrations that are defined in the migration wave and perform post-migration activities.

To begin the process of migrating mobility groups with migrations waves, see Import and Commit Migration Waves.

Import and Commit Migration Waves

VCF Operations HCX provides two mechanisms for adding migration waves for migration: direct syncing from the VCF Operations for networks component, and manually by importing the exported CSV files from VCF Operations for networks.

Follow this process to manually import migration waves from CSV files for migration. For an overview of the migration process using migration waves, see About Migrating Mobility Groups from Migration Waves

> **Note:**
>
> Whether you import a migration wave manually or sync it directly in VCF Operations HCX from the VCF Operations for networks component, the process requires an initial validation step. The status of the validation appears in the Migration Planning interface. If validation fails for migration waves synced directly from the VCF Operations for networks component, you must fix the issues in VCF Operations for networks component. If the validation step for migration waves synced from VCF Operations for networks is successful, these migration waves are automatically committed for migration and you can continue to Migrate Mobility Group Workloads in Migration Waves.

1. In the VCF Operations HCX UI, navigate to **Migration**.

   The Migration Planning screen appears.

   > **Important:**
   >
   > Migration waves synced directly from VCF Operations for networks appear in the display in Committed or Validation Failed status depending on the outcome of the validation process. For these entries, skip the rest of this procedure and go directly to Migrate Mobility Group Workloads in Migration Waves.

2. Click `Import Migration Wave`.
3. Select the destination HCX Manager that is the target of the mobility operations.
4. Import the CSV files associated with the migration wave.

   Exporting a migration wave from VCF Operations for networks generates a zip file with two CSV files that you extract and upload to VCF Operations HCX.

   a) Import the Workload Data File.

      This data file contains the information for each workload in the migration wave: vCenter hosting the workload, migration wave name, and mobility Ggroup for the workload.

      The imported file must be from the same vCenter Server where VCF Operations HCX is deployed. For example, if the source VCF Operations HCX is deployed on vCenter Server 1, the imported wave file must include workloads from vCenter Server 1. This means workloads in a wave file from vCenter Server 2 will be reported as non-existent on the VCF Operations HCX.

   b) (Optional) Import the Network Dependencies File.

      The network dependencies file lists the networks that the workloads connect to in the vCenter Server. The Network Dependencies file is optional for initially importing the migration wave workloads and can be imported later in the process.

5. After uploading the files, click **Import Migration Wave**.

   As part of this step, VCF Operations HCX performs various validation checks of the imported migration wave data against the actual workloads in the vCenter Server.

   Following validation, the Migration Planning Screen appears. For migration waves that are imported manually, the Wave Source column displays CSV.

   For migration waves that sync directly from VCF Operations for networks, the Wave Source column displays VCF Operations for networks.

6. Review the Status column.

   The status column can have the following values following the import operation. For a complete list of status values, see Monitoring Migration Waves.

Status	
Validating	Indicates that validation checks are in progress. For large migration waves, validation can take several minutes.
Validation Failed	VCF Operations HCX performs a number of validations on the migration wave information in the CSV files against the contents of the source vCenter Server. A mismatch in the information can cause a failure. For example, if a workload included in the migration wave, or a network listed in the network data file are no longer present in the vCenter Server, a failure can occur. If a failure occurs, you can modify the migration wave in VCF Operations for networks and re-import the files.
Ready	Indicates that all groups in the migration wave have passed the validation checks and the wave can be committed for migration.
Committing/Committed	Indicates that the migration wave has passed validation, mobility groups have been created in VCF Operations HCX, and those groups are ready for configuration. migration waves synced directly from VCF Operations for networks are automatically set to Committed state after passing validation. For migrations waves manually added in VCF Operations HCX and that are in Ready state, you must take action to commit the migration wave for migration.

7. For entries in Ready state, click the vertical ellipsis next to the migration wave name and select **Commit Migration Wave**.

   With this action, mobility groups are created in Draft state once the migration wave commit is successful.

The imported migration wave appears in the Mobility Planning screen. For more information, see Monitoring Migration Waves.

Next, follow the procedure to configure and schedule the mobility group workloads included in the migration wave and, when ready, run the migrations. See Migrate Mobility Group Workloads in Migration Waves.

Migrate Mobility Group Workloads in Migration Waves

This section describes the procedure for configuring and migrating mobility groups in migration waves.

- The migration service is activated in both the source and destination site compute profile.
- The migration service is activated in the Service Mesh.
- Information regarding each migration type is reviewed and understood. For more information, see Migration Types.

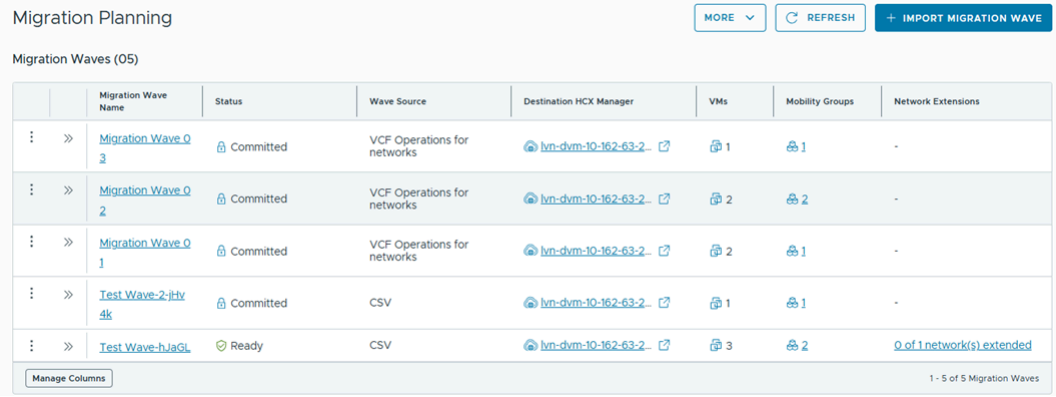
The process for migrating mobility groups in migrations waves has three high-level steps:

1. For each group in each migration wave, configure the mobility groups.
2. Run the pre-migration checks.
3. Run the migrations.

Follow this procedure to configure and run the migrations.

1. In the VCF Operations HCX UI, navigate to **Migration**.

   The Migration screen appears. Synced migration waves appear in the Wave Source column as VCF Operations for networks. Manually imported migration waves appear in the Wave Source source column as CSV.



2. Review the Status column for each migration wave.

   Synced migration waves that have passed validation appear with the status Committed. Manually imported migration waves that have passed validation appear in the display as Ready.

   For a complete list of migration wave states, see Monitoring Migration Waves.

3. For entries in Ready state, click the vertical ellipsis next to the migration wave name and select **Commit Migration Wave**.

   A pop-up screen prompts you to commit the wave. With this action, the migration wave state changes to Committed.

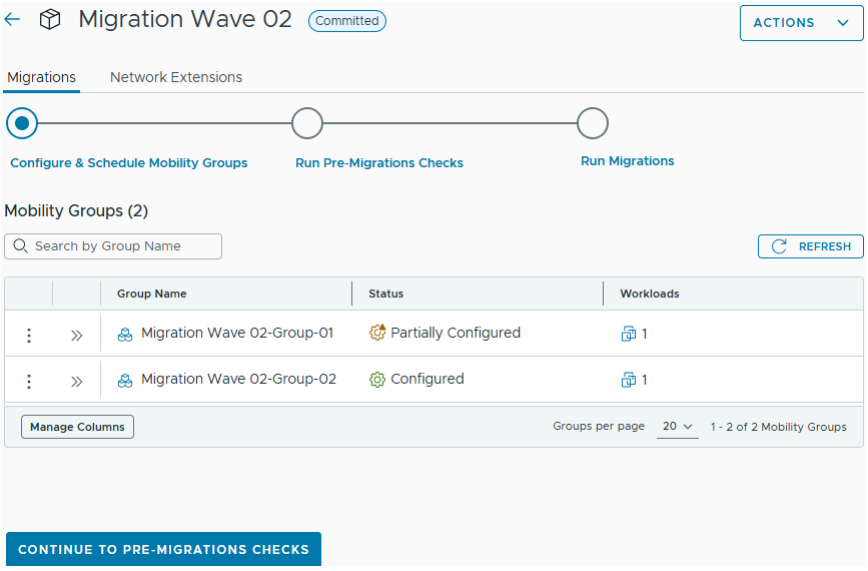
   For either synced or manually imported migration waves, committing these waves creates the mobility groups for migration and validates they are ready for configuration.

4. (Optional) To review the mobility group and network information for each migration wave, click the verticial ellipsisnext to the migration wave name and select **View Details**.

   The specific migration wave screen appears with tabs for viewing Migration and network extension data. For more information, see Monitoring Migration Waves.

5. To configure and schedule mobility groups for migration, click the name of a committed migration wave.

   The system displays the list of mobility groups in the migration wave along with the current status of each group and the number of workloads in each group.

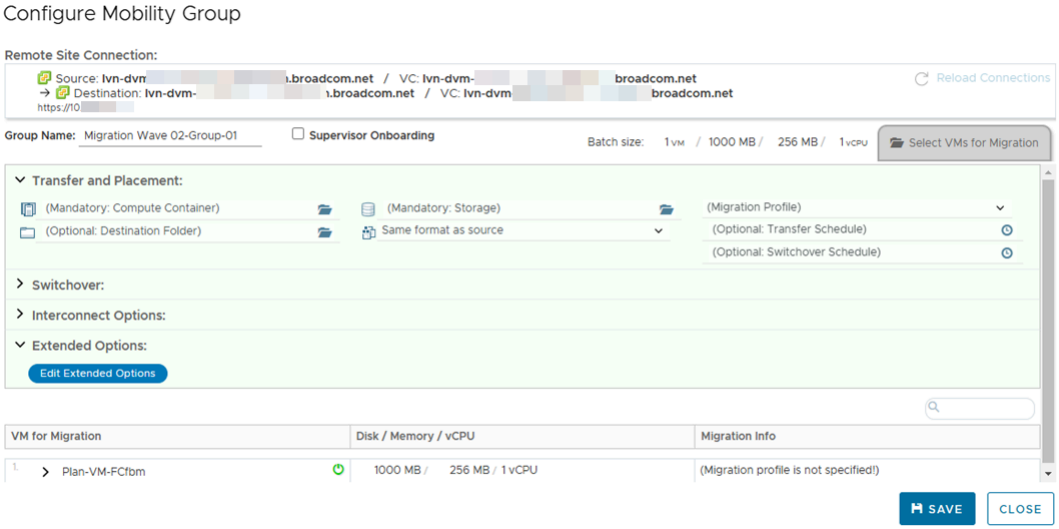


   For newly committed migration waves, all groups will appear in Not Configured state.

Note:

Mobility groups can appear in various states if those groups were subject to prior configuration attempts. For more information about the mobility group workload status, see Monitoring Mobility Group Activity for Migration Waves.

6. For mobility group entries in Not Configured or Partially Configured state, click the vertical ellipsis and select **Configure Mobility Group**.



Note:

The **Select VMs for Migration** button is not available for migration waves. Once a migration wave is committed for migration, mobility groups are created and the associated workloads cannot be changed.

7. (Optional) If Supervisor Onboarding is selected, you must configure the Supervisor Namespace and Storage Class.
8. Enter your group Transfer and Placement, Switchover, and Interconnect service mesh selection, along with any Extended options.

   For a group to move to Configured state, all mandatory fields must be completed:

   - Destination compute container
   - Destination storage
   - Migration type (profile)
   - Disk provisioning format (Virtual Disk Format)
   - Destination network for each VM for migration

   These settings, except for the destination network, apply to all members of the group by default. To override the default settings for specific virtual machines in the group, select and expand the virtual machine entry and set different options.

   For additional information regarding the Extended options, see Additional Migration Settings.

   Note:

   - For HCX Assisted vMotion (Direct), the Interconnect service mesh selection is not available.
   - You can set the Switchover and Transfer schedule for Bulk, Replication Assisted vMotion (RAV), or OS Assisted Migration (OSAM) migrations. Transfer scheduling is not available for vMotion, HCX Assisted vMotion (Direct), or Cold migration.

     The Transfer schedule identifies the date and time when base synchronization to the destination site begins. The Switchover schedule defines the date and time when the final data synchronization occurs and the workload is switched over to the destination site.
   - Extended Options provide additional settings.

9. Select the destination network for each virtual machine to be migrated:

   In most cases, the extended network between the source and destination sites is automatically selected. You can change this selection as needed.

   Important:

   VMs connected to standard port groups must be explicitly mapped to the target networks.

   a) Expand each virtual machine selection.
   b) Next to each guest virtual machine NIC name, click the folder for a list of available target networks.
   c) Select the checkbox next to the network that you want the guest virtual machine to map to, and then click **Select**.
   d) (Optional) To specify a new guest OS IP address for the virtual machine at the target network, expand the NIC entry and enter the new IP address, gateway, and subnet mask.

10. To complete the mobility group configuration, click **Save** or **Close**.

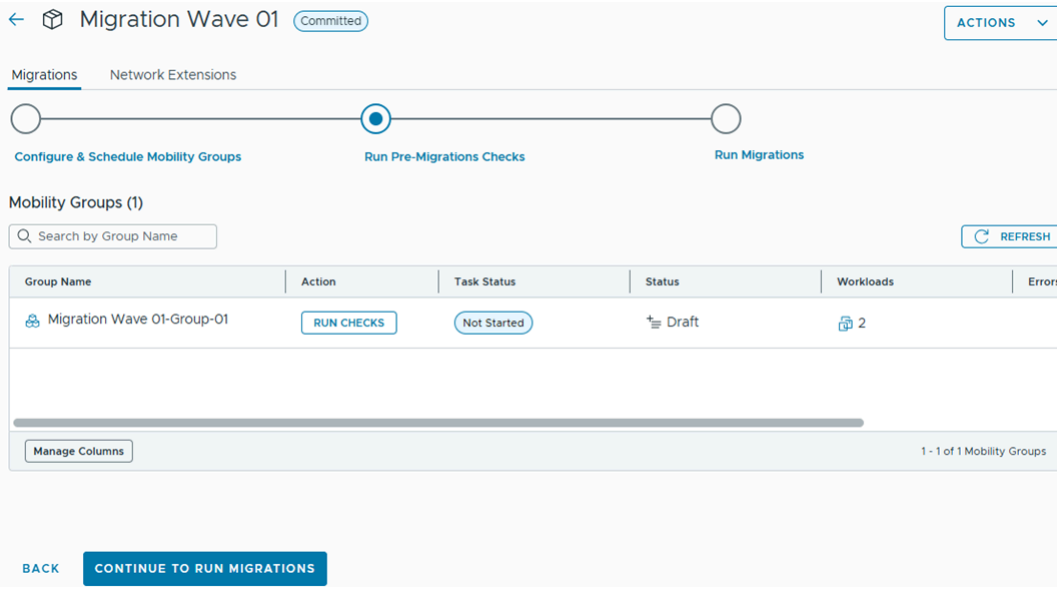
    Saving the configuration changes the group status to the Partially Configured or Configured state. Groups in Configured state are ready for validation checking and migration.

    Attention:

    You do not have to configure every group in the migration wave, but only groups in the Configured state can be migrated. Other groups in the migration wave can be configured later.

11. When one or more groups in the migration wave is configured, click **Continue to Pre-Migration Checks**.

    The Run Pre-Migration Checks screen appears. The display includes the list of mobility group names and group status, and number of workloads in each group.



12. For groups in Configured state, click **Run Checks**.

    The system validates the configurations and reports any errors in the Error column along with specific information about the error. Errors must be resolved before the migration can proceed.

13. When the validation succeeds, click **Continue to Run Migrations**.

    The Run Migration screen appears. The display includes the list of mobility groups, group status, and number of workloads. If a migration is in progress, it displays information regarding the syncing and Switchover status. For completed migrations, the screen shows the number of workloads that migrated successfully and the number that encountered an error.

    Mobility groups displaying the **Run** action, have passed pre-migration checks and are ready for migration.

14. To start a group migration, click **Run**.

    The system prompts you to validate the group information.

15. If validation succeeds, click **Start Migrations.**

    The migrations begins and the mobility groups status changes to In-progress.

This completes the process for migrating mobility groups in migration waves.

For information about monitoring mobility groups and migration activity, see Monitoring Mobility Group Activity for Migration Waves. For information about managing mobility groups, including available group actions, see Managing Mobility Groups for Migration Waves.

Monitoring Mobility Group Activity for Migration Waves

All mobility group and individual workload migration information for a migration wave is available through the Migration Planning interface.

Each migration wave is composed of one or more mobility groups. At any point, you can check the status of mobility group migrations, the status of each group, and the status of individual VMs.

> **Note:**
>
> Migration waves must be in the Committed state before mobility groups are available for migration. For more information, see Import and Commit Migration Waves.

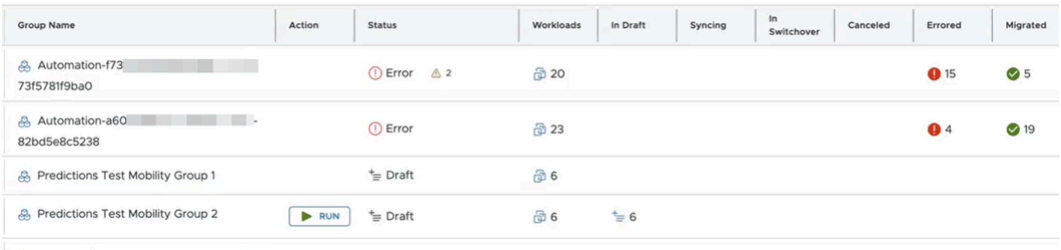
For a list of actions available for managing mobility groups in a migration wave, see Managing Mobility Groups for Migration Waves.

Monitor Mobility Group Migrations

To display a summary list of mobility group migration activity for a migration wave, select a migration wave name in the migration planning screen, and then click **Run Migrations**.

The display includes the list of mobility groups, group status, and number of workloads. If a migration is in progress, it displays information regarding the Syncing and Switchover status. For completed migrations, the screen shows the number of workloads that migrated successfully and the number that encountered an error.

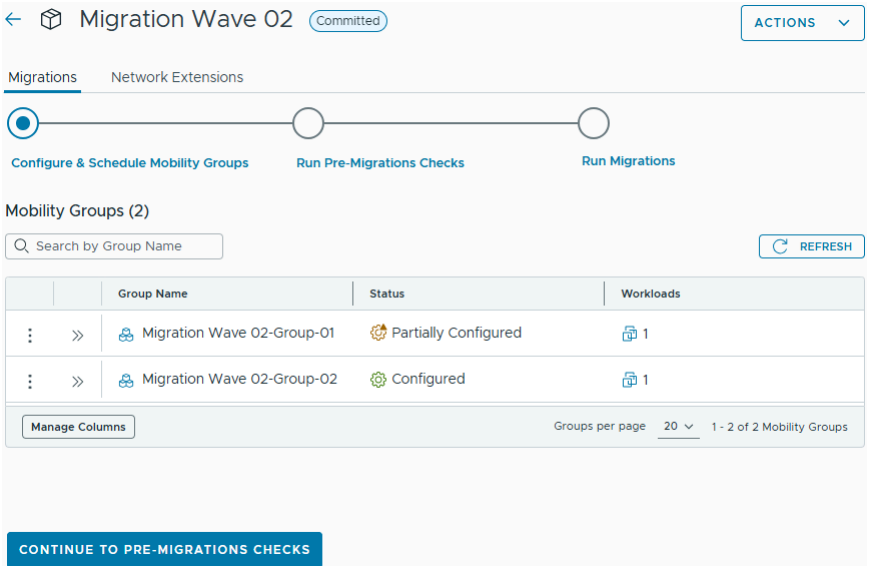
Mobility groups displaying the **Run** action, are fully configured, have passed pre-migration checks, and are ready for migration.



To view detailed information regarding migrated workloads that appear in the Errored column, expand the mobility group and review the VM details. For workloads in a failed state, click the information icon to see details about the failure. For more information regarding workload status, refer to "Monitor Mobility Group Workloads" in this section.

Monitor Mobility Group Status

To display the list of mobility groups for a committed migration wave, select a migration wave name in the Migration Planning screen. The list of mobility groups appears in the display along with the group status and number of workloads in each group of the migration wave.



Mobility groups can be in one of the following states:

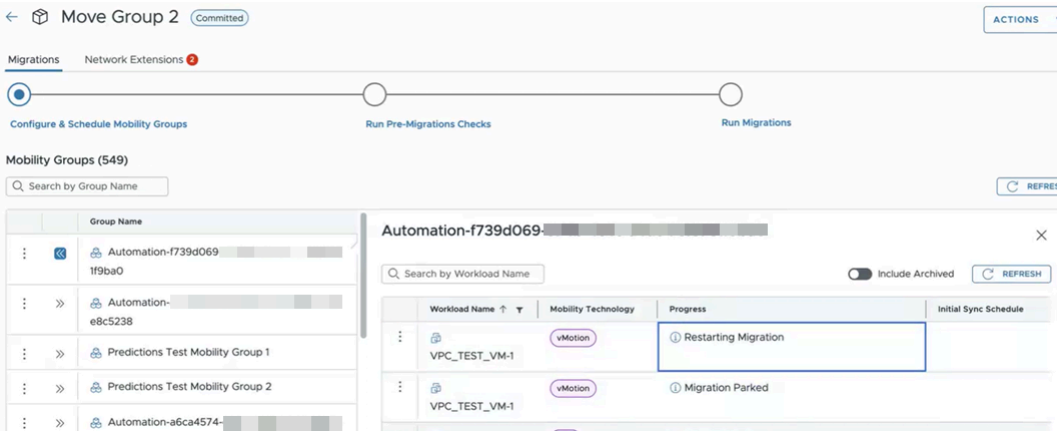
Group Status	Description
Not Configured	Indicates that no group information has been configured. This is the initial (draft) state for mobility groups in newly committed migration waves.
Partially Configured	Indicates that the configuration effort for workloads in the mobility Group has started but is not yet complete, and that the configuration has been saved for later editing.
Configured	Indicates that the mobility group information has been fully configured.
Queued	Indicates that the migration has started, and that scheduled migrations are pending the start time.
In-Progress	Indicates migrations for the group have started.
Completed	Indicates that migration is completed for that group.
Error	Indicates that one or more workloads in the mobility group have configuration errors which were detected during the pre-migration checks. You must edit the mobility group configuration before migrations can proceed.
Canceled	Indicates that the migration operation has been canceled.

To display the migration status for a mobility group, select **Run Migrations** and expand the mobility group. Then select the **Mobility Group Details** tab to see the overall group migration information. This information includes the number of workloads in the group and the total resource requirements. These details also include the transfer percentage for the entire group and the number of workloads migrated.

Monitor Mobility Group Workloads

For each mobility group in the migration wave, you can view the current status of each workload in the group. To view the workload status, expand each mobility group.

The workload display includes the workload name, migration technology (if configured), workload status (Progress), configured initial sync (transfer) schedule, and switchover schedule.



Mobility group workloads can be in one of the following states:

Group Workload Status	Description
Not Configured	Indicates that workload configuration has not started.
Partially Configured	Indicates that the workload configuration requires additional setup.
Configured	Indicates that the mobility group configuration is complete and the workloads in the group are ready for migration. Indicates that the workload configuration is complete. If all the workloads of the group are configured, then the group status is configured and ready for migration.
Queued	Indicates that the workload is scheduled for migration and pending the Initial Sync time.
Migration Started Migration Parked Migration Parked with Errors	Indicates the workload migrations for the group have started. If the migration is waiting for other migration workloads or tasks, it moves to Migration Parked state until resources are available for the migration to continue. If some migrations in the mobility group are parked and some migrations have failed, then the overall status for group becomes parked with errors.
Migration Failed	Indicates that the specific workload migration did not complete. This can occur due to maintenance at the destination site or connectivity issues. You can retry the migration at another time.
Migrated	All the migrations in the mobility group have finished successfully. The migration is completed.
Error	Indicates that workloads in the mobility group have configuration errors, which were detected during Run Pre-Migration Checks. You must edit the mobility group configuration before migrations can proceed.
Canceled	Indicates that the migration operation has been canceled.

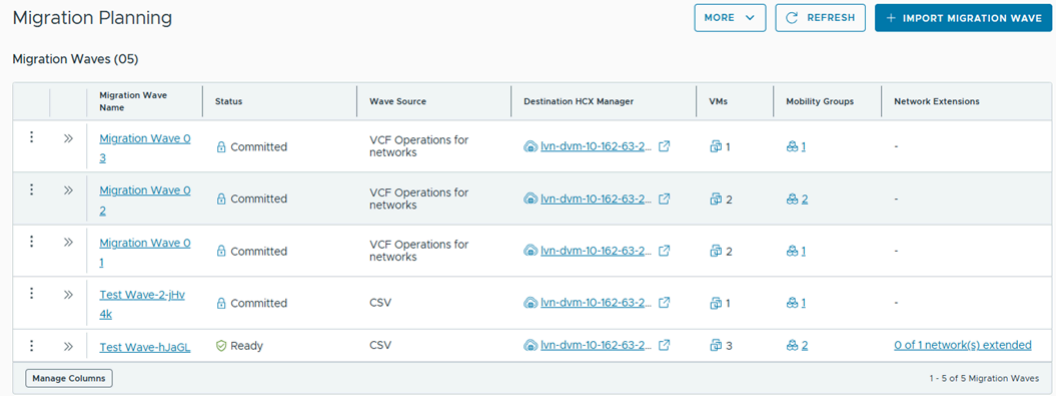
Monitoring Migration Waves

The migration wave status reflects overall wave status for both pre-commit and post-commit activity.

Migration waves are initially imported either manually as CSV files exported from VCF Operations for networks, or directly synced from VCF Operations for networks. Migration wave data is validated against the vCenter Server.

If the validation is successful, manually imported migration waves enter a pre-commit, or Ready state, after which these waves can be committed for migration. Migration waves that are directly synced from VCF Operations for networks can go to either Committed or Validation Failed state depending on the outcome of the validation process.

In the post-commit state, information from the imported migration waves is used to create the mobility groups and associated workloads for migration.

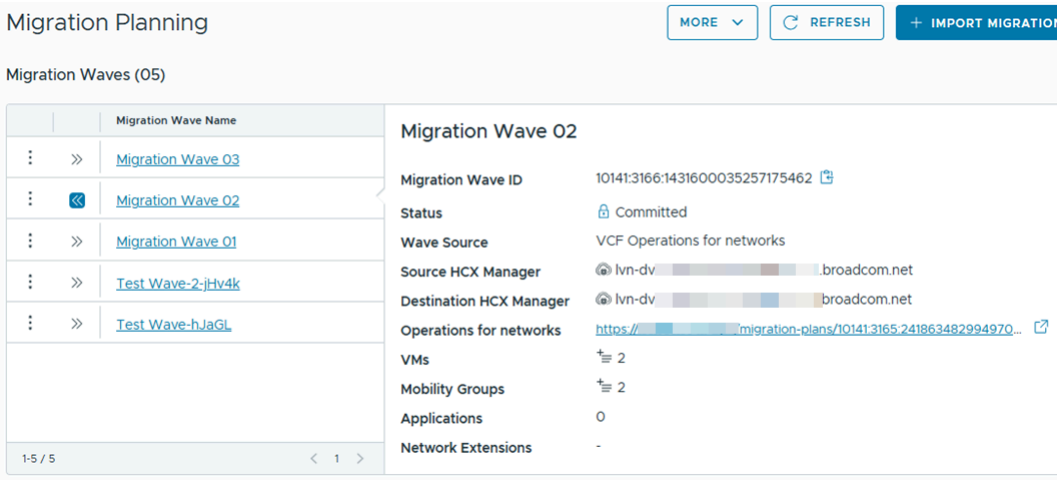


Each migration wave can have multiple mobility groups. While each mobility group can be configured and migrated independently of other groups in a migration wave, the status of the migration wave itself is dependent on the combined status of all mobility groups in the migration wave. For example, the status of a migration wave is not considered complete until all mobility groups in the wave have reached a terminal state, meaning successful, failed, or canceled migrations.

Migration waves can have the following status:

Migration Wave Status	Description
Validation Failed	(Pre-commit) Indicates that the imported migration wave did not pass validation. For manually imported migration waves, the error must be fixed in the CSV file and then re-imported. If validation fails for migration waves that are synced from VCF Operations for networks, you must fix the error in VCF Operations for networks.
Ready	(Pre-commit) Indicates that validation was successful for the manually imported migration wave files and the wave can be committed.
Commiting/Commited	(Post-commit) Indicates that the migration wave is committed and the migration groups have been prepared for configuration.
Execution In Progress	(Post commit) Indicates the state of migrations in the mobility groups for the migration wave.
Concluding/Concluded	(Post commit) Indicates that all mobility groups operations for the migration wave have terminated to some state. The migration wave is no longer available for migrating workloads and is is now in a read-only mode.
Deleting	(Post commit)

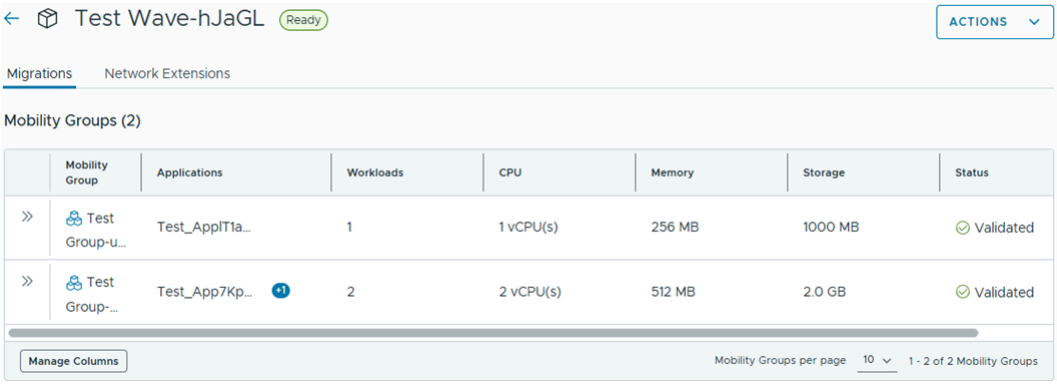
Expand a migration wave to see additional details:



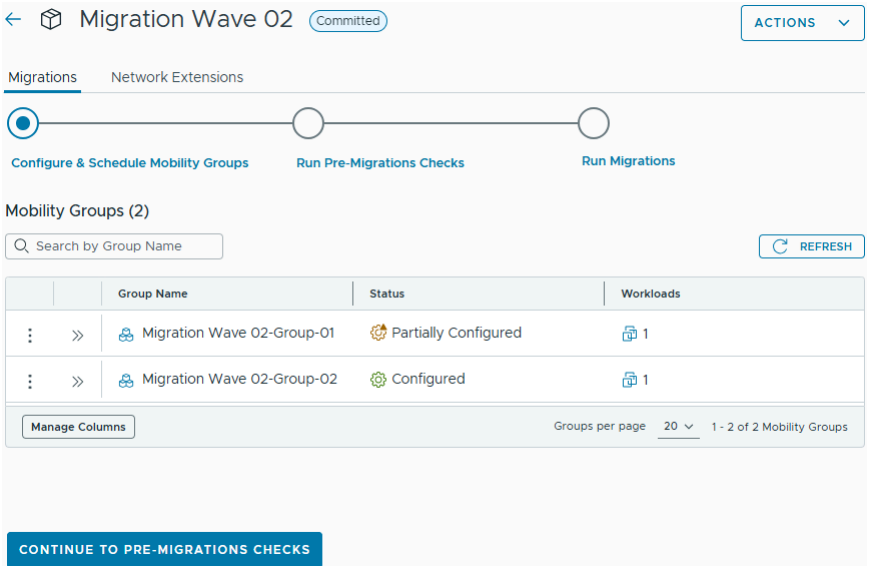
To view mobility group and and network extension information for each migration wave, click the vertical ellipsis next to each migration wave name and select **View Details**.

For migration waves in a pre-commit state, the **Migrations** tab provides summary information about each group in the wave, including the number of workloads, resource requirements, and validation status. To view information about each workload in the group, expand the mobility group.

> **Note:**
>
> If the mobility group is in a Validation Failed state, you can expand the group to display workload information that is related to the failure.



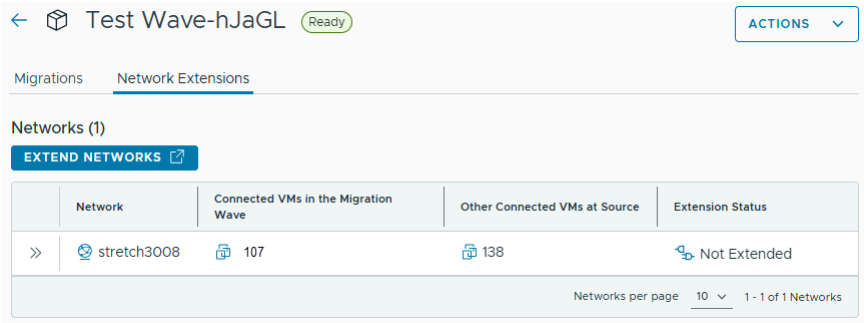
For migration waves in a post-commit state, the **Migrations** tab displays mobility group information at each point in the migration wave timeline. In Configure & Schedule Mobility Groups, you can expand a group to see individual workload information or click the vertical ellipsis to take action on the mobility group. In Run Pre-Migrations Check, you can expand the group to see summary migration status information for the group and VMs, or click the vertical ellipsis to take action on the mobility group.



To display network information for the migration Wave, click the Network Extensions tab. As part of migration wave validation, the system checks whether the network is present in the vCenter Server and then lists whether the network is extended or not. If the network validation fails, the system displays a message about the error.

Note:

If validation fails, click the information icon to get details about the issue.



Managing Migration Waves

For migration waves in either a pre-commit or post-commit state, VCF Operations HCX provides various actions for managing waves.

From the Migration Planning interface, you have access to a number of migration wave actions. In the Migration Planning screen, select the vertical ellipsis next to the migration wave name to view the available actions:

Note:

Committing a migration wave locks the content of that wave and no further changes can be made. This limits the number of available actions for committed migration waves.

Migration Wave Action	Description
View Details	Displays mobility group and network extension information. For additional information, see Monitor Migration Waves.
Commit Migration Wave	For migration waves in a pre-commit state, this action commits those waves for migration and prepares the mobility groups in each migration wave for configuration.

Migration Wave Action	Description
Edit Migration Wave	For migration waves in a pre-commit state, this action provides options for changing the destination VCF Operations HCX, imported Workload data CSV file, and imported Network Dependencies CSV file.
Delete Migration Wave	This action applies to either manually imported or synced migration waves. This action removes the migration wave from VCF Operations HCX, permanently deleting these objects: • Network extension recommendations • All mobility groups, configurations, and schedules **Note:** Migration waves synced from VCF Operations for networks are removed from VCF Operations HCX but remain available in VCF Operations for networks in an uncommitted state.
Validate Migration Wave	For migration waves where the status is Validation Failed, or if you make changes to a migration wave, this action runs the validation again.
Edit Destination VCF Operations HCX	For migration waves in a pre-commit state, this action allows for changing the destination VCF Operations HCX.
Go to Migration Planning	For migrations waves synced directly from VCF Operations for networks, this action takes you to the VCF Operations for networks console.

Managing Mobility Groups for Migration Waves

For committed migration waves, you can select from a number of actions to manage mobility groups and individual workloads.

Managing Mobility Groups for Migration Waves

To display the list of mobility groups for a committed migration wave, click a migration wave name from the Migration Planning screen. The system displays the list of mobility groups along with the three high-level steps comprising the migration timeline:

- Configure and schedule mobility groups.
- Run pre-migration checks.
- Run the migrations.

The list of actions for a mobility group can vary based on the migration timeline and the status of the group.

Mobility Group Action	Description
Configure Mobility Group	With this action, you create the initial configuration for a mobility group or edit the configuration for a partially configured mobility group.
Edit Migration Schedule	This action modifies the transfer and switchover schedules for the group.
Remove Initial Sync Schedule	This action removes the transfer schedule set for all eligible migrations in the mobility group. If any of the migrations in the group are waiting for the transfer start window to kick off, then for those migrations the initial sync starts immediately.
Remove Switchover Schedule	This action removes the switchover schedule set for all eligible migrations in the mobility group. If any of the migrations in the group are waiting for the switchover start window to kick off, then for those migrations the switchover starts immediately.

Mobility Group Action	Description
Delete Mobility Group	This action deletes the mobility group.
Retry Migration(s)	For running migrations, if any of the workloads in the mobility group are canceled or fail to migrate, this action creates a new mobility group for the failed or canceled migrations in the group. This action opens the Migration Wizard with the configurations populated from the previous group. At this point, you can modify the settings and retry the migration. For more information, see Retry a Failed or Canceled Migration
Cancel Mobility Group	This action cancels the execution of the mobility group.

Managing Mobility Group Workloads for Migration Waves

To review the actions available for individual workloads, select **Run Migrations** from the migrations timeline, expand a mobility group, and click the **VM** tab. Select one or more workloads and review the available actions:

Workload Action	Description
Force Cleanup	Following a failed or canceled migration, you can use the **Force Cleanup** option to clear internal operations and processes from source and destination sites. For information, see Managing Failed or Canceled Migrations.
Cancel Migration	This operation cancels a migration that is in progress. For information about the effects of canceling an OSAM migration, see Canceling a Migration. For information about cleaning up failed or canceled migrations, see Managing Failed or Canceled Migrations.
Archive	This operation clears the migration entry from the display. Use this action to clear failed, canceled, and completed migration activity. Clearing the migration history updates the VCF Operations HCX dashboard migration counters but does not remove the migration-related details from the VCF Operations HCX log files. **Note:** To view previously archived workloads in the display, select`Include Archive`. Archived workloads appear grayed out in the list.
Force Power-off	For eligible migrations, this operation allows for powering off a source virtual machine if the guest shutdown fails during switchover.
Start Initial Sync	For workloads that are scheduled for migration but have not reached the maintenance window, this operation starts the migration transfer phase immediately.
Edit Migration Schedule	Use this operation to change the Initial Sync or Switchover schedule for an individual workload.
Switchover Now	For workloads that are scheduled for a future switchover time and have completed the initial sync, use this operation to start migration switchover to the destination site operation immediately.

Migrating Virtual Machines with Mobility Groups

You can assemble one or more virtual machines into logical sets, for execution and monitoring of migrations as a group.

- General:
  - The migration service is activated in both the source and destination site compute profile.

- The migration service is activated in the service mesh.
- Information regarding each migration type is reviewed and understood. For more information, refer to Migration Types.
- VMware Operations HCX migration of Photon OS-based VMs is not supported.

Note:

For information related to migrating mobility groups that are created as part of a migration wave, see About Migrating Mobility Groups from Migration Waves.

With mobility groups, you have the flexibility to manage migrations for sets of virtual machines by application, network, pod, or other aspects of your environment.

Review the following considerations for migrating workloads:

- For paired vSphere-based sites, migrations are always started from the source site. For non-vSphere site pairs, migrations are always from the non-vSphere site to the destination site.
- The VCF Operations HCX system automatically detects virtual machine disk additions or removals and reconfigures running migrations to accommodate these changes. These disk changes are honored only if the changes occur before the migration switchover phase. If disk changes occur during the switchover phase, the changes are not recognized, which can affect the success of the migration operation. Support for adding or removing disks is available only with Bulk and Replication Assisted vMotion migrations.
- Taking snapshots of a VM during migration, either manually or using a third-party backup solution, can disrupt the migration process. To prevent any impact, it is required to stop those services that may create or remove snapshots during migration. Refer to KB79220 for more information.
- For Bulk, vMotion, and RAV migration types, you can select a service mesh to use for migration operations. For each site pair, you can have one or more service meshes. Each service mesh configuration has a specific resource and network configuration. In some cases, you might choose a specific service mesh to use for migrations based on the mesh configuration. If no service mesh is selected for migration, VCF Operations HCX automatically selects the service mesh to use. To add a service mesh for a site pair, see Configuring the Service Mesh.
- VCF Operations HCX refreshes the VM inventory every five minutes. When selecting VMs for the mobility group, VCF Operations HCX provides filters for listing the VMs by cluster, VMs, storage, and networking. The networking filter displays VMs by the network to which they are attached, including DVPG-backed networks, NSX segments, and VPC subnets.
- When selecting the destination network in the migration configuration wizard, note that VCF Operations HCX allows migrations from NSX VPC subnets at the source site to VPC or non-VPC subnets at the destination site. Likewise, you can migrate VMs from non-VPC subnets at the source site to VPC subnets at the destination site.

Note:

The selectable service mesh option requires that both the source and destination sites run HCX 4.8 or later.

For the operational limits supported with VCF Operations HCX migrations, see Configuration and Service Limits.

1. In the VCF Operations HCX UI, navigate to **Migration**.
2. Click **Create Mobility Group**.
3. Select a source site and a destination site for the migration.

   VCF Operations HCX displays the inventory of virtual machine workloads that are available for migration and that can be added to a group. For OS Assisted Migration, select an entry with Non vSphere Inventory. These entries provide the inventory of the guest virtual machines on which you installed HCX sentinel software.
4. Enter a mobility group name.

   If no group name is provided, the system automatically assigns a five-character identifier as the group name. You can change this name later by editing the group information. See Manage Mobility Groups.

5. Select the set of virtual machines to include in the migration group and click **Add to Selection**.
6. Click **Continue to Destination Settings**.
7. If you are migrating to a supervisor cluster, select **Show destination supervisor resources** to specify Namespace and Storage class options.
8. (Optional) Select the group level compute, storage, folder, and disk format options.

   The settings you provide are applied to all members of the group by default. To override the default settings for specific virtual machines in the group, select and expand the virtual machine entry, and set different options. For additional information regarding the Extended options, see Additional Migration Settings.

   Note:

   - For HCX Assisted vMotion (Direct), the Interconnect service mesh selection is not available.
   - You can set the Switchover and Transfer schedule for Bulk, Replication Assisted vMotion (RAV), or OS Assisted Migration (OSAM) migrations. Transfer scheduling is not available for vMotion, HCX Assisted vMotion (Direct), or Cold migration.
     The Transfer schedule identifies the date and time when base synchronization to the destination site begins. The Switchover schedule defines the date and time when the final data synchronization occurs and the workload is switched over to the destination site.
9. Select the destination network for each group of virtual machines to be migrated.
10. **Important:**

    VMs connected to standard port groups must be explicitly mapped to the target networks.

    You can override the mapping of any extended source networks, and select the required destination network.
11. Click **Continue to Migration Settings**.
12. Select the migration type. For more information about migration types, see Migration Types
13. (Optional.) Select the service mesh you want to use for the migration.

    If both the source and target HCX Manager version is greater than 9.1.0, and there are multiple service meshes existing with the same service boundaries, then the migration workload gets distributed among each of these service meshes in a round robin algorithm. You can choose not to specify a service mesh at the group or VM level in which case VMware Operations HCX will distribute migrations.
14. Continue to specify the options to complete the migration settings for the migration type you selected.
15. Click to**Continue to Customize VM Settings**.

    In the **Placement Options**, **Network Mapping**, **Migration Options**, and **Migration Schedule** tabs, you can override group level configurations, and can specify missing information.
16. In the **Placement Options** tab, specify the required Compute, Folder, Storage, and Disk Format options.
17. Click **Customize** to map the available hard disks to data sources. You can map each hard disk to a separate data source.
18. Complete any settings that are marked with pending. You can also change the group settings, if required.
19. When you have provided all the required information, the **Validate** button is active, and you can validate your migration settings.
20. Click **Continue to Review & Submit**.
21. Review the migration settings and click **Go** to validate your selections and start the migration.

VCF Operations HCX provides multiple options for monitoring and managing migrations. To review the activity, progress, and history of your migrations, refer to these topics:

- Monitor Mobility Group and Workload Migrations

- Manage Mobility Groups
- Manage Mobility Group Workloads
- Viewing Migration Event Details

Monitor Mobility Group and Workload Migrations

All mobility group and individual workload migration information, regardless of the source of that migration (migration wave or non-wave) is available through the Migration interface.

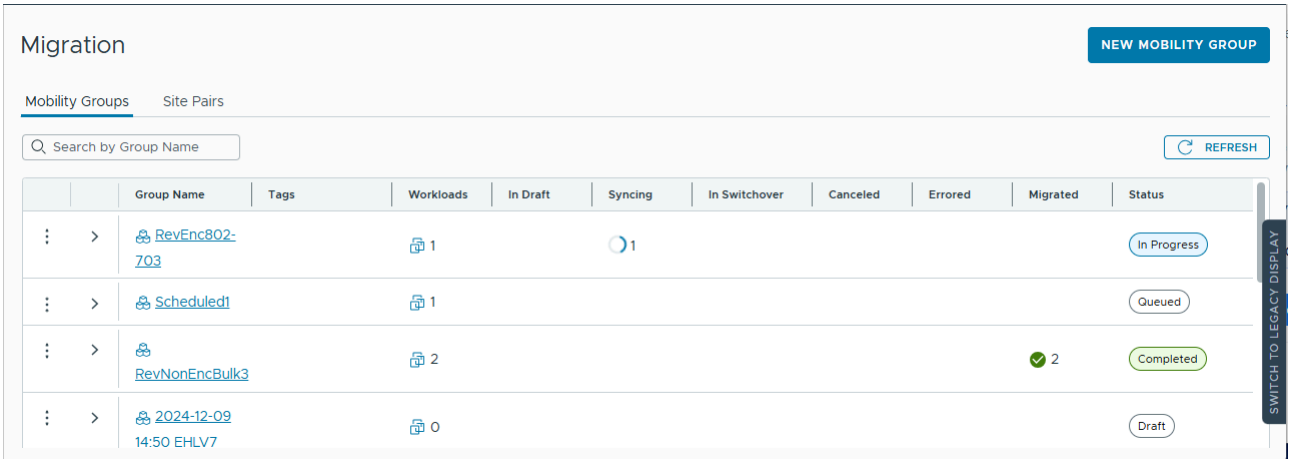
From the Migration interface, you can monitor summary group information, individual group information, and workload-specific configuration and migration details.

For information related to managing mobility groups, including editing, deleting, canceling and retrying migrations, see Manage Mobility Groups. For information related to managing group workloads, see Manage Mobility Group Workloads.

> **Note:**
>
> The Migration interface includes an **Auto Refresh** option which updates mobility group and individual workload migration status every 30 seconds. This setting persists across all migration management screens and resets only when moving to other VCF Operations HCX Infrastructure, Services, or Administration operations.

Monitor Mobility Group Information

The Migration interface provides an at-a-glance view of both migration wave and non-wave generated groups.



Group information is displayed across many columns. To modify the column display, click **Manage Columns** at the bottom of the screen.

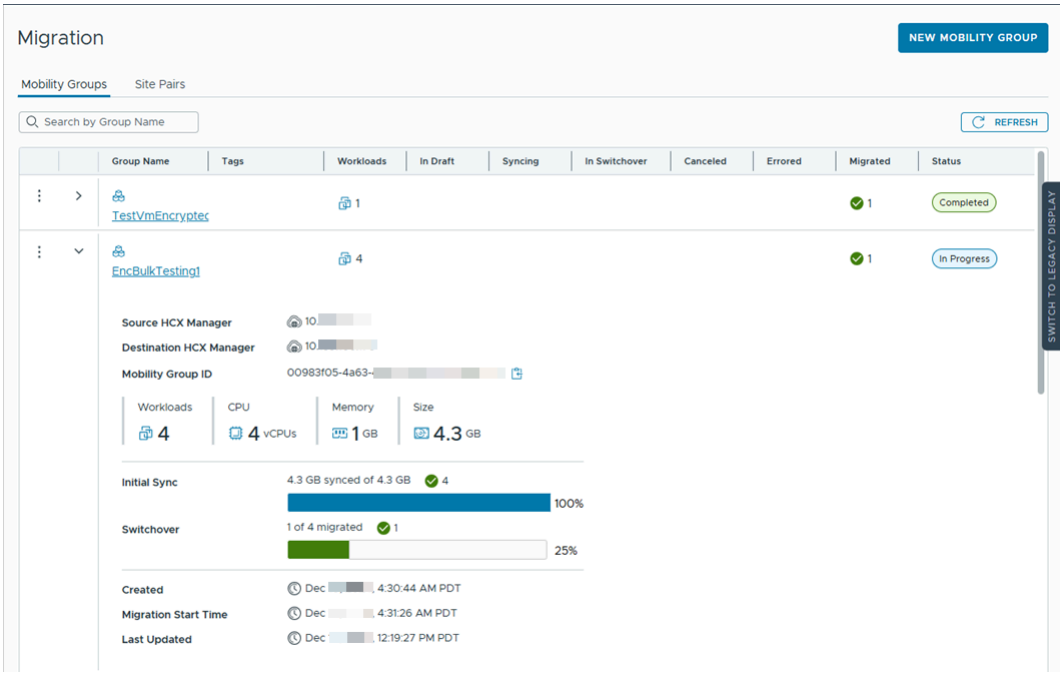
Group information includes the following details:

Summary Detail	Description
Tags	Displays any tags added as part of the group configuration. If the group is part of a migration plan created in VCF Operations for networks, the Plan tag appears in this column.
Workloads	Indicates the number of workloads associated with the group.
Drafts	Indicates the number of workloads that are in Draft state.
Syncing	Displays the number of workloads in the initial sync phase with the destination site.
In Switchover	Indicates the number of workloads that have completed syncing with the destination site and have started the switchover phase of the migration.
Canceled	Identifies the number of workloads that have been canceled.

Summary Detail	Description
Errored	Identifies the number of workloads that have failed migration. See Managing Failed or Canceled Migrations.
Migrated	Indicates the number of workloads that have completed migration.
Status	Displays the overall group status. A group can be in one of the following states: Draft, In Progress, Completed, Canceled, or Error.
Queued	Displays the number of workloads waiting to start migration.
Waiting Switchover	Displays the number of workloads waiting to start switchover.

Monitor Group Workload Information

To view additional details associated with a specific group, use the arrow next to the group name to expand the group information.



Specific group information includes the following details:

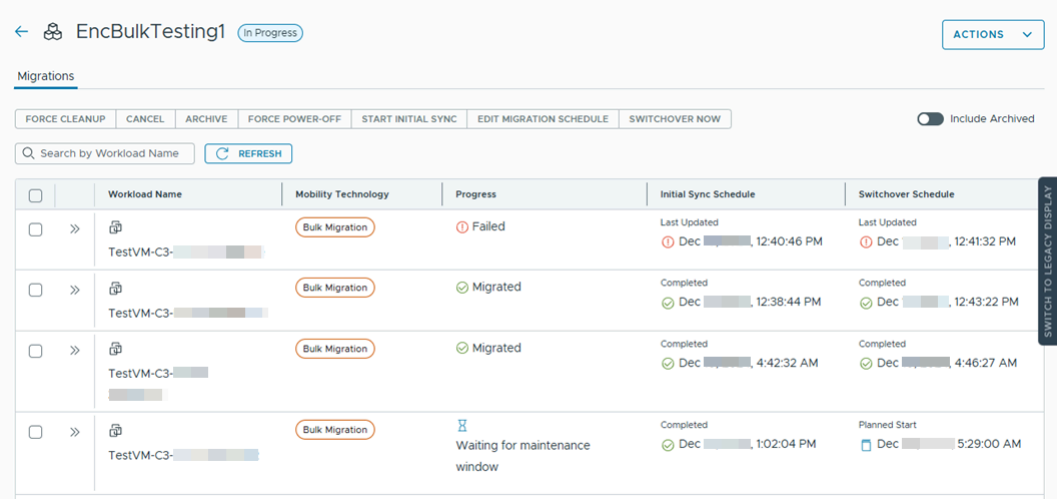
Group Detail	Description
Source VCF Operations HCX	Indicates the source HCX Manager in the site pair.
Destination VCF Operations HCX	Indicates the destination HCX Manager in the site pair.
Mobility Group ID	Unique ID associated with the group.
Migration Plan ID (if mobility group is part of a migration wave)	Unique ID associated with groups that are part of a migration plan created in VCF Operations for networks.
Total workload resource utilization	Specifies the total number of CPU resources required by workloads and the amount of memory and storage associated with the group workloads.
Initial Sync	Displays a bar graph indicating the percent progress for the initial synchronization phase for all workloads in the group.
Switchover	Displays a bar graph indicating the percent progress for the migration switchover phase for all workloads in the group.
Created	Displays the date and time of the group creation.

Group Detail	Description
Migration Start Time	Displays the date and time when the migration started.
Last Updated	Indicates the last time the group information was updated. To see the latest information, click **Refresh** at the top right of the screen.

Monitoring Individual Group Workloads

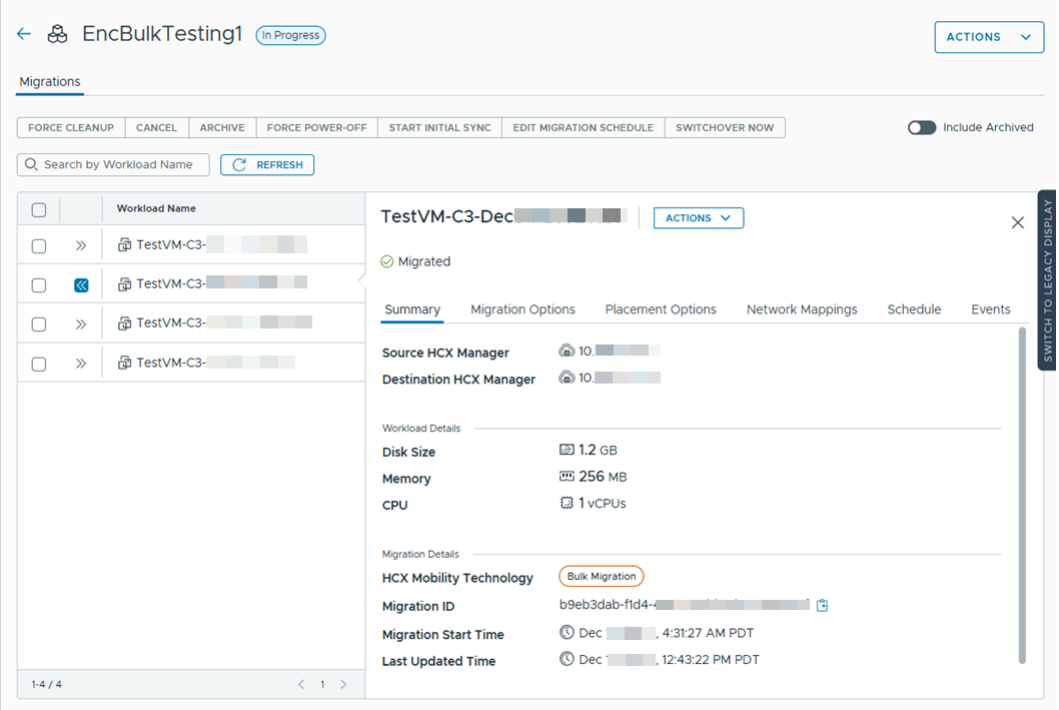
For each mobility group, you can get individual summary progress information for the group workloads along with detailed configuration and migration status information on individual workloads.

To view details associated with the workloads in a specific group, click the group name in the Migration screen.



Group Workload Detail	Description
Workload Name	Identifies the name of the workload in the group.
Mobility Technology	Indicate the migration type configured for the workload.
Progress	Indicates the migration status for the workload. Workloads can be in one of the following states: • Draft • Queued • Initial sync started • Initial sync completed • Waiting for maintenance window • Switchover started • Switchover in progress • Switchover complete • Migrated • Migration canceled • Failed • Migration parked • Migration parked with errors
Initial Sync Schedule	Displays the configured time when the workload is scheduled to start the initial sync phase of migration to the destination site.
Switchover Schedule	Displays the configured time when the workload is scheduled to start the switchover phase of migration to the destination site.

To view individual workload configuration and migration status information, expand each workload entry.



Expanding an individual group entry displays both summary configuration information for that entry and a set of actions that you can perform on the workload. For more information about these workload actions, see Managing Migration Groups and Workloads.

To view the summary configuration settings for the workload, select one of the information tabs:

- Summary
- Migration Options
- Placement Options
- Network Mappings
- Schedule
- Events

To view a comprehensive list of migration events associated with each migration workload, click the **Events** tab. Reviewing the Events information can help you troubleshoot issues related to mobility group migrations. For more information regarding migration events, see Viewing HCX Migration Event Details.

Manage Mobility Groups

From the Migration interface, you can take various actions to manage mobility groups, including editing configuration and schedule information, starting migrations, and deleting groups.

> **Note:**
>
> For information on managing individual workloads within a mobility Group, see Manage Mobility Group Workloads.

1. Navigate to **Services** > **Migration**.
2. In the list of mobility groups, for a chosen group, select the vertical ellipsis next to the group name.
3. Click the action to perform:

Group Action	Description
Run Migrations	Starts migrations for groups in draft state where the configuration has been validated, and for groups scheduled for migrations but not yet in the maintenance window.
Edit Mobility Group	Allows editing of the migration group. For example: • Add additional virtual machines to the group. • Change the default migration profile for the group. • Change the migration profile of individual virtual machines. • Delete a specific virtual machine from the group. • Restart failed or canceled migrations.
Edit Migration Schedule	Provides an option to reschedule the transfer or switchover for Bulk, Replication Assisted vMotion (RAV), or OS Assisted Migration (OSAM) migrations. For vMotion, HCX Assisted vMotion, and Cold migrations, you can reschedule the switchover setting.
Retry Migration(s)	Retries all failed or canceled migrations in the group. This action opens the Migration Wizard with the configurations populated from the previous group. At this point, you can modify the settings and retry the migration.
Delete	Deletes a group entry. You can delete a group only when all entries in the group are in the Draft state, or when all migrated workloads for the group have been archived. See Manage Mobility Group Workloads.

4. To see updated results, click `Refresh`.

Manage Mobility Group Workloads

You can manage individual mobility groups workloads, including editing the migration schedule information, starting and cancel individual migrations, and archiving migrated workloads information.

For information on managing mobility groups, see Manage Mobility Groups.

1. Navigate to **Services** > **Migration**.
2. For a chosen group, click the group name.

   The system displays a list of virtual machine workloads in the group.
3. Select one or more workloads and review the available operations:

Workload Operation	Description
Force Cleanup	Following a failed or canceled migration, you can use Force Cleanup to clear internal operations and processes from source and destination sites. For information, see Managing Failed or Canceled Migrations.
Cancel	This operation cancels a migration that is in progress. For information about the effects of canceling an OSAM migration, see Canceling a Migration. For information about cleaning up failed or canceled migrations, see Managing Failed or Canceled Migrations.

Workload Operation	Description
Archive	This operation clears the migration entry from the display. Use the Archive option to clear failed, canceled, and completed migration activity. Clearing the migration history updates the VCF Operations HCX Dashboard migration counters but does not remove the migration-related details from the VCF Operations HCX log files. **Note:** To view previously archived workloads in the display, toggle the `Include Archive` switch. Archived workloads appear grayed out in the list.
Force Power-Off	For Bulk migration only, this operation allows for powering off a source virtual machine if the guest shutdown fails during switchover.
Start Initial Sync	For workloads that are scheduled for migration but have not reached the maintenance window, this operation starts the migration transfer phase immediately.
Switchover Now	For workloads that are scheduled for a future switchover time and have completed the initial sync, this operation starts migration switchover to the destination site operation immediately.
Defer Switchover	Selecting this option keeps the migration in data sync phase until the switchover is explicitly opted for by specifying a schedule in future or an immediate switchover.
Edit Migration Schedule	Use this operation to change the Initial Sync or Switchover schedule for a workload.

4. Select an operation and complete any required information.
5. Following an operation, click `Refresh` to update the page information.

To review the mobility group status, go back to the Migration screen and click `Refresh`.

Estimate Migrations with Mobility Groups

Using the VCF Operations HCX predictive estimation feature, you can estimate the migration phase transfer for Bulk and Replication Assisted vMotion (RAV) migration types, and the relocation phase transfer for vMotion migrations.

Predictive estimation has the following requirements:

- At least 50 successfully completed migrations. Predictive estimates for both forward and reverse migrations requires at least 50 successful migrations in each direction.
- The mobility group is in Draft state.
- The mobility group has been validated successfully.
- Migration prediction is available only for datastores and HCX-IX appliances that have participated in the previous migrations to train the machine learning model.

For Bulk and Replication Assisted vMotion (RAV) migrations, VCF Operations HCX can generate real-time estimates regarding the migration transfer phase. For vMotion migrations, this includes real-time estimates regarding the relocation phase. Real-time estimation gathers information from the current migration environment. This estimate is based on sampling the underlying metrics of the environment, such as bytes transferred, rate of transfer, network throughput, and number of disks. Changes in the underlying metrics can impact the estimate. The interval between estimates varies with the size of the migrated virtual machine:

Virtual Machine Size	Estimate Interval
Less than 50 GB	Every 1.5 minutes
Greater than or equal to 50 GB but less than 1 TB	Every 5 minutes
Greater than or equal to 1 TB	Every 15 minutes

Note:

The estimate interval begins after the system has gathered the underlying metrics and completed calculations, meaning there may appear to be a delay in presenting the initial estimate. For relatively small transfers, the transfer may complete before providing an estimate.

For Bulk and RAV migrations, VCF Operations HCX provides predictive estimates for mobility groups in Draft state. Predictive estimation uses historical data from completed migrations in combination with machine learning (ML) to estimate the transfer phase.

To generate and view migration estimate, review these steps:

1. Navigate to **Services** > **Migration**.
2. For groups with Bulk, RAV, or vMotion migration types, click a group name.

   VCF Operations HCX displays the list of workloads in the group.
3. Expand the **Actions** menu, and select `Get Predictions`.

   For Bulk and Replication Assisted vMotion migrations, the Progress column provides a best-effort, real-time estimation of the time required to complete the transfer phase for a specific virtual machine. For vMotion migrations, the Progress column provides a real-time estimate to complete the relocation phase.

For each estimate in the Progress column, an information icon provides the prediction details and the estimated time in minutes and seconds for the most recent prediction.

Viewing Migration Event Details

When migrating content between paired sites, VCF Operations HCX performs a detailed set of actions that are visible in the system as migration events.

Viewing these events provides diagnostic information about the migration. This information provides the detailed migration workflow, the state of the migration, how long the migration remains in a certain state, and whether the migration has succeeded or failed. Understanding what is happening at any point in the migration can provide insight into what infrastructure or configuration changes might be necessary to address any migration issues that might occur.

HCX displays event information for all migration types.

Events are listed in the order performed. Migration events are separated by the migration phase:

- Transfer Events
- Switchover Events
- Cleanup Events

During the transfer phase, a base synchronization of source VM data takes place with additional transfers over time. The transfer phase can take time based on the amount of data being transferred and the network performance.

During the Switchover phase, HCX halts mobility operations on the source VM, and performs a final synchronization of the source data to the destination site.

During the Cleanup phase, HCX completes post migration actions at the source and the destination sites as necessary.

Note:

For migrations older than 30 days, VCF Operations HCX clears the history of events from the system automatically. For migrations that you archive using the Archive operation, the migration data is saved but the event history is cleared from the system.

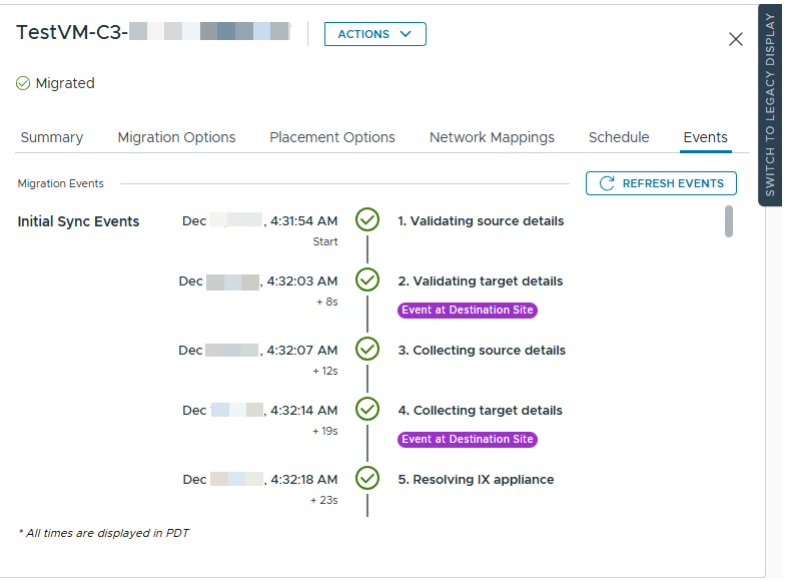
1. From the Migration interface, select a mobility group.
2. To view a list of workloads, click on a group name.

   The interface displays a list of workloads associated with the group.
3. To view workload-specific settings, expand an entry.

   The interface displays the configuration settings and migration events for that virtual machine.

4. Click the **Events** tab.

   The interface displays the migration events for that virtual machine.



5. Scroll through the events portion of the screen display for details about the migration.

   Use the refresh button to update the list of events. Destination events are highlighted. Each event provides an offset time from the Start event. Events that are more than 24 hours old include a date stamp.

   Note:

   While the migration is in progress, the system provides options for managing the migration, including scheduling the migration switchover, canceling the migration, or forcing a power off. For more information about the available actions, see Managing Migrations.

When migration is complete, cleanup event messages indicate that the source and destination sides of the migration have been cleaned up and the migration is complete.

Additional Migration Settings

The HCX migration interface provides a set of options that can be used to tailor the behaviors and conditions of the virtual machine before or after the migration operation.

HCX has several types of optional settings for use when migrating virtual machines: Switchover, Virtual Machine, and Extended.

Note:

The available options depend on the selected migration type.

Switchover Options

Switchover options are applied during the switchover phase of the migration. Except for the Force Power-Off VM setting, which can be set on an individual virtual machine (VM) migration, these options are set globally for a group of selected VMs.

Force Power-Off VM	By default, HCX attempts to shut down the virtual machine guest gracefully during the HCX Bulk migration operation. If the OS interrupts the termination process, the migration operation fails. Checking this option causes HCX to force the power-off.
Remove Snapshots	Causes HCX to consolidate snapshot files during migration of the virtual machine. For more information, see Understanding Virtual Machine Snapshots During Migrations.

Force Unmount ISO Images	Causes HCX to remove mounted ISO images before migrating the virtual machine.
Expedited Switchover (applied during bulk migration only.)	Reduces migration downtime by bypassing the final offline sync, and immediately recovering the VM at the destination after the VM at the source is powered off. Drag the slider to indicate an acceptable data loss. For example, the last 30 minutes of data.
Guest Shutdown Timeout	Indicates the wait interval for the guest shutdown for bulk migration switchover. Shut down a VM gracefully before migrating by specifying an interval of between 5 - 30 minutes.

Virtual Machine Options

The virtual machine (VM) options are available only for individual VM migration.

Enable Seed Checkpoint	If a migration is unsuccessful or canceled, Seed Checkpoint retains the target disks that are created at the target site. Without Seed Checkpoint, the HCX rollback process cleans up the target disks created during the migration and all transferred data is lost. Seed Checkpoint is available with HCX Bulk Migration and Replication Assisted vMotion. **Note:** To resume from the migration checkpoint, select **Enable Seed Checkpoint** when you restart the migration. If the Seed Checkpoint option is not selected, the migration creates new disks at the target site and ignores the existing checkpoint. To use this option, select a specific datastore for migration and not a datastore cluster. Restarting a Seed Checkpoint migration to a new datastore creates additional target disks on that datastore, and saves new checkpoint information related to that datastore with the virtual machine data in HCX.

Extended Options

Extended options customize the characteristics of the migrated virtual machine without having to manually update the settings after migration. Extended options can be set globally for a group of virtual machines migrations or for an individual virtual machine.

Retain MAC	Causes a virtual machine to keep its current MAC address during HCX Bulk migration operation, allowing communications to resume gracefully, and honors MAC-based security policies. This option is selected by default for vMotion and Replication Assisted vMotion migration types and cannot be changed.
Upgrade Virtual Hardware	Allows HCX to upgrade virtual machine hardware to the latest supported version as part of the migration operation, making current virtual machine hardware features immediately available to the migrated virtual machine.
Upgrade VMware Tools	Allows HCX to upgrade VMware Tools to the latest supported version as part of the migration operation, making current VMware Tools features immediately available to bulk migrated virtual machine.
Deactivate Per Virtual Machine EVC	vSphere Enhanced vMotion Compatibility (EVC) ensures that workloads can be live migrated, using vMotion, between ESX hosts in a cluster that is running different CPU generations. EVC is a cluster level setting that supports virtual machine mobility within a cluster. For virtual machines implementing per-VM EVC, the EVC mode becomes an attribute of the virtual machine rather than the specific processor generation it happens to be booted on in the cluster. If this option is selected then this virtual machine might not be able to vMotion back to the source site.
Host Name	Sets the host name of the migrated virtual machines at the destination site.
Domain Name	Sets the domain name for the virtual machine at the destination site. This option is available only with Linux virtual machines.
Personalization Script	Uploads a customization script to a migrated virtual machine. The script runs before and after guest customization. The customization script cannot exceed 1500 characters. For Linux VMs, Guest Customization using the Personalization Script option requires VMware Tools version 10.1.0 or later. See VMware KB 74880.

Note:
In the VMware Tools configuration, the **enable-custom-scripts** option is deactivated by default for security and must be enabled manually to run the custom script. For information about enabling this option, see VMware KB 74880.

DNS Customization Sets the Primary and the Secondary DNS servers for the migrated virtual machine.

Generate a new Security Identifier (SID) Generates a new Security Identifier (SID) for the migrated virtual machine. This option is only for Windows virtual machines. Make sure that the virtual machine has an active "local administrator" present. If the virtual machine is connected to the domain, it is moved to workgroup, and no DNS suffix is set.

Resize CPU Changes the number of vCPUs for the migrated virtual machine. For more information about VMware virtual machine vCPU limitations, see the VMware vSphere product documentation.

Resize Memory Changes the memory size for the migrated virtual machine. For more information about VMware virtual machine memory limitations, see the VMware vSphere product documentation.

Replicate Security Tag Replicates tags that are associated with a virtual machine undergoing migration. The tags are replicated from the source site to the destination site.

Migrate Custom Attributes Migrates VMware Custom Attributes associated with a virtual machine undergoing migration. The Custom Attributes are replicated from the source site to the destination site.

Migrate vCenter Tags Migrates vCenter tags that are associated with a virtual machine undergoing migration. The vCenter Tags are replicated from the source site to the destination site.

Migrate vSphere Datasets (available with bulk migration only) vSphere datasets are key value pairs that are shared between the vSphere management layer and the guest operating system. Enabling this option ensures that these datasets are migrated along with the virtual machine.

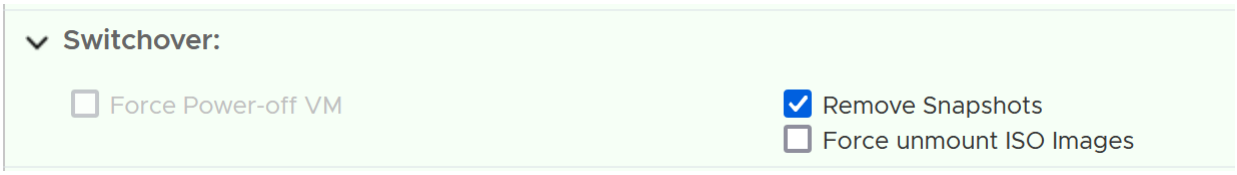
The following table summarizes the options that apply to each migration type.

Option	Cold Migration	vMotion Migration	Bulk Migration	Replication Assisted vMotion Migration	HCX Assisted vMotion (Direct)	OS Assisted Migration
Force Power-off VM	No	No	Yes	No	No	N/A
Remove Snapshots	No	Yes	Yes	Yes	No	N/A
Force Unmount ISO Images	No	Yes	Yes	Yes	No	N/A
Seed Checkpoint	No	No	Yes	Yes	No	No
Retain MAC	No	Yes (default setting; not selectable)	Yes (not supported for single vCenter operation)	Yes (default setting; not selectable)	Yes (default setting; not selectable)	Yes (default setting; not selectable)
Upgrade Virtual Hardware	Yes	Yes (upgrades on reboot)	Yes	Yes	Yes	N/A
Upgrade VMware Tools	No	Yes	Yes	Yes	Yes	N/A
Deactivate Per-VM EVC	No	Yes	Yes	Yes	No	N/A
Hostname	No	No	Yes	No	No	No
Domain Name	No	No	Yes	No	No	No
Personalization Script	No	No	Yes	No	No	No
DNS Customization	No	No	Yes	No	No	No

Option	Cold Migration	vMotion Migration	Bulk Migration	Replication Assisted vMotion Migration	HCX Assisted vMotion (Direct)	OS Assisted Migration
Generate a new Security Identifier (SID)	No	No	Yes	No	No	No
Resize CPU	No	No	No	No	No	Yes
Resize Memory	No	No	No	No	No	Yes
Replicate Security Tags	Yes	Yes	Yes	Yes	Yes	N/A
Migrate Custom Attributes	Yes	Yes	Yes	Yes	Yes	N/A
Migrate vCenter Tags	Yes	Yes	Yes	Yes	Yes	No
Expedited Switchover	No	No	Yes	No	No	No
Guest Shutdown Timeout	No	No	Yes	No	No	No
vSphere Datasets	No	No	Yes	No	No	No

Understanding Virtual Machine Snapshots During Migrations

Virtual machine snapshot handling varies slightly by migration type. The migration interface includes an option to explicitly remove virtual machine snapshots during the migration operation.



HCX vMotion, RAV, or Cold migration

As part of the migration validation, VCF Operations HCX detects and warns if snapshots are present for the selected VM.

If snapshots are not present, the migration proceeds.

If snapshots are present, check the **Remove Snapshots** option to proceed. VCF Operations HCX will delete snapshots and consolidate changes before proceeding with the migration's relocation.

Not selecting the **Remove Snapshots** option can have the following effects on migration:

- Interruption of migration streams where any of the VMs being migrated have snapshots.
- Failure of pre-migration validation checks or of the migration attempt.

Bulk migration

For Bulk migrations, VCF Operations HCX always removes virtual machine snapshots prior to migration to the remote site. As part of the migration validation, VCF Operations HCX detects whether snapshots are present for a VM. If present, VCF Operations HCX consolidates the LWD streams, and the replicated VMs will have no snapshots.

> **Note:**
> For Bulk migrations, the **Remove Snapshots** option has no effect. Bulk migration always removes snapshots, and migrations continue without interruption due to snapshots.

Managing Failed or Canceled Migrations

For a failed or canceled migration, VCF Operations HCX offers you several options for what to do next.

Depending on the state of the migration, responding to a failed or canceled migration can include the following options:

- Retry a migration.
- Use the Force Cleanup option to clear failed or canceled migration processes.
- Archive the migration to clear the migration history.

Retrying a Migration

For a migration in a failed or canceled state, you can use the **Retry** option at the mobility group level. The retry operation populates all the failed and canceled migrations in the Migration Wizard. You can exclude workloads not to retry. This operation creates a new mobility group.

By default, the **Retry** option enables the **Enable Seed Checkpoint** feature for Bulk and RAV migration types. If the **Enable Seed Checkpoint** option was selected during the initial migration attempt, the retry operation re-establishes the migration effort using the already migrated data. Enabling seed checkpointing prevents having to restart a migration from scratch due to a failure or cancelation.

> **Note:**
> If a cancelation or failed migration rollback does not complete, use the **Force Cleanup** option before the **Retry** option.

To retry a migration, refer to Retry a Failed or Canceled Migration.

Using Force Cleanup to Clear a Failed or Canceled Migration

Following a failed or canceled migration, the system attempts to clean up migration-related processes that started but did not complete. These processes occur on both the source and the destination sites. If the migration clean-up does not succeed, new migration or migration retry operations can fail. To view detailed information regarding the state of a migration failure or cancelation, expand the specific migration entry.

To clear any incomplete processes related to the migration, use the **Force Cleanup** option. If **Enable Seed Checkpoint** was selected for the migration, **Force Cleanup** displays an option to remove the checkpoint data.

To clean up processes for a failed or canceled migration, see Force Cleanup for a Failed or Canceled Migration .

Archiving a Failed or Canceled Migration

To clear migration activity that requires no further action, use the **Archive** option. Archiving migrations removes them from the **Migration Management** and the **Migration Tracking** screens but does not remove the migration-related details from the HCX log files. Clearing the migration history using the **Archive** option updates the HCX Dashboard migration counters.

To archive migration activity, see Clearing the Migration History.

Canceling a Migration

The Migration interface includes an option for canceling in-progress migrations.

For OSAM, the effect of canceling a migration depends on the state of the migration when selecting the **Cancel** option:

- Canceling a migration while an HCX appliance is replicating data to the destination site deletes the associated resources created at the destination site with no effect on the source VM.
- Canceling a migration when the source system is in the final sync phase reboots the source system and deletes the associated resources.
- Canceling a migration after the target VM has been created deletes the destination virtual machine and the associated resources.
- Canceling a migration after the source virtual machine is powered down requires you to restart the virtual machine at the source site. HCX deletes the associated resources at the destination site.

For RAV, vMotion and Cold migration, you can cancel a migration at any point and the source VMs retain their original state.

For Bulk migration, the replication is canceled on the source site VM, and replicated data is deleted from the destination site. Canceling a migration after the target virtual machine is powered on prevents the source virtual machine from returning to a powered-on state. Ensuring only one instance of a virtual machine is powered on prevents duplicate active instances of a virtual machine on both the source and the destination sites.

If Seed Checkpoint is enabled for RAV or Bulk migration, VCF Operations HCX retains any transfered data at the destination site.

Retry a Failed or Canceled Migration

For a migration in a failed or canceled state, you can use the **Retry** option at the mobility group level to retry the original migration operation.

The mobility group status must be in a failed or a canceled state before retrying a migration. In the case where the cancelation or failed migration rollback is not complete, as a best practice, perform **Force Cleanup** before using the **Retry** option.

1. Navigate to the Migration interface.
2. Click the mobility group that has an error status or has been canceled, click the vertical ellipsis, and select **Retry Migrations**. The Migration Wizard displays all workloads in a `Canceled` or `Errored` state.
3. (Optional) You can deselect the workloads that you do not want to migrate.
4. To start the migration, click `Go`.

   The system creates a new mobility group. To view the status of each migration go to the mobility group details page.

Force Cleanup for a Failed or Canceled Migration

The **Force Cleanup** option provides the method for manually cleaning up failed or canceled migration processes that were not automatically cleaned up by the system.

Migration processes for a failed or canceled migration must be cleaned up before retrying a migration.

1. Navigate to the Migration interface.
2. Review the `Canceled` or `Errored` columns for failed or canceled migrations.
3. Review system and event messages to make sure that indications related to the migration failure or cancellation were cleaned up.
4. For any migrations that did not clean up properly, click **Force Cleanup**.

   If there are multiple migrations requiring clean up, you can select all that apply.

   A pop-up window appears prompting you to confirm the clean-up operation and offering the option to remove seed checkpoint data.

   CAUTION:
   Removing the seed checkpoint data discards any previous migration progress, and that progress cannot be recovered.
5. (Optional) Select **Remove Seed Checkpoint data**.

   You might select this option if the seed data at the destination site is not required for future use.
6. To proceed with the clean-up operation, click Force Cleanup.

   Observe the specific migration event messages related to the migration to verify that the cleanup was successful.

If the clean-up operation did not succeed, repeat the **Force Cleanup** operation.

Clearing the Migration History

You can clear the migration activity for a site using the **Archive** option.

To clear Failed, Canceled, or Completed migration activity that requires no further action, use the **Archive** option. Clearing the migration history updates the HCX Dashboard migration counters but does not remove the migration-related details from the HCX log files. Also, archiving migrations removes them from the Migration screens in the UI.

> **Note:**
>
> For Failed, Canceled, or Completed migrations older than 30 days, VCF Operations HCX clears the history of events from the system automatically. For migrations that you archive using the Archive operation, the migration data is saved but the event history is cleared from the system.

1. Navigate to the HCX Dashboard.
2. Select the **Services** > **Migration**.
   The Migration interface displays a summary of the Mobility groups for a site pair.
3. Click the group name that contains the workloads to archive.
   The system displays the list of workloads that are associated with the group.
4. Select the migration entries that you want to clear from the display.

   Note:
   You cannot clear migrations that are in progress.
5. Click **Archive**.
   A pop-up screen appears prompting you to acknowledge the request to archive the migration entries.
6. Click **Archive Migration**.
   The system clears the selected entries from the migration history for the group.

Repeat the process for workloads in other groups as necessary.

Integration with VCF Operations for Networks

You can export waves of VCF Operations for networks discovered applications to VCF Operations HCX for migration as mobility groups. VCF Operations HCX integration with VCF Operations for networks is available through API calls.

Public APIs are available for exporting VCF Operations for networks Application Groups to HCX as mobility groups. To view the VCF Operations HCX API for creating mobility groups, log in to access the VCF Operations HCX API documentation: https://<*hcx_ip_or_fqdn*>/hybridity/docs. Navigate to **Mobility** > **Migration Group APIs** in the documentation.

VCF Operations for networks uses Application Discovery and Dependency Analytics to identify migration waves. From this information, VCF Operations for networks defines Application Groups that are then exported using public APIs to VCF Operations HCX as established mobility groups.

After VCF Operations HCX creates the mobility groups, you prepare for migration using the mobility group configuration procedures.

> **Note:**
>
> All limitations and requirements for migrations and migration types apply to mobility groups created from VCF Operations for networks.

1. Import the VCF Operations for networks discovered application groups into VCF Operations HCX using API calls.
2. Navigate to **Services** > **Migration** and verify that VCF Operations HCX created the mobility groups.

   Note:

   Mobility groups created by VCF Operations for networks have a VCF Operations for networks label and a timestamp to differentiate them from other mobility groups created by VCF Operations HCX admins.

3. Configure the mobility groups for migration.

   All mobility group operations are available for configuration, including setting the migration type, scheduling the migration, and editing the group information. See Migrating Virtual Machines with Mobility Groups.

4. Complete the migration.

The workloads included in the API are created and migrated as mobility groups in VCF Operations HCX.

Managing System Settings

Use HCX Manager Appliance Management for viewing, configuring, and managing system-level functions.

The Appliance Management is reached by navigating to the management port: https://*hcx-ip-or-fqdn*:9443. This interface uses the system administration credentials set up during the OVA deployment.

The Appliance Management provides the access to the system Dashboard, Appliance Summary, Configuration, and Administration information.

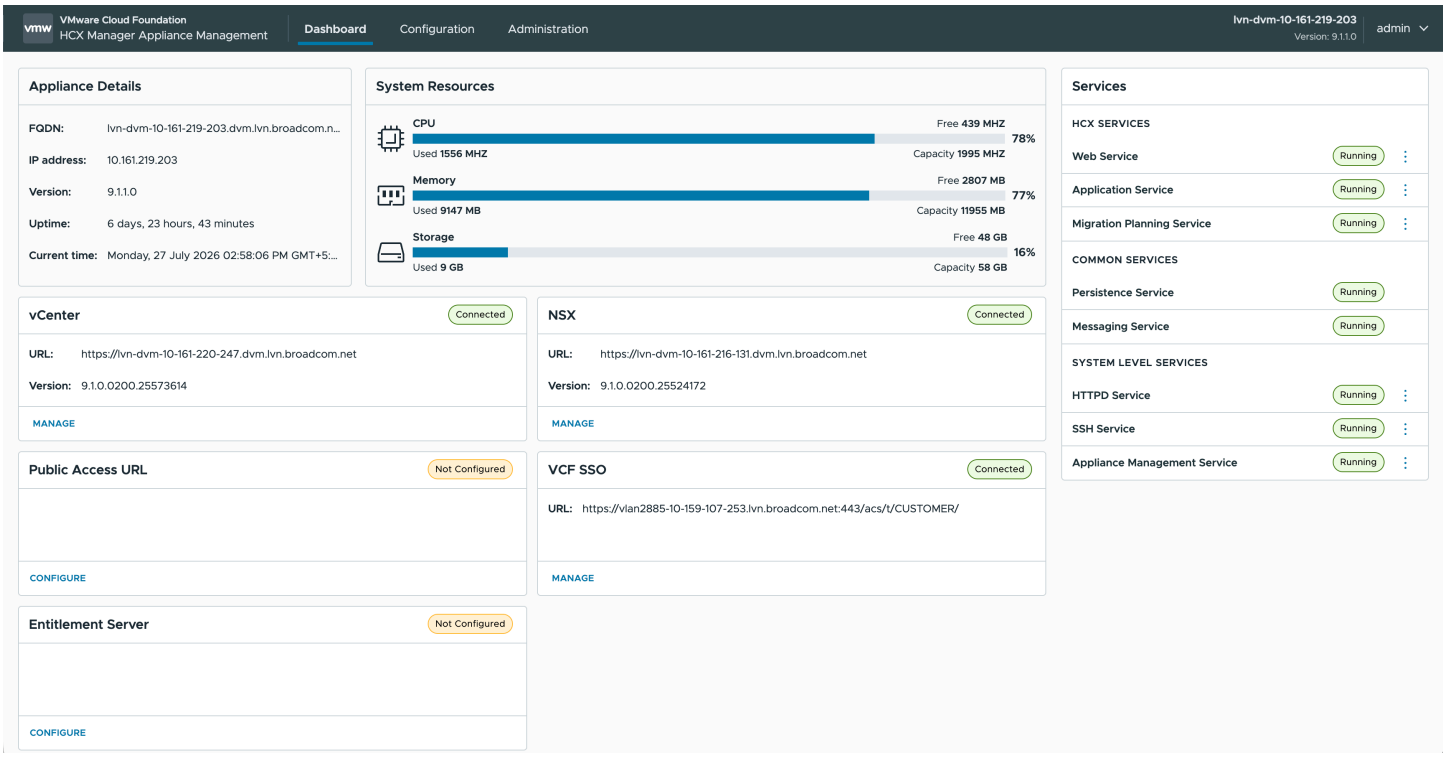
See, Monitoring Services from the System Interface.

> **Note:**
>
> VCF Operations HCX system operations might be performed by your cloud service provider.

Understanding the System Management Dashboard

The system dashboard provides the access to status and services, configuration settings, and system-level administration tasks.

The dashboard is the first screen that appears after you log in to HCX Manager Appliance Management.



The dashboard provides the access to various system management settings through a set of tabs at the top of the display.

Note:

For installations where the vCenter Servers are in linked-mode, the dashboard includes information from vCenter Servers that are registered to a system.

Tab	Description
Dashboard	Displays the appliance status as a set of summary panels: • System information and resource usage • NSX • vCenter • SSO • VCF SSO • Public Access URL • vCenter Entitlement Server The tiles that are visible in the display depend on the installation and services. To change the configuration settings for a panel, click **Manage**. The system redirects you to update the settings.
Configuration	Displays the list of service configuration settings. • Licensing • vCenter Server • NSX Manager • VCF SSO • Public Access URL • HCX Role Mapping • Location • CEIP (Customer Experience Improvement Program) To display and edit the current settings, click an item in the list.
Administration	Displays the list of system-level configuration settings. • General Settings – Time Settings – Syslog Server – System Name – Change Password • Network Settings – General Network – DNS Servers – Proxy Configuration – Static Routes • System Management – Backup and Restore – Upgrade – Technical Support Logs • Certificates – Trusted CA Certificates – Server Certificate To display or edit the settings, click an item.

Updating the Time Settings

The system provides initial NTP Server settings during the OVA deployment in the vCenter Server. These settings can be updated in the Appliance Management.

CAUTION:

If you edit the NTP Settings, you must restart the Appliance Management Service.

Editing and Removing the NTP Server Configuration

VCF Operations HCX requires a valid NTP server-synchronized time for integrated systems operations.

1. Log in to the Appliance Management.
2. Select the **Administration** tab.
3. Navigate to **General Settings** > **Time Settings**.
4. Click **Edit** or **Unconfigure NTP Servers**.
5. Enter the name of the NTP server.
6. Multiple servers can be specified using a comma-separated list.
7. Select the **Dashboard** tab, locate the appliance management service in the Services tile, click the ellipsis, and select **Restart**.

Updating the System Name

The initial hostname is provided during the OVA deployment. The system name can be updated in the Appliance Management.

Editing the System Name

1. Log in to the Appliance Management.
2. Select the **Administration** tab.
3. Navigate to **General Settings** > **System Name**, and click **Edit**.
4. Enter the system name, and click **Save**.

Changing the Admin Password

You can change the user admin password from the Appliance Management.

You can also change the root or admin password using the CLI. See Updating System Passwords using the CLI.

1. Log in to the Appliance Management.
2. Select the **Administration** tab.
3. Navigate to **General Settings** > **Change Password**.
4. Respond to the prompts for the current and new password.
5. Click **Set Password**.

Adding a Remote Syslog Server

Configure a remote syslog server to gather logs for both HCX Manager and any data plane appliances that are deployed for VCF Operations HCX. Your changes take effect in both VCF Operations HCX and all appliances.

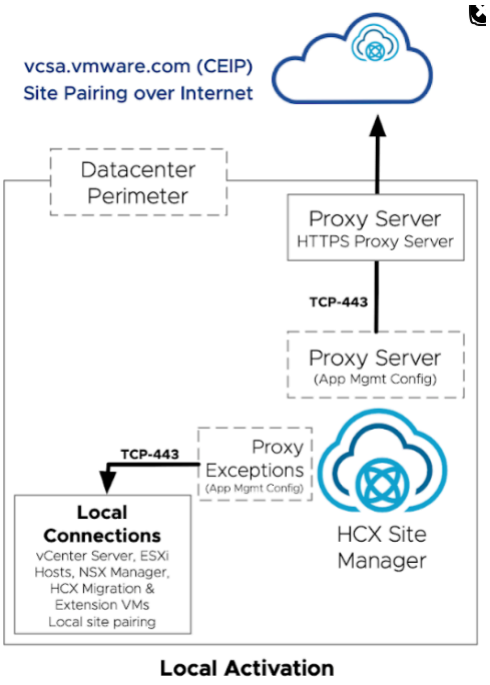
By default, VCF Operations HCX and deployed appliance (HCX-IX, HCX-NET-EXT, HCX-SGW, HCX-SDR, HCX-SRG) log messages are available locally. Configuring a remote syslog server sends logs that are generated on both VCF Operations HCX and appliance virtual machines to that local server.

1. Log in to the Appliance Management.
2. Select the **Administration** tab.
3. Navigate to **General Settings** > **Syslog Server**, and click **Edit**.
4. Enter the server information and click **Save**.
5. For non-enhanced service meshes, perform a Redeploy or Update Appliance action for the logs to start streaming to the syslog server. This action is not required for enhanced service meshes.

For existing data plane appliances, perform a Force Sync operation from the VCF Operations HCX UI for the changes to take effect. Appliances deployed after adding the external syslog server automatically forward log messages.

Configure a Proxy Server

A proxy server is an intermediary “message-forwarding agent” selected by a client through its local configuration for outbound HTTPS requests for security or shared caching. Add a proxy server configuration in HCX Manager to send HTTPS requests to a proxy server in the environment.



Local Activation

When configuring a proxy server, refer to the following considerations and best practices:

- HCX Manager systems make various HTTPS requests during normal operation.

  Outbound Connections (Local Mode):

  - HCX Manager to `vcsa.vmware.com` (for sites participating in the Customer Experience Improvement Program [CEIP])
  - HCX Manager to Remote HCX Manager `(for site pairing)`

  Local Connections:

  - HCX Manager to Registered vCenter Server
  - HCX Manager to Registered vCenter ESX Hosts
  - HCX Manager to Registered NSX Manager system
  - HCX Manager to Migration and network extension service mesh appliances deployed by this HCX Manager
- A proxy server is usually intended to handle internet-bound conections from internal systems (to endpoints that resolve to public IP addresses).
- Use the **Proxy Server** field to enable proxy operation.

- For HCX to function correctly when a proxy server is configured, local connections must be explicitly excluded from proxy operation. Use the **Proxy Exclusions** field.
- The destination HCX Manager for site pairing must be configured with the Local Connection when the IP address is internally reachable without traversing the proxy.
- Use the **Proxy Exclusions** field for broad or granular configurations.
  A simple way to restrict Local Connections is to enter one large subnet that includes all internal IP address space for the data center in the **Proxy Exclusions** field. Alternatively, restrict Local Connections by specifying granular subnets in the **Proxy Exclusions** field.
- Kerberos and Windows NTLM Proxy Servers are not supported.

  Important:

  Configuring a proxy server without local exclusions typically results in migration failures and errors during HCX operation. See VMware KB 89180.

  Note:

  The service mesh does not support proxy server configuration.

1. Log in to the Appliance Management.
2. Select the **Administration** tab.
3. Navigate to **Network Settings** > **Proxy Configuration** and click **Edit**.
4. Enter the proxy server settings:
   a) Proxy Server IP address or FQDN.
   b) Proxy Server Port.
   c) Proxy Server User.
   d) Proxy Server Password.
   e) Proxy Exclusions.

   Using a comma-separated list to define all related proxy server exclusions, enter any IP, subnet, host, and/or domain names. Use * for wildcard values and do not include complete URLs (no https://).
5. To verify the configuration, click **Test Connection** and enter the test URL.
6. Click **Save**.
7. Restart the Appliance Management services for the proxy exclusions to take effect. See Monitoring Services from the System Interface.

Managing CA and Self-Signed Certificates

Add or remove certificates from the system certificate store using HCX Manager Appliance Management.

VCF Operations HCX uses self-signed certificates for the Appliance Management (port 9443) and VCF Operations HCX UI (port 443) appliances.

When upgrading from earlier HCX releases, VCF Operations HCX detects and rotates self-signed certificates nearing expiry.

If the HCX certificate is set to expire in less than one year, a new self-signed certificate is generated and shared between the Appliance Management and HCX service appliances.

HCX deployments never rotate the customer-imported certificates.

If your environment uses a certificate monitoring system, that system must accept the new self-signed certificate.

Non-Disruptive Certificate Rotation

VCF Operations HCX works with VMware Cloud Foundation certificate rotation policies.

VCF Operations HCX relies on connections with vCenter and NSX for management and operational activities. In VCF 9.0 and later, when vCenter or NSX certificates are rotated, those changes are automatically detected, and the new certificate

is used without disruption to network connectivity or migration services. VCF Operations HCX also supports importing modified certificates of these services: VCF SSO, vCenter acting as a vCenter Entitlement Server, external vSphere SSO.

After the initial VCF Operations HCX configuration, there is no further requirement for managing system certificates.

VCF Operations HCX provides the flexibility to manage certain certificates sometimes. For example, when importing certificates from a remote site.

Importing Trusted Certificates from a Remote Site

You can import and trust certificates from remote systems on HCX Manager.

1. Log in to the Appliance Management.
2. Select the **Administration** tab.
3. Navigate to **Certificates> Trusted CA Certificates**.
4. Click **Import**.
5. Select the certificate import option: **File**, **URL**, or **Content**.
6. Enter the information for the selected option.

   For example, when selecting the URL option, enter the IP or FQDN that the source site uses to reach the destination site.
7. Click **Apply**.

Updating the Local Server Certificate

This procedure allows you to manually update the HCX Manager certificate.

1. Log in to the Appliance Management.
2. Select the **Administration** tab.
3. Navigate to **Certificates** > **Server Certificate**.
4. Enter the certificate and the key information, and click **Apply**.

Removing Certificates

You can view the certificate information and can remove certificates in the Appliance Management.

> **CAUTION:**
>
> Deleting non-user imported certificates can impact signing chains or otherwise impact VCF Operations HCX appliances.

1. Log in to the Appliance Management.
2. Select the **Administration** tab.
3. Navigate to **Certificates** > **Trusted CA Certificate**.
4. Click the vertical dots menu next to the certificate you want to delete, and then click **Delete**.

Backing Up and Restoring the System

You can back up and restore the appliance from HCX Manager Appliance Management.

Backup and restore operations are available in the Appliance Management except when restricted by a cloud service provider. You first use the Appliance Management to generate a configuration file and then use that file to restore to a healthy system.

The HCX service appliances, which include HCX-IX and HCX-NET-EXT, do not require individual backups. A restored VCF Operations HCX reconnects to existing service appliances that were created within the backup time frame. If the

service appliances are no longer functional, HCX Manager deploys new appliance virtual machines based on the backed-up configuration.

Backing Up VCF Operations HCX

You use HCX Manager Appliance Management to create a backup file.

This operation backs up the following information:

- Inventory data
- Configuration files
- Certificates
- System UUID

The backup file is saved in tar.gz format.

1. Log in to the Appliance Management.
2. Select the **Administration** tab.
3. Select **Backup and Restore**.
4. Set up an SFTP server for uploading the backup file:
   a) Click the **SFTP server setting** tab.
   b) Click **Add**.

      Note:
      The best practice is to use a Linux-based OpenSSH host for file transfer operations.
   c) Enter the SFTP server information and click **Save**.
5. Configure a backup schedule and the backup scheduling by clicking **Backup Settings** in the top-right corner of the page.

   Note:

   The best practice is to schedule daily backups. Restoring from backup files that are more than two days old is not supported due to potential inventory changes from the backup time to present.

Restoring the System

You use HCX Manager Appliance Management to restore the system from a backup file. The restore operation is used in cases where the system has become corrupt or unusable due to resource or system failures.

You have deployed a replacement system that is clean of prior configuration settings.

The replacement system has the same software version and IP address as the original system.

The Backup file being used for the Restore operation cannot be older than 2 days.

Note:

A clean system deployment requires only the minimum configuration to be manageable and that the system is network reachable from the operator or client system.

This operation restores the appliance to the state it was in at the time of the backup. The contents of the backup file supersede configuration changes made before restoring the appliance.

Restoring from backup files that are more than two days old is not supported due to potential inventory changes from the backup time to present.

Note:

A restored VCF Operations HCX cannot connect to HCX service appliances that were created after the backup file was generated.

1. Log in to the Appliance Management.
2. Select the **Administration** tab.
3. Navigate to **Backup and Restore**.
4. Click **Backup Settings** in the top-right corner of the page
5. Verify that the system is operating properly:
   a) Select the **Dashboard** tab and confirm that the NSX and vCenter status is green.
   b) In the Services tile, verify that the Hybridity Services, Common Services, and System Level Services are running.

Manage Alerts

VCF Operations HCX generates alerts with different severity levels to flag events such as migration, network stretch failure, or reachability issues connecting to an endpoint or paired site.

Based on the type of alert, you can perform various actions to acknowledge or reset the alert.

Alert Type	Action	Action Description
Critical	Acknowledge	The alert has been noted and the corrective actions will be taken. HCX records the user who acknowledged the alert, including a time stamp for the entry.
	Reset to Green	Action has been taken to correct the alert. HCX records the user who resets the alert, including a time stamp for the entry. The alert is removed from the display.
Warning	Acknowledge	The alert has been noted and the corrective actions will be taken. HCX records the user who acknowledged the alert, including a time stamp for the entry.
	Reset to Green	Action has been taken to correct the alert. HCX records the user who resets the alert, including a time stamp for the entry. The alert is removed from the display.
Info	Suppress	Signifies that the alert has been reviewed and can be ignored. HCX removes the alert from the display.

1. Log in to HCX Manager.
2. Select **Alerts** and review the list of alerts for action.
   a) (Optional) Expand an entry to see more information regarding an alert.
   b) (Optional) To sort the list by status, filter the list by Severity and select **Info**, **Warning**, or **Critical**.
   c) (Optional) To sort the list by **Entity Name** or **Creation Date**, select the respective filter and enter the search information.
3. To take action on an alert, click the ellipsis next to the alert entry, and select the action.

Upgrade VCF Operations HCX

The information includes step-by-step instructions for upgrading VCF Operations HCX components.

> **Important:**
>
> For details on the support and upgrade requirements for VCF Operations HCX releases, review the release notes specific to the update.

About Upgrades

VCF Operations HCX upgrades may include new features, software fixes, and security patches.

Upgrades are published periodically for VCF Operations HCX.

Overview of VCF Operations HCX Upgrades

The following high-level process applies when upgrading VCF Operations HCX sites:

1. VCF Operations HCX admins review current release information for the latest upgrades on support.broadcom.com.
2. The admin identifies site paired VCF Operations HCX client systems, and applies the upgrades to the paired VCF Operations HCX systems. The admin can upgrade VCF Operations HCX source and destination during separate maintenance windows, but for optimal compatibility upgrade both systems together.
3. After upgrading HCX Manager at the source and destination sites, the admin navigates to **Interconnect** > **service mesh** and upgrades the service mesh appliances.

Planning for Upgrades

As part of upgrade planning, and to ensure that VCF Operations HCX components and appliances are upgraded successfully, review the service upgrade considerations and requirements.

- The VCF Operations HCX upgrade bundle must be downloaded manually from `support.broadcom.com`.
- Site pairs must display healthy connections before applying the upgrade.
- Unless directed by Broadcom Support to perform an upgrade to resolve a known issue, HCX appliances reporting a degraded state must be restored to a healthy state before the upgrade.

Service Upgrade Considerations

- Apply service upgrades during a maintenance window where no new VCF Operations HCX operations are queued up.
  - HCX Manager and service mesh appliances can be upgraded independently, during separate maintenance windows.
  - The upgrade window accounts for a brief disruption to the network extension service, while the appliances are redeployed with the upgraded code.
  - Apply service upgrades during a maintenance window where no new mobility operations are queued up. During the window, the service mesh appliances are upgraded to the new release.
  - Appliance upgrades are triggered for each service mesh at the source VCF Operations HCX, but run in parallel at each site.
- The VCF Operations HCX upgrade bundle can be uploaded to HCX Manager before the upgrade to reduce the time of the maintenance window.
  - If Site A is paired with Site B, and Site A is also paired with Site C, plan the upgrades for Site A, B and C for the maximum compatibility across all environments. The environments can be upgraded in separate windows.
- Applying a service upgrade causes HCX Manager to reboot:
  - Existing network extensions continue to work during the reboot. New network extensions cannot be configured while the system is rebooting.
  - Because upgrading VCF Operations HCX does not disrupt the Interconnect service mesh, installing updated releases when they become available is recommended to ensure that systems have the most recent fixes and security patches.
- HCX Interconnect (Migration, network extension) service appliance upgrades are performed independently of the manager upgrades:
  - Upgrade the service mesh appliances only after the site-paired HCX appliances are upgraded.
  - Updating the Interconnect service appliances disrupts those services while the upgrades are being applied.
    - Ensure that no migrations are running or new migrations are scheduled when upgrading the appliances.
    - Updating the HCX network extension (HCX-NET-EXT) appliances disrupts connectivity that crosses the network extension path. The tunnel state re-converges in less than one minute after triggering the upgrade. Upgrade the network extension components during a maintenance window.

- Upgrading from HCX 4.11, you must remove the following services from the compute profile and service mesh for all paired HCX appliances: WAN Optimization, V2T Migration, Disaster Recovery.

Service Upgrade Sequence

1. When an upgrade is available:
   – Identify the environments connected by the site pairing. The paired systems are displayed in the two tables in the Administration tab.
     - Download the VCF Operations HCX bundle from `support.broadcom.com`.
     - Ensure that no new migrations or network extensions are configured during the upgrade.
     - Ensure that all ongoing migrations have finished.
     - Ensure that there are no failovers scheduled during the upgrade.
2. You can perform an upgrade from either the VCF Operations HCX UI or the Appliance Management (:9443) on all paired systems:
   – The system reboots during the upgrade procedure.
   – Allow the system several minutes to complete the initialization process.
   – Click System Updates to verify that the current version is upgraded.
3. The service mesh can be upgraded once all paired VCF Operations HCX systems are upgraded and all services have returned to a fully converged state.
   – The Interconnect (HCX-WAN-IX, HCX-NET-EXT) service appliances can be upgraded from the source system. Use the service mesh interface to redeploy or upgrade the Interconnect service appliances:
     - Upgrade or redeploy the Interconnect (HCX-WAN-IX) appliances.
       - Verify that the required tunnels are functional before resuming services or proceeding to the next component.
     - Upgrade or redeploy the network extension (HCX-NET-EXT) appliance.
       - Verify that the required tunnels are functional before resuming services or proceeding to the next component.
     - If the topology has multiple source sites that are paired to a destination environment, the components upgrade has to be triggered at each source site.

Upgrade VCF Operations HCX

This procedure describes the process for upgrading HCX Manager at the source and destination sites.

- Verify the system reports healthy connections to:
  – vCenter
  – NSX Manager (if applicable)
- Verify that the VCF Operations HCX reports that there are healthy connections to the HCX Interconnect (service mesh) appliances.
- Verify that site pair configurations are healthy.

In all cases, upgrading HCX Manager is a manual operation.

It is a best practice to create a backup prior to upgrading VCF Operations HCX. See Backing Up and Restoring the System. This backup option may not be available in some environments where HCX is managed by the cloud service provider.

In addition to backing up HCX Manager, optionally use the vSphere Snapshot feature to take a snapshot of the source and the destination sites. If necessary, you can use snapshots to roll back the VCF Operations HCX version. See Roll Back an Upgrade Using Snapshots.

This procedure describes how to perform upgrades in the Appliance Management. You can also perform upgrades in HCX Manager by selecting **System Updates**.

> **Note:**
>
> An automatic snapshot is taken by HCX Manager upgrade workflow (only if the HCX is deployed within the registered vCenter Server.) This snapshot is retained only for a day.

1. Download the latest VCF Operations HCX Upgrade bundle from `support.broadcom.com`.

   The upgrade bundle includes all software necessary for upgrading VCF Operations HCX, including HCX Manager, service mesh appliances, OS Assisted Migration appliances (SGW, SDR, SGR), and individual VMs installed with OSAM sentinel software.
2. Log in to the Appliance Management.
3. Select **Administration**.
4. Navigate to **System Management** > **Upgrade**.
5. Click **Begin Upgrade**.
6. Select an upgrade bundle or upload a new bundle, and click **Continue**.
7. View the upgrade details, and click **Confirm & Start Upgrade** to continue to upgrade.

   Upgrade details such as version are displayed in the Upgrade pane when the upgrade is complete.

Next, upgrade the component service appliances in the service mesh. See Upgrade the HCX Service Mesh Appliances.

Upgrade the HCX Service Mesh Appliances

The service mesh appliances are upgraded independently of HCX Manager. service mesh appliances are flagged for available upgrades anytime VCF Operations HCX has newer software installed.

- You have upgraded site-paired VCF Operations HCX to the same version on both the source and the destination site.

HCX has both system manager and service mesh appliances. You deploy VCF Operations HCX in the vCenter Server. When creating the service mesh, HCX Manager deploys the service mesh appliances based on the site entitlements and compute profile configuration. The service mesh initiates a secure pipeline and activates the mobility and network extension operations between sites.

When upgrading service mesh appliances, consider the following best practices:

- Upgrade HCX Manager at each site prior to upgrading the service mesh appliances.
- service mesh appliances should be upgraded to the same version as HCX Manager.
- If mobility operations are in progress or networks are currently being extended using NE appliances, you can delay upgrading service mesh appliances until the operations have been completed. You can plan the appliance upgrade during a separate maintenance window.

Important:

If you are upgrading from HCX 4.11.x or earlier, VCF Operations HCX automatically changes any Interconnect appliance password that does not meet the password criteria for enhanced security:

- Minimum length: 15 characters
- Maximum length: 20 characters
- At least one lowercase letter, one uppercase letter, a number, and one of the following special characters: ! @ # $ ^ *
- Must NOT include:
  - A dictionary word (e.g., 'password', 'admin')
  - A palindrome
  - More than four monotonic character sequences
  - Three of the same consecutive characters

To retrieve an appliance password, see Retrieve an Interconnect Appliance Password.

Note:

If service mesh appliances are running a version that is lower than HCX Manager, review the release notes for each HCX release to identify resolved or known issues associated with mobility or network extension operations and upgrade the appliances accordingly. Also, review any Knowledge Base (KB) articles posted for VCF Operations HCX.

1. Log in to HCX Manager at the source site.

   You initiate service mesh appliance upgrades only from the HCX source site. For non-vSphere site pairs, you log in to whichever VCF Operations HCX is paired with the non-vSphere site.

2. Navigate to **Interconnect** > **Service Mesh** > **View Appliances**.

   If an upgrade is available, Interconnect appliances show a green flag in the Available Versions column.

   Note:
   If you are running OSAM in your environment, the upgrade notification does not appear for the Sentinel Gateway, Sentinel Data Receiver and Sentinel Resource Gateway appliances. Continue to follow the steps in this procedure to upgrade sentinel appliances.

3. Select each HCX service mesh appliance that must be upgraded.

4. Click **Update Appliance**.

   The **Update Appliance** option is not displayed when there are no available upgrades.

   Note:
   Network extension appliances are available for in-service upgrades. For more information, see In-Service Upgrade for Network Extension Appliances.

5. Verify the Current and Available versions are valid.

6. To confirm the operation, click **Update**.

   The selected component and any peer component at the paired site are upgraded at the same time. Click the **Tasks** tab to view the upgrade progress details.

7. (Optional) If you are running OSAM in your environment, redeploy the sentinel appliances:
   a) Navigate to **Service Mesh** > **View Appliances**.
   b) Select the OSAM appliance.
   c) Click **Redeploy**.

      Redeploying the SGW appliance automatically upgrades both the SGW and the SDR appliances at the source and destination sites.

When the service mesh appliances converge to a Tunnel Up state, the upgrade is complete.

The service mesh appliances apply the upgrades, reboot, and become operational in less than two minutes after rebooting.

Roll Back an Upgrade Using Snapshots

You can roll back a VCF Operations HCX upgrade using VMware snapshots, which preserve the state and data of the HCX virtual machine at a specific point in time.

- Take and apply a vSphere snapshot for all VCF Operations HCX site pairs. For the rollback procedure, a snapshot of the source and destination VCF Operations HCX systems must exist.

If a rollback is necessary due to an unexpected issue in the new version that cannot be resolved or workaround in a timely manner, follow this procedure.

1. Roll back the HCX systems:

   Note:

   Use snapshots to roll back only HCX Manager versions. After rolling back the versions, restore the HCX appliances through the service mesh configuration in VCF Operations HCX Interconnect UI. Do not attempt to restore snapshots of any data-path appliances.

Rollback Scenario	Process
HCX Manager version has been upgraded but the servicr appliances have not been upgraded.	1. Roll back HCX Manager versions using the snapshots. 2. To validate the service mesh for the roll back version, navigate to **Interconnect** > **Service Mesh** and click **Resync**.
HCX Manager version has been upgraded, and the service mesh appliances also have been upgrades.	1. Roll back HCX Manager versions using the snapshots. 2. For HCX Manager appliance, navigate to **Interconnect** > **Service Mesh** > **View Appliances**. 3. Select all the appliances. 4. To restore the service mesh appliances for the rollback version, click **Redeploy**. 5. For each appliance, confirm that the tunnel status is **Up** and the appliance versions are reverted.

2. To verify the rollback, click System Updates and confirm that the rollback version appears.

Removing VCF Operations HCX

You can remove VCF Operations HCX from your environment by uninstalling the site-paired HCX appliances.

VCF Operations HCX supports on-going operations, including application migration and workload rebalancing while servicing the deployment. If VCF Operations HCX must be removed, stop all ongoing operations.

The procedure for uninstalling VCF Operations HCX can vary based on your environment and privileges, but has the following general workflow:

1. Stop all migrations.
2. Unstretch the network to remove network extension appliances. See Removing a Network Extension.
3. Delete the service mesh to remove the Interconnect appliances.
4. Remove VCF Operations HCX.

For uninstallation details, see Uninstalling VCF Operations HCX.

Uninstalling VCF Operations HCX

Uninstalling requires removing the VCF Operations HCX service from both source and destination site.

All migrations and replications are finalized.

A graceful uninstall of HCX appliances is always initiated from the source side. The process requires that HCX is fully functional, including site pairings and communication between source and destination site appliances.

1. In the VCF Operations HCX UI.
2. Verify that no migration operations are running.
3. Remove all network extensions from source-site data centers. See Removing a Network Extension.
4. In the HCX system containing the service mesh configuration, complete the following substeps to delete all service mesh instances:
   a) Navigate to **Interconnect** > **Service Mesh**.
   b) For each service mesh, click **Delete**.

   Note:

   Removing the service mesh from the source site also deletes it from the destination site.

5. To disconnect all HCX site pairings, complete the following substeps:
   a) From the HCX Console, navigate to **Site Pairs**.
   b) For each site pair, click **Delete Site Pair**.
6. To remove VCF Operations HCX, complete the following substeps:
   a) At the destination site, navigate to the vCenter **Hosts and Clusters** tab.
   b) Expand the cluster where HCX Manager is deployed and locate the virtual machine.
   c) Right-click the HCX entry and power off the selection.
   d) Right-click the HCX entry and select **Delete from Disk**.
   e) Repeat this procedure at the source site.
7. If required, unregister vCenter and NSX gracefully. Remove users created in vCenter SSO for HCX.

Troubleshooting

The following sections contain common HCX Manager troubleshooting scenarios, troubleshooting methodology, general information collection, and how to use built-in diagnostic tools like the HCX Manager Central CLI.

Starting the SSH Service

This section describes how to start the **SSH Service** on VCF Operations HCX to access to the command-line interface.

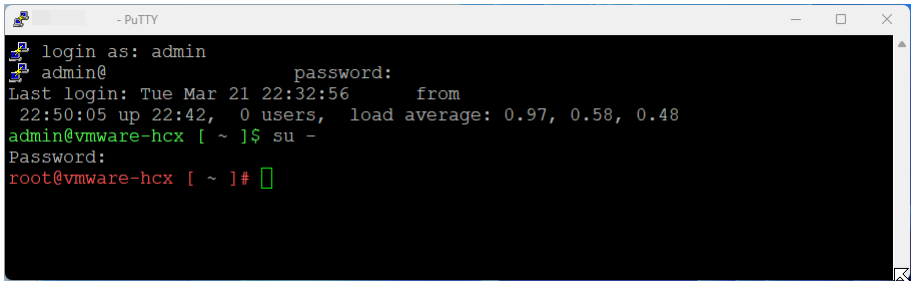
To access to the VCF Operations HCX shell, use a VMware Remote Console (VMRC) session in the vSphere Client or establish an SSH session. If the **SSH Service** was not activated during the initial VCF Operations HCX installation, you must first activate it:

1. Log in to the Appliance Management.
2. Select the **Dashboard** tab.
3. In the Services tile, locate the **SSH Service**, click the ellipsis next to the service, and select **Start**.

Logging in to HCX Manager Shell

Log in to the VCF Operations HCX shell by using VMRC or an SSH session. The first-level access uses the admin account that is created during the initial installation of HCX Manager. If requested to do so by Support, you can switch the user to root once you log in with the admin account.

1. Connect to HCX Manager using VMRC or SSH.
2. When prompted for credentials, enter admin as the user name and then enter the password.
3. Switch to root by typing `su -` and providing the root password.



Locating the System IDs Using VCF Operations HCX Shell

When working with support, you might have to provide the VCF Operations HCX System IDs.

1. Connect to the shell using VMRC or SSH.
2. Switch user to root: `su`
3. Type `cat /common/location`
4. Note the System ID.

```
[root@usphcxmgr01 ~]# cat /common/location
20180214233611705-9242afbe-985a-44b0-b282-49f381aee0e2
[root@usphcxmgr01 ~]#
```

Using Central CLI to Connect to Service Mesh Appliances

From the Central CLI (CCLI), you can connect to the various services for troubleshooting or gathering information.

1. Access the CCLI. See Using the Central CLI.
2. Type `list` to view a list of nodes.
3. Identify the Node ID for the service to which you want to connect.
4. Type `go #` where # is the node **ID**.
5. Type `ssh`.

```
10.197.131.23 - PuTTY
login as: admin
admin@                password:
Last login: Wed Mar 22 00:29:59      from
 00:34:04 up  5:04,  0 users,  load average: 0.24, 0.52, 0.43

26 Security notice(s)
Run 'tdnf updateinfo info' to see the details.
admin@vmware-hcx [ ~ ]$ su -
Password:
root@vmware-hcx [ ~ ]# ccli
Welcome to HCX Central CLI
[root@vmware-hcx] list
|-----------------------------------------------------------------------------------------------------------|
| Node                                                  | Id | Address           | State     | Selected |
|-----------------------------------------------------------------------------------------------------------|
| SM-NSXT_localEndPoint-b35d0_remoteEndPont-be1eb-IX-I1 | 0  | [redacted]        | Connected |          |
|-----------------------------------------------------------------------------------------------------------|
| SM-NSXT_localEndPoint-b35d0_remoteEndPont-be1eb-NE-I1 | 1  | [redacted]        | Connected |          |
|-----------------------------------------------------------------------------------------------------------|
| SM-NSXT_localEndPoint-b35d0_remoteEndPont-be1eb-NE-I2 | 2  | [redacted]        | Connected |          |
|-----------------------------------------------------------------------------------------------------------|
| SM-NSXT_localEndPoint-b35d0_remoteEndPont-be1eb-NE-I3 | 3  | [redacted]        | Connected |          |
|-----------------------------------------------------------------------------------------------------------|
[root@vmware-hcx] go 0
Switched to node 0.
[root@vmware-hcx:SM-NSXT_localEndPoint-b35d0_remoteEndPont-be1eb-IX-I1] debug remoteaccess enable
Remote Access Enabled
[root@vmware-hcx:SM-NSXT_localEndPoint-b35d0_remoteEndPont-be1eb-IX-I1] ssh
Welcome to HCX Central CLI
Last login: Wed Mar 22 00:32:51 2023 from 127.0.0.1
root@SM-NSXT_localEndPoint-b35d0_remoteEndPont-be1eb-IX-I1 [ ~ ]#
```

6. Use the `help` command to display available commands.

Gather Technical Support Logs

This procedure describes how to collect and download the Technical Support logs from the Appliance Management.

You can also collect and download the logs in HCX Manager by selecting **Administration** > **Support**.

1. Log in to the Appliance Management.
2. Select the **Administration** tab.
3. Navigate to **System Management** > **Technical Support Logs**.
4. Select the checkbox next to one or more log bundles that you want to generate.
5. Click **Generate** to prepare the log file bundles.

Gathering Technical Support Logs: Script Access

This section describes the process for gathering logs for Technical Support using a standalone script.

Root user privilege is required for this procedure.

In some cases, gathering system logs through the UI or API calls might be unavailable. For example, if HCX Manager or web engine stops or fails to start, you cannot generate the log bundles. You can generate and download the DB Dump and Core VCF Operations HCX log bundle using the standalone script included in this procedure.

> **Note:**
>
> The VCF Operations HCX Data-Path Appliances IX/NE and OSAM sentinel logs export option is not available.

1. Using SSH, log in to VCF Operations HCX as an admin and change the user to root.
2. Change to the `/opt/vmware/tools/` directory.
   ```
   # cd /opt/vmware/tools/
   ```
3. Run the utility to review the script usage and options:
   ```
   # ./export_tech_support_bundle.sh
    Utility script to export Core HCX-Manager Logs and/or the Database Dump
    Syntax: sh export_tech_support_bundle.sh [-l|d|h]
    options:
     l Export Core HCX Manager logs
     d Export Database Dump
     h Usage help
   ```
4. Run the script with Core HCX Manager and Database Dump options:
   ```
   # ./export_tech_support_bundle.sh -l -d
   ```
   The system checks for sufficient disk space, and then exports the bundle under the `/tmp/techsupport` directory. The bundle filename and location are displayed on the screen.

   > **Note:**
   >
   > Use any file transfer protocols like SFTP/SCP or WinSCP to upload the bundle directly from VCF Operations HCX to an external server.
5. To avoid unnecessary utilization of the disk, delete the bundle from the `/tmp/techsupport` directory once support bundle has been successfully downloaded to an external server.
   ```
   # cd /tmp/techsupport
   # rm <file name>
   ```

Viewing Logs in the Shell

Service logs are useful when troubleshooting failures.

There are two key logs in VCF Operations HCX that can be reviewed and used when troubleshooting problems or to monitor system activities. Both are located in `/common/logs/admin` and they are the Application log (`app.log`) and the Web log (`web.log`). The Application log logs all activities for the app-engine service. The Web log logs all activities for the web-engine service.

1. Use VMRC or SSH to connect to the HCX Manager shell.
   See Logging in to HCX Manager Shell.
2. Switch user to root: `su -`.
3. Change directory to `/common/logs/admin`.
4. From within this directory, you can open the relevant logs using standard Linux text commands.
   When troubleshooting failures, search using keywords such as Fail, ERROR, exception, or migration.

Monitoring Services from the System Interface

You can monitor or restart the services from HCX Manager Appliance Management dashboard.

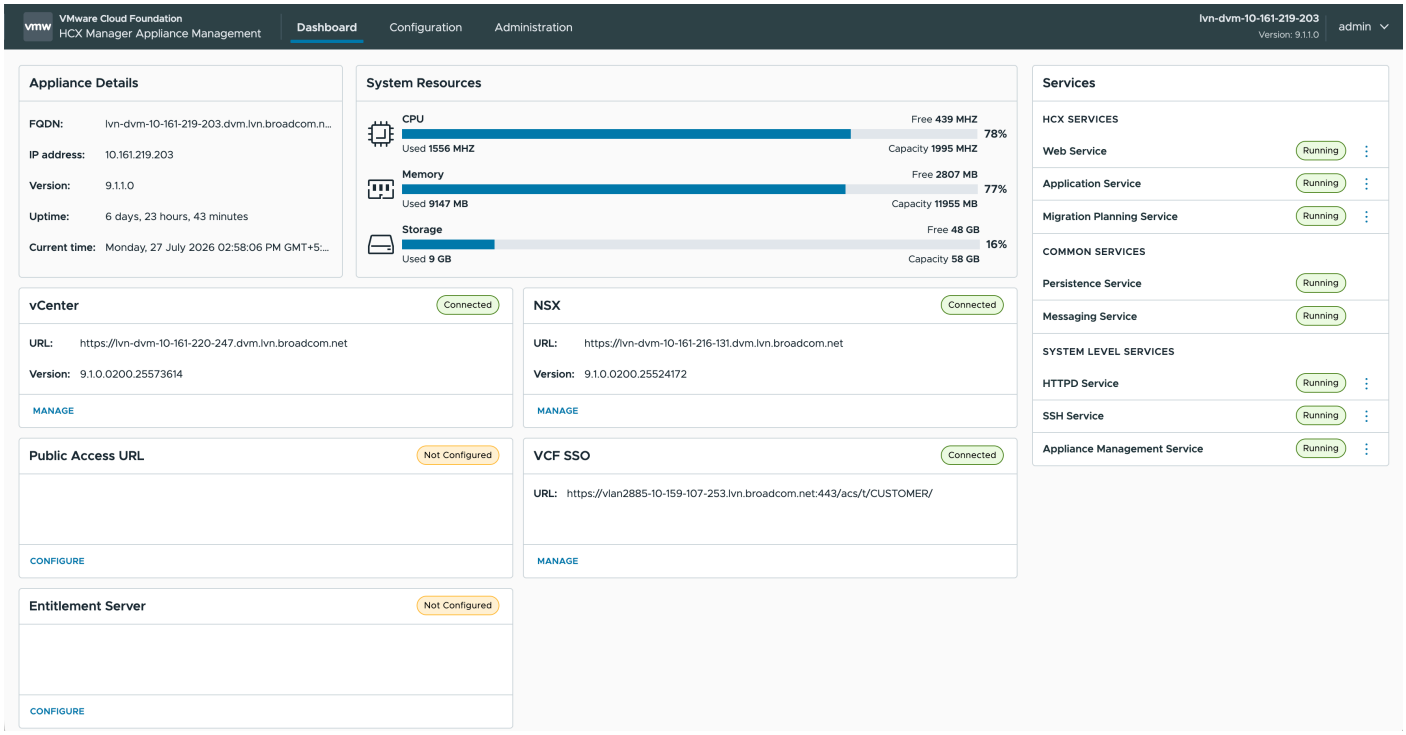
There are several services critical for VCF Operations HCX operations. Two key services to observe are the Application Service and the Web Service.

When working with a support team at Broadcom, you might have to confirm that these services are running or might have to restart them. You can view and restart the HCX Manager services in several places.

> **Important:**
>
> Do not restart services unless directed to do so by Broadcom Support.

1. Log in to the Appliance Management.

   The dashboard is displayed.
2. In the dashboard, you can view all services and monitor or restart them. The only service that is optional is the SSH service. All others must always be running.



Manage Services from the CLI

You can manage the VCF Operations HCX services using the command line interface.

1. Use VMRC or SSH to connect to the shell.
   See Logging in to HCX Manager Shell.

2. Switch user to root: `su -`.
3. Type `systemctl action service_name`.
   - You can choose from one of the following actions: `status`, `stop`, `start`, `restart`.
   - The service name can be the web-engine or the app-engine.

```
#systemctl status web-engine

#systemctl status app-engine

#systemctl stop web-engine

#systemctl restart web-engine
```

Viewing the System State

You can view the VCF Operations HCX system state from the VCF Operations HCX interface dashboard.

For VCF Operations HCX to run properly, it is important that it has sufficient available resources. You can view the key system resources such as CPU, memory, and storage from the dashboard in HCX Manager Appliance Management. The dashboard provides useful information such as the version running on HCX Manager, the uptime, its IP address, and current time. All useful information when reviewing logs or required by Support.

1. Log in to the Appliance Management.
2. In the dashboard, review the CPU, Memory, Storage, Uptime, and Version.

Viewing Entries in the vSphere Task Console

Most operations such as the initial appliance deployment, extending a network, or a migration can be monitored from the vSphere Web Client Task Console.

1. Open the vSphere Client and navigate to **Home**.
2. Navigate to **Tasks**.
3. In the Task Console, filter the results by using `HCX` in the search filter.
4. Look for any failures or errors. If you see an error, you can review the logs to find additional details.

Using the Central CLI

The HCX Manager Central CLI is used for diagnostic information collection and secure connections to the service mesh.

The Central CLI (CCLI) on HCX Manager allows you to run commands available centrally on the site manager to view the run time state for HCX Manager services. The CCLI reduces troubleshooting time by providing centralized diagnostics and improves the security posture of the service mesh appliances by eliminating the need to run the SSH service on them. To use it, first you must activate the CCLI on the HCX Manager.

1. Use VMRC or SSH to connect to the HCX Manager shell.
2. Type `ccli`.

   The CCLI is now available.
3. To begin exploring the CCLI, use the `help` command.

Password Recovery

You can reset the admin or the root password on Photon OS based HCX appliances.

HCX 4.4.0 and later releases support VMware Photon OS. If an account password is lost or forgotten, a Photon OS password recovery procedure is available to reset it.

A password reset can impact ongoing migration and configuration workflows. Allow those operations to complete before proceeding to recover the password.

The recovery procedure, which requires a reboot of HCX Manager, does not impact the following scenarios:

- Network extensions actively forwarding.

To reset the root or admin password, see the VMware KB article 89212.

Upgrading System Passwords using the CLI

Upgrade the system admin or the root passwords from the command line interface.

> **Note:**
>
> For information about recovering a password, see Password Recovery.
>
> To change the admin password from HCX Manager Appliance Management, see Changing the Admin Password.

1. Log in as the admin user: `ssh admin@`*`IP address`*.
2. Enter the admin password.
   The system prompt appears.

   ```
   [admin@hcxmgr ~] $
   ```
3. Enter the `passwd` command.

   ```
   [admin@hcxmgr ~]$ passwd
   ```

   Follow the prompts to complete the password change.
4. (Optional) Change the root user password:
   a) At the system prompt, elevate to root user using the `su root` command:
      The root prompt appears.

      ```
      [root@hcxmgr /home/admin]#
      ```
   b) At the root prompt, enter the `passwd` command.
      Follow the prompts to complete the password change.

Log out and log back in to confirm the change.

Unlock an SSH Admin Account

After three failed login attempts, VCF Operations HCX locks the user account.

To unlock the account, you must access HCX Manager through the Web Console in the vCenter Server.

1. Log in to vCenter Server where HCX Manager is deployed.
2. From the vSphere Client interface, selectHCX Manager.
3. From the vSphere **Client Summary** tab, select **Launch Web Console**.
4. Log in as root user using the credentials set up during VCF Operations HCX installation.
5. At the root prompt, enter the following command:

   ```
   root# faillock --user admin --reset
   ```
6. Follow the prompts to reset the password.

   The procedure is complete.

Try using SSH to log in again.

Retrieve an Interconnect Appliance Password

You can find the password for any of the service mesh interconnect appliances (HCX-WAN-IX or HCX-NET-EXT) using the `retrievePassword` utility.

1. Log in to HCX Manager with SSH and enter the admin password.
2. Switch to the root user and enter the root password.

   ```
   # su root
   ```
3. At the root prompt, enter the command to retrieve the list of appliances:

   ```
   root# /opt/vmware/bin/interconnectDebugUtils.sh retrieveAppliances
   ```

   The list of service mesh appliances appears with the ApplianceId and ApplianceName.
4. At the root prompt, enter the command to retrieve the appliance password:

   ```
   root# /opt/vmware/bin/interconnectDebugUtils.sh retrievePassword <ApplianceId>
   ```

   The system displays admin and root passwords for the appliance.

To change an appliance password, see Manage Service Mesh Appliances.

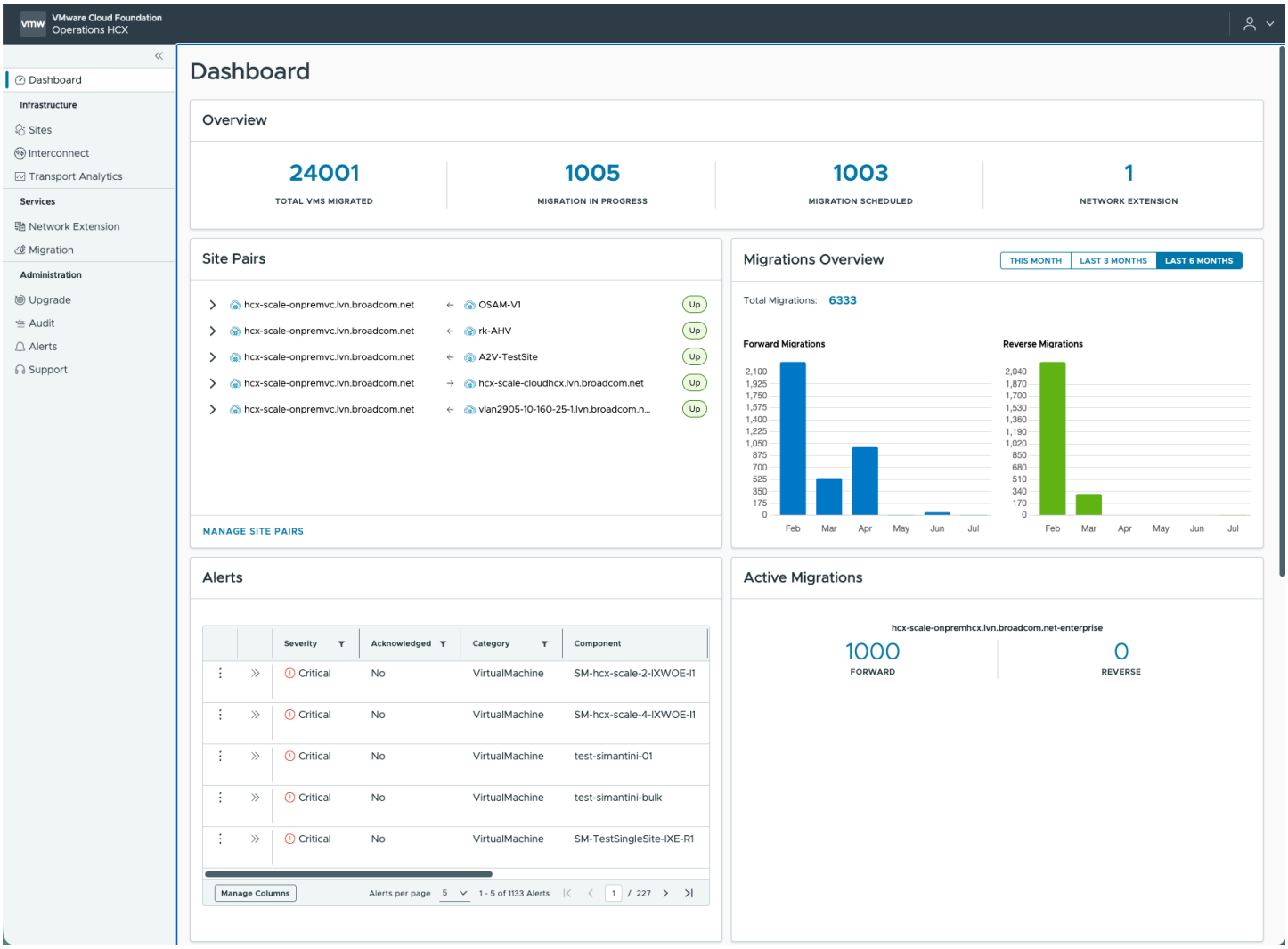
Monitoring Systems

VCF Operations HCX native tools and views can be used to collect the current state of the system and general system health.

Understanding the VCF Operations HCX Dashboard

The Dashboard provides a summary of operations, data center locations, resource usage, status, and activity.

The Dashboard is the first screen that appears when you open the UI.



The Dashboard highlights various HCX functions in a set of panels. You can change settings that are related to those panels.

Note:

For HCX installations where the vCenter Servers are in linked-mode, the Dashboard includes information from all vCenter Servers registered to an HCX system.

Panel	Description
Overview	Lists HCX operations: • Number of virtual machines migrated • Number of migrations in progress • Number of scheduled migrations not started • Number of extended networks
Site Pairs	Displays connected site pairs and lists the pair status, Up or Down. To create a site pair, click New Site Pairing. To view detailed instructions for adding a site pair, see Add a Site Pair.
Active Migrations	Displays ongoing migrations for the selected HCX system. Use the pull-down menu to change the source site.

Panel	Description
Migrations Overview	Summarizes completed migrations for the selected HCX system. Use the date pull-down menu to display migrations for a specified period: • Last 6 Months • Last 3 Months • This Month
Alerts	Provides a comprehensive the list of logged Alert messages: Critical, Warning, Info. For a description of Alert messages and available actions, see Manage Alerts.